Producing Prosperity

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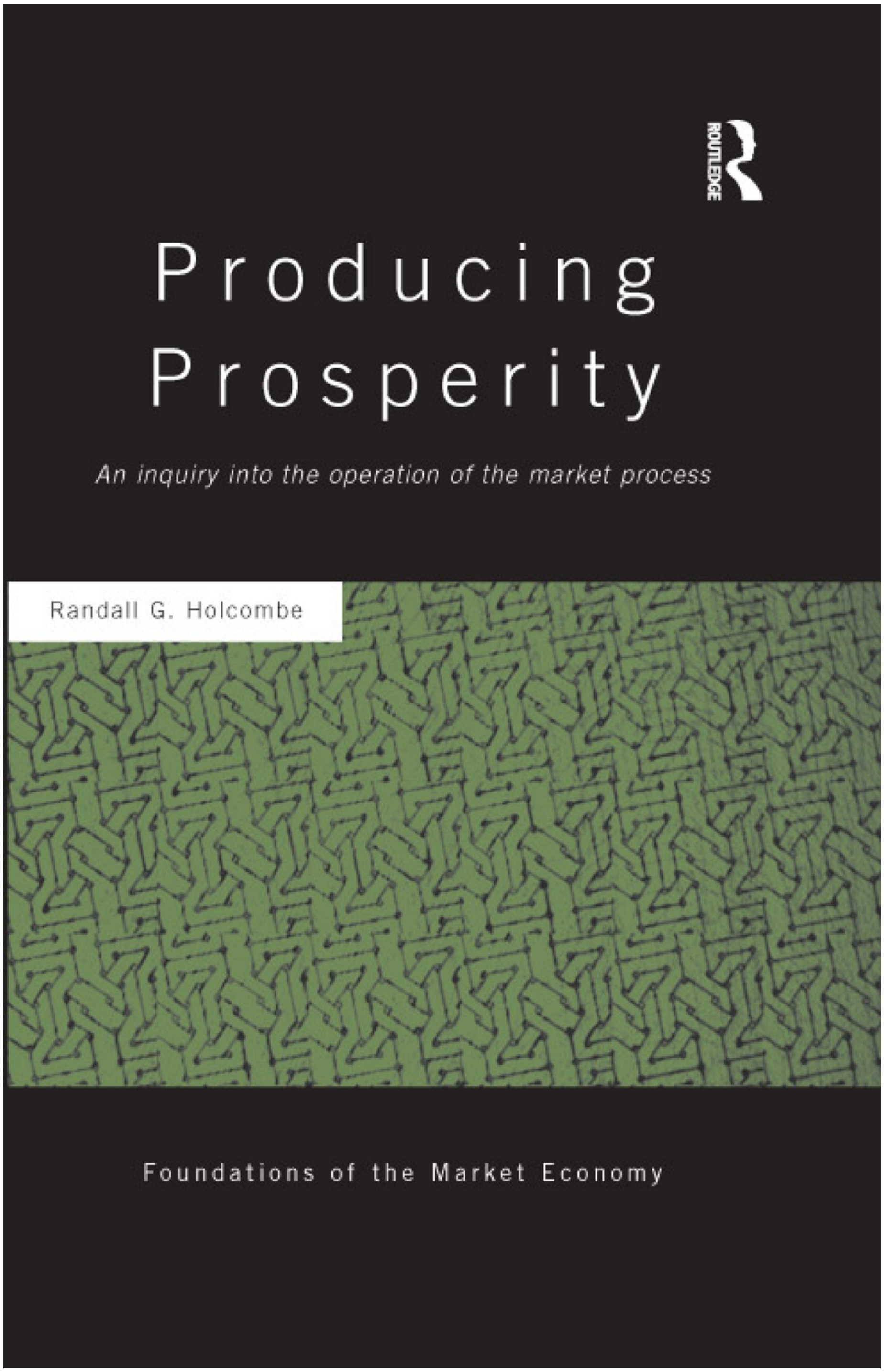
ROUTLEDGE
Producing
Prosperity
An inquiry into the operation of the market process
Randall G. Holcombe
Foundations of the Market Economy

Producing Prosperity

The substantial prosperity that characterizes market economies at the beginning of the twenty-first century is relatively recent in human history. Prior to the Industrial Revolution, economic progress was so slow that people would not have been able to recognize it in their lifetimes, whereas today, economic progress is so much a part of people's lives that they take it for granted.

In this new volume, Randall G. Holcombe argues that economic analysis, as it developed through the twentieth century, relies heavily on concepts of economic equilibrium, and is not descriptive of the dynamic real-world economy that is characterized by economic progress. Even in dynamic settings, economic models focus on income growth, leaving out the entrepreneurial forces that generate economic progress, resulting in the introduction of new goods and services and new production processes. Economic analysis focuses on the forces that lead to an economic equilibrium, not the forces that produce prosperity.

This characterization of economic analysis describes a substantial component of economics as it has developed over the past century. However, there are also economists who have analyzed the factors that lead to an entrepreneurial and innovative economy, generating progress rather than equilibrium. This volume does not question the value of past research, but argues that, looking ahead, economics should build on its past to focus on factors that create an entrepreneurial and innovative economy that is characterized by progress and prosperity. This would make economic analysis more consistent with the remarkable progress and prosperity that characterizes the modern economy. This volume lays out a framework for economic analysis that consistently incorporates the real-world factors that produce prosperity.

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Routledge foundations of the market economy
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A central theme in this series is the importance of understanding and assessing the market economy from a perspective broader than the static economics of perfect competition and Pareto optimality. Such a perspective sees markets as causal processes generated by the preferences, expectations and beliefs of economic agents. The creative acts of entrepreneurship that uncover new information about preferences, prices and technology are central to these processes with respect to their ability to promote the discovery and use of knowledge in society.

The market economy consists of a set of institutions that facilitate voluntary cooperation and exchange among individuals. These institutions include the legal and ethical framework as well as more narrowly "economic" patterns of social interaction. Thus the law, legal institutions and cultural and ethical norms, as well as ordinary business practices and monetary phenomena, fall within the analytical domain of the economist.

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For Lora, my wife and colleague

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Preface

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The remarkable prosperity that people in capitalist economies enjoy at the beginning of the twenty-first century has to rank at the very top of all achievements of human civilization. This prosperity is all the more remarkable when one realizes that the economic progress that has produced it dates back only a few hundred years. One can look at the state of civilization in ancient Rome, or in China around the same time, to see that civilization had already advanced well beyond a primitive existence, but the standard of living, the goods and services people consumed, and the methods of production advanced relatively little from those ancient civilizations until the Industrial Revolution began in the 1700s. At that point economic progress accelerated, but that progress, which people take for granted today because they have known it their whole lives, is very recent in the course of human history.

Economics as an area of inquiry has advanced along with the development of the economy, but in many ways the economy that is described by economic theory is not the same as the real-world economy that the theory should describe. The theme of this volume is that the features of the real-world economy that have produced prosperity are too often omitted from the economic models economists use to describe the economy. One consequence is that the policy recommendations economists derive from their models are often counterproductive. They work against prosperity, rather than working to further it, because the models they use do not represent accurately the factors that produce prosperity. This volume lays out a framework for analysis that takes account of those factors that produce prosperity as an outline for a productive course for the future development of economic analysis.

I would stop short of saying that economic analysis has taken a wrong turn. Much insight has come from the economic models of the twentieth century, which primarily have been built around trying to better understand the properties of economic equilibrium. However, this volume argues, first, that the economic progress that has produced today's prosperity is not well understood within that equilibrium framework, and second, that because of the development of equilibrium economics in the twentieth century there is not much left to learn about the properties of equilibrium in the real-world economy. Too much economic research is devoted to discovering equilibrium properties of models which have little to do with real-world economic phenomena.

Economics has changed its emphasis in the past. Chapter 1 argues that in the eighteenth century the focus of economic analysis was on how wealth is created, as revealed in the title of Adam Smith's An Inquiry Into the Nature and Causes of the Wealth of Nations. In the nineteenth century the focus shifted toward the development of a theory of value, and to distributional issues. The focus of economic analysis in the twentieth century was on the properties of economic equilibrium. So, one would not be surprised to see a similar shift in the focus of twenty-first century economics. There is already a substantial amount of work pointing toward an increased interest in the factors that produce prosperity, and this volume organizes those ideas into a framework for economic analysis. The book is an inquiry into the operation of the market process, focusingfirmly on developing a framework that explains those factors that produce prosperity.

This volume is a product of a research history that goes back into the 1990s, when I became more interested in the relationship between entrepreneurship and economic progress. Over the years I have written a number of articles and an earlier book, Entrepreneurship and Economic Progress (Routledge, 2007), looking at the implications of entrepreneurship, not only as a process of discovering and implementing innovations in an economy, but also as a process that creates additional entrepreneurial opportunities as a by-product of entrepreneurship.

This volume takes that same approach to entrepreneurship, to evaluate the ways that economic analysis depicts the activities of firms, the people who run those firms, and the implications for the broader economy. It is an analysis of economic analysis.

I have the good fortune of being married to an economist who has been very supportive not only of this project but of all I have undertaken in life, both within and beyond my career as an economist. I am happy to have the opportunity to dedicate this book to my wife, Lora.

1 Producing prosperity

]1 Producing prosperity

The subject matter of economics

Since its beginning, the primary subject matter of economics has been what enables economies to produce prosperity. Adam Smith, often cited as the father of economics, titled his great 1776 treatise An Inquiry Into the Nature and Causes of the Wealth of Nations. The title reveals that Smith's primary interest was to explain how economies produce prosperity. That topic remains at the forefront of economic analysis up into the twenty-first century, but economists' views on how prosperity can be produced have changed substantially from the late twentieth century to the early twenty-first. Until the fall of the Berlin Wall in 1989, followed by the dissolution of the Soviet Union in 1991, conventional wisdom among economists leaned toward the idea that economic planning by government was the surest road to prosperity. Despite the fact that market economies were more prosperous during the twentieth century than centrally-planned economies, much of the economics profession perceived central planning as a more rational way to organize an economy. A belief common among professional economists was that centrally-planned economies would grow faster and eventually overtake market economies. By the beginning of the twentieth century, the overwhelming evidence on the failure of central planning reversed that conventional wisdom. Market economies, not central economic planning, are now seen as the way to produce prosperity.

This change in the conventional wisdom of economists came about because of the real-world evidence that markets work better than central planning, not because of any advances in economic theory. In fact, economic theory had much to do with the tendency of economists to see the merits of central planning. Economic theory had developed in the twentieth century in a way that supported the organization of an economy through central planning, and that ignored those aspects of market economies that actually were responsible for the increasing prosperity they enjoyed. Economists developed complex mathematical models of the economy, and thinking that the models captured the essential elements of the real-world economy, looked for policy changes that could improve the performance of their model economies in theory. Then they recommended to real-world policy-makers that they implement those same policies. In the real world, however, those policies did not give the same results as in the theoretical model economies. While that would seem to indicate that, at best, those models did an incomplete job of understanding the causes of prosperity, at the beginning of the twenty-first century the theory remained the same, and had not yet caught up to correspond with the evidence on the way that market economies work to produce prosperity.

The purpose of this volume is not to add any evidence to the argument that market economies pave the road to producing prosperity. In addition to the real-world evidence, there is also a substantial amount of literature – much of it written since 1990 – that makes a solid case. So, this volume takes that conclusion as given. This book examines the development of economics in the twentieth century and suggests that it developed in a way that supports the idea that central planning can work better than markets, and left out of its analysis the key factors that actually contribute toward producing prosperity. The market economy is an engine of economic progress, and this volume examines economic theory to see how it can be developed to more closely correspond with the real-world advances in prosperity that have been occurring for more than two centuries.

The payoffof this exploration, at its most ambitious, would be to point the way toward the development of economic theory that more closely corresponds to the real-world economy, and that can lead to better policy recommendations. Even with more modest ambition, if such an exploration shows the shortcomings of applying twentieth-century economic theory to the issue of economic progress, it might prevent some poor policy decisions from being made.

The remarkable rise of prosperity

The substantial increase in world prosperity since the beginning of the Industrial Revolution is nothing short of remarkable; yet it is easy to take for granted because it is so much a part of everyday modern life. Prior to about 1750, when the Industrial Revolution was beginning, prosperity was a privilege of the very few, and people became prosperous not because of their own productivity, but rather because a lot of workers had no choice but to turn over a share of their productivity to the prosperous classes. Prior to the Industrial Revolution, prosperity was a function of class. People were born into a certain station in life, and the institutions of the day did not allow the mobility that would permit those of modest birth status to rise to the level of the prosperous classes. Wealth and status were inherited, and the transfer of resources toward the prosperous classes did not occur as a by-product of voluntary exchange, but rather as a forced transfer, enforced through political and military power. That changed with the Industrial Revolution, which allowed people born into modest circumstances to build businesses, and through those businesses to build wealth rivaling and even surpassing those with inherited wealth.

Opportunities for the masses to become prosperous did not come instantaneously with the Industrial Revolution. Indeed, John Stuart Mill, writing in 1848, a century after the Industrial Revolution began, noted the very uneven – and in his view unfair – distribution of income produced by the Industrial Revolution. Discussing the prospects for communism, Mill observed:

If ... the choice were to be made between Communism with all its chances, and the present state of society with all its sufferings and injustices; if the institution of private property necessarily carried with it as a consequence, that the produce of labour should be apportioned as we now see it, almost in an inverse proportion to the labour – the largest portions to those who have never worked at all, the next largest to those whose work is almost nominal, and so in a descending scale, the remuneration dwindling as the work grows harder and more disagreeable, until the most fatiguing and exhausting bodily labour cannot count with certainty on being able to earn even the necessaries of life; if this or Communism were the alternatives, all the difficulties, great or small, of Communism would be as dust in the balance.[1]

Ultimately, Mill decided that the present state of society could be modified, so the choice was broader than just the present state of society of communism, but Mill's discussion of income inequality is a telling observation of the consequences of capitalism in his time.

There was one substantial difference in the inequality Mill observed and the inequality that had characterized the world just a century before. The prosperity of the most prosperous in Mill's day came from their direction of productive activity, and that production laid the foundation for an increasing standard of living for everyone. Within a century and a half, the wealthy nobility of 1750 was being displaced by the new industrialists, and the wealth of John D. Rockefeller, Andrew Carnegie, and Cornelius Vanderbilt far eclipsed the wealth of nobility.

By the middle of the twentieth century, just two centuries into the Industrial Revolution, prosperity had trickled down to the common worker. In 1798 Thomas Robert Malthus had famously predicted that most people would be consigned to life at a subsistence level of income, because population growth would always outstrip the increase in available resources to support that population. By 1950, ordinary citizens in the most prosperous nations were not only well-fed, but enjoyed indoor plumbing, a wide variety of electrical appliances to make their lives easier, and could often afford their own automobiles. By the beginning of the twenty-first century, even most people who were classified below the poverty line in developed economies owned automobiles, mobile telephones, refrigerators, televisions with VCRs, microwave ovens, and lived in housing with indoor plumbing and central heating. By many measures, the standard of living of most people classified as living in poverty at the beginning of the twenty-first century was above the average standard of living just half a century before.

This prosperity has not been shared throughout the world. While growing prosperity continues in many of the world'swealthier countries, people in the world's poorest countries remain at a subsistence level of income, or worse, as they starve because of an inadequate food supply. As much of the world enjoys an increasing standard of living, the standard of living has fallen in some of the world's poorest countries. Obviously, prosperity is not inevitable. Some have argued that the prosperity of people in some nations is a result of their exploiting those in poor nations. This possibility alone provides a good reason for investigating the origins of prosperity. The analysis below shows that for the most part, this is not true. Progress in the most prosperous nations benefits those who live in the poorest, but exceptions to this conclusion are noted later in this chapter. Overall, prosperity in one part of the world provides advantages to people in less-prosperous places. The existence of more and less prosperous places points toward two important issues, however.

One issue is what makes the difference between prosperous and poor societies. If one can understand how prosperity is produced, the lesson can be applied to make everyone more prosperous. A second issue is that prosperity is not inevitable, meaning that there is the risk that societies characterized by prosperity and progress could find themselves stagnating or even declining if they do not maintain those conditions that produced prosperity to begin with. There are good reasons for understanding the underlying factors that produce prosperity.

Two functions of the economy: market clearing and progress

To lay a foundation for understanding where this analysis is heading, consider two functions that a market economy undertakes: market clearing and producing progress. Every day the economy produces a vast array of goods and services, and the economic activities of people are coordinated so that, in general, those goods and services are consumed by the people who value them the most. While this idea can be debated and discussed in detail, it can be safely glossed over at this point, because most of twentieth century economics was devoted to analyzing just this idea. As the forces of supply and demand work, prices adjust in an economy so that the quantity supplied in each market just about equals the quantity demanded, and markets clear with goods and services going to those who are willing to pay the most for them. This describes the market-clearing function of the economy.

This market clearing has been analyzed from two related perspectives: equilibrium and efficiency. In twentieth-century terms, when the quantity supplied equals the quantity demanded in all markets, the economy arrives at a general equilibrium, and the equilibrium notion is that as long as the underlying conditions do not change, market forces always tend to pull the economy toward this equilibrium state. Under certain conditions this equilibrium state also allocates resources as efficiently as possible, and a substantial amount of twentieth-century economics was devoted to understanding what the conditions of efficiency are, and how the economy can be nudged toward a more efficient allocation of resources when they are violated.

As depicted by twentieth-century economics, both of these perspectives – equilibrium and efficiency – are static in nature. Equilibrium is static almost by definition, because the economy tends toward a particular outcome, and if that equilibrium is disturbed, economic forces tend to pull the economy back to the equilibrium. The fundamental nature of models of dynamic equilibrium is not much different; the economy just tends toward a determinate equilibrium path. Efficiency also is defined in static terms, as a Pareto optimum. Once an economy arrived at an efficient allocation of resources, there was no way, in theory, to improve on that allocation. In a competitive equilibrium, resources would be allocated as efficiently as possible, and markets would continually clear by allocating resources to their most highly valued uses.

This market-clearing function of the economy is awe inspiring: something to marvel at, to be sure. It is well worth studying and understanding, and it was the focus of attention of the economics profession throughout the twentieth century. Models were developed in an equilibrium framework, and within the twentieth-century equilibrium framework progress was left out of the analysis. Even when models turned to topics of economic growth, they did not capture the process of economic progress as it actually occurred in market economies. Economics as a discipline should certainly retain the lessons it learned about the way that markets clear, but in the twenty-first century there is a strong argument to be made that the focus of economic analysis should turn to economic progress – the topic that most interested Adam Smith – rather than the properties of economic equilibrium that dominated twentieth-century economic analysis.

Looking at the real world, market economies are characterized by markets that clear, and by continual economic progress. Economists understand the market-clearing aspects of the economy well, and this volume looks toward developing economic analysis so that it reflects those characteristics of the economy that generate progress and prosperity.

This section deliberately refers to market clearing rather than equilibrium, because they are different things. However, within an equilibrium framework, they can easily appear to be the same. As such, this distinction deserves some analysis at the outset. In an equilibrium model, the quantity supplied equals the quantity demanded, so the market clears. It appears, therefore, that market clearing is evidence of equilibrium. Within the equilibrium framework, it is the case that in an equilibrium the market clears. However, markets might clear without there being any underlying equilibrium. What is required for market clearing is that the price adjusts to equate the quantities supplied and demanded. If the market is disrupted, in an equilibrium framework market forces bring it back to equilibrium. However, if disruptions occur that change the underlying conditions markets can still clear (prices move to equate the quantities supplied and demanded) without there being any underlying equilibrium toward which the economy is moving.

For those who have been schooled in the equilibrium framework (the present writer included) there is a temptation to preserve that framework by depicting such disruptions as changing the underlying equilibrium, so that the economy moves from one equilibrium to another, or in a dynamic framework, to depict the economy as maintaining some equilibrium trajectory. However, if the disruptions are due to the nondeterministic choices of market participants – and many of them are – then there is no equilibrium that underlies the trajectory of the economy. Where it goes tomorrow depends on the choices individuals make today, and there is no way to depict this path dependence as determinate, or as an equilibrium. That idea will be developed further in Chapter 3, but to start on this line of analysis it is valuable to point out that market clearing and equilibrium are two distinctly different economic phenomena. Equilibrium implies market clearing, making them appear equivalent when viewed through the lens of equilibrium economics. However, the economy can be changing daily, in a non-deterministic fashion, never returning to conditions that existed before, and still prices can adjust so that markets continually clear. Thus, the distinction is important within the context of twentieth-century equilibrium economics, and the marvel of market clearing does not imply the economy is characterized by any sort of equilibrium.

It is important to understand how and why markets clear, and those insights of twentieth-century equilibrium economics remain an important part of economic wisdom. However, the process that generates economic progress is not the same as the process that clears markets. Starting from a hypothetical condition of equilibrium, the analysis that follows shows: why the economy can never remain in equilibrium; that when one is analyzing the causes of prosperity it is misleading to view the economy in equilibrium; and that many of the implications for economic growth that emanate from equilibrium models misrepresent the actual process that produces prosperity.

Equilibrium versus progress: a brief history of economic ideas

To place the ideas in this volume within the context of contemporary economic theory, it is worth considering the background within which economic analysis became so firmly grounded in an equilibrium framework. Economics as an area of inquiry was not always so focused on the properties of economic equilibrium. That fixation on equilibrium only began in the twentieth century, and prior to the this economists would not have thought of an economic system tending toward some equilibrium outcome. Rather, until the twentieth century economists had the idea that the economic activities of today were leading toward economic conditions that would be different in the future.

Economics as a discipline is only about three centuries old. Ludwig von Mises begins his treatise, Human Action, by saying, “Economics is the youngest of all sciences.”[2] His argument is that while people have discussed economic phenomena since ancient times, it was only recently that “the inescapable interdependence of market phenomena”[3] was discovered. That discovery – the foundation of economics – was only embodied in an independent discipline beginning roughly in the 1700s. That would make economics about three centuries old. In each of its three centuries economists have focused on different issues.

In the eighteenth century, economics focused on the question of how wealth is created. The conventional wisdom of the time, embodied in mercantilism, was that accumulating gold and silver made nations wealthy, and the mercantilists advocated policies that led to the accumulation of precious metals.[4] Arguing against this conventional wisdom, the physiocrats said that wealth came from the land, and supported policies that promoted agriculture and mining. Manufacturing, the physiocrats argued, was just rearranging things people already had, whereas agriculture brought new wealth into existence. Adam Smith entered this intellectual environment with his The Wealth of Nations, whose title clearly reveals Smith's emphasis on wealth creation.

The major economic ideas of the eighteenth century revolved around the issue of producing prosperity, and Smith, rather than looking at the economy in an equilibrium framework, was very process oriented in his analysis. He began by attributing the wealth of nations to the division of labor (in the book's very first sentence), and then went on to explain how people have an incentive to discover new ways to increase the division of labor. The division of labor is limited by the extent of the market, Smith said, but as markets grow, the division of labor causes productivity to increase, which increases income and causes markets to grow. There is a self-reinforcing mechanism that can lead to ever-increasing prosperity. In contrast to the twentieth-century equilibrium view of economics, Smith saw an economy where an ever-increasing division of labor produced continuing economic progress, so economic conditions in the future would be different from those today. In Smith's view the economy was characterized by progress, not equilibrium.[5]

For present purposes, the nineteenth century began in 1798, with the publication of Malthus's Essay on Population.[6] Malthus's essay, followed up by David Ricardo's more rigorous Principles of Political Economy in 1817,[7] argued that population growth would always outstrip the resources available to support the population. People would never become prosperous – indeed, most people would always be condemned to live at a subsistence level of income, and any temporary increases in living standards that might occur would eventually be eroded by population growth. For obvious reasons, this line of reasoning put an end to the eighteenth-century focus that economics placed on wealth generation. In the Malthusian/Ricardian framework, most people will always remain in poverty.

Two issues emerged as focal points for the development of economic analysis in the nineteenth century. One was the distribution of income. If most people are condemned to live at a subsistence level of income, who gets what becomes a bigger issue. As Ricardo argued it, landlords owned a scarce factor of production that would increase in value as population grew, resulting in an increase in the demand for food. Meanwhile, labor is stuck at a subsistence level of income, and as food prices rise because of population growth and increasing demand, the profits of capital owners also get squeezed. Before profits fall to zero, Ricardo argued, "the very low rate of profits will have arrested all accumulation, and almost the whole produce of the country, after paying the labourers, will be the property of the owners of land and the receivers of tithes and taxes."[8] But, John Stuart Mill, who further developed Ricardo's framework, argued, "The laws and conditions of the production of wealth, partake of the character of physical truths. There is nothing arbitrary about them. ... It is not so with the distribution of wealth. That is a matter of human institutions solely. The things once there, mankind, individually or collectively, can do with them as they like."[9] Mill also pessimistically viewed the economy as moving toward a steady state in which economic progress would cease, following Ricardo's line of reasoning.

It makes sense that the distribution of income would be a bigger issue in an economy that held out little hope for economic progress for most people, but from a theoretical standpoint another substantial issue in the nineteenth century was the origin of value. Ricardo promoted the labor theory of value, which was picked up by Karl Marx,[10] and was refined by John Stuart Mill in what was the ultimate development of the Ricardian system. In direct opposition to this theory, which set an objective standard for value (labor) and argued that value was determined on the supply side of the market, utility theories of value argued that value is determined subjectively by the utility people received in consumption, and that value was determined by the demand side of the market. While early nineteenth-century supporters of the utility theory of value can be found, such as Jules Dupuit and Hermann Heinrich Gossen, the strongest supporters were William Stanley Jevons and Carl Menger.[11] Jevons explicitly said that he intended to overthrow "the noxious influence of authority" of Mill and Ricardo.

In the nineteenth century, in view of the dismal prospects for people's economic future laid out by Malthus and Ricardo, the issue of producing prosperity (which dominated eighteenth-century economics) shrank in importance and was displaced by the policy issue of understanding the causes of the distribution of income, and the theoretical issue of the determinants of value. However, this dismal view of most people's economic prospects, carried forward by Malthus, Ricardo, Marx, and Mill, did not rest on any notion of equilibrium. Ricardo argued that as population grew and agricultural activity was forced to utilize increasingly less fertile land, food costs would rise, squeezing profits and lowering investments, even as land rents rose. Marx, building on Ricardo, viewed change as inevitable and forecast the collapse of the capitalist system because of the inherent contradictions upon which it was built.

Economic analysis in the nineteenth century did not depict an economy moving toward an equilibrium, but rather one that was continually evolving, although in a direction quite different from the one Smith envisioned. While it is true that both Ricardo and Mill envisioned the economy as moving toward a situation in which progress would come to a halt, their analysis focused on the process by which this would occur rather than the end-state that would eventually result.

From the standpoint of pure theory, the major issue was what determines value, with the objective, cost of production, supply-side theories of Ricardo and his intellectual heirs vying against the subjective, utility-based, demand-side theories most prominently advocated by Menger and Jevons.

The issue of value was resolved by Alfred Marshall in 1890, whose equilibrium framework came to dominate early twentieth-century economics, and depicted value in use as consumer surplus and value in exchange as the market price. Contrasting supply-side versus demand-side theories of value, Marshall said, "We might as reasonably dispute whether it is the upper of the under blade of a pair of scissors that does the cutting, as whether value is governed by utility or cost of production."[12] Marshall's dominant influence over the economics discipline at once ended the debate over value theory and turned the focus of economics to the properties of economic equilibrium.

Reviewing the history of economic ideas prior to Marshall, it is apparent that economics as an area of inquiry was not oriented toward analyzing the economy as a social system that tends toward equilibrium. The orientation of economics toward equilibrium began with Marshall, and more than just Marshall himself, with the Marshallian framework that dominated the first half of twentieth-century economics – only to be nudged aside by the Walrasian general equilibrium framework in the second half. The key point is that until the twentieth century, economists did not depict the economy as being in equilibrium, or tending toward equilibrium. They did discuss the tendency of prices to gravitate toward a natural price, but even here they discussed the process by which prices would adjust rather than focusing on some ultimate price. They recognized that the economic processes going on today were leading the economy so that the future economy would be different in fundamental ways from its present state. Viewing an economy as being in equilibrium, or tending toward equilibrium, is an innovation of twentieth-century economics.

Twentieth-century equilibrium economics

Following this line of reasoning, twentieth-century equilibrium economics begins in 1890 with the publication of Marshall's Principles of Economics. Leon Walras's general equilibrium analysis predates Marshall,[13] but it did not have its main influence until well into the twentieth century, whereas Marshall's framework dominated the economics profession in the first half of the twentieth century, and still provides the foundation for introductory economics courses. While Marshall had a nuanced understanding of economic phenomena, and often referred to evolutionary ideas in his text, his most lasting influence was his development of a rigorous partial equilibrium framework for economic analysis. The fundamental issue addressed by twentieth-century economics was the refinement and development of this framework to better understand the properties of economic equilibrium.

Welfare economics, following Marshall's student A.C. Pigou,[14] was built on this Marshallian framework, and the theory of market structure was refined by Chamberlin and Robinson.[15] Walras's general equilibrium framework moved into the mainstream in the very influential treatises of John R. Hicks and Paul Samuelson,[16] and following Arrow and Debreu's proof of the uniqueness and stability of competitive equilibrium,[17] that benchmark of Pareto optimal competitive equilibrium was used to evaluate the performance of the economy.

This is an extremely significant development in economic thought. From its earliest development up until the twentieth century, economists would have judged economic welfare based on economic progress – how much living standards were improving. But with the equilibrium frameworkfirmly in place, twentieth-century welfare economics created a different standard that was completely static. Once a Pareto optimum was achieved, in this framework, welfare was maximized, and it would not be possible to make one person better offwithout making someone else worse off. The invisible hand, once depicted by Adam Smith as leading an economy toward increased prosperity, was now envisioned as holding an economy in place, and pushing it toward that static Pareto optimal competitive equilibrium, where welfare was maximized.[18] If "market failures" prevented the economy from reaching this optimum by itself, government policies might be used to move toward Pareto optimality. The point is, equilibrium analysis had so taken over microeconomic analysis that economic theorists developed their theories as if there was a unique stable equilibrium under which welfare would be maximized, and if the economy arrived there, it would not be possible to improve welfare further.

Meanwhile, macroeconomics, following John Maynard Keynes's General Theory,[19] and even more so Hicks's interpretation of Keynes,[20] retained the Mashallian notion of equilibrium, developing a framework within which the equilibrium towardwhich the economy tendedwas not necessarily a full-employment equilibrium that made effective use of the economy's resources. Macroeconomic analysis looked for reasons why an economy might find itself at an equilibrium with less than full employment, or with excessive inflation, and developed policies that could produce a macroeconomic equilibrium with full employment and low inflation. While macroeconomics following Keynes was significantly different from microeconomics following Marshall, the two retained one key element of commonality: their focus on equilibrium. That focus extended beyond just being concerned with methods of analysis, all the way to policy goals. Just as welfare is maximized in microeconomics at a Pareto optimal competitive equilibrium, the policy goal of macroeconomics was to arrive at a macroeconomic equilibrium with full employment and low inflation. Once those objectives were attained, equilibrium economics saw no further prospects for improving economic welfare.

The properties of equilibrium remained the focus of economics throughout the twentieth century. In macroeconomics, the issue was to understand the properties of an equilibrium which exhibited full employment and low inflation, while in microeconomics the policy goal was to allocate resources efficiently, using the unique and stable competitive equilibrium as a benchmark representing the best possible outcome. The research paradigm proceeded under the idea that the more we know about the properties of economic equilibrium the better equipped we are to design welfare-maximizing policies.

Toward the end of the twentieth century issues of economic growth resurfaced, led by Nobel laureate Robert Lucas,[21] Paul Romer,[22] and other scholars working firmly within this equilibrium framework. The issue was reminiscent of the eighteenth-century focus on wealth, but the methods remained firmly within the twentieth-century equilibrium paradigm. Certainly, there were alternatives to this equilibrium approach to growth, such as Nobel laureate Douglass North's institutional analysis,[23] but even a renewed interest in growth theory toward the end of the twentieth century took the approach that growth was an equilibrium phenomenon.

This brief recounting of the history of economic ideas shows that only in the twentieth century, beginning with Marshall, was the economics profession so firmly focused on analysis within an equilibrium framework. Within that framework, the old question of how economies produce prosperity, which was a key question at least since Adam Smith's influential treatise, was ignored. In macroeconomics the issue was how to produce full employment and low inflation, not growth. In microeconomics the issue was arriving at a Pareto optimal allocation of resources, not growth. In the 1990s growth returned as a prominent issue in economics, but in that equilibrium framework the issue was to try to understand the growth of income, not economic progress more generally.

Two observations that come from this analysis are that, first, equilibrium was not always the focus of economic analysis – this came only in the twentieth century – and second, that as economics focused more on equilibrium, it focused less on economic progress until, by the mid-twentieth century, economic progress was almost completely eliminated from economic analysis. These two observations are directly linked. In the equilibrium framework, if an economy arrived at a Pareto optimal allocation of resources, with full employment and low inflation, nothing could be improved. Equilibrium models had assumed away not only the factors that cause economic progress, but even the prospect of economic progress itself.

The market system and economic progress

Prior to the 1990s a common belief among reputable economists was that central economic planning was a more rational way of allocating economic resources thanusing the market system. After the collapse of the Berlin Wall in 1989, followed by the demise of the Soviet Union in 1991, the turn of events changed the thinking of economists, and by the early twenty-first century the near-universal belief among economists was that markets do a better job of allocating resources than central planning. The idea has become so generally accepted among economists that it will not be argued here. Many other works make the argument persuasively.[24]

The argument in support of the market allocation of resources is not quite that clear-cut, however. While in principle there is general agreement that markets allocate resources more effectively than government planning, the conventional wisdom among economists is far from arguing that economic planning by government should be eliminated. First, there is the argument that markets work only within a system of clearly defined and enforced property rights, and government is necessary to do the defining – through courts and legislatures – and enforcing of those rights. Further, there is the argument that, unfettered, businesses will act against the public interest, so business activity should be regulated. There are also arguments for government steering economic activity through incentives, subsidies, regulation, government research and development activities, and more. So, while in principle the argument that markets allocate resources better than government has broad support among economists and among the general public, in practice there is no clear line, in either the public's view or the views of economists, about the appropriate role of government in the economy.

Would a clear-cut change in expert opinion sway popular opinion toward the experts? In a famous passage in The General Theory, John Maynard Keynes suggests so. Keynes argued that "the ideas of economists and political philosophers, both when they are right and when they are wrong, are more powerful than is commonly understood. Indeed, the world is ruled by little else. Practical men, who believe themselves to be quite exempt from any intellectual influences, are usually the slaves of some defunct economist."[25] If Keynes is right,[26] one might look toward the framework within which expert opinion is formed on the role of markets versus government to see how that line should be drawn.

Twentieth-century equilibrium economics creates a theoretical environment within which government planning appears to be far more effective than it actually is in the real world, for several reasons. One is that it tended to ignore the political environment – and indeed, the institutional environment more generally – because of its focus on the properties of economic equilibrium. A second more substantial problem is that the equilibrium framework left out of its analysis the key elements of a real-world economy that generate economic progress and prosperity. A good way to see this is to review the socialist calculation debate that occurred during the first half of the twentieth century.

The socialist calculation debate

The socialist calculation debate began when, shortly after the 1917 Russian revolution that created the Soviet Union, Austrian economist Ludwig von Mises presented a paper before the Vienna Economics Society claiming that market prices were necessary for rational economic planning, and for that reason a socialist economy could not work and was doomed to failure. Mises published his ideas in a book 1922, sparking dissent among economists who supported central economic planning.[27] Defenders of central economic planning were put on the defensive because, while the new Soviet Union claimed to be designed on Marxist principles, in fact Marx never did explain how a socialist economy would work. Marx's writings were critiques of the capitalist system, not blueprints for the operation of its socialist replacement.

After a period of debate, what appeared to many economists to be the fatal blow to Mises's arguments appeared in a 1938 book by Oskar Lange and Fred M. Taylor, On the Economic Theory of Socialism.[28] Lange and Taylor's answer to Mises was a straightforward development of twentieth-century equilibrium economics. They argued that one way a centrally-planned economy could operate would be for the central planners to set up a system of administered prices for all goods in the economy, creating an accounting system that would mimic the operation of a market economy. The central planners would ask suppliers and demanders of all goods how much they would want to buy and sell at various prices. If the quantity supplied was greater than the quantity demanded at that price, the planners would lower the price, and if the quantity supplied was less than the quantity demanded they would raise the price. In this way, they could iteratively adjust prices so that the quantity supplied would equal the quantity demanded in all markets. Thus, Lange and Taylor argued, one way that a socialist economy could rationally allocate resources would be to have central planners mimic what a market economy did. Central planning could then produce results identical to those of the market.

Lange and Taylor's answer to Mises was viewed by the economic mainstream as providing a definitive answer, and the general view was that Mises was wrong, and lost the debate. Lange and Taylor pointed out that this was not the only way that central planners could guide an economy, and probably not the best way either. But, what they argued was that if central planners acted this way, they could exactly replicate a market allocation of resources, so central planning, in theory, could do at least as well as the market in allocating resources.

In this view, the advantage of central planning was that planners could deviate from a market allocation to improve the allocation of resources. One way they might do that is to favor the production of some goods over others, but probably the most significant advantage central planning could offer in this framework is that it could devote more of an economy's resources toward investment, resulting in more growth and more productivity in the future. In a market economy, saving and investment are left to the preferences of individuals, and at least since World War II there has been a constant policy issue based on the idea that Americans do not save enough. That would be a non-issue in a centrally-planned economy where the division between the production of investment goods and consumer goods would be determined by the economic experts, not by the aggregation of every individual's decision on how much to save and how much to consume.

Even though the consensus of the economics profession was against him, Mises never did concede the issue, and until his death in 1973 continued to maintain that central planning could not work to rationally allocate economic resources. As an illustration of the conventional wisdom, Nobel laureate Paul Samuelson, in the 1973 edition of his best-selling introductory economics textbook – published the year Ludwig von Mises died – noted that while per capita income in the Soviet Union was about half what it was in the United States (Samuelson likely overestimated Soviet per capita income here), because the Soviet Union had had a more efficient economic system their economic growth rate was higher. Because of the increased economic growth in the Soviet Union, Samuelson projected that Soviet per capita income would catch up to per capita income in the United States perhaps as soon as 1990, but almost surely by 2010.[29] This argument remained up through Samuelson's 1980 edition, with the years pushed up to 2000 and 2020, but disappeared by the next edition, which was published in 1985.

Keep in mind that Samuelson was not the maverick here: Mises was. Samuelson was merely reporting in his introductory textbook the conventional economic wisdom of his time. It was not until after the collapse of the Berlin Wall that the conventional wisdom changed, and that change was due to the events of history, not advances in economic theory.[30] The arguments in favor of central planning are summarized well within the socialist calculation debate, and within this framework one can see that they are based on the foundations of twentieth-century equilibrium economics. In this view, what a market economy does is bring markets into equilibrium, equating the quantity supplied to the quantity demanded. The Lange and Taylor argument is that central planners can do that too, but central planners also have the option to reallocate resources to improve on market outcomes.

This chapter noted earlier that market economies perform two functions: market clearing and the production of economic progress. But, the argument went, the twentieth-century equilibrium framework focuses on that first function – market clearing – and leaves out of its analysis the key features of the economy that generate progress and prosperity. The Lange and Taylor argument for "market socialism," where central planners mimic what the market does, falls short because even in theory, their market socialism only addresses the market-clearing function of the economy. But, if the model of the economy from which economists are working only depicts the market-clearing function, then the very nature of the model leads economic analysis to ignore those aspects of the economy that lead to progress and prosperity. Twentieth-century equilibrium economics depicts the economy as stagnant, not moving forward.

Making a static economy grow

Out of this equilibrium approach to economic analysis came neoclassical growth theory, which, sticking with the established framework, depicted growth as an equilibrium phenomenon. Based on Robert Solow's neoclassical model of economic growth,[31] neoclassical growth theory depicts an equilibrium growth path of an economy. In this model, output, Q, is a function, f, of capital, K, and labor, L, or Q = f(K,L). To have output, Q, grow, this framework shows three possibilities. Economists have long-recognized the importance of investment to growth, so increasing K is an obvious path to growth. Another path would be to increase L, and Robert Lucas has insightfully demonstrated that L is more than simply the number of laborers, or hours of labor, and is better represented by some measure of human capital.[32] But looking at growth in this framework, economists from Solow on recognized that increases in K and L fell far short of explaining increases in Q. Looking back at the framework, that implies that changes in f are primarily responsible for economic growth.

Changes in f have been viewed as the result of technological change, pushing the neoclassical growth framework to try to incorporate it into the model. Paul Romer offered some of the more innovative developments along these lines,[33] but the model retained its equilibrium foundations. Research and development, in this framework, is produced just like other inputs into the production process, changing the nature of the production function to allow greater productivity from the resources, K and L, that go into it. The equilibrium framework has been extended to allow for growth, but by design it has remained an equilibrium framework.

The chapters that follow will make the argument that economic growth is not an equilibrium phenomenon, and that growth is better analyzed as a by-product of economic progress than as a phenomenon on its own. Consider the neoclassical framework in the context of twentieth-century economic growth, where measured Q per capita increased by about seven times over the course of the century. It is inconceivable that people at the end of the century would have wanted to produce and consume seven times as much of the same goods they were consuming at the beginning of the century. Output went up largely because the composition of Q changed. Automobiles replaced horse-drawn carriages, email replaced paper mail, movies and television replaced books and live concerts. The old products remained – at least, many of them did – but growth occurred because new consumption opportunities were available, encouraging more production and more consumption.

If it is the case that output – measured Q – would not have gone up as it did without changes in the composition of Q, then a model that depicts output simply as a homogeneous Q cannot possibly depict the actual process of growth, because by assuming a homogeneous output, it has assumed away one of the key causal factors that propelled growth. The phenomenon is better seen as progress, and must explain changes in the composition of Q in order to explain increases in Q over time. Equilibrium growth models do not fare well in explaining how prosperity is produced.

Capitalism and prosperity

The short answer to the question "What has produced the remarkable progress and prosperity of the past several centuries?" is capitalism. A market economy, with private ownership of the means of production, strong protection of property rights, rule of law, and minimal interference with private economic activity, produces prosperity. In the twentieth century, a strong case would have had to have been made for this claim in the face of the challenges raised by the proponents of central economic planning. By the twenty-first century, after the collapse of the centrally-planned economies and the continued strong performance of capitalist economies, the argument is uncontroversial, and also well-documented.[34] This leaves open the question about what role government plays in supporting the market system. Some would argue that markets would perform best with no government at all;[35] others would argue that a market economy could not function without some minimal government;[36] and many others would argue that markets fail to allocate resources efficiently without government intervention on many margins.[37]

These arguments on the appropriate scope of government are important, but beside the point in the present investigation. Capitalism has generated the most substantial economic progress in the history of mankind, and the task at hand is to understand how this has happened, from a theoretical point of view, for the purpose of gaining a more fundamental understanding of the operation of a market economy.

The economic theory of markets and prosperity

A substantial part of this book deals with issues related to economic theory, for two reasons. First, most of the economic analysis of market economies that took place in the twentieth century has painted a picture that is, at best, incomplete in describing the features of the economy that produce prosperity. Second, with a better theoretical understanding of the causes of prosperity, economic policymakers will be in abetter position to maintain existing economic institutions that produce prosperity, and to design new ones that can further enhance prosperity.

The main focus of this volume is to extend the theoretical analysis of the market economy to better encompass those features of the economy that produce prosperity. That motivation is theoretical in nature, because economic theory has not done a good job of explaining the remarkable economic progress that has been occurring since the beginning of the Industrial Revolution. While this volume is oriented primarily toward economic theory, ultimately the foundations of theory are used to create public policy. As noted earlier, John Maynard Keynes, at the end of his General Theory, wrote "Practical men, who believe themselves to be quite exempt from any intellectual influences, are usually the slaves of some defunct economist. ... I am sure the power of vested interests is vastly exaggerated compared with the gradual encroachment of ideas."[38]

For practical people who want continued prosperity, it is important to develop and understand the theoretical foundations that produce it.

When one looks across the nations of the world, some are prosperous and growing, while some are poor and stagnant, or even declining. Simple observation shows that nations that adopt market institutions prosper; those that do not stagnate.

The United States, Western Europe, and Japan, are prosperous market economies. China and India, two poor economies in the twentieth century, are among the fastest-growing at the beginning of the twenty-first. They are leaving behind their centrally-planned economies with big government and moving toward market economies, and their growth rates show the result.

Empirical evidence indicates that for each 10 percent increase in government spending as a share of income a nation's economic growth rate falls by 1 percent.[39] When one considers that a 3 percent rate of economic growth is relatively high, the effect is substantial. For example, if a country increases its government spending as a share of income from 40 percent to 50 percent, that would result in a 1 percent decline in economic growth. A fall from, for example, 3 percent growth to 2 percent will have a huge impact. A 3 percent annual growth rate would result in a doubling of income in 24 years. A 2 percent annual growth rate would result in a doubling of income every 37 years, and a 1 percent annual growth rate would result in a doubling of income every 71 years. The more nations rely on markets rather than government, the more prosperous they become. That is why it is important to understand how markets work.

Through the process of exchange, nations that participate in the global economy benefit from the prosperity of other nations both because they can buy goods from producers who are more efficient at producing them and because they can sell the goods they produce to consumers in other nations.

Prosperous nations sometimes hold back the less prosperous

While prosperity in some nations overall aids the citizens of less prosperous nations, there are some policies that prosperous nations pursue that work to the detriment of the less prosperous. While this must be admitted up front, it also must be recognized that it is not the prosperity of the prosperous per se that holds back the less prosperous, but rather the policies of governments in prosperous nations that work against citizens in poorer nations.

Through the twentieth century and into the twenty-first, Western European nations and the United States have subsidized their agricultural industries, sheltering them from foreign competition, and enabling them to produce surpluses that have spilled over into world markets. With cheap food available from these prosperous nations, agricultural producers in less developed nations have faced competition serious enough to hold them back. Poor African farmers find it difficult to compete with cheap subsidized food from Western nations, and agriculture in developing nations has been stunted as a result. One reason poor countries cannot feed themselves is that they cannot compete against the subsidies that rich countries give their farmers.

Foreign aid has also had a detrimental effect. Foreign aid tends to go to the governments of poor countries, not to the poor citizens who need it. The aid enables the leaders of those countries to support themselves and their armies by relying on foreign aid rather than the productivity of their own economies. The result has been stagnation and poverty for many citizens, even as their nations' political leaders have lived comfortably on the charity of wealthier nations.[40] With no incentive to improve the economic performance of their countries, leaders have focused on their own wellbeing. Indeed, if the aid is contingent on being poor, an improvement in a country's standard of living could reduce its foreign aid revenues, hurting the nation's leaders in the short term.

Much foreign aid during the twentieth century was not really intended to help those in need anyway. During the Cold War era, both the United States and the Soviet Union were trying to buy political alliances and support, which was done by providing foreign aid in the way that best benefitted the nation's political leaders who could offer that support, not by helping the truly needy. Just as domestic welfare programs can produce welfare dependence among recipients, so foreign aid can lead poor countries to an economy that is dependent on continued aid to maintain even its meager standard of living.

The problem is not one produced by market economies, however, but rather by their governments, who can afford to indulge in policies that benefit special interests in their constituency, to the detriment of poorer nations.

Conclusion

The evidence that market economies produce prosperity is persuasive enough, and is documented sufficiently in other places such that it will not be defended here. The issue under consideration in this book is why market economies produce prosperity. Economists have been interested in this question at least as far back as 1776, when Adam Smith published his monumental The Wealth of Nations, yet an argument can be made that contemporary economic analysis offers less insight into the issue than did Smith. The subject matter of this volume is the economic analysis – the theoretical foundation – that lends insight into the way that market economies produce prosperity.

2 Economic models

]2 Economic models

A framework for analysis

The real-world economy is too complex to understand simply by observing it. Were that not the case, there would be no reason to study economics, because one could just observe economic phenomena and understand them through simple observation. Because of the complexity of economic phenomena, economic analysis is undertaken by developing models that are simplified depictions of reality. The idea is that interactions among the components of the model are analogous to interactions among the components of the economy. Thus, if one can understand the model, which is simpler than reality, by analogy one can understand the more complex economy.

David Ricardo has been called the first great model-builder in economics. In his Principles of Political Economy and Taxation he laid out a model of the economy that is very similar to those used in contemporary economic analysis.[1] In Ricardo's framework output is produced by land, labor, and capital, much like the production function presented in Chapter 1. Among other insights Ricardo derived from his model, he observed that with afixed amount of land, adding additional quantities of labor and capital to produce more output would result in increasing costs per unit of agricultural production, and the rising food prices accompanying a growing population would keep most people at a subsistence level of income.[2] It turned out that Ricardo's conclusion was at odds with reality because he did not anticipate the substantial increases in agricultural productivity that would occur over the next two centuries. Even there, the failure of Ricardo's conclusion to materialize can be analyzed within the context of his model. Where output is a function of land, labor, and capital, he failed to anticipate the change in the functional form of the production function.

Whereas Ricardo included three factors of production in his production function, contemporary economics, within the framework presented in Chapter 1, typically depicts the production function as $Q = f(K,L)$, and includes only capital and labor as factors of production, leaving out land. In Ricardo's framework land was a vitally important factor for understanding his conclusions. Of course, land remains an important factor of production, but for many purposes it can be assumed away, simplifying the theory and the analysis and making it easier to understand. Whether this simplifying assumption is acceptable depends on whether the economic phenomenon one is trying to understand is affected by the assumption.

All models employ simplifying assumptions. No model can be as complex as the real world, because the real world is too complex to take every detail into account. Furthermore, the fact that a model is simpler than the real world is a virtue because it makes it easier to understand. If the real-world phenomena under examination could be understood without the model, the model would serve no purpose. The simplifications introduced by models are assets that aid in understanding, not liabilities. The simpler the model the better, unless it is too simple and leaves out key elements of the phenomena it is trying to describe.

Different types of assumptions

Assumptions that are employed in models are of three types. First, models assume some features of the real world away. They are not included in the model. Second, models make unrealistic assumptions about some of their components, or about some of the relationships among components. In addition, implied in a model are assumptions about its applicability. Models may be applicable under some circumstances but not under others. Alan Musgrave has referred to these types of assumptions as heuristic assumptions, negligibility assumptions, and domain assumptions.[3]

In economic analysis, a clear case of assuming away some features of the real world (Musgrave's negligibility assumption) is the use of partial equilibrium analysis to depict the way that prices and quantities are determined in markets. Partial equilibrium models focus on what happens in one market, assuming away any effects that might spill over from other markets. Often, this offers the clearest vision of a particular phenomenon that one wants to depict. For example, in a partial equilibrium framework one can clearly show the effect of a price ceiling in a competitive market. The price ceiling increases the quantity demanded and lowers the quantity supplied, resulting in a shortage. Of course, money that would have been spent in that market without the price control will be diverted to other markets, resulting in secondary effects, so one might argue that a general equilibrium analysis would be necessary to fully capture all the effects of the price ceiling. Yet, if one wants to understand why a price ceiling causes a shortage, a strong argument could be made that the lesson is best illustrated in a partial equilibrium framework, and secondary effects that spill over into other markets only detract from what the partial equilibrium model shows. The secondary effects are negligible, and can safely be assumed away.

If one is analyzing the effect of a price ceiling on gasoline, for example, a simple supply and demand framework depicting the gasoline market as competitive will clearly illustrate the point. Is the market for gasoline really competitive? If it was, why do retailers spend so much on branding and advertising their products, and why does gasoline at different stations sell for different prices? These facts suggest that the market for gasoline is not perfectly competitive (in the neoclassical sense anyway), but it is close enough that making an unrealistic assumption that depicts it as a competitive industry helps focus attention on the fact that holding the price below the market equilibrium level increases the quantity demanded and reduces the quantity supplied, resulting in a shortage. To illustrate that key point clearly, an unrealistic assumption − Musgrave's heuristic assumption − is employed to make the model simpler. If the goal is to understand why price ceilings lead to shortages, unrealistic assumptions can help focus the model on the key relationships one wants to understand, making the model better for the purpose at hand.

The purpose of the model is to help its users understand reality, so the ideal model leaves out those things that are irrelevant to the problem at hand to focus attention only on those factors that are important to what one wants to understand. Simplification is a virtue, as long as the model's assumptions do not leave out or unrealistically depict factors necessary to understand the phenomena being analyzed.

Maps as models

Baumol and Blinder use a map as an example of a model when introducing the use of models in their introductory textbook.[4] The example is a good one, because a map is, in fact, a model: a geographic rather than an economic one.

Imagine that someone wants to drive from Atlanta to Los Angeles, but had never made the trip before. How could that driver get there? Looking that the heading from this section, all readers are thinking: “Use a map!” The map would depict the roadways along the route, so it is likely that a driver who has never made the trip before could get there just as efficiently as someone who has made the trip many times. The map is a model of the real world.

To help convey information to its user, the map is designed using the same types of simplifying assumptions discussed above. The map assumes many features of the real world away by not depicting them on the map. Roadways and major cities will be depicted on the map, but the map will leave out many features of the real world the driver will see. Water towers, bridges, radio towers, billboards, and a host of other real-world features will not be depicted. Far from being detrimental, these simplifying assumptions help the user understand the map. By omitting features of the real world that are irrelevant tofinding the best route for the journey, the user can better understand which routes to take without having to wade through a clutter of irrelevant information. The model is improved by assuming somethings away.

The map also makes many unrealistic assumptions. The first thing the user will notice is that the map is much smaller than the real world. Imagine how difficult it would be to read a map if it was not scaled down! (Maps of other things – for example, the tracings on a microprocessor – will aid understanding if they are scaled to be larger than in the real world.) Roads on the map will be drawn unrealistically larger than to actual scale, to emphasize the information that is most important to the map’s user. Interstate highways might be depicted as parallel blue lines. Are interstate highways really blue? No, but to depict them that way, rather than as a nondescript grey, helps highlight the information the user wants to know.

Small towns may be depicted on the map just as little round dots with the town’s name adjacent. Are those towns really round? Of course not, but depicting them that way helps users understand their position relative to the towns, which is the only information about the towns that is relevant as the traveler undertakes the journey. Those dots representing the towns are likely to be smaller in diameter than the width of those parallel blue lines representing the interstate highways. Are those towns really not as wide as a highway? Probably not, but depicting them that way helps emphasize the information that is important to the user. The map is purposefully designed to depict the world unrealistically, because depicting it that way emphasizes the information that is important to the user, and downplays (or assumes away altogether) features of the real world that are unimportant.

Maps, as models of the real world, make unrealistic assumptions and assume away features of the real world to make it easier for users of those geographic models to understand the real world. Economic models use those same types of assumptions to make the models more useful in understanding the real world.

No model is ideal for every purpose

Once the driver in the above example reaches Los Angeles, the map that provided information about how to get from Atlanta to Los Angeles will be of little use in finding specific locations in Los Angeles. A different map of the Los Angeles area would be more helpful. The map of the local area will show Los Angeles streets and landmarks in more detail, but will not show the location of Los Angeles relative to Atlanta, or other places outside the local area. Because maps are simplified models of the real world, no one map can be ideal for every purpose. Things that are best left out or unrealistically depicted for one purpose may be essential for another purpose, in which case the best map for the first purpose will be of little help in the second.

If one were to fly from Atlanta to Los Angeles rather than drive, the pilot would use a map quite different from a road map. The road map will show few if any terrain elevations, whereas the aviation map will show terrain elevation over the entire area of the map. Elevation is a detail that can be safely assumed away when driving, but is essential information when flying. Similarly, radio towers and other obstructions sticking into the air will appear on aviation maps but not on road maps. If one examined aviation maps in more detail, it would also become apparent that pilots use different maps for different purposes. One type of map is used for instrument flying, another for visual flying, and yet another for making instrument approaches to airports.

Maps are models that are simplified representations of geographic reality to help the user understand where geographic features are in relation to one another. Similarly, economic models are simplified representations of economic reality which help the user to understand the operation of economic phenomena. Just as no one map is ideal for every use, because of the simplifying assumptions that go into the design of the map, no one economic model is ideal for every use either. Because economic models make unrealistic assumptions, and assume some things away, the model is likely to be uninformative, and even misleading, if it is used to try to understand phenomena related to aspects of reality that are assumed away or depicted unrealistically.

No model is ideal for every purpose. An important part of economic analysis is understanding which models are best employed to analyze specific phenomena, because a model that is ideal for one purpose will be completely inapplicable for others. Economic models help understand economic reality, but because they present a simplified depiction of reality, they necessarily leave some factors out. The model cannot help the user understand phenomena that it leaves out of its analysis. Think about the map example again. How useful would a map be if one depicted everything every map user could want to know? The map would have the interstate highways for long-distance travel, every local street for people looking for a specific location, the elevations and aerial obstructions for pilots, and more. The map would be too cluttered and complex to be understandable and useful. The same is true of economic models.

Economic models must be judged in relation to the issues they analyze. Models are not good or bad in isolation, but rather range from appropriate, to less appropriate, to inappropriate for analyzing certain economic phenomena. Criticisms of particular models or approaches do not imply that a model or approach is worthless for any purpose (although one could imagine models or approaches where this is the case), but rather that the particular model or approach is inappropriate for a particular purpose. The model must be evaluated in the context of its application.

Applicability of a model

Unlike physics, where the objects being modeled strictly obey physical laws, economics models the result of individuals interacting within a social system, and individual behavior can change. A model that applies in some situations may be completely inapplicable in others.

The chapters that follow will argue that models used to understand how prices and quantities are determined in markets are often inapplicable to understanding the factors that produce economic progress. But to see that models which lend understanding in one context may be inapplicable in other contexts, it is not necessary to go so far as to say that models which explain some phenomena do not explain others. Models that work well to explain some phenomena in certain contexts may fail to explain those exact same phenomena in others.

A good example is the development of Keynesian macroeconomics to explain inflation and unemployment in the macroeconomy. Keynes, of course, in his General Theory,[5] argued that the classical theory, appropriate under certain situations, was inappropriate in others. When A.W. Phillips published his famous 1958 article showing a trade-off between inflation and unemployment over a long period of time,[6] Nobel laureates Paul Samuelson and Robert Solow criticized Phillips for not going far enough in developing a theoretical foundation for his empirical finding.[7] Samuelson and Solow argued that there was a trade-offbetween inflation and unemployment, and that economists should identify and present this "menu of choices" to policymakers, who could then choose the best combination of those two variables.

When policymakers actually did this, the trade-off that had existed for decades disappeared within a few years. As Robert Lucas explained in what has come to be known as the Lucas critique,[8] the trade-off existed when individuals in the economy could not anticipate the inflation policy of policymakers, but once policymakers announced a policy that could be anticipated, individuals changed their behavior to respond to the policies they anticipated, causing the long-standing Phillips-curve relationship to disappear. A framework that worked well to explain the co-movements in inflation and unemployment over a period of decades did not work at all in a different policy regime. Models that are applicable for some purposes and under some conditions are not applicable for other purposes or for other conditions. This is more obvious when a model is employed to explain phenomena that rely on some part of reality that is assumed away, but the Lucas critique shows that it really is more general than this. One cannot simply mechanically apply a model that has worked well in the past to a new situation and hope to have the same success. Models that are ideal for some purposes are misleading and uninformative in others.

Positivism

Perhaps the most famous methodological article in all of economics is Milton Friedman's "The Methodology of Positive Economics."[9] In it, Friedman argues that, "Viewed as a body of substantive hypotheses, theory is to be judged by its predictive power for the class of phenomena which it is intended to 'explain.'"[10] Friedman recognizes that economic models employ unrealistic assumptions, and assume some things away, but argues that the appropriateness of those assumptions should be judged by how well the model predicts.

To take an example from physics, falling objects accelerate at 9.8 meters per second per second in the absence of any other forces. Assuming away any resistance from the air, one can use this physical constant to predict how long it will take a rock dropped from any height to hit the ground. Can the effect of air resistance be assumed away? Empirical tests with a rock and a stopwatch indicate that gravity can be safely assumed away, and that the rock does accelerate that fast. Would the same acceleration be expected if one pushed an automobile off the top of a tall building? (This test will be left to the reader to perform.) What about dropping a feather from the top of that same building?

Assuming away the resistance of the air, the automobile and the feather would be expected to impact the ground at the same time if dropped from the tall building at the same time. However, in the real world the automobile will impact the ground well before the feather. Air resistance can be assumed away in the case of the automobile, with little effect on the outcome, but not in the case of the feather. Assumptions that are appropriate under some circumstances do not work well in others. Friedman is clear that assumptions are to be judged in the context of the phenomena the model is trying to explain, and the reader can intuitively understand why air resistance can be assumed away when looking at the effect of gravitational acceleration on a rock but not on a feather.

Intuition only takes one so far in judging the role of assumptions in economic models, however, as the Lucas critique example from the previous section shows. Nobel laureates Samuelson and Solow followed Friedman's positivism in accepting a model that conformed well to empirical data, only to find that model breaking down shortly after they advocated it to policymakers. The Lucas critique shows that in economics assumptions must be judged on more than just their ability to predict.

One can argue about the degree to which Samuelson and Solow actually followed the positivist methodology. Following that methodology, one first develops a theory with testable hypotheses, and then applies the theory to data to test whether the data are consistent with the hypotheses. In Samuelson and Solow's case, they saw the data first, in the form of Phillips' article, and then applied a theory to it after the fact. (Although, perhaps not. They applied the already-existing Keynesian macroeconomics to the data. One might argue that Phillips was actually testing that theory even though his article did not present the data that way.) But, once the hypothesis was made, other data, including data presented by Samuelson and Solow, supported it.[11] In any event, this episode in the history of economics shows the peril involved in uncritically accepting the assumptions in a model because the model predicts well under certain circumstances.

One cannot be too critical of Samuelson and Solow for developing a theory to explain data they had already seen. Economic analysis must proceed this way except in rare circumstances, because there is only one economy, and economic analysis is intended to help understand phenomena in that economy. Thus, economists observe the phenomena they hope to explain in almost every case before they develop their models to explain those phenomena. As McCloskey has observed, despite the reverence economists pay to the positivist methodology, they rarely ever actually undertake their own research that way, which is a good thing.[12] This by itself should call into question the uncritical acceptance of assumptions on the basis that they have worked well in past applications.[13]

Unrealistic assumptions, and assuming away aspects of economic reality, are a key part of the art of economic model-building, but those assumptions must be critically analyzed every time a model is applied. Assumptions that work well under certain circumstances mislead in others, so economic analysis must constantly be attuned to the relevance of assumptions in every application of every model.

Ptolemaic economics: the calibration of macroeconomic models

At the beginning of the twenty-first century the calibration of macroeconomic models to real-world data has become a standard tool of macroeconomics. Following the pioneering work of Finn Kydland and Edward Prescott,[14] a model is developed and its parameters are calibrated such that the movement of data in the model corresponds with the movement of real-world macroeconomic data. The idea is that if the calibrated model accurately represents past movements of real-world macroeconomic data, the underlying model represents macroeconomic reality to a sufficient degree, such that policy conclusions can be drawn from it. The model can be used, for example, to examine the effects of policies such as tax changes or stabilization policy measures, with the idea that the model is sufficiently accurate to lend insight into the effects that such changes would have if implemented in the real world.

Such calibrated models are not immune to the Lucas critique, but macroeconomists are well aware of the possibility that policy changes could result in behavioral changes that will alter the underlying relationships in the model. A potentially more significant problem is that there is no assurance that just because the model mimics the underlying macroeconomic data, that therefore the processes in the model mimic macroeconomic processes in the real world.

Consider the Ptolemaic model of the universe, conjectured by Claudius Ptolemy (83–168 ad). Ptolemy had designed a model of the universe with the Earth at the center, and with the heavenly bodies rotating around the Earth on concentric spheres. The spheres had different axes, with some spheres revolving around others, to produce a remarkably accurate model of the movements of the heavenly bodies. Modern planetariums are designed based on the Ptolemaic model of the universe because it can be so accurately calibrated to the actual movements of the heavenly bodies. Ptolemy's model of the universe, with the Earth at the center, did not receive a serious challenge until Nicolaus Copernicus (1473–1543 ad) presented an alternative model with the sun at the center of the solar system, and with the apparent movements of heavenly bodies in the sky due to the Earth's movement, not due to the movements of those other bodies.

When Copernicus's work first appeared, the Ptolemaic model of the universe, which had been refined for centuries, provided a more accurate depiction of the actual movements of the heavenly bodies. Yet today we know that the Copernican model of the universe more accurately depicts the actual underlying processes that generate those movements. Perhaps it makes little difference if one wants to simply understand where one might expect to see particular heavenly bodies in the future, but if one wants to launch rockets into space to explore those heavenly bodies, the Ptolemaic model of the universe will provide an inadequate model for understanding how to do that.

Calibrated macro models are to some degree like the Ptolemaic model of the universe. Clearly they are unrealistic in some sense, if they rely on representative agents rather than modeling a large number of economic actors, and if they aggregate capital into an aggregate K in the model rather than allowing for a heterogeneous capital structure. How important these simplifying assumptions are cannot be determined by how closely the model conforms to real-world macroeconomic data. In different settings the real world may perform significantly differently from the model.

While the calibrated model follows real-world data closely, it does not follow the positivist methodology which might be used to justify its assumptions. With positivism, the theory is developed and then tested against data; with calibrated macro models the models are adjusted to conform to the data.

Calibrated macro models have been used with some success to model economic growth, and they are not without their insights. For example, these models illustrate that macroeconomic fluctuations are a part of a larger process that determines the trajectory of the economy over time. Business cycles and growth are not independent of each other. This insight can be traced back at least to Joseph Schumpeter's Business Cycles, published in 1939,[15] but it is an insight that was lost through the Keynesian episode of macroeconomics. So, the point is not that one should completely dismiss these calibrated general equilibrium macro models, but rather that one must recognize that just because they are calibrated to accurately represent movement in macroeconomic data does not in any way imply that they accurately represent the underlying processes that generate that movement, any more than the Ptolemaic model of the universe which accurately represents the movements of the heavenly bodies accurately represents the processes that generate them.

Epistemology versus ontology

Epistemology studies the scope and nature of knowledge. It deals with understanding what knowledge is, how it is acquired, and how people know what they know. Ontology examines the nature of reality. One might be tempted to view the discussion of this chapter as an epistemological one. What type of model is most appropriate for understanding the causes of prosperity? The epistemological case against the use of equilibrium models is that this framework is not the most conducive to understanding the process of economic progress. But the issue is deeper than simply equilibrium models not being the best tools for the job, along the lines of thinking that a carpenter might better finish a job by using screws rather than nails. Rather, if economic progress is not an equilibrium phenomenon, then trying to represent it as one leads to a fundamental misunderstanding of the process. It is not an epistemological issue – that is, an issue about how one comes to understand the facts about the process of economic progress – but rather an ontological issue. If economic progress is not an equilibrium phenomenon, then one cannot understand it with the use of equilibrium models.

Tony Lawson has been an insightful critic of neoclassical economics, emphasizing its ontological problems.[16] In the neoclassical framework, given initial conditions, maximizing behavior leads to a deterministic equilibrium, and in a dynamic framework, to a deterministic growth path. In the real world, innovators introduce new products and processes into an economy that disrupt existing economic arrangements, causing individuals to react to the changes, and in turn creating additional disruptions. The market system provides an environment which allows people to adjust to the disruptions caused by such innovators, and by outside influences such as the weather and government interventions. But people will not always react the same way to an uncertain future, and the actions of some people in the economy will alter the trajectory of the economy in a non-deterministic way.

Using deterministic models to depict a process that is fundamentally nondeterministic is a recipe for misrepresentation and misunderstanding. To be useful, models must depict processes that are analogous to those they purport to model in the real world, and equilibrium models, by their very nature, cannot depict the non-deterministic process of economic progress. Equilibrium models cannot be salvaged by adding stochastic elements to them, because the non-deterministic components of economic progress are not stochastic or random. People make choices for good reasons in the face of an unresolvable uncertainty about the future.

The future can never be known because it depends upon the choices of a multitude of people who are making their own decisions about their future courses of action. The decisions made by some people affect the conditions faced by others. In hindsight, some of these decisions will produce good outcomes and others will not. The aggregation of all these individual decisions, including those that in hindsight people might want to reconsider, determines the trajectory of the economy. This trajectory is not deterministic, and it is not an equilibrium. For this ontological reason, economists must move beyond equilibrium models to understand economic progress.[17]

This argument does not imply that all of equilibrium economics must be discarded, or that neoclassical economics has no useful lessons. It does imply that equilibrium models cannot capture the essential processes underlying the generation of economic progress, because those processes are not equilibrium processes. If one wants to understand how markets clear, equilibrium models do a good job. If one wants to understand the process of economic progress, equilibrium models are inappropriate because they assume away the underlying causes of economic progress.

Conclusion

Economic analysis takes place by using models to depict economic processes. If the relationship among the components of the model is analogous to the relationships among the real-world components that parallel those in the model, then by understanding the model one can understand something about the nature of the real world. All models are simplified depictions of reality. They leave some components of reality unrepresented, and unrealistically represent others in order to emphasize certain economic relationships. The real world is so complex that no model could include every aspect of the real world. Even if it did, the model would not be useful for understanding the real world, because the model would be just as complex as the real world. If the real world could be understood without the aid of a simplified model, there would be no reason to develop the model. One could just observe the real world and understand how it works, without recourse to a model.

Simplification in economic models is an asset. The model should be as simple as possible while still retaining the relevant components of the process it hopes to depict. The simplicity of a model helps users understand how the real world works; complexity obscures the workings of the real world. But a model can be too simple, if it leaves out some aspects of the real world that are essential for understanding the phenomena the model is trying to explain.

Because every model leaves out and unrealistically represents some aspects of reality, no model can be ideal for explaining everything. Thus, in what follows, when certain models are criticized as inappropriate for the understanding of economic progress, that does not imply that those models do not have their uses. It simply means they are inappropriate for the particular situation under analysis. The analysis in this book is not intended to argue that neoclassical economics has no use, or lends no insight into economic phenomena. Neoclassical models are very useful for certain purposes. Rather, the argument is that neoclassical models have been extended beyond those phenomena that they explain well to areas where they do not fit. In particular, the neoclassical equilibrium framework is not conducive to explaining economic progress.

The reasons behind the inappropriateness of equilibrium models to the analysis of economic progress are ontological rather than epistemological. It is not that there is a better way to understand economic progress than through equilibrium models; it is that economic progress is not an equilibrium phenomenon, and therefore equilibrium models will necessarily misrepresent the underlying processes.

One purpose of this chapter is to make it clear that the line of analysis in this volume does not suggest that neoclassical equilibrium analysis should be abandoned in favor of another approach. Rather, despite the insights it offers, while the neoclassical framework is appropriate for analyzing some economic phenomena, economic progress is not one of them.

3 The market process

]3 The market process

Economic analysis tends to take place from two different perspectives. The most natural perspective – because it perceives actual economic activities that take place – is the market process perspective. In the market process perspective, economic actors undertake activities of production and exchange with the intention of improving their perceived well-being, and economic analysis seeks to understand why production and exchange take place, and examines factors that facilitate or hinder production and exchange. In contrast, the equilibrium approach offers a different perspective, looking at what the outcome will be after all profitable exchanges have taken place. Process versus outcome are the two main perspectives from which economists view economic activity.

The equilibrium approach to economics is so ingrained in economic analysis that economists have a difficult time thinking outside it, and even economists firmly focused on a market process approach often undertake their analysis with reference to some underlying equilibrium concepts. For example, Ludwig von Mises, whose work almost defines the foundations of contemporary Austrian economics, relies on concepts of a "state of rest," an "evenly rotating economy," and a "stationary economy" as points of reference. Mises says:

action tends toward a state of rest, absence of action. . . . The notion of a plain state of rest is not an imaginary construction but the adequate description of what happens again and again on every market. In this regard it differs radically from the imaginary construction of the final state of rest.[1]

The final state of rest occurs when all market forces have played themselves out, which Mises observes can never happen. Yet he notes, "What makes it necessary to take recourse to this imaginary construction is the fact that the market at every instant is moving toward a final state of rest."[2] Mises goes on to define an evenly rotating economy: "The evenly rotating economy is a fictitious system in which the market prices of all goods and services correspond with the final prices."[3] The difference between this and the final state of rest is that an evenly rotating economy adds a temporal element. Economic data remain the same, period after period. Although this is a fictitious state of affairs, Mises says "However, in order to analyze the problems of change in the data and of unevenly and irregularly varying movement, we must confront them with a fictitious state in which both are hypothetically eliminated."[4]

Murray Rothbard, another giant in the development of the Austrian school, employs these same concepts, and employs the more familiar terminology of final equilibrium rather than Mises's reference to a state of rest.[5] The point is that even economists who are very committed to a process-oriented approach to economics look back to the foundation of equilibrium concepts to build their theories. Indeed, equilibrium concepts are useful in economics, and essential to understanding certain economic phenomena. But they may get in the way of understanding others, including the factors that produce prosperity.

While much of economic theory is explicitly built on equilibrium foundations, a substantial – and surprising – amount also takes a market process approach. The amount is surprising, because academically trained economists naturally think in terms of equilibrium concepts, so are unlikely to realize that equilibrium concepts lie outside the structure of many analyses.

Transaction cost economics

An obvious place to start in considering the market process in economic analysis is transaction cost economics, and Ronald Coase's "The Problem of Social Cost" provides a good frame of reference as a seminal article in the area.[6] Coase's article uses a series of legal cases to illustrate the role of transaction costs. In one case, a doctor and a confectioner work in adjoining buildings, peacefully co-existing until the doctor adds an examination room to his building. The doctor's examination room is adjacent to the confectioner's building and is situated such that vibrations from the confectioner's machinery make the doctor's new examination room unusable. While this is a British case, the doctor takes the typical American solution: He sues the confectioner, asking the court to order the confectioner to cease and desist from the activities that cause the vibrations.

Among the many insights in Coase's article, he emphasizes the reciprocal nature of the problem, to use his terminology. Traditionally, economists would analyze this situation as a classic case of an externality, where one party (the confectioner) is imposing a cost (the vibrations) on another party (the doctor). Coase points out that there is really one right involved, for if the confectioner has a right to use the vibrating machinery the doctor has no right to have an office free of vibrations, and if the doctor has a right to an office free of vibrations the confectioner has no right to use the machinery. The issue, then, is who should have the right.

Coase argues that if transaction costs are low, resource allocation will be the same regardless of who wins the case. Transaction costs include anything that impedes the undertaking of a mutually advantageous exchange. Assume, to make a concrete example of the case, that the doctor can make $100 an hour using the examination room while the confectioner can make $80 an hour using the vibrating machinery. For simplicity, assume away any intermediate solutions, such as the doctor using the examination room in the morning and the confectioner using the machinery in the afternoon, or buying soundproofing to stop the vibrations. If the doctor wins the case, he will be able to use the examination room and the confectioner will have to stop using the vibrating machinery. But, if the confectioner wins the case, Coase observes, the right is worth more to the doctor, so the doctor could offer to pay the confectioner $90 an hour to forego using the machinery. The confectioner will be better off getting $90 an hour rather than $80, and the doctor will earn $100, leaving him $10 an hour better off than not using the examination room. Of course, if the numbers were reversed in this example, the use of the machinery would be worth more to the confectioner, and the confectioner would buy the rights from the doctor if the confectioner lost the case.

Coase's article offers many interesting legal cases illustrating the same essential point: if there are no transaction costs – that is, no impediments to mutually advantageous exchange – resources will go to those who value them the most, because they will purchase them from those who place a lower value on them. Coase is not implying that it does not matter who wins the case. The right is valuable to both parties, so both of them want to own that valuable right. However, regardless of whom the court assigns the right to, if transaction costs are low the person who places the highest value on it will buy it.

Coase also is not implying that transaction costs are low, or that resources are allocated to their highest-valued uses. Rather, by illustrating what they would be with zero transaction costs, he is identifying transaction costs as being the impediment to mutually beneficial exchange. Coase is illustrating that in the absence of transaction costs resources will flow toward those who value them the most, but the more significant implication is that because transaction costs exist in the real world, some mutually beneficial exchanges that could be made in the absence of transaction costs will not be made because transaction costs stand in the way. Transaction cost economics looks for ways in which transaction costs can be reduced to facilitate mutually advantageous exchange.

The transaction-cost framework for analyzing exchange lies squarely within the market process approach to economic analysis. The object of analysis is the exchange, or an exchange that hypothetically could be made if transaction costs did not stand in the way. If there is a way to lower the transaction cost, transacting parties can benefit. Even when exchanges are made, transaction costs are borne by the parties to the exchange.

For example, to buy groceries at a grocery store requires the purchaser to have the purchases checked out by the store so their prices can be added up to determine how much the purchaser will pay. At one time this was done by attaching prices to all the items and having a checker read the prices and key them into a cash register which would total the amount owed. The cost of putting prices on items, of having the checker read them and key them in, and the possibility that an error would be made in the process, are all transaction costs that make the cost to the purchaser higher than the return to the seller. Now it is common for goods to have barcodes that are scanned at checkout so they do not have to be individually priced, and so the checker does not have to read the prices and key them in, which lowers the transaction cost. Some stores are moving toward having customers scan their own purchases, eliminating the need to employ a checker, further lowering transaction costs. The reduction in transaction costs means higher benefits to the transacting parties.

Sometimes transaction costs are so high that they prevent an exchange that would be mutually beneficial in the absence of transaction costs. Transaction costs are real costs, of course, so if they are larger than the combined gains from trade to the buyer and the seller the transaction is not economically worthwhile. However, if the transaction costs can be lowered, a mutually beneficial exchange can take place, enhancing the well-being of both parties.

In all of this analysis what is being analyzed is the process of exchange. There is no reference to equilibrium, and no implication that the exchange is leading to an equilibrium. The market process is being analyzed, not the outcome. One might conjecture that if all mutually beneficial exchanges that could be made were made, the market would arrive at some type of equilibrium, but that really goes beyond analyzing the process of exchange that is the focus of transaction-cost economics. Some exchanges might disrupt a stable pattern of economic activity, as, for example, when a new shopping-center is built that competes with long-established local retailers, or when a manufacturer introduces a new good that reduces the demand for goods of pre-existing competitors. But, these conjectures are not relevant to transaction cost economics and go beyond its scope. Such conjectures push to make transaction cost economics consistent with equilibrium economics -and it may well be. But the point is that whether or not it is consistent with equilibrium economics, it is something different. It is an analysis of the market process, and whether there is some stable outcome at the other end goes beyond the analyzing of the transaction.

The new institutional economics

Transaction-cost economics is a component of the new institutional economics. One might think of institutions, following Douglass North, as "the humanly devised constraints that shape human interactions."[7] This definition incorporates a lot of what constitutes the framework for human interactions. Institutions might be partitioned into those that are purposefully designed versus those that arise spontaneously, and might be partitioned into the way that behavior within those institutions is enforced.[8] For example, when doing business, shaking hands with those involved in a transaction would be an institution that arose spontaneously (but, in Japan, bowing might be more appropriate than shaking hands), whereas a formal contract would be purposefully designed, and supported by a long history of purposefully designed contract law. Of course, contract law can also evolve "as a result of human action but not of human design,"9 as legal decisions in specific cases influence future contracts in ways not foreseen when those decisions are made. Enforcement of institutions might be first-party enforcement (I should act this way in this particular situation), second-party enforcement (if I do not fulfill my part of the bargain, the individual I am working with might not be willing to deal with me again),[10] or third-party enforcement (if I do not fulfill my part of the bargain, I could be sued and ordered by the court to pay damages).

Institutions shape all human interaction, and as a subset, all economic activity. They may be basic institutions, as when someone stops by a roadside vendor and tenders money in exchange for goods. Even here there are complex institutions at work, starting with the medium of exchange that the vendor is willing to accept in exchange for the goods, the amount of bargaining that might go on regarding the terms of the exchange, and the degree to which parties can trust each other not to misrepresent what is being exchanged. (Is the money counterfeit; are the apples at the bottom of the bag the same quality as those visible at the top?) For more involved transactions, such as a firm contracting a construction company to build a factory, the institutions are extremely complex, ranging from financing institutions, institutions to determine title to the land on which the factory is being built, and so forth. Consider the impact of only one institution -the limited liability corporation – on economic progress and prosperity. By limiting investors' liability to the amount they have invested in the firm, financing opportunities are opened up, but perhaps there are also increased opportunities for those who run corporations to behave irresponsibly or opportunistically.

In neoclassical equilibrium models, institutions are implied in the framework but except in rare instances are not explicitly incorporated into the models. For example, neoclassical models typically assume that the only way individuals can acquire resources is through production and exchange. Obtaining resources through theft or fraud is ruled out by assumption, but there is no explicit modeling of the institutions that prevent these real-world activities from happening, or from becoming more prevalent. Similarly, the transaction costs that create impediments to mutually advantageous exchanges tend to be binary in models:

either an exchange can take place and is modeled as having no transaction costs, or the transaction cost is an impediment to exchange that prevents the exchange from occurring. In general equilibrium models there must be an implied institutional framework that facilitates exchange and that deters theft and fraud, but except in rare instances those institutions remain outside the scope of analysis.

Oliver Williamson provides a nice taxonomy of the new institutional economics, which he also refers to as contractual economics, and differentiates it from what he refers to as non-contractual, or technological economics.[11] Non-contractual economics includes neoclassical economics, and a host of other economic theories including evolutionary economics, X-efficiency and psychological theories. Williamson's view of institutional economics is that it examines the interactions among individuals who want to accomplish various ends. Transaction cost economics is the part of institutional economics that deals with incomplete arrangements. Individuals look for ways to lower transaction costs to facilitate exchange. Agency theory deals with complete arrangements, as for example when parties try to write contracts that contain incentives that work to further the interests of all parties. Public choice, constitutional economics, and property rights theory deal with the environment within which people interact. Property rights define the rights of individuals and public choice theory describes how people act collectively to achieve ends they might not be able attain to when acting as individuals.

Williamson's taxonomy thus divides institutions into those that deal with the environment and those that deal with arrangements individuals make. Under the environment Williamson includes law and government (probably informal institutions should also be included). The environment applies to everyone and cannot be escaped, except perhaps by explicit agreement among individuals in specific interactions. Under arrangements Williamson includes those institutions that individuals work out among themselves -contracts and agreements – that do not necessarily apply to others who are not party to those contracts and agreements.

The emphasis in institutional economics is always on how people interact to achieve their ends, and not on an end state that may lie at the conclusion of those interactions. Whether such an end state exists is not directly relevant to the institutional approach to economics, which examines the process by which people interact and not some equilibrium that may or may not result. In this way, institutional economics takes a market process approach to economic activity rather than an equilibrium approach.

Does the market process lead to equilibrium?

Economic analysis is so thoroughly steeped in equilibrium thinking that it is tempting to think that the market process leads to an equilibrium. Consider again the transaction cost framework discussed earlier. Using Coase's example, one can see that if transaction costs are absent, the doctor and the confectioner can bargain to an outcome that allocates the resources at their disposal to their highest-valued uses. It is tempting to think that if all exchanges take place in a mutually advantageous manner, then all resources will be allocated to their highest-valued uses, and the economy will find itself at a Pareto optimal equilibrium. This idea, going back at least to Leon Walras,[12] and extended by Arrow and Debreu,[13] is true under some assumed circumstances, but is it true in an actual market economy?

When one looks at the process, an exchange made by one person affects the exchange opportunities available for others. To take some dramatic examples from the computer industry, the development of the microprocessor opened the opportunity to produce microcomputers, and the development of the graphical user interface created a market for computer mice and other pointing devices. As exchanges take place over time, innovations by some create opportunities for others, so the economy never comes to an equilibrium. The economic environment is always changing, and in the evolving economy new exchange opportunities are constantly appearing, and when those exchanges take place they modify the environment so that exchange opportunities not previously available are created. If one thinks of an equilibrium, as in physical systems, as a steady-state environment toward which the system tends to move, and as a condition that, if disturbed the system tends to return to, there is no equilibrium in economic systems analogous to equilibria in physical systems. The economy is continually evolving, never returning to its earlier state. The transactions that occur at one point in time modify the opportunities that individuals will have to make future transactions.

One might extend the equilibrium notion to think of an equilibrium growth path of an economy, but this implies deterministic forces that lead the economy in some determinate direction. This more complex notion of equilibrium will be considered further in the following chapters, but it does not hold up if the decisions of some economic actors in the present alter the future course of the economy. When one simply looks at the course of economic activity, it appears that market economies are characterized more accurately as exhibiting continual economic progress rather than moving toward equilibrium.

Information and uncertainty

The new institutional economics provides an example of a market process approach rather than an equilibrium approach to economic analysis, but it too has tended to look at issues related to static inefficiencies rather than the process that generates economic progress. Reducing transaction costs can facilitate exchanges, enhancing people's well-being, but the story stops once the exchange is enabled. Similarly, agency theory looks at principal–agent problems that can stand in the way of mutually advantageous exchanges,[14] but once these agency problems are overcome, the story ends. This is not a problem if one simply wants to analyze how institutions might be modified to enhance efficiency, or how transaction costs stand in the way of an improved allocation of resources. When analyzing what produces prosperity, however, the issues analyzed by the new institutional economics are the beginning of the story, and the market process builds on any institutional enhancements to enable economic progress.

When one looks at how to overcome an agency problem, or lower a transaction cost, the analysis presents a clear depiction of the costs and benefits. If the doctor and the confectioner in Coase's example can come to terms, those resources will be allocated to their highest-valued uses. This analysis provides insight in cases where the highest-valued uses of resources are apparent, but in most real-world instances this will not be the case. As James Buchanan has pointed out, when one chooses one option, there is no good way to assess what would have happened had another option been chosen.[15] When the doctor–confectioner example was used above, for sake of illustration the doctor was assumed to be able to earn $100 a day from the use of the examination room while the confectioner was assumed to be able to earn $80 a day from using his vibrating machinery. In the real world, while the parties might make estimates of future earnings, there would be no way to know whether the estimates would are accurate.

The confectioner may have had a history of $80 a day earnings using that machinery, but there is no way to know that those earnings would continue if the confectioner kept using the machinery as before. There could be an increase or decrease in demand for the confectionary, affecting those earnings in the future. Similarly, there could be a change in the confectioner's costs – sugar prices might rise, for example – which would lower the amount the confectioner could earn using the machinery. So, those numbers used for sake of illustration earlier must be viewed as purely hypothetical, and an estimate of what the confectioner believes he can earn.

The case would be even more speculative in the case of the doctor, who has built the examination room but has been unable to use it because of the vibrations from the confectioner's machinery. The doctor could estimate an additional $100 in earnings, but this would be purely speculative, because the doctor has no experience using the room. Perhaps once the room entered use, the doctor might discover that it added much less to his income. Perhaps it would add more. The doctor may have good estimates, but no way to know for sure.

More to the point, when considering the Coase theorem, assume the doctor wins (or purchases, should he lose the case) the right to use the examination room, and does indeed clear an extra $100 a day from using it. Does that mean that resources were allocated to their highest-valued use in this case? One can only speculate. The example assumed the confectioner would have earned $80 a day, but if the confectioner is prevented from using his machinery, there is no way to tell what the confectioner actually would have earned. Perhaps due to some change in either supply or demand conditions the confectioner would really have earned $120 a day, but there is no way to know, because the confectioner is not actually using the machinery. The doctor ended up with the right because the doctor's estimates of the earnings that could come from using the examination room were above the confectioner's estimates of the earnings that could be generated from using the vibrating machinery. But, once one option is taken, there is no way to know for sure what the other option would have generated had it been taken. And, one cannot rule out the possibility that the doctor obtained the right only because the doctor was less risk-averse than the confectioner.

The market acts as a discovery process to reveal the likely results of various actions. If the confectioner ceases his production in this case, there is no way to know what earnings he might have generated had he kept using his machinery. However, some indication can be revealed if confectioners in other locations are using similar processes to produce similar confections. If confectioners in other locations realize rising profits, this confectioner may try to buy out the doctor to try to replicate the success other confectioners are having. The success of McDonald's in the 1950s and 1960s, for example, resulted not only in the expansion of that company but encouraged others to enter the fast-food business to compete with McDonald's for some of those profits. Similarly, the success of Starbucks in the 1990s resulted not only in the expansion of that company, but in other companies (including McDonald's) entering the boutique coffee market.

The market reveals information about what might be profitable, but one can never know, even after the fact, if profits are being "maximized," in the sense that there would not have been another course of action that would have been even more profitable. The success of one business can never be duplicated exactly. A store in one location will face a different market than an "identical" store in a different location, and the personnel who work in one location will be different in difficult-to-observe ways from the personnel in another. What makes one store more successful than another? A better location? A better manager? There are, of course, a host ofother factors that also weigh in.

Markets are always changing, and firms must continue to evolve to keep up with changes in the market, or be left behind by competitors. This has implications for firms that are explored in more detail in Chapter 5.

The Coase theorem

Because the new institutional economics is used as an example of a market process approach rather than an equilibrium approach, and because Coase's article on social cost was used as a specific example of a market process approach to economics, it is worth also considering the Coase theorem in that context. While Coase's article provides a good example, a reader of the article will notice that it contains no theorems. The Coase theorem was coined by George Stigler to describe implications he (and others) drew from that article. The Coase theorem might be stated, "In the absence of transaction costs, the allocation of resources is independent of the assignment of property rights," or more normatively, "In the absence of transaction costs resources are allocated to their highest-valued use." From the examples above, the reader can see that this follows because if there are no impediments to mutually advantageous transactions, the party who values a good or a right the most will buy it.

If a transaction cost is defined as anything that prevents a mutually advantageous exchange from taking place, then the theorem is tautologically true. But, consider things that might prevent a mutually advantageous exchange from taking place. One could be loss aversion.[16] Experimental evidence appears to show that people tend to feel a bigger utility loss from a given dollar loss than the utility gain they would get from the same gain. Thus, people are reluctant to take fair bets when those bets can affect their material well-being. For example, most individuals would not gamble $1,000 on the flip of a coin. Consider again the doctor and the confectioner. If both behave this way, then it is possible that if the doctor gets the right the confectioner will be reluctant to buy it, even with a feeling it would likely be profitable, and if the confectioner gets the right the doctor would be reluctant to buy it, again even with a feeling it would likely be profitable.

One might tautologically say that loss aversion constitutes a transaction cost, but even with no external impediments to exchange – no institutional impediments – resources would be impeded from flowing to their highest-valued uses simply because of the way individuals weigh the costs and benefits of the choices they make.

Setting aside many possible complications, how would one ever know if the allocation of resources is independent of the assignment of property rights? There would be no way to tell, because if property rights are assigned one way, they are not assigned in an alternative way, and there is no way to be sure that the allocation of resources is the same under the current assignment of property rights as it would be under some hypothetical alternative. Similarly, how could one ever know if resources are allocated to their highest-valued uses? One can only observe the actual use of resources, and there is no way to know that some alternative allocation would or would not produce a higher value.

Think about this in terms of the Coase theorem. Actually, Coase's article is showing that if impediments to mutually advantageous exchange are removed, people can make the exchange and all parties to a voluntary exchange will view themselves as being better off, revealed by their consent to the exchange. This is an observation about the market process. The Coase theorem attempts to extend this observation about the market process and apply it to describe a market equilibrium. If all transaction costs were eliminated, the allocation of resources would be independent of the assignment of property rights and resources would be allocated to their highest-valued uses. This extension of Coase's observation about the market process to the characteristics of some end state does not necessarily follow. Transaction cost economics analyzes the market process, but the Coase theorem is an attempt to extend a market process analysis to draw conclusions about a market equilibrium.

Value and exchange

In an insightful article with the same title as this section head, Meir Kohn describes the difference between the equilibrium approach to economics, which he calls the value paradigm, and the process approach, which he refers to as the exchange paradigm.[17] Kohn argues that in the second half of the twentieth century economic theory developed based on the equilibrium approach laid out in mathematical form by John R. Hicks in Value and Capital,[18] and by Paul Samuelson in Foundations of Economic Analysis,[19] to the point where "adherents of the Hicks–Samuelson research program came to see the theory of value as being economics: they saw the two as identical and indistinguishable."[20] Kohn notes that the proponents of equilibrium economics have recognized its shortcomings and have tried to patch things up, and that other economists have taken different approaches. Two alternative approaches he notes are empirical – the use of applied econometrics to understand the underlying relationships in the economy – and institutional economics that focuses on the market process rather than on equilibrium end states.

The equilibrium approach to economics assumes an equilibrium, and "To a large extent, everything else is just a corollary of this fundamental assumption."[21] The discussion of the Coase theorem in the previous section shows how it is possible to combine the market process and equilibrium approaches to economics, but Kohn argues that this leads to confusion because the equilibrium approach to economics eliminates any role for institutions. In the real world – and in the market process approach to economics – individuals act and make choices. In the equilibrium approach individuals are assumed to be maximizers, and their actions are dictated by the characteristics of the equilibrium rather than the other way around. The equilibrium approach requires sufficient assumptions about individual behavior that if even some people deviate the equilibrium no longer exists. Institutions cannot influence people's choices in such a setting.

Kohn characterizes the equilibrium approach to economics as a top-down approach, in that everything in the theory is dictated by the nature of the resulting equilibrium, whereas the process approach is a bottom-up approach, where outcomes are the result of individual behavior. "Exchange opportunities are not given, but must be found or created. Prices are not provided by magic, but must be set by someone. Exchange involves interaction not with an impersonal market but with other individuals."[22] In such a setting, individuals can act heterogeneously. They may act with imperfect information so that resources do not always flow to their highest-valued uses. Under the same setting, with imperfect information, different individuals might make different choices. Contracts are not always honored; promises not always kept. In the market process approach people look for ways to reduce transaction costs and make exchanges that are too costly under the status quo. In the equilibrium approach, either the exchange is made in equilibrium or it is too costly and is not made.

The key distinction between the approaches is that there is no requirement for an equilibrium in the market process approach. People engage in production and exchange to try and enhance their well-being, but whether this leads to an equilibrium is an open question that the market process approach does not have to answer. The equilibrium approach starts with an equilibrium outcome as a benchmark, and draws inferences from the nature of the assumed equilibrium.

Kohn sees merit in using the equilibrium approach to understand how relative prices are determined. As noted in Chapter 2, all models make simplifying assumptions to focus in on particular phenomena the model is trying to depict, and equilibrium models do a good job of explaining how prices are determined in markets. Indeed, Kohn calls this the value paradigm precisely because it does a good job of explaining how markets reveal the value of goods and services.[23] The problem lies in extending that framework to apply it to aspects of economics in which the underlying causal factors are assumed away.

One key difference in the two approaches is the scope of the analysis. In the equilibrium approach, conclusions require assumptions about every individual and everyfirm in the economy. In the case of general equilibrium, all economic actors are explicitly included in the model; in the case of partial equilibrium an assumption is made that those outside the model have a negligible impact on those included in the model. The market process approach focuses on interactions among individuals and does not need to consider whether a market, or the entire economy, is in equilibrium. Indeed, unlike in general equilibrium models, in the real world many exchanges may be "disequilibrium" exchanges. The degree to which disequilibrium exchanges occur in the real world is relevant to equilibrium models, but is of little concern to the market process approach to economics.

Behavioral foundations of economics

The equilibrium framework for economic analysis makes strong assumptions about the behavior of individuals in an economy. Individuals are assumed to be rational utility maximizers, where rationality is described by a list of behavioral assumptions. More is preferred to less, preferences are transitive, and individuals exhibit diminishing marginal rates of substitution in consumption.[24] Preferences are assumed to be stable over time, although in static models time is one of the factors that is assumed away. Among the many assumptions that must be met for the assumed equilibrium to actually be realized, these assumptions about the behavior of individuals are crucial. The model's equilibrium is dependent upon all individuals conforming to these assumptions of utility maximization.

The development of behavioral and experimental economics in the last half of the twentieth century calls into question these neoclassical assumptions of utility maximization. Following Herbert Simon's framework of bounded rationality,[25] behavioral economics has performed experiments on individuals that often demonstrate systematic deviations from the neoclassical assumptions of utility maximization.[26] Vernon Smith has echoed similar sentiments about the role of experiments in economics to discover behavioral patterns that may be at odds with the neoclassical assumptions.[27] Because equilibrium models assume this utility maximizing behavior, the foundations of those models crumble if real-world behavior does not correspond to the assumptions that drive the models.

George Ackerlof argues that if one looks at actual economic behavior, a number of results that are commonly accepted in macroeconomics in the early twenty-first century disappear.[28] He takes issue with: the conclusion that consumption depends on wealth, not income; the Modigliani-Miller theorem; the concept of the natural rate of unemployment; rational expectations; and the equivalence of tax versus debt financing of government expenditures. Ackerlof argues that those conclusions of macroeconomics are not supported by macroeconomic data, and that a behavioral approach to economics explains why. Ackerlof substitutes one model of economic behavior for another, but his message is that if the behavioral assumptions in the new classical macroeconomics are replaced by more realistic assumptions about behavior, the analysis arrives at different conclusions.

In some cases the neoclassical assumptions might be descriptive, and one might, following Milton Friedman,[29] argue that the neoclassical behavioral assumptions are close enough that the models work as if people are neoclassical utility maximizers. But, as discussed in Chapter 2, unrealistic assumptions in economics raise a host of other issues. The models may work well for some purposes (for example, describing how prices and quantities are determined in the market) yet not work well for others (for example, understanding the causes of economic progress and prosperity).

The market process approach to economics requires much less stringent behavioral assumptions than the equilibrium approach. All that is required is that people act to further what they perceive to be their own interests. As Ludwig von Mises describes it:

Human action is necessarily always rational. . . . When applied to the ultimate ends of action, the terms rational and irrational are meaningless. The ultimate end of action is always the satisfaction of some desires of the acting man. Since nobody is in a position to substitute his own value judgments for those of the acting individual, it is vain to pass judgment on other people's aims and volitions. No man is qualified to declare what would make another man happier or less discontented.[30]

Mises goes on to say:

It is a fact that human reason is not infallible and that man very often errs in selecting and applying means. An action unsuited to the end falls short of expectation. It is contrary to purpose, but it is rational, i.e., the outcome of a reasonable – although faulty – deliberation and attempt -although an ineffectual attempt – to attain a definite goal.[31]

Rather than making extensive assumptions about individual behavior, the market process approach only requires the simple assumption that people act to further their interests. Any action is utility maximizing in the view of the actor, because the action would not take place if the actor was not trying to accomplish some goal by undertaking the action. The market process approach is much less demanding in the behavior that it assumes about individuals for the analysis to be descriptive of real-world economic processes.

Conclusion

Formal economic analysis, from the beginning introductory courses in economics up through graduate training and professional research, starts with an equilibrium approach to economic analysis. The equilibrium approach is so ingrained that it can be difficult for those trained in economics to escape, and it can be difficult to see that the market process approach is fundamentally different. Much economic analysis is based on the market process approach, including the new institutional economics, transaction cost economics, agency theory, much of law and economics, a substantial amount of contemporary growth theory (to be discussed in more detail in Chapter 6), and the Austrian school approach to economic analysis. Yet for economists steeped in the equilibrium tradition, it is often difficult to see beyond it.

Consider the section on the Coase theorem in this chapter. Coase was talking about the market process, and the way in whcih transaction costs stand in the way of people's abilities to make mutually advantageous exchanges. Yet Coase's insights on the market process are often extended to the Coase theorem's conclusions about some end state toward which that process leads. The Coase theorem does not necessarily follow from Coase's analysis of transaction costs, but this may be difficult to see for economists who want to place all economic analysis within the context of equilibrium.

4 Market clearing forces in the economy

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Chapter 3 found much to criticize in the equilibrium approach to economic analysis, but did note that it offers substantial insight into the way that prices and quantities are determined in a market economy. This chapter begins by looking into the determination of prices and quantities as a foundation not for discovering the properties of equilibrium in an economy, but as a foundation for understanding the market process.

Market clearing and equilibrium are not the same things, although they have been depicted as such at least since Adam Smith described the process in The Wealth of Nations. An equilibrium implies that if that equilibrium is disturbed, there are forces that re-equilibrate and bring the system back to its former equilibrium state. That notion has some merit when examining the market clearing forces in an economy, but has little merit if it is extended to an examination of the forces that produce economic progress and prosperity. The economic forces that work to clear markets are not the same forces that work to generate progress.

Adam Smith's discussion of market clearing

Adam Smith, often referred to as the father of economics, discusses market clearing in a manner that conforms closely to the way it is treated in modern economics. Smith begins by introducing the concept of natural price, which was a common concept in Smith'sday:

When the price of any commodity is neither more nor less than what is sufficient to pay the rent of the land, the wages of the labour, and the profits of the stock employed in raising, preparing, and bringing it to market, according to their natural rates, the commodity is then sold for what may be called its natural price.[1]

The natural price, then, is the price that just covers the cost of production, and corresponds to Alfred Marshall's competitive equilibrium. After discussing the influence of time on prices, Marshall notes, "In a stationary state then the plain rule would be that cost of production determines value."[2] And, noting the existence of inframarginal firms:

When different producers have different advantages for producing a thing, its price must be sufficient to cover the expenses of production of those producers who have no special and exceptional facilities ... When the market is in equilibrium, and the thing is being sold at a price which covers these expenses, there remains a surplus beyond their expenses for those who have the assistance of any exceptional advantages.[3]

So, Smith's natural price is the price that just covers the cost of production, which corresponds to Marshall's equilibrium price, which covers the cost of production for the marginal firm. Smith then goes on to say, "When the quantity of any commodity which is brought to market falls short of the effectual demand,... the market price will rise more or less above the natural price,"[4] and "When the quantity brought to market exceeds the effectual demand, . . . The market price will sink more or less below the natural price ... The natural price, therefore, is the central price, to which the prices of all commodities are continually gravitating."[5]

Marshall notes:

When demand and supply are in stable equilibrium, if any accident should move the scale of production from its equilibrium position, there will be instantly brought into play forces tending to push it back to that position; just as, if a stone hanging by a string is displaced from its equilibrium position, the force of gravity will at once tend to bring it back to its equilibrium position.[6]

If one reads Marshall, it is clear that he is concerned about disequilibrium phenomena, and about changes that occur over time. However, the part of

Marshall that has survived in contemporary economics is the partial equilibrium framework that Marshall so elegantly developed, and Marshall's notion of equilibrium is captured in the pendulum analogy just quoted.

What one sees from these passages is that Adam Smith had a notion of an equilibrium price, called the natural price, which is the price that just covers the cost of production. While one should be cautious about interpreting older scholars through the lens of contemporary economics, Smith's natural price seems consistent with Marshall's equilibrium price, and Marshall's example of the stone clearly shows the idea that if a market is disturbed from an equilibrium position, market forces will work to return it to its equilibrium state.

What Smith describes is how changes in supply and demand can disrupt a market, and the way that the market responds by temporarily moving the price away from the natural price to clear the market. Smith describes, in other words, the market clearing function of prices, even while noting that market forces tend to pull prices back to their natural prices. Marshall adds some sophistication to this, describing four periods of production. In the market period the supply of a good is fixed and prices adjust to clear the market. In the short period the quantity supplied can be adjusted within the constraints of a given "stock of plant," and in the long period all factors are variable. Marshall also notes a longer secular period, "caused by the gradual growth of knowledge, of population and of capital," which has not been clearly characterized in the neoclassical framework.[7]

The point of this digression into the history of economics is to note that the concepts of market clearing and equilibrium have always been associated with each other in economic analysis, going back at least to Adam Smith. Yet the two are not necessarily the same. If a stable current state of affairs is disrupted by some factor that will recede so that underlying conditions will return to what existed prior to the disruption, one can expect a return to the earlier stable state of affairs. If the current state of affairs is disrupted by some factor that permanently changes the underlying conditions, there will be no return to the earlier state of affairs. New conditions will produce a new outcome, meaning that the old stable state of affairs cannot be thought of as an equilibrium.

Market clearing versus equilibrium

Imagine for example an abnormally cold winter that produces a freeze damaging a substantial portion of the Florida orange crop. This might be depicted as a shift inward in the supply curve for oranges, raising the price and reducing the quantity exchanged. If the weather returns to normal the next year, the market reproduces its former "equilibrium." The market system has worked to temporarily increase the price to clear the market. Now imagine the situation in the early decades of the twentieth century, when technological advances introduced an increasing number of lower-priced automobiles into the market, causing a reduction in the demand for horses and horse-drawn wagons. Those markets were disrupted, much like the orange market in the previous example, but by factors that changed market conditions forever. Those markets would never return to their previous states, and it would be misleading to think of those previous states as any sort of equilibrium.

While superficially these two examples have some similarities, they are fundamentally different, and illustrate how economic analysis can be brought to bear to answer different types of questions. In the case of the orange harvest, the market responds to a temporary disruption with price adjustments that keep the quantity supplied equal to the quantity demanded. This is the market clearing function of the economy. In the second example, prices will also work to clear the market, but the example itself is fundamentally different because it is an example of an irreversible change. The market will never go back to its previous state, so it would be incorrect to think of its previous state as an equilibrium.

The nature of economic progress lies in the fact that irreversible changes continue to occur in an economy, and the economy progresses, never returning to its former state. It is possible that as progress takes place temporary disruptions will also occur, and the market clearing forces of the economy will come into play. It is also the case that as permanent changes take place in an economy the same prices that reflect the new underlying conditions will be working to clear markets. So, economic forces are always working to clear markets, and a market economy cannot be properly understood without understanding how economic forces clear markets. If one wants to understand how markets clear, equilibrium models work well. As Chapter 2 noted, different models are appropriate for different purposes. If one wants to understand economic progress and prosperity, equilibrium models do not work well, because economic progress is not an equilibrium phenomenon.

The examples of disruptions in the markets for oranges and automobiles were introduced to illustrate that while some disruptions in markets are temporary and will be reversed, others are permanent and the previous underlying conditions will never again exist. The dichotomy here is somewhat artificial in that permanent changes are always occurring in all markets. The hypothetical disruption described in the orange market was deliberately designed as temporary and reversible, but in fact, even as the weather has a short-run and temporary effect on the market, new harvesting methods are being developed, new fertilizers are being designed, and through grafting, selective breeding, and genetic engineering the productivity of the orange trees themselves are being improved. So, while some changes of a temporary nature do occur, other changes of a permanent nature are occurring at the same time.

The changes considered above are all on the supply side of the market, but changes occuron the demand side too, as goods may come in and out of fashion. Orange juice is relatively high in carbohydrates, so when low-carbohydrate diets came into fashion the demand for orange juice fell. Similarly, transportation costs can affect the demand for goods in locations that are far away from where the goods are produced.

At any point in time, supply and demand factors are constantly changing, sometimes in ways that will reverse themselves in the future and sometimes in ways that will permanently affect the market. As a result, when looking at the way that the forces of supply and demand determine prices and quantities in markets, it is more accurate to think of the resulting prices as the prices that clear the market, so that the quantity supplied equals the quantity demanded. Only in particular unrealistic cases will these be equilibrium prices and quantities, in the sense that if they are disturbed the market will tend to return to those prices and quantities. As economic progress occurs permanent changes affect all markets, and markets do not return to their former states, so they are not equilibrium prices and quantities.

In equilibrium models prices and quantities tend to return to their equilibrium values because the models assume that the underlying conditions do not change; that is, they assume that there is no economic progress. One might then refer to these prices and quantities as equilibrium values under those assumptions. But note that to do so assumes away fundamental aspects of the market process. Rather than referring to those prices and quantities as equilibrium values under certain assumptions, it is more accurate to think of the prices as market clearing prices, and the quantities as the resulting quantities that equate the quantity supplied with the quantity demanded at that point in time.

Natural price and the subjective nature of cost

While Adam Smith did not actually use the concept of equilibrium, his concept of natural price lays a foundation for thinking of markets in this way, and there is a clear similarity between Smith's natural price and Marshall's equilibrium price. As Smith framed it, the natural price of a good is the price that just covers the cost of production. However, this formulation takes the cost of production to be an objective point of reference. If one knows the cost of production, one knows the good's natural price. In fact, causality runs in the other direction. A good's price determines its cost of production, not the other way around.

Inputs into the production process do not have intrinsic value, but are valued because they produce final products that are valued by consumers. If consumers place a high value on the final goods, that in turn makes the inputs that produce the goods valuable. If for some reason the value of a final good decreases – for example, if computers displace typewriters – then the resources that produce the less-valuable final goods will become less valuable as a result. Those inputs will retain value to the extent that they can be used to produce other goods, but to consider the example just given, the decline in the value of typewriters will result in a decline in the value of the resources used to produce typewriters, signaling to those resources that they should shift to the production of other goods if feasible. Resources that can be used only for the production of typewriters, such as the molds that are used to form the typebars that strike the typewriter ribbon, may become completely worthless.

William Stanley Jevons makes this point with regard to the value of different types of labor, saying, “I hold labour to be essential variable,so that its value must be determined by the value of the produce, not the value of the produce by that of the labour. I hold it to be impossible to compare a priori the productive powers of a nawy, a carpenter, an iron-puddler, a schoolmaster, and a barrister.”[8]

Carl Menger makes the same point, referring to first-order goods as consumption goods and higher-order goods as inputs into the production process. Menger notes that the value of first-order goods is determined by the value consumers place on them, and that higher-order goods derive their value from the value of what they produce. He says:

The law that the goods-character of the corresponding goods of lower order in whose production they serve must not be regarded as a modification affecting the substance of the primary principle, but merely as a restatement of that principle in a more concrete form.[9]

Smith refers to the cost of production as if it is objectively given, and then determines the natural price of goods on the market. Yet it is the market value of the final goods that determines the value of the inputs that produce them, and therefore the cost of production. Consider the incomes paid to professional athletes, for example. Football and basketball players make huge salaries because sports fans place a high value on watching the best athletes compete. Salaries for sports stars in the early twenty-first century are substantially higher than they were in the 1950s, because back then there was little television coverage of the games. The games had a market value only to those who bought tickets and attended in person. With the advent of widespread television coverage, the games – the output – became more valuable because more people could watch, and sponsors became willing to pay substantial amounts to advertise during the televised games. As the output became more valuable the inputs became more valuable, and the salaries of sports stars rose.

The value of the inputs is determined by the value of the output they produce, not the other way around. Why is the market value of a professional basketball player higher than the market value of a professional lacrosse player? Because people place a higher value on watching basketball than on watching lacrosse. If those preferences were reversed, lacrosse players would earn more and basketball players would earn less.

The value of the output is subjective. Some consumers may value a good more than others, and consumers in general may change their subjective evaluations of goods. For example, convertible automobiles tend to be popular body styles, but partly due to safety concerns their popularity waned in the 1970s in the United States, to the point where Cadillac marketed its 1976 Eldorado as “The last convertible in America.” The popularity of convertibles has risen since then, and 40 percent of Volkswagen Beetles in 2003 were convertibles.[10] Similarly, shark fin soup is a delicacy in the orient, but has little popularity with consumers in the United States. The subjective value of a good can vary over time, as the example of convertible automobiles illustrates, and can vary among different consumers and markets at the same time, as the example of shark fin soup illustrates. As the subjective value of output varies, so will the subjective value of the inputs that produce the output.

If inputs can easily be shifted to produce more valuable goods, they may be insulated from large changes in their market values. For example, automobile factories that make convertibles may be shifted toward producing hard-top automobiles. However, when rising gasoline prices lowered the demand for large sport utility vehicles (SUVs) in 2008, General Motors responded by closing plants rather than converting them to produce more fuel-efficient automobiles.[11] A once-valuable automobile factory that used to produce SUVs found its value reduced to approximately the value of the land it was built on, because the value of its output fell reducing the value of the factory that produced that output. In contrast, the increased popularity of shark fin soup, especially in China, has increased the value of one of its inputs – sharks – to the point that there is concern that shark populations are threatened by overfishing.[12] SUV factories become worthless because the value of their output to consumers declines, while sharks increase in value because they are an input into shark fin soup, which has increased in value. The value of inputs is determined by the value of the outputs they produce. Because the value of the final goods is determined subjectively by consumers, the value of the inputs is also determined subjectively, and therefore costs are subjective.

The forces of supply and demand clear markets

Imagine that a grocery store has a stock of perishable fresh peaches, and that at its normal rate of sales the produce manager determines that half the peaches will be unsold at current sales rates. The quantity the store has on hand exceeds the quantity consumers will demand at the price the store is currently charging, according to the manager's projections. The produce manager will likely mark the peaches down in price, and the lower price will increase the quantity demanded so that the peaches will sell before they spoil and their value is reduced to zero. One does not need to use much imagination for this example, because one can observe this type of activity at grocery stores on a regular basis. When the supply of a good is very inelastic, as in the grocery store with an inventory of perishable fruit, sellers will adjust their price so they can sell their goods rather than have them spoil and lose their value altogether.

Supply may be less inelastic if goods are more durable, but even here if demand falls off sellers are likely to mark their prices down to move excess inventory. As summer wears on, stores mark down their swimwear to make room for warmer clothing, even though it would be possible to store an inventory of swimwear until the following summer. Both of these cases illustrate how suppliers adjust their prices to clear markets.

In the Marshallian model of competitive markets, competitive firms are price takers that have to accept the market price. But as in the examples above, somebody has to determine the price of goods. They are not just given. Typically sellers set a price, and in some markets that price is subject to negotiation as buyers attempt to negotiate to get a better price. Nevertheless, the forces of supply and demand guide both buyers and sellers, who adjust the quantities they supply and demand based on prices for the same or similar goods for sale nearby; that is, based on the forces of supply and demand. But, somebody has to set the price, and sellers typically take the lead because they deal in a larger volume of the good than buyers, and because they need to manage their inventory. When inventory shades toward the low side, they will be in a position to charge a higher price; when it shades toward the high side they will be pressured to charge a lower price. This is a process where somebody sets the price, however, not one in which everybody is a price taker.

If one looks at the stock market, where there are a large number of buyers and sellers both can choose to take the market price by putting in a market order. Other buyers and sellers will put in orders to buy or sell as a specific price, and although each individual buyer and seller may be a small part of the market, their orders set the market price, dictated by their unwillingness to sell at a lower price, or buy at a higher price. Somebody sets the price of every transaction. In this way the forces of supply and demand act to clear the market.

The market clearing function of the market works so well that people in market economies tend to take it for granted. When one goes into a grocery store intent on buying a particular product, it is usually there (except, of course, for seasonal goods that are not available all year round), and shoppers are even irritated when the specific product they want is not there. But, remarkably enough, even in a grocery store that is likely to stock tens of thousands of items, most of them are in stock most of the time, and shoppers take for granted that they will be. There is no need to keep a large inventory of goods at home just in case; when one wants a particular food item it is almost surely available at the grocery store. Market clearing works so well that the typical consumer does not even notice, let alone marvel that the market economy can work so effectively to make just the right amount of all goods available just about all the time.

Market clearing prices and quantities are not equilibrium prices and quantities, and in general there is no tendency for prices and quantities to return to some level that they were in the past. As peach season passes, the price of peaches rises and the quantity exchanged falls. If the weather cooperates and the next year brings a big harvest of peaches, next year's price will be lower and the quantity exchanged will be higher, while unfavorable weather will bring a higher price and a lower quantity exchanged. Conditions change, and as they do prices and quantities change with them to clear markets, but there is no natural price, and no necessary tendency for prices to return to some level that existed in the past.

Characteristics of goods also tend to change. One reason a retailer may not keep an inventory of swimwear from one year to the next is to make room for inventory that will sell better during cold weather, increasing profits. Another reason is that styles change, and swimming trunks that would have sold well in the past give way to board shorts that are now preferred swimwear. Styles may reappear, but there is no reason to think that the market next summer will be a repeat of the market this summer. Similarly, new features appear on automobiles that make old models obsolete, and purchasers look for different characteristics in automobiles. In the 1990s pickup trucks and SUVs increased in popularity, partly due to lower gasoline prices, but SUVs were not even a product category in the 1970s. Higher gasoline prices reduced the demand for larger vehicles in the 2000s, which as noted above closed some previously valuable factories and led to the bankruptcy of SUV-producers Chrysler and General Motors in 2009.

Market conditions do not tend toward equilibrium, in the sense that conditions that existed before will tend to reappear if the market is disturbed. Rather, as economic progress occurs, conditions in markets continually evolve in a process Schumpeter called creative destruction. Producers who cling too long to old models of profitability in the face of evolving markets lose market share and their profits turn into losses, as the cases of Chrysler and General Motors show. People still want automobiles, but markets change in subtle but permanent ways and producers who do not accurately perceive those changes become casualties in that process of creative destruction.

Markets clear by equating the quantity supplied with the quantity demanded, but markets do not equilibrate; if by equilibrate one means tending to return to a state that existed in the past.

Entrepreneurship, market clearing, and the law of one price

Consider an example where apples are selling for \$0.50 in one city and for \$0.75 in a nearby city. An entrepreneur who spotted the price discrepancy could be in a position of making a profit by buying apples for \$0.50 in the first city and selling them for \$0.75 in the second. Before looking into the actions of the potential entrepreneur, first consider the question of whether the apple market in this hypothetical example is in equilibrium.

From the argument of the previous section one could answer that the equilibrium concept does not apply to markets the same way it does to physical systems, and while markets may clear, they do not equilibrate. If the quantity supplied equals the quantity demanded in both cities, then the markets clear.

Different economists define equilibrium in two subtly different ways, and the above example fits one definition of equilibrium but not the other. Sometimes equilibrium is defined as a situation in which everybody's plans are mutually consistent so that everyone's plans can be realized. As the quantity supplied equals the quantity demanded in both cities, everyone who wants to buy can, and everyone who wants to sell can, so the market is in equilibrium by this definition. Another definition is that equilibrium exists when there are no unexploited profit opportunities. In this case, even though everyone's plans can be realized, if apples can be bought in one city for \$0.50 and sold in another for \$0.75, then it appears that there is an unexploited profit opportunity, so even though markets clear, they are not in equilibrium.[13] In neoclassical general equilibrium, both of these conditions are satisfied. Everyone can realize their plans, and there are no unexploited profit opportunities, so the distinction between these two concepts of market equilibrium is sometimes overlooked.

The entrepreneur in this example is looking at an arbitrage opportunity. A good can be bought for less in one market and sold for more in another. So, we continue with the example and assume that the entrepreneur does indeed buy apples for \$0.50 in thefirst city and take them to the second city to sell them.

Two things can interfere with the apparent opportunity for the entrepreneur to realize a profit on the transaction. First, moving apples from one location to another is not costless. The entrepreneur may need a truck to carry the apples, and some apples may be damaged in shipping so that the entrepreneur ends up buying more apples than can be offered for sale. For example, it may turn out that it costs \$0.15 an apple to ship them from one city to the other, so the entrepreneur will have to spend a total of \$0.65 an apple, not \$0.50. Clearly, an apple in the first city is not the same as an apple in the second. The second factor is time. The entrepreneur cannot instantaneously transfer apples from one city to the other, and it may turn out that after buying them in the first city, by the time they get to the second city the price has changed so that the seller can only get \$0.60 an apple. If both of these factors come into play, what initially appeared to be a profit opportunity turned out not to be.

The two factors that stand in the way of the entrepreneurial profit are production and time. In the case of the apples, the production was the simple process of moving apples from one city to another. Consider a more complex example of an entrepreneur who believes he can make a profit by manufacturing sports cars with stainless steel bodies. The entrepreneur projects that after the cost of materials, paying auto workers, building a factory, and so forth, the cost of resources used to build a car will be less than the price the finished car will be able to bring in the market. Conceptually, this is the same as the apple example. The entrepreneur perceives that something can be bought in one market, and after production and time can be sold for more in another.

Even activities that appear to be pure arbitrage still have the factors of production and time standing between the profit anticipated from purchase and the sale. In financial markets, where arbitrageurs look for discrepancies in prices of securities, currencies, and other financial assets in various markets around the globe, the arbitrageurs buy in the low-priced market and sell in the high-priced market to make a profit. Because arbitrageurs are always looking for such discrepancies, they tend to be small and disappear fast, and even in computerized trading where trades are executed in less than a second, financial arbitrageurs are looking for faster computers and faster algorithms for spotting discrepancies, because prices can also change in less than a second. If traders are too slow, the profit opportunity vanishes. Similarly, the traders use offices, computers, and other inputs in their businesses. From one end of the spectrum to the other, whether one is talking about currency arbitrage or the production of automobiles, entrepreneurs operate by spotting profit opportunities where inputs can be purchased and, after production and time, sold for more than their purchase price.

The actions of entrepreneurs, who spot and act on profit opportunities, increase the demand for goods that can be profitably used, thus pushing up their price and increasing the supply of goods that can be profitably produced, and subsequently pushing their price down. Prices across markets then tend to equalize so that differences are no greater than the cost of moving them from one market to another. Thus, the actions of entrepreneurs result in prices for the same goods and services being approximately the same from one market to the next. This is the law of one price.

The informational content of prices

Prices convey information to market participants. Consider again the potential entrepreneur who sees apples selling for $0.50 in one location and $0.75 in another. The entrepreneur can improve the allocation of resources by transporting apples from the first location to the second, and the entrepreneur (as Adam Smith notes) seeking only his own gain, can improve aggregate welfare by taking something worth only $0.50 and transforming it into something worth $0.75. The only information the individual needs to know to identify this value-enhancing action is the prices of apples in the two locations. If it turns out that the transportation cost is $0.15 per apple, each apple transported from the first market to the second enhances the value of total output by $0.10.

If one were to think about this example in an equilibrium framework, one would expect that, over time, as the supply of apples was reduced in the first location and increased in the second, prices would adjust so that perhaps in the first location the price of apples would increase to $0.55 and in the second would decrease to $0.70, so the difference would just cover the $0.15 in transportation costs. However, in order to entice the entrepreneurial action to begin with, there had to be a profit opportunity, meaning that prices were not at levels that would clear both markets once apples began moving from one market to another. In the terminology of equilibrium economics, the markets were characterized by disequilibrium prices.

This notion of disequilibrium prices is set in a particular context, for if there is no trade in apples between the two locations, $0.50 is the equilibrium price in the first location and $0.75 is the equilibrium price in the second. The market for apples clears in both locations. The action of the entrepreneur reduces the supply of apples in the first location and increases it in the second, which then causes the market clearing price to change. In the first situation, each market is in equilibrium when viewed individually, because everyone's plans can be realized, but not when viewed together, because there is an unexploited profit opportunity.

In this example, the particular structure of prices offers the entrepreneur a profit opportunity, and once the entrepreneur acts on that opportunity the price structure changes. Prices provide information to market participants about the opportunity cost of goods and services, about inputs into the production process, and about the value of output in a market. Entrepreneurs see the opportunity of buying in some markets and selling in others – with production and time standing between the buying and selling – as signals of profit opportunities, but their actions change the prices they use as signals. Actions that appear profitable at an observed structure of prices may not be after those actions have taken place.

One might view this as the result of people trading at disequilibrium prices, but disequilibrium prices, in the neoclassical sense of prices that do not equate the quantity supplied with the quantity demanded, are rare enough that they do not require detailed analysis. Rather, market clearing prices that exist at one point in time may not be a good reference for prices that can be expected over a longer period of time. Prices convey a substantial amount of information to market participants about the activities of other market participants, as Hayek emphasized,[14] but prices are subject to change and entrepreneurial actions are one reason why prices do change.

Consider again the example of the apple seller who profits from transporting apples from the lower-priced to the higher-priced market. Other people observe the entrepreneur's actions and imitate the entrepreneur, and once several people try to exploit that profit opportunity prices may change so that all of them end up taking losses. In the late 1990s the internet boom led entrepreneurs to foresee a substantial increase in the demand for fiber-optic cable to transport internet data. The result was overinvestment and substantial losses for those companies who invested in fiber-optics. Entrepreneurs spotted a profit opportunity, but by the time they were able to act on it the market had changed so that their actions resulted in losses rather than profits.

Prices convey a substantial amount of information to market participants about the economic activity of other market participants. Makers of computers, such as Apple and Dell, can look at the prices for microprocessors, hard drives, LCD screens, and so forth, as an indicator of the opportunity cost of manufacturing those components and do not have to understand the details of the manufacturing processes of those components. Similarly, they look at the market prices of computers of various characteristics and match components that cost a certain amount and perform to a certain specification to produce an output they believe will sell for more than the cost of inputs. Much of what Apple and Dell need to know about Intel's business is summarized in Intel's price list for the various microprocessors Intel sells.

Because production and time always stands between the entrepreneur's purchases and sales, those prices are subject to change and can never provide a sure indicator of the availability of profit. While one might think about trades at disequilibrium prices as providing misleading information, prices are always changing, and are rarely disequilibrium prices in the sense that they do not clear markets. Rather, as factors in markets are changing, prices will change so that today's market clearing price will be different from market clearing prices in the future.

The market clearing process

Analyzing the properties of equilibrium leaves out an analysis of the forces that clear markets, but those forces are by definition a part of the equilibrium framework. Without them, the economy would not equilibrate and markets would not clear. Thus, even if one remains firmly within the equilibrium framework of economic analysis, the market clearing process must tacitly be lurking behind the framework in which all markets clear.

The Marshallian model of competitive equilibrium assumes away this process because all firms are price takers. Their marginal revenue remains the same regardless of the quantity they produce, so they must take the market price as given. But if all firms take the market price as given, no firms will change their price when market conditions change. Somebody must initiate the change. Similarly, to find a general equilibrium vector of prices Walras relied on the fiction of an auctioneer who called out a vector of prices and registered excess supplies and demands in all markets. For goods in which there was an excess supply the auctioneer would lower the price and for goods in which there was in excess demand the auctioneer would raise the price until by this process of groping, the auctioneer eventually arrived at the equilibrium vector of prices. All the while, no trade would occur until the equilibrium vector of prices was calculated.

In the real world, nobody knows the market clearing vector of prices, or even one market clearing price, so people trade at prices they think offer them good value. If sellers find their inventories getting low, they will raise their prices; if they are building up inventories they will hold the line on prices, or drop them, depending on how desperate they are to sell. Despite his emphasis on the concept of natural price, Adam Smith did a good job in The Wealth of Nations describing the way that real-world sellers respond to changes in market conditions by changing their prices. Similarly, demanders buy at prices they believe are attractive and provide them with more utility than the opportunity cost of the purchase, and may hurry to buy if they foresee prices rising in the future, or wait to buy if they anticipate price declines. In the context of equilibrium economics, all this means that many trades occur at disequilibrium prices.

Israel Kirzner does a good job of describing how entrepreneurial action leads to market clearing prices.[15] Entrepreneurs see a profit opportunity, buy in one market or several markets, and sell in others to make that profit. The result is that profits are competed away, enforcing the law of one price, and resulting in market clearing prices that have competed away profit opportunities.

Progress and entrepreneurship

This process of entrepreneurship that spots profit opportunities and acts on them to produce market clearing prices is the same entrepreneurial process that produces economic progress. Returning to the example of the entrepreneur who sees that apples are sold for $0.50 in one location and $0.75 in another, the entrepreneur buys in the first location and sells in the second, which pushes the price up in the first market and down in the second, which enforces the law of one price and clears both markets. It is a short step in theory – but a difficult step in practice – for the entrepreneur to see that inputs can be bought in one set of markets to produce a new or improved good that can sell in other markets, leading to economic progress.

When Steve Jobs and Steve Wozniak started Apple Computer, they were doing something little different, conceptually, from the apple vendor in the example used throughout this chapter. They saw that they could purchase microprocessors, disk drives, CRT monitors, and other components, and combine them to produce personal computers that could be sold for more than their components cost. In seeing and acting on that profit opportunity, they took a step in the direction of economic progress. This is the theme underlying this book.

Market outcomes versus market processes

Within the equilibrium framework of analysis, economic forces lead to the outcome of equilibrium. When looking at the market as an ongoing process, there is no ultimate outcome. Rather, the process is ongoing and is not leading toward some specific outcome.

The market process is more analogous to a biological system than a physical system. In a physical system, forces work to balance each other as the system approaches a steady state. The movement of the planets in the solar system is an example where the planets remain in orbit, tracing out the same orbits over time. As another example, heat is applied to a pot of water on a stove, and the water approaches an equilibrium temperature, which depends on the amount of heat applied, the amount of water in the pot, and other factors. In a biological system, forces are also at work as the system differentiates itself, new characteristics or species emerge, and old ones are displaced. Whether one is thinking of the evolution of a particular species or the evolution of an entire ecosystem, the process of evolution involves continual change, but there is no equilibrium outcome toward which the system is evolving.

Market processes work this way. As entrepreneurs introduce new goods and new production processes into the economy, the economy evolves. The evolution is one of continual change, but that change is not a movement toward some particular outcome. There is no equilibrium, in the sense of an outcome toward which the system is moving. The market process is not leading toward some determinant outcome. There is no outcome at all, but rather a continual process, because the system continues to evolve.

Market clearing and producing prosperity

The great advances in economics in the twentieth century were primarily advances in understanding the nature of market clearing forces in the economy. Twentieth-century economics was developed within an equilibrium framework where those market clearing forces were depicted as leading toward an equilibrium. In depicting market clearing in this way, the forces that lead to market clearing were separated from the forces that lead to economic progress.

Entrepreneurial actions in the economy underlie both the market clearing forces and those that produce economic progress, but in the equilibrium framework the market clearing forces are depicted while the forces that generate economic progress are assumed away. Economic progress occurs when entrepreneurs bring new goods to market, when they improve existing goods, and when they develop new production processes that lower the cost of production. Economic analysis recognizes these entrepreneurial actions to a limited degree, but not within the equilibrium models that describe markets and economies. In competitive equilibrium, the Marshallian framework assumes that firms produce a homogeneous product, which means that the product characteristics are fixed and innovation is assumed away. In general equilibrium models, even with n goods, the characteristics of those n goods are given. While one might introduce good n+1 as a new good, the model does not have a mechanism by which new goods are introduced. Similarly, production functions are given in the model, so when output is a given function of inputs, such as Q = f(K,L), there is no mechanism for an entrepreneur to alter the production function to bring innovation to market. Thus, the model, by design, assumes away entrepreneurial innovation.

Equilibrium models depict markets in equilibrium, by design, so they also leave out the innovative actions of entrepreneurs, even though the idea does lurk in the background. While the model depicts the equilibrium outcome, the notion remains that if the equilibrium is disturbed then "market forces" will return the economy to equilibrium. Those market forces are the actions of entrepreneurs who spot profit opportunities and act on them to equilibrate markets, as Kirzner describes.[16]

In fact, entrepreneurs, in their search for profit opportunities, are alert not only to arbitrage opportunities when prices deviate from market clearing prices, but also to opportunities to introduce new products and production methods. As this chapter has noted, both apparently different types of entrepreneurship really amount to the same thing: buying inputs and, after production and time, selling output. Successful entrepreneurship results when the sale of outputs generates more revenue than the cost of the inputs. The entrepreneurial actions that clear markets also produce prosperity, but the equilibrium framework assumes away aspects of the real world that generate increasing prosperity. This is not abad thing in itself – as Chapter 2 noted, all models make assumptions to focus on some aspects of reality to the exclusion of others. Nevertheless, it is important to recognize when features of reality crucial to the analysis of the problem at hand have been assumed away.

To consider further the relationship between market clearing and producing prosperity, consider an analogy of a bicycle rider who travels from his home to a store. How does he do it? One way to describe it is to explain how the rider is able to stay balanced on the bicycle. If the bike starts leaning toward the left, the rider needs to turn the handlebars to the left to keep the bike upright. To turn, the rider needs to lean into the direction of the turn, and as the bike leans that way move the handlebars in that direction enough to keep the bike from falling while maintaining the turn. Riding over gravel, the rider will need to be careful to keep the bike close to vertical or it can slide out from under him. To be successful the rider needs to maintain this balancing action the whole way. Another way to describe how the rider gets to the store is to describe the route of travel. Leaving home, the rider makes the first left turn, and then two blocks later at the bottom of the hill makes a right. After traveling three kilometers the rider turns left onto the gravel path which leads directly to the store.

Both of these descriptions explain how the bicycle rider makes the trip, but one explains how the rider is able to keep the bike upright without describing the route, while the other describes the route without describing how the rider is able to keep the bike balanced on two wheels. Both descriptions are correct, but they are describing different aspects of the trip. In this analogy, the equilibrium framework describes how the bike remains upright, and how little corrections by the rider maintain that equilibrium upright position. This is something the rider needs to know to make the trip, so it is valuable information, but it is not information that needs to be transmitted to someone who already knows how to ride a bicycle. Similarly, it is valuable to know how market forces keep markets close to "equilibrium," but this is something markets do on their own. The understanding is valuable knowledge, but does not provide any information about where the economy is going.

The bicycle is not just remaining balanced on two wheels, it is moving forward on a journey. To really understand the journey, information about the route of travel is at least as valuable as information about how the rider remains upright on the bike. The equilibrium framework looks at the stability of the bike, or the economy, at a particular point in time. One might think of the bike, or the economy, as always tending toward equilibrium, as the bike remains upright and the economy remains close to general equilibrium. The equilibrium framework offers substantial insight into that process. But if one is interested in understanding how market economies generate ever-increasing prosperity, the equilibrium framework assumes the underlying factors away. It is like studying how the bike remains balanced on two wheels without studying where it is going. This volume is more interested in the question of where it is going.

Conclusion

The equilibrium framework within which economics developed in the twentieth century offers important insights into the way that market forces adjust so that prices move to clear markets and equate the quantity supplied with the quantity demanded in each market. The equilibrium framework depicts markets in equilibrium, under the assumption that market forces are at work to keep it there. The process of adjustment is not depicted in equilibrium models, as if adjustment occurs instantaneously, but no matter how rapid the adjustment process, it would appear that some exchanges must occur at disequilibrium prices to provide signals that adjustments need to take place. Israel Kirzner's discussion of the role of entrepreneurship in spotting profit opportunities in disequilibrium shows the essential role of entrepreneurship in that equilibrating mechanism.

To depict the result as an equilibrium, however, is to take a very static view of economic activity. In fact, the same process of entrepreneurship that helps clear markets also works to generate economic progress, so it is better to think of the vector of prices that results from the market process as market clearing prices rather than equilibrium prices. They are not an equilibrium because factors that disturb the existing constellation of prices and quantities often are permanent changes, and when markets adjust they are leaving behind the old conditions, never to return, and moving along a trajectory of new conditions in the future. That creative destruction produces new prices and quantities, and also new and improved goods, and new and improved production processes. So, it is more accurate to describe markets as clearing rather than equilibrating.

Entrepreneurship plays an essential role in the market process, and much entrepreneurial activity takes place within firms. Thus, it is a natural progression to move from a discussion of market clearing to analyze the role of firms in that process that clears markets.

5 The innovative nature of firms

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In a general equilibrium approach to economic analysis, the economy is characterized as being in equilibrium and the economic activities that facilitate a transition from one equilibrium state to another, or that keep the economyon an equilibrium trajectory in a dynamic setting, are of secondary importance. Taking a lesson from the methodological issues discussed in Chapter 2, this might be justified as assuming away features of the real world that are irrelevant to the issues the models wish to depict, to focus on issues the user of the model wishes to analyze. To take the analogy from the previous chapter, if one wants to understand how the bicycle rider remains upright on two wheels, one can ignore the path that the bicycle is taking from one location to another because it is irrelevant to understanding how the bicycle rider balances on two wheels.

One cannot hope, however, to come to a deeper understanding about the path over which the bicycle travels by studying ever more intensely how a bicycle is able to stabilize itself on two wheels. This chapter examines the role of firms in an economy to gain some insight as to how they influence the path through which the economy travels.

Here we are not so interested in understanding the role firms play in maintaining market clearing prices, but it turns out that this process is integral to the role firms play in producing economic progress. The neoclassical theory of the firm depicts firms as profit-maximizing machines that take given inputs and given production functions to combine the optimal quantities of inputs to produce the profit-maximizing quantity of output. In the context of an economy characterized by economic progress, this depiction falls short in a serious way: it depicts the managerial role of those who are making decisions for the firm, but not the entrepreneurial role. This neglect of the entrepreneurial role of the firm is more than just a minor oversight, or something that can safely be assumed away. When one analyzes the economy in the context of continual economic progress, entrepreneurship is the fundamental function of the firm, and determines the way that firm activities are aggregated to provide insight into the functioning of markets.

Planning within the firm

The socialist calculation debate, discussed in Chapter 1, was over the merits of central planning versus the market mechanism as a way of allocating economic resources. That chapter argued the merits of the decentralized market mechanism in which outcomes emerge as a spontaneous order, rather than a centrally planned order. Yet even in the most decentralized market system, individuals make plans and attempt to carry them out by interacting with others in the market. Sometimes the cost of entering the market for every transaction entails more cost than making a long-term contract that creates more predictability and, perhaps, lower transaction costs, as Ronald Coase noted in is famous article, "The Nature of the Firm." Firms fulfill this role by extending the planning that individuals do, not only to their own activities, but to some of the activities of others.[1]

Consider an individual who wants to engage in manufacturing. Once the scale of activity expands beyond what the individual can achieve alone, the individual could enter the labor market every day to hire assistants, but if the capital involved in the manufacturing operation requires a certain amount of labor to be used at its peak efficiency, it will make sense for the capital owner to contract with workers and pay them a fixed wage in exchange for their labor, making the capital owner the residual claimant. This provides the firm's owner with predictable prices and quantities for these factors of production. Similarly, rather than enter the market for raw materials as inputs into production every day, the owner of the capital might contract with suppliers for long-term agreements to deliver raw materials. Such contracts, with workers, suppliers, and others, increase the predictability of those aspects of the business, albeit by reducing flexibility. Thus, following Jensen and Meckling's well-known ideas, a firm can be viewed as a nexus of contracts.[2]

Typically, the firm's owners own the capital and hire labor, but this could be reversed and labor could be the residual claimant and rent their capital. Law firms tend to operate this way. The same principles hold, however. By signing a lease on office space, for example, law firms do not need to enter the market every day to find locations to work, and so can make plans on how to use capital owned by others, but at a cost of giving up some flexibility. Knowledge is incomplete and fragmented, as Brian Loasby notes,[3] and the creation of firms produces some certainty and control in economic activities. Firms have more certainty (though not complete certainty) about those activities for which they have contracted, so they place under their control those things they want to be able to count on, and engage in market transactions in cases where they view the flexibility of market exchange to be advantageous.

Firms would prefer to be able to count on everything, leaving no uncertainty about the profitability of their actions, but the market mechanism itself prevents this. Prices will fluctuate, and demand for both inputs and outputs will change as the economy evolves. Entrepreneurs look for opportunities in these changes, but may be able to anticipate changes in some areas better than others. Contracts give firms a way of holding some things constant, and of reducing transaction costs.

Management versus entrepreneurship

In the neoclassical theory of the firm, where output, Q, is a function f of inputs capital, K, and labor, L, firms work within very narrow parameters. The production function, Q = f(K,L), is given to the firm, so the type of output being produced, Q, is given, the production process, f, is given, and the factors of production used to produce output, K and L, are given. The people who run the firm are only able to choose the quantities of K and L they will use in production, and to ensure that they are used efficiently so that, given the amount of K and L they choose, they get the maximum possible amount of Q.[4] The people who run firms are managers. They choose the profit-maximizing amount of K and L, see that inputs are efficiently employed and, if the industry is a competitive industry, earn a normal profit just sufficient to keep them in business.

Even within this framework, management is not a trivial task. The production process, f, may not be obvious, and there may be differences in laborers or capital equipment, but in the neoclassical formulation, once the parameters of the production function are known by the manager, management is simply a matter of combining them by using the optimal quantity of inputs to produce the optimal quantity of output. There is a challenge involved in acquiring the knowledge the manager needs to make the optimal choice, and there is an issue of compensating suppliers of labor and other inputs such that they have the incentive to supply the optimal amount without shirking,[5] but the actual choices managers make are, in theory, straightforward, and there is in fact an optimal choice that can be determined with full information.

In the real world none of the elements of the production function are given to the people who run firms. The characteristics of the output, Q, that firms produce is determined by those who run the firms. Businesses are continually looking for ways to improve their products, to make them more desirable than those of their competitors, and to introduce new products into markets. The neoclassical theory of competitive firms assumes that companies produce a homogeneous product, but this is an assumption made so that the output offirms can be added up to determine the quantity of output in the market. You can't add apples and oranges, which is why neoclassical economics assumes homogeneous output in competitive markets. In fact, those who run firms attempt to improve the characteristics of products they produce to make them more attractive to purchasers as a competitive strategy. In the real world, the characteristics of Q are not given, but are choice variables for those who run firms.

Similarly, production functions, f, are not given to those who run firms. Rather, those in charge are always looking for ways to produce more efficiently. Henry Ford's decision to produce automobiles on an assembly line is a dramatic example, but firms are changing their production processes continually. One might attempt to build this into the neoclassical framework by assuming a production process g that is more efficient than f, so that Q* = g(K,L) > Q' = f(K,L), but this does nothing to explain how those who run firms identify or choose new production processes. Typically, it is assumed that the production process isfixed and given to thefirm anyway, particularly when one is looking at equilibrium in markets.

In the same way, K and L are not given. Firms can employ different types of capital, and different types and qualities of labor. Consider again the cash-register example from Chapter 3. Decades ago cash registers only totaled the amount owed by purchasers, whereas modern cash registers scan prices and calculate the change due, reducing errors, but also reducing the amount of human capital required to be a cashier. Different types of capital and labor can be used in different types of production processes to produce the same output. Sometimes people who run firms will want to employ people with more human capital, but other times – as the cash-register example illustrates – they will want to change their inputs such that they can get more productivity from workers with less human capital.

The people who run firms are always looking for ways to improve their products by altering the characteristics of Q, to lower their costs by altering their production technology f, and by using more efficient or less expensive inputs K and L. New products and new production processes are outside the bounds of current knowledge or experience. The people who make these choices are entrepreneurs, who are looking for better ways of producing better output. The degree to which they succeed is determined by the market test: profit. Entrepreneurs who are successful make profits; those who are not take losses.

There is a crucial distinction between management and entrepreneurship. The neoclassical theory of the firm depicts them as being run by managers who work within a given production function. In a dynamic economy firms are run by entrepreneurs who choose every parameter within the production function.[6] Managerial aspects of the firm are important to understand – especially for managers – but if one wants to use the theory of the firm to understand the way firms influence markets and the overall economy, the entrepreneurial function of the firm is more significant than its managerial function. When considering the role offirms in producing prosperity, entrepreneurship drives firms to add to an economy's prosperity.

Firms must be entrepreneurial

In a market economy all firms must be entrepreneurial as those who run them look for better ways to produce better products. If they are not, they will continually lose ground to those who are. Entrepreneurs in any market can profit from altering product characteristics so that consumers prefer their new products to those previously available. Entrepreneurs in any market can profit from altering their production technology in order to produce goods at a lower cost than their rivals. As some entrepreneurs find better methods of production and introduce superior products into the market, the profits of their rivals will be eroded. This means that all firms must be entrepreneurial, that all firms must continually improve their products and their production processes, or they will be left behind by their competitors.

If those who run firms were to take the neoclassical model of the firm as a model on which to base their behavior, they would continually fall behind the rest of the market, their normal profits would turn into losses, and they would be driven out of business. Firms cannot survive by relying only on management, by minimizing costs within a given production process to produce a given product. Because their competitors are always moving ahead, all firms must be entrepreneurial to survive. Good management helps firms to be profitable; but successful entrepreneurship is necessary for firms to survive. Firms must continually innovate to keep up with the market. Profit maximization means successfully innovating, not minimizing costs.

Entrepreneurial production of market power: the most effective competitive strategy

Innovation takes many forms and entrepreneurs are always on the lookout for ways to improve their products and production methods. One of the key elements of entrepreneurial innovation is product differentiation. In the neoclassical theory of the firm, product differentiation enhances consumer welfare by offering consumers greater variety, but that benefit is offset by the higher average cost of production for monopolistically competitive firms. In fact, entrepreneurs do not differentiate their products just to make them different from the products of other firms, or to give consumers more variety, but to make their products better. By improving product quality and bringing new products to market, product differentiation is the engine of economic progress.

The neoclassical theory of the firm differentiates competitive firms from monopolies (and considers many variants in between), but assumes that the market structure is exogenous, even while demonstrating the advantages to firms of having market power. In contrast to this theoretical world, real-world firms can take action to produce market power for themselves, so market structure is not exogenous. The most effective competitive strategies for firms consist offinding ways to differentiate their products and otherwise give themselves some degree of market power, not minimizing the cost of producing a homogeneous product, as the neoclassical competitive framework suggests. This entrepreneurial element of the competitive firm – which is necessary for the firm to survive – is missing from the neoclassical framework.[7]

The profit-maximizing strategy that neoclassical theory describes for competitive firms will not maximize profits in a real-world competitive environment. Seeking economic profits by gaining market power is the profit-maximizing competitive strategy. This has been recognized in managerial economics textbooks at least since Michael Porter's 1980 book on business strategy,[8] but product differentiation as a competitive strategy has not migrated into the mainstream of neoclassical microeconomics. Managerial economics focuses on profit-maximizing strategies for firms rather than on implications for the economy as a whole. In microeconomic theory textbooks, pure competition, which is the market structure that maximizes social welfare, rules out product differentiation by assumption.

Consider the inconsistency in the theory of the firm from the standpoint of the entrepreneur, where firms in competitive industries are assumed to produce homogeneous products, take as given the output they produce, and must be content to earn only normal profits barely sufficient to keep them in business. Meanwhile, monopolies have differentiated products and earn above normal profits. The textbook conclusion is that competitive firms maximize their profits by choosing the optimal quantities of inputs to produce the optimal amount of output. If only the managers of these competitive firms read a few chapters ahead in the textbooks where their behavior is being described, they would see that in fact the profit-maximizing strategy is to differentiate their products so they can gain some market power and earn above-normal profits. The most effective competitive strategy is always to look for ways to gain some monopoly power.

Product differentiation: the engine of economic progress

As entrepreneurs look for ways to differentiate their products to make them more attractive to customers, they also provide economy wide benefits by producing economic progress. Neoclassical economics recognizes the advantages of product variety that product differentiation brings with it, but also argues that markets with differentiated products do not produce at the minimum average total cost. There is a trade-off of higher cost for more variety. But this literature looks at product differentiation in a static framework, ignoring a larger advantage of product differentiation: product differentiation brings improvements in products that generate economic progress. In the real world, product differentiation results in innovation, which generates economic progress and leads to lower average costs, not higher costs as the neoclassical model indicates. While it is true that the product variety generated by product differentiation increases welfare, the economic progress that results from product differentiation is much more important to social welfare than the variety that product differentiation produces.[9] The largely unrecognized fact is that product differentiation is the engine of economic progress.

Profit maximization and market structure in the neoclassical framework

In the neoclassical theory of the firm, firms are run by managers who maximize profits by finding the optimal combination of inputs and producing the optimal quantity of output. In their widely used intermediate microeconomic theory textbook, Pindyck and Rubinfeld say, "To maximize profit, the firm selects the output for which the difference between revenue and cost is the greatest."[10] After explaining short-run profit maximization in detail, Pindyck and Rubinfeld discuss long-run profit maximization, noting that "the long-run output of a profit-maximizing competitive firm is the point at which long-run marginal cost equals the price."[11] In another widely-used textbook, Besanko and Breautigam echo Pindyck and Rubinfeld, saying, "a price-taking firm maximizes its profit when it produces a quantity Q* at which the marginal cost equals the market price."[12] Note that Besanko and Breautigam refer to the firm as a price-taker, in contrast to Pindyck and Rubinfeld's characterization of a competitive firm.

Pindyck and Rubinfeld further note that "in competitive markets economic profit becomes zero in the long run."[13] Further explaining, they say:

Why does a firm enter a market knowing that it will eventually earn zero profit? The answer is that zero economic profit represents a competitive return for the firm's investment of financial capital. With zero economic profit, the firm has no incentive to go elsewhere because it cannot do better financially by doing so. ... The idea of an eventual zero-profit long-run equilibrium should not discourage the manager – it should be seen in a positive light, because it reflects the opportunity to earn a competitive rate of return.[14]

Besanko and Breautigam say, "we have seen a key implication of the theory of perfect competition: Free entry will eventually drive economic profitto zero. This is one of the most important ideas in microeconomics."[15]

Meanwhile, those same textbooks describe monopolists as being able to price above marginal cost and being able to potentially earn economic profits as long as the firm's monopoly profits can be sustained. Besanko and Breautigam note that “the monopolist’sprofit-maximizing price exceeds the marginal cost of the last unit supplied”[16] and that “the monopolist’s economic profits can be positive. This is in contrast to a perfectly competitive firm in long-run equilibrium”[17]

Right away, one should notice a major inconsistency in the way that these textbooks outline profit-maximizing strategies for competitive and monopolistic firms. Why should a manager be content to earn a competitive rate of return when, if the manager can find a strategy to gain some monopoly power, the firm can earn a higher rate of return? Contrary to the textbook advice, the profit-maximizing competitive strategy is not to minimize average total cost and be content with normal profits in the long run, but to look for ways to gain monopoly power, because those same textbooks say that monopolists can earn positive economic profits. The only way this advice to those who run competitive firms could be correct is if market structure is exogenous, and firms in competitive industries have no way to gain any monopoly power. But, as explained in more detail below, this is not the case. Competitive firms can gain some monopoly power by differentiating their products.

Besanko and Breautigam go on to note that “profit maximization implies cost minimization.”[18] In another microeconomics textbook Hal Varian echoes this observation, saying “It turns out to be convenient to break the profit-maximization problem into two stages: first we figure out how to minimize the costs of producing any desired level of output y, then we figure out which level of output is indeed a profit-maximizing level of output.”[19] In yet another textbook, Perloff says:

To maximize profit, all firms (not just competitivefirms) must make two decisions. First, the firm determines the quantity at which its profitis highest. Profit is maximized when marginal profit is zero, or, equivalently, when marginal revenue equals marginal cost. Second, the firm decides whether to produce at all.[20]

This neoclassical theory of profit maximization assumes that the firm is given a production function in which the nature of inputs and output are not under the control of the people who run the firm. The firm is not even able to alter aspects of the production process, which is given by the parameters of the production function. All the firm can do, given its production function, the inputs it will use, and the output it will produce, is choose the quantities of inputs that will maximize profit.

In the real world the people who run firms can decide to purchase different inputs and to produce different outputs. They can decide to adopt different production processes. In short, firms can innovate by changing their production processes, including developing lower-cost methods themselves, but more importantly, in the real world firms can choose to change the characteristics of their products in ways that they believe will make them more appealing to customers. The profit maximizing strategy is not to minimize costs, but to create a differentiated product that gives the firm some market power. The differentiated product may even cost more to produce, but can increase profits if customers are willing to pay more to purchase it. As obvious as this seems, it is at odds with the way the neoclassical model depicts competitive markets, as illustrated by the quotations from leading microeconomics textbooks above.

This is not a minor point. As Joseph Schumpeter notes:

The essential point to grasp is that in dealing with capitalism we are dealing with an evolutionary process. ... Capitalism, then, is by nature a form or method of economic change and not only never is but never actually can be stationary.[21]

If Schumpeter is correct in his description of a market economy, it is an inadequate defense of the neoclassical equilibrium framework to argue that it describes the way things will settle out in equilibrium, because a static equilibrium outcome will never be descriptive of the real-world economy. Whereas the neoclassical framework assumes that competitive markets are characterized by homogeneous products, thus assuming away product differentiation, in the real world product differentiation is one of the most significant competitive strategies used by firms in competitive markets.

Product differentiation in the neoclassical framework

In the neoclassical framework, product differentiation is depicted as creating downward-sloping demand curves for monopolistically competitive firms. Product differentiation imposes a cost on the economy because firms do not produce at minimum average total cost – instead there is a benefit of greater product variety available to consumers. A welfare analysis of product differentiation involves weighing the higher cost of production in monopolistically competitive firms against the benefit of greater variety. Discussing product differentiation, Pindyck and Rubinfeld say:

Any inefficiency must be balanced against an important benefit that monopolistic competition provides: product diversity. Most consumers value the ability to choose among a wide variety of competing products and brands that differ in various ways. The gains from product diversity can be large and may easily outweigh the inefficiency costs resulting from downward-sloping demand curves.[22]

Echoing that, Perloff says that "Although differentiation leads to higher prices, which harm consumers, differentiation is desirable in its own right. Consumers value having a choice, and some may greatly prefer a new brand to existing ones."[23] Varian notes that firms

may find it profitable to enter an industry and produce a similar but distinctive product. Economists refer to this phenomenon as product differentiation – each firm attempts to differentiate its product from the other firms in the industry. The more successful it is at differentiating its product from other firms selling similar products, the more monopoly power it has.[24]

In a graduate-level text, Mas-Colell, Whinston, and Green note that "in the presence of product differentiation, equilibrium prices will be above the competitive level."[25] Another microeconomics textbook by Browning and Zupan considers product differentiation as a source of monopoly power, saying:

Consumers may perceive the product sold by an incumbent firm to be superior to that offered by prospective rivals. Based on this perception, consumers are willing to pay more for the incumbent firm's product. For example, Ray-Ban sunglasses may be sufficiently differentiated in consumers' eyes to give the company some pricing latitude over potential competitors – even though the competitors have access to the same production technology.[26]

This quotation squares well with a subjective theory of value, because Browning and Zupan note that the differentiation is based on the perception of a difference, even though firms have access to the same production technology.

Robert Frank offers more insight into the phenomenon of product differentiation, noting:

An additional factor that limits the usefulness of the excess-variety conclusion is that it is based on a purely static model of the world. ... the process that leads to these varieties has stimulated extraordinary technological innovation ... The fruits of this innovation would have to be accounted for in a more complete comparison of optimal and equilibrium amounts of variety.[27]

Frank correctly notes that product differentiation does not just produce variety, it produces innovation and progress, but he notes that this "would have to be accounted for in a more complete comparison," rather than actually undertaking that comparison.

Discussing monopoly profits, Pindyck and Rubinfeld explain in a standard way the inefficiency that results from monopoly, saying:

Even if a monopolist's profits were taxed away and redistributed to the consumers of its products, there would still be an inefficiency because output would be lower than under conditions of competition. The deadweight loss is the social cost of this inefficiency.[28]

But if the monopoly profits are a result of an innovation that makes purchasers want that firm's product more than the products of its competitors, the profit is a return to innovation and not an inefficiency. In the purely static neoclassical framework any price above minimum average cost generates an inefficiency, but when differentiated products leading to monopoly profits are the result of innovation, they are a sign of efficiency – the extra value consumers receive because of the innovation – not inefficiency. Those profits would be the reward to firms that were able to provide more value at a lower cost to their customers. Without those profits there would be no incentive to bring innovations to market, and – equally significantly – there would be no market indicator that the innovations actually added value.

Varian says that "consumers will typically be worse off in an industry organized as a monopoly than in one organized competitively. But, by the same token, thefirm will be better off!"[29] Again, this holds true in the static setting of neoclassical equilibrium economics, but in the real world monopoly profits go to firms that are able to differentiate their products so that customers want those products rather than the products of other firms. Varian's conclusion holds if market structure is somehow exogenously determined, or if government restrictions on competition create monopolies. It would not hold if innovations generated by the firms themselves when trying to maximize their own profits produced the monopoly power.

Regarding competitive equilibrium, Frank states that:

Both long-run and short-run equilibrium positions are efficient in the sense that the value of the resources used in making the last unit of output is exactly equal to the value of that output to the buyer. This means that the equilibrium position exhausts all possibilities for mutually beneficial exchange.[30]

If this were true, in the long-run equilibrium economic progress would cease. In fact, firms are always looking for new opportunities to provide differentiated goods and services to customers, and, Frank's description of competitive equilibrium notwithstanding, it is inconceivable that "all possibilities for mutually beneficial exchange" will ever be exhausted. Competitive behavior on the part of firms involves finding those previously undiscovered possibilities for mutually beneficial exchange before competing firms do. [31]

Product differentiation in the Austrian framework

The Austrian school of economics has emphasized the dynamic nature of competition and depicted competition as a process rather than an outcome, so it is worth noting that the Austrian school's analysis of product differentiation mirrors the neoclassical analysis. For example, Murray Rothbard has a lengthy analysis of imperfect or monopolistic competition, mostly devoted to discussing why the arguments that monopolistically competitive firms have excess capacity and price above minimum average total cost are irrelevant or wrong. He devotes only a sentence to recognizing Schumpeter's views on progress, saying "In addition, Schumpeter has stressed the superiority of the 'monopolistic' firm for innovation and progress, and Clark has shown the inapplicability, in various ways, of this static theory to the dynamic real world."[32] But he offers no further explanation, and from there goes on to explain that cost curves are relatively flat, so pricing above minimum average total cost will mean a negligible increase in price anyway, and that a theory in which all firms have excess capacity is self-contradictory.

Later, criticizing Galbraith's theories of affluence and consumption, Rothbard again notes the advantages of variety offered by product differentiation, and does offer one sentence:

F.A. Hayek has pointed out the important function of luxury consumption of the rich, at any given time, in pioneering new ways of consumption, and thereby paving the way for later diffusion of such 'consumption innovations' to the mass of consumers.[33]

There are only a few sentences in Rothbard that reference the work of Hayek and Schumpeter suggesting the importance of product differentiation to economic progress, and Rothbard, like most of the mainstream, emphasizes the role of product differentiation in offering consumers a greater variety of consumption opportunities.

In the passage Rothbard cites, Hayek is discussing the importance of the growth of knowledge to economic progress, rather than product differentiation as a competitive strategy. His point is that:

The rapid economic advance that we have come to expect seems in a large measure to be the result of this inequality and to be impossible without it. ... If today in the United States or western Europe the relatively poor can have a car or a refrigerator, an airplane trip or a radio, at the cost of a reasonable part of their income, this was made possible because in the past others with larger incomes were able to spend on what was then a luxury. ... Many of the improvements would indeed never become a possibility for all if they had not long before been available to some.[34]

Hayek offers great insight into the nature of economic progress, but without tracing it back to the firm's strategy of product differentiation.

Armentano discusses monopolistic competition in static terms, concluding that "Once it is acknowledged that differentiated goods have subjective value that buyers are willing to pay for, perfect competition with homogeneous products can no longer be considered universally optimal or efficient."[35] Similarly, Frank Machovec discusses product differentiation in terms of "finding the best mix" of products, and concludes that the market economy, rather than government planning, can "freely coordinate the small pieces of information possessed by each agent so as, at each iteration, to experimentally adapt the current mix of goods in a way that improves the return to consumers."[36] These Austrian scholars couch product differentiation within the framework of the advantages of avariety of goods for consumers, rather than discussing the role that product differentiation plays in generating economic progress. Similarly, Ludwig von Mises discusses imperfect competition as the ability of someone with a differentiated product to charge a higher price, but does not discuss product differentiation as a contributor to economic progress.[37]

Kirzner is critical of the theory of monopolistic competition because of its equilibrium framework. He notes that "we can expect this disequilibrium constellation of product qualities, styles, sizes, color, packagings, and so on to change systematically under the influence of the market forces set in motion by the state of disequilibrium."[38] He notes, with emphasis, that "a variety of product qualities may be produced for no other reason than that equilibrium has not yet been reached."[39] He goes on to note that product differentiation "may, once equilibrium has been reached, come to be shaken down into product uniformity."[40]

Kirzner goes on to contrast his vision of entrepreneurship with Schumpeter's – whose ideas are very consistent with those presented here – noting:

For Schumpeter the entrepreneur is the disruptive, disequilibrating force that dislodges the market from the somnolence of equilibrium; for us the entrepreneur is the equilibrating force whose activity responds to the existing tensions and provides those corrections for which the unexploited opportunities have been crying out.[41]

While Schumpeter's notion of creative destruction is very consistent with the notion of product differentiation presented here, Kirzner presents product differentiation as a potential sign of disequilibrium that may vanish once an economy reaches equilibrium.

The concept of product differentiation as the engine of economic progress is a Schumpeterian notion. But when one surveys the Austrian literature on product differentiation, the bulk of it is similar to the neoclassical literature in that it emphasizes the advantages to consumers of a variety of products, and overlooks the role of product differentiation as the engine of economic progress. Schumpeter is the notable exception.

The advantage conveyed by product differentiation

The neoclassical model notwithstanding, no competitive advantage is conveyed merely by differentiating one's product. In the neoclassical equilibrium for monopolistically competitive firms, each firm just earns a normal economic profit, which is the same profit competitors earn, so there is no apparent payoff in profits from firms differentiating their products.[42] In fact, undertaking any expense to differentiate one's product may put the firm at a competitive disadvantage, because after the expense of product differentiation, product A is no more differentiated from B than B is from A. Thus, all firms share in any benefits from the product differentiation of other firms, and it would seem foolish for one firm to incur those costs. In the neoclassical framework all product differentiation does is give consumers more choices, but no one firm has any incentive to raise its costs above those of its competitors to convey this benefit to consumers.

Perhaps a market with differentiated products might be larger in total than one with homogeneous products. If the market offers a variety of soft drinks the total quantity of soft drinks sold in the market may be larger than if only one type were available. Nonetheless, in the neoclassical framework firms in competitive and monopolistically competitive firms still only earn normal profits in the long run, giving no individual firm an incentive to undertake the cost of designing a differentiated product.

In fact, the reason firms differentiate their products is not to make them different, it is to make them better. What "better" means in this context is that consumers will be willing to pay an increased amount for the differentiated product that more than compensates the firm for its expenses incurred in differentiating the product. Because it costs something to change the characteristics of a product, the product must convey a greater utility to the purchaser to make the product differentiation worthwhile. Of course, the expense might be undertaken to lower production costs, making the product less expensive for consumers while providing greater profit to the producer. Over time small changes in products create major improvements, and old product characteristics fall by the wayside, to be replaced by new and improved products, thus generating economic progress. This is the "creative destruction" that Schumpeter argued is a fundamental characteristic of markets and competition.

Product differentiation does not give firms any advantage because they have made their products different. Even in the neoclassical model, this just generates the same normal profits that characterize firms producing homogeneous products in competitive markets. Rather, competitive advantage through product differentiation comes from making a product that purchasers want more than the products of competitors – by improving their products to try to gain a competitive advantage, product differentiation becomes the mechanism by which firms generate economic progress.

Competitive strategy and product differentiation

The behavior that the neoclassical theory of the firm describes as profit maximizing will not, in fact, maximize profits. The profit-maximizing strategy of competitive firms is to differentiate their products to allow them to price above average total cost. The neoclassical theory of pure competition makes the assumption that firms in the industry produce homogeneous products so that the output of all the individual firms can be added up to get the total industry's output. But one must recognize that the homogeneous output that characterizes competitive firms in this framework is an assumption that is made to allow aggregation within the model, it is not a conclusion about competitive behavior. To emphasize, the neoclassical model assumes that competitive firms are characterized by homogeneous products, it does not conclude that competition results in firms producing homogeneous products.

In fact, product differentiation is an important competitive strategy: a strategy that firms use to gain a competitive advantage over their rivals. Product differentiation makes markets more competitive, not less competitive, as the neoclassical model implies. In an industry with homogeneous products, all firms must do to remain "competitive" is minimize their costs, much as the neoclassical model describes. However, if one of the competitors differentiates its product to make it more desirable to consumers, other firms will lose customers, revenue, and profits until they catch up. Even if they do catch up they will only, once again, be earning normal profits. This is not a strategy for long-term success. Such a firm will either be earning normal profits when caught up, or less than that if trying to catch up, and so will always lag behind its rivals and may not be able to remain in business. In the real world of competition, a competitive firm will always be looking for ways to make its output more desirable than the output of other firms, differentiating its product to stay ahead of its competitors and earn profits. Through product differentiation firms attempt to increase their market shares, and try to create an environment in which they can set their prices above average total cost to earn economic profits. To the extent that they are successful, their competitors will lose customers and profits.

Unlike the neoclassical model, where firms produce homogeneous products by assumption, actual market competition is characterized by firms always looking for ways to differentiate their products to make them more desirable to their customers. This product differentiation leads to innovation in the market, and this innovation produces the creative destruction described by Schumpeter. In real-world competitive markets, firms that do not innovate will continue to fall further behind their competition, until what once was a profitable business eventually becomes unviable.

Because markets are continually evolving as innovators find ways to produce products that are more appealing to demanders, a competitive firm that simply follows the neoclassical profit maximization strategy will eventually fail, being put out of business by competitors that differentiate their products and find other ways of increasing profits. Innovation – not cost minimization – is the optimal strategy for profit maximization. When one realizes that market structure is not exogenous, it becomes clear that profit maximization in competitive markets does not consist of minimizing costs, but rather of pursuing strategies that will gain competitive firms some monopoly power. The profit-maximizing strategy in the neoclassical competitive model works only because of the assumption of homogeneous products. But that assumption is not descriptive of real-world competitive markets in which product differentiation is an important competitive strategy.

In the neoclassical model people who run firms are managers, and they manage their firms by finding the cost-minimizing combination of inputs to produce the profit-maximizing level of output. As Peter Klein notes, the neoclassical theory of the firm is told from the perspective of the manager rather than the owner-entrepreneur.[43] In the real world, profit-maximizing firms are run by entrepreneurs who look for new and better ways to produce output by lowering their costs, and by finding more desirable product characteristics to entice buyers to pay a price above average total cost for their products. A firm cannot remain in business by following the neoclassical profit-maximizing strategy, because as entrepreneurial competitors find better production methods and improve product characteristics so that customers are more satisfied with their products, the neoclassical profit-maximizing firm will fall further behind its competitors. Good management and effective entrepreneurship both helpfirms increase their profits, but over the long run an entrepreneurial firm with less effective management can remain profitable by staying ahead of its competitors, while a well-managed firm without entrepreneurship is doomed to fail as it is overtaken by its more entrepreneurial rivals.

Sometimes product differentiation implies changing consumer perceptions as much as actual product characteristics. Advertising can help do this. Are "Butterball" turkeys, or "Perdue" or "Tyson" chickens really that different from their competitors' products? Are there any real differences in brands of gasoline? If consumers did not think so, there would be no advantage to creating the brand. But sometimes small differences in products make big differences to consumers. Why not buy a particular brand if it has a reputation for quality control, is perhaps more convenient to buy, or the seller is willing to stand behind its product in the way that a "generic" firm might not? Perhaps more significantly, if such slightly differentiated products gain one firm a competitive advantage, that gives other firms the incentive to create more value-enhancing differentiation to regain market share.

Is Wal-Mart a counter-example to the argument that product differentiation leads to profits, because it just sells the same products that can be purchased elsewhere? Wal-Mart's innovative methods of inventory control and store operation create a business model that differentiates its product by giving consumers low prices and a large selection of goods. Wal-Mart's innovations in the production of retail sales makes it similar to Ford's innovative assembly line in this regard. People could buy automobiles before they could buy Fords, just not as cheaply.

By assuming that competitive firms produce homogeneous output, the neoclassical theory of the firm leaves out a key component of competition. Product differentiation is, in fact, necessary for competitive firms to survive. It should be an integral part of the theory of competitive markets, not assumed away. Furthermore, it has important implications for the aggregate economy.

A broader look at the literature

The analysis that microeconomic theory has done on product differentiation stands in contrast to work that has been done barely outside the confines of academic economics. It is interesting to contrast, for example, a managerial economics textbook written for MBA students by Besanko, Dranove, Shanley, and Schaeffer,[44] with Besanko and Breautigam's microeconomic theory textbook discussed earlier in the chapter, because the two books have an author in common. The former book discusses strategies to deter entry, creative destruction, incentives to innovate, and a host of other competitive strategies that can enable firms to gain market power and enhance their profits. Yet all of this is assumed away in the latter book, in which the optimal strategy for competitive firms is depicted as minimizing costs and producing the profit-maximizing level of output.

In Besanko and Breautigam's microeconomic theory book, "Free entry will eventually drive economic profit to zero. This is one of the most important ideas in microeconomics."[45] Yet Besanko et al.'s MBA textbook says that "market forces are a threat to profits, but only up to a point. Other forces appear to protect profitable firms. ... A firm may prosper indefinitely in an industry with intense pricing rivalry and low entry barriers."[46] Why the contradiction? One would be reluctant to say that Professor Besanko is ignorant of information that is in his other book. The reasonable explanation is that Besanko and Breautigam's microeconomics textbook reports the accepted wisdom in microeconomic theory at the beginning of the twenty-first century, which does not incorporate the entrepreneurial aspects of decision making within firms discussed in Besanko et al.'s MBA book. Similarly, Hal Varian, in a book with Carl Shapiro,[47] discusses competitive strategies such as differentiating products and innovating to add value to products that are competing against similar products on the market, whereas Varian's microeconomics textbook discusses competitive strategy in the neoclassical setting of undifferentiated products.

Despite the fact that there is a broader literature that treats product differentiation as entrepreneurial and innovative, it has not found its way into the core of microeconomic theory, so firms in that theory are limited in their activity to choosing the optimal mix of inputs and the optimal quantity of output. This more limitedfirm is thenused as abuilding block for general equilibrium models, for macroeconomic analysis, for growth theory, and for welfare economics. Certainly the world, in general, is not ignorant of the benefits of improvements in product characteristics over time. However, this knowledge has not yet penetrated the core of neoclassical microeconomics.

Conclusion

The primary role of the theory of the firm in economics has not been to try to explain the decision-making process within a firm, or how firms actually operate. Rather, the theory of the firm presents firms as constructions that purchase inputs to produce outputs for the purpose of understanding the workings of markets, not the workings of firms. Thus, it would be unfair to critique theories of firms by suggesting that they do not represent what actually goes on inside a firm. The issue is whether a theory of the firm depicts firms in such a way that the firm's effects on markets are accurately depicted.

In important ways, the answer to this question is no. The theory of the firm has been designed around explaining how markets generate equilibrium prices and quantities, and for this purpose the neoclassical theory serves us well. However, when one looks at the actual functions of markets, the market clearing aspect of markets, while important, is only part of what markets do. The market economy has had a more significant impact on people's lives as it has generated economic progress and increased people's well-being, and the neoclassical theory of the firm has not done a good job of explaining how firms contribute to economic progress.

The neoclassical theory depicts the people who run firms as managers who efficiently allocate resources within the given parameters of a production function. From the standpoint of generating economic progress, the more important function of those who run firms is entrepreneurial. The entrepreneurs who run firms introduce innovations into the economy that improve people's well-being.

Thus, an entrepreneurial theory of the firm is required to understand the firm's role in producing prosperity.

This chapter has argued that product differentiation is the mechanism by which entrepreneurs introduce innovations into the economy, and that the institutional structure of the firm is a key feature of the economy that enables the calculation of the costs and benefits of entrepreneurship, and so facilitates entrepreneurial innovation.

6 The role of firms in the market

]6 The role of firms in the market

Firms serve many purposes in a market economy. Frederic Sautet lists four reasons why economic activity is undertaken within firms:[1] (1) firms exist for technological reasons, because certain types of production processes cannot be separated; (2) they are a manifestation of a certain type of market contract, or are a nexus of contracts;[2] (3) they exist because sometimes it is less costly to bypass markets to allocate resources in a hierarchical manner;[3] and (4) they exist to concentrate the capabilities of economic actors who are more efficient if they work together as a team.[4]

Another reason, considered later in this chapter, is that firms act as knowledge repositories, because sharing knowledge with those within the firm's boundaries increases profits for the sharing group, whereas sharing knowledge with competing firms lowers the firm's profits. David Harper has emphasized the institutional prerequisites that are required for firms to process knowledge, and more generally, has depicted firms as institutions that foster the growth of knowledge.[5] Firms, as economic institutions, exist for many reasons. When looking at their role in the market, they are the organizational structures entrepreneurs use to control resources and to turn the innovations they bring to market into profits. Firms are, at their foundation, entrepreneurial and innovative. Chapter 5 noted this role, and the current chapter picks up at that point to look at the relationship between firms and markets.

Entrepreneurship and knowledge

Entrepreneurship, following Kirzner, means spotting and acting on unexploited profit opportunities.[6] While at first it might appear that entrepreneurial opportunities are available to anyone who is alert enough to notice them, in fact it takes a considerable amount of knowledge to observe aprofit opportunity in an entrepreneurial economy where many others are searching for profit opportunities. As Richard Langlois notes, a key role of the firm is to enable entrepreneurs to solve coordination problems that arise because of change and uncertainty.[7] Look at all the entrepreneurial opportunities that have arisen in the electronics industry over the past several decades. Someone not intimately familiar with the industry would have no chance of spotting a profit opportunity there.

The same is true of any industry. Someone not intimately familiar with the agriculture industry would have little chance of spotting a profit opportunity from reallocating crop land, using a different seed variety, and so forth, and it would take a substantial amount of technical knowledge to develop different seed varieties. Someone not familiar with the auto repair business would have little chance of spotting a profit opportunity in opening an auto repair shop. The requisite knowledge would include not only technical knowledge about how to repair cars, but also knowledge about local markets and knowledge about business and personnel management.

Much of the required knowledge will be specific local knowledge, and tacit knowledge that would be difficult to explain to someone unfamiliar with the very specific conditions related to a particular opportunity. A human resources director at a large corporation will have a great deal of knowledge about personnel management, but probably not the right knowledge to be able to manage a group of carpenters at a local construction site. The entrepreneur overseeing the construction project will likely have obtained that knowledge from previous experience in that line of work, seeing how it was done and understanding from first-hand experience how carpenters can be kept happy and productive in their jobs.

Tacit knowledge, as the term is used here, means knowledge that cannot be articulated and explained to others, but that can be absorbed by people in close proximity who observe that knowledge being used. This appears consistent with Hayek's idea that every person has "knowledge of the particular circumstances of time and place ... which use can be made only if the decisions depending on it are left to him or are made with his active cooperation."[8] If the knowledge were easily communicated to others, the holder of the knowledge could communicate it and would not have to be actively involved in decisions relying on it. As Pierre Desrochers persuasively argues, although it is difficult to articulate, tacit knowledge can often be transferred to those in close proximity when they observe that knowledge being applied.[9]

When tacit knowledge exists the people who have that knowledge have an incentive to share it with those within their firms, because it increases the firm'sprofits, and therefore the earning potential of those within the firm. They have an incentive to keep that knowledge from those in competing firms, because sharing that knowledge would erode the firm's ability to profit from it. The firm is then an institution that breaks down a barrier to the transmission of tacit knowledge, because those within the firm's boundaries benefit from sharing it with others within the firm.[10] They bear a cost (in terms of lost profits) if that knowledge spreads to other competing firms. In this way, the firm acts as a knowledge repository.

Schumpeter made the distinction between invention and innovation.[11] Invention means developing new techniques, products, and processes,whereas innovation means introducing them into the market. R&D produces inventions, but inventions provide no benefit to firms unless then can be transformed into innovations.

If entrepreneurship entails noticing a profit opportunity, the function of R&D is to create an environment within which profit opportunities are more likely to be spotted, with the hope that someone in the firm doing the R&D can spot the opportunity first and turn it into an innovation. As a well-known example, the graphical user interface used on computers like the Apple Macintosh and those using Microsoft Windows was invented at Xerox, but Xerox never produced a profitable product using the interface they invented. The inventors were at Xerox, but Apple and Microsoft were the innovators that spotted the profit opportunity and brought profitable products to market.

The effect of firms on markets

In the neoclassical theory of the firm, other firms have a minimal impact on the activities of any particular firm. In competitive industries firms produce homogeneous products and are price takers, so while the actions of all firms taken together are significant, the actions of any one firm are insignificant. Similarly, in monopolistic markets there is only one firm in the market, so other firms are assumed to have minimal effects on the monopolist, because there are no close substitutes for the monopoly's output. Even in monopolistic competition in the style of Chamberlin and Robinson other firms only serve the role of entering or exiting the market so that in equilibrium the monopolistically competitive firms only earn the same rate of return as they would earn in a competitive industry.

Firms in an entrepreneurial economy must be considerably more cognizant of the activities of other firms than firms in a neoclassical equilibrium setting. Entrepreneurs who run firms are always looking for advantages they can gain over competitors by lowering their costs, or by introducing product innovations that will make their products more desirable to purchasers. The entrepreneurship of some firms forces all firms to be entrepreneurial, for if they are not they will be left behind by those that are. So, as Chapter 5 noted, firms cannot take their production processes or the characteristics of their final products as given. They must always be aware of what their competitors are doing, and respond either by matching them, or ideally by coming up with better innovations sooner than their competitors. Innovations by one firm force its competitors to change some aspects about the way they do business.

Firms that supply intermediate goods must not only be aware of the activities of other firms competing with them, but of the firms they are supplying. If a firm can come up with a way to give its customers a competitive advantage it can take market share away from its competitors. One way the computer industrygrew in the last halfof the twentieth century was for computer manufacturers to not only produce computers for their customers, but also to show their customers how using their computers could enhance their business operations. At the beginning of the twenty-first century IBM, which was accused of monopolizing the computer hardware industry in the 1960s, has become more of a software and services company focused on helping its customers use computers more productively.

Innovation creates new markets and new profit opportunities. Using the computer industry as an example again, the invention of the microprocessor led to the development of the personal computer, a use toward which the microprocessor's inventors did not anticipate. That in turn created profit opportunities not only in computer manufacture, but in all types of goods and services associated with computers, ranging from software to monitors and pointing devices.[12] Profit opportunities are not limited to related products, however. For example, the development of the computer industry in silicon valley opened the opportunity for entrepreneurs to establish car dealerships, restaurants, and indeed all types of retail establishments. The entrepreneurial insight in these cases was not that there was a profit opportunity in a new product or production process, but rather that existing products and production processes could be profitably expanded in particular locations.

Innovations by entrepreneurs ripple through the entire economy as some markets expand, others contract, and products that once were popular fall from favor. Firms cannot help but be entrepreneurial if they want to survive, because the economic landscape is continually changing. Firms that have identical business plans may be in a position to reap substantial profits and expand in San Jose, but may be losing business and even be forced into closing in Detroit. Firms are affected by the innovations of their suppliers, of their customers, and even to a major extent by unrelated businesses. Those that are able to anticipate opportunities will thrive; those that are unable to foresee changes that will have adverse consequences will be the victims of creative destruction.

Schumpeterian and Kirznerian entrepreneurship

As noted in Chapter 5, Kirzner went to some lengths to distinguish his views on entrepreneurship from those of Schumpeter, saying:

For Schumpeter the entrepreneur is the disruptive, disequilibrating force that dislodges the market from the somnolence of equilibrium; for us the entrepreneur is the equilibrating force whose activity responds to the existing tensions and provides those corrections for which the unexploited opportunities have been crying out.[13]

When one looks at the role of the entrepreneur in both cases, the entrepreneur is acting the same way, and hoping for the same result: there is an unexploited profit opportunity and in both cases the entrepreneur acts to try to profit from it.[14] Still, the distinction Kirzner makes isworth exploring, because it is important when examining the effect that entrepreneurship has on markets.

To envision the creative destruction of Schumpeterian entrepreneurship one can picture an economy in general equilibrium such that everyone's plans are mutually consistent and nobody sees an advantage to be gained from changing their economic behavior from the status quo. Then an entrepreneur introduces an innovation into the economy which makes a new or improved product more attractive to consumers. That entrepreneur reaps profits while other competitors suffer losses, requiring a reallocation of resources from their old uses toward new ones. The entrepreneur disrupts the old equilibrium. With the economy in disequilibrium, additional profit opportunities arise to reallocate resources, and Kirznerian entrepreneurs exploit these profit opportunities to bring the economy back to equilibrium.[15] Schumpeterian entrepreneurs disrupt an existing equilibrium and Kirznerian entrepreneurs re-equilibrate an economy in disequilibrium.

In the real-world economy the distinction between the two types of entrepreneurship is not as great as it first appears. Entrepreneurship is a constant feature of the economy — it does not start from a steady state —and as a result the economy is never in an equilibrium in the sense that economic actors can continue their present activities into the future. They must always be anticipating the future direction of the economy. In another sense, however, if everybody is insightful enough to correctly anticipate the economy's future direction, all plans will be mutually consistent even as they are constantly changing. The economy will be in equilibrium in the sense that everybody's plans for the future can be realized.[16] In such a setting, Kirznerian entrepreneurs spot those profit opportunities as they arise, so Schumpeterian entrepreneurs can never be too disruptive.

One might view Schumpeterian entrepreneurs as introducing innovations into the economy to disrupt the status quo and Kirznerian entrepreneurs as seeing the disruption and acting to bring the economy back to a new status quo. While this makes sense within an equilibrium framework (which Kirzner explicitly references), in an economy where things are always changing the Schumpeterian entrepreneur is acting to create a new equilibrium just as much as the Kirznerian entrepreneur is disrupting the status quo, even if that status quo contains unexploited profit opportunities.

The distinction makes the most sense when considering an equilibrium to be a condition in which everybody's plans are mutually consistent, so everybody's plans can be realized. In that setting, an entrepreneurial innovation can disrupt people's plans, and can make people's plans that would have been realizable without the innovation unrealizable with it. Given the disruption, entrepreneurs now must discover how to reallocate resources so that their plans can in fact be realized. The disruption is likely to be less disruptive on the consumption side of the market, because entrepreneurs profit from giving purchasers options they prefer over their previous options, and more disruptive on the production side of the market, where those who produced the old options may no longer find that course of action profitable.

If entrepreneurs are unable to immediately adjust to the new circumstances, resources may be unemployed for some time before they are reallocated to new uses. Thus, creative destruction has macroeconomic implications, which will be taken up in Chapter 9.

Entrepreneurship, innovation, and progress

Entrepreneurs introduce innovations into an economy with the hope of earning profits. Indeed, Kirzner defines entrepreneurship as spotting a profit opportunity. This makes profit the measure of the success of entrepreneurial action. Some innovations will add value to the economy, some will not. Profit is the measure of this value added, because if the entrepreneur profits from the innovation that means that the cost of implementing it is less than the value of the innovation to those who benefit from it, with the benefit being measured by their willingness to pay for it. If the innovation does not add value to the economy, the cost of implementing it will be as great or greater than the amount those who benefit are willing to pay, and the innovator will take a loss. Is an entrepreneurial innovation beneficial? Does it constitute economic progress? Profit is the measure that answers this question.

Peter Lewin notes that firms play a crucial role in measuring the benefits produced by entrepreneurial innovation, because firms are institutions that consolidate those costs and benefits.[17] Imagine that an entrepreneur starts a firm to produce a single product. The creation of Apple Computer by Steve Jobs and Steve Wozniak to produce the Apple II can serve as an illustration. The new firm incurs costs to produce the product and receives revenue from selling it. The profitthefirm makes is a measure of the value of the firm's product over the opportunity cost of producing it. That is, the profit measures the degree to which the innovation constitutes economic progress. Had those entrepreneurs been overly optimistic about their new product, either overestimating the market for it or underestimating the cost of producing it, the firm would have taken losses signaling that the innovation did not add value to the economy, and the production of the unprofitable product would have ceased.

The firm plays a crucial role in economic calculation here, because it aggregates all of the costs and benefits of the product. If the single-product firm is viable its product constitutes economic progress. In markets with imperfect foresight the firm may not be viable even with a product that contributes to economic progress. For example, the entrepreneur may be unable to raise sufficient revenue to run the firm until revenues start to come in. But the firm cannot be viable in the long run unless its product contributes to economic progress.[18]

Most firms produce a variety of products, so the accounting is not so straightforward. Firms must allocate their costs to various product lines, account for synergies across product lines, and allocate resources accordingly. Accounting conventions are designed to pull part of the load here, so firms can allocate their costs across product lines and decide which are profitable and which are not. Product lines that appear unprofitable in that their direct revenues do not cover their direct costs may still in fact be profitable to the firm. For example, retailers may carry high-end products that do not pay their costs directly, but draw customers into the store and increase the sales of lower-end products. Likewise, manufacturers may produce flagship products that do not pay their direct costs but enhance their brands'reputations.

Multiple-productfirms may be able to realize economies by sharing R&D efforts, advertising, and by lowering the volatility of revenues if revenues of all product lines do not rise and decline together. However, as firms increase their product lines it becomes increasingly difficult to account for their costs and benefits, increasing the likelihood that firms will not be maximizing the profits from each of their lines because they are unable to do a precise calculation. This puts a limit on the size of potentialfirms, because if larger firms have more difficulty calculating the profitability of various lines, smaller firms can compete in areas where larger firms miscalculate. Thus, firms play a crucial role in the calculation of the costs and benefits of entrepreneurial innovation, and in guiding innovators to add value to the economy and generate economic progress. One big firm would be unable to undertake such calculations as effectively as several smaller firms.

Firms as repositories of knowledge

One can view firms as institutional structures that calculate costs and benefits to ensure that resources are being used efficiently, but that is not the purpose of the firm; it is a means to an end. The end is for the firm to make profits, which is done by combining inputs to produce outputs that sell for more than the inputs cost. The calculation is undertaken for the purpose of ensuring that the revenues do, in fact, exceed the cost in the various product lines the firm produces. Of course, those in firms have many purposes in mind other than profits. They want to produce good products for their consumers, they enjoy the challenges of their jobs, the prestige of their positions, and the sense of purpose and accomplishment they get from working. But all these other purposes hinge on the firm's profitability. If it is profitable, thefirm can continue to function and support any number of other motivations that those who run it and work there may have. If it is not, the firm cannot continue.

This analysis has already established the necessity of continual entrepreneurship on the part of those who run firms, for if they are not entrepreneurial they will fall behind other firms that are. The fact that some are entrepreneurial means that all have to be. The quest for profits means looking for ways to combine inputs into outputs more efficiently, and looking for ways to improve the desirability of outputs, perhaps by producing entirely different outputs. This is done within firms because firms are repositories of knowledge, and the knowledge within firms help those within them to find profitable opportunities.

Adam Smith noted in The Wealth of Nations[19] that many inventions that increased the efficiency of production were developed by people directly involved in production. As Smith explained it, those are the people who have the most detailed knowledge and insight into the production process. Hayek noted that everybody has some specific knowledge of time and place that may be difficult to communicate to others, and that an important function of a market economy is to coordinate all this knowledge so that people can make the best use of this tacit knowledge possessed by others.[20] While the market mechanism does coordinate people's activities, the tacit knowledge people have can be transmitted to others who are in close proximity and can observe the use of that knowledge directly. This gives rise to agglomeration economies.[21]

Agglomeration economies explain why concentrations of firms in specific types of industries tend to arise, such as financial firms in New York and computer firms in silicon valley. Individuals in some firms can observe first-hand what those in other firms are doing, and factors ranging from production methods and personnel policies can be imitated by those who observe them first. Even if eventually those things might be observed by people in different cities, and different countries, economic progress implies that those who are able to observe and adopt them first have an advantage, and late adopters may be too late to capitalize on ideas and methods that have become outdated. Similarly, people in close proximity to others will be more aware of job opportunities, and will be better able to explore the possibilities of ideas that are not fully framed, than people who are not in close proximity to one another.

All of the arguments regarding agglomeration economies and the development of cities apply even more strongly to firms. The boundary of the firm is a knowledge boundary, in that people in one firm have incentives to keep their knowledge from those in other firms, but have an incentive to share their knowledge with those within their own firms. Knowledge that escapes the boundary of a firm enables other firms to be more profitable, and competition from other firms erodes the profit of the firm whose knowledge has escaped. Meanwhile, knowledge shared within the firm's boundaries can increase the profits of the firm, thus benefitting all of those within it.[22] Entrepreneurs look for ways to take advantage of the knowledge within their firms to find profit opportunities, and to share that knowledge with those within the firm who can make the best use of it. Meanwhile, they have an incentive to conceal that knowledge from those in other firms who could use the knowledge to compete with their own firm.[23]

Agglomeration economies result from people being able to absorb the tacit knowledge of those in close proximity. People in differentfirms have an incentive to hide this knowledge from each other, because when other firms absorb their tacit knowledge it can erode their profits, whereas people in the same firm have an incentive to share this knowledge with each other because when others within their firms absorb their tacit knowledge it can increase their profits.

The role of the firm is to remove the boundaries so that people have an incentive to share their tacit knowledge. Firms are repositories of knowledge within which people have an incentive to share and enhance each other's productivity, and beyond which people have an incentive to hide that tacit knowledge and make it as proprietary as possible.

This analysis overlooks competition among people within a firm, and assumes they all have the common goal of increasing the firm's profits. For example, a sales manager may oversee a dozen sales people in the firm, and individual salespeople are compensated by measuring their sales as a percentage of the whole firm's sales. In this case salespeople may have an incentive to keep their successful sales methods from the other salespeople in their firm. A better compensation scheme might be to compensate them on the volume of their sales without comparing their sales to the other sales people. Even in this case people with more sales are likely to be looked on more favorably, and the lowest sellers might find their jobs in jeopardy. These organizational problems can multiply when firms are divided into competing divisions.

Such organizational problems can limit the size of firms, but the principle remains that firms exist as repositories of knowledge, and serve the purpose of reducing barriers to the productive sharing of knowledge. The transmission of tacit knowledge is sometimes referred to as a knowledge externality, and agglomeration economies are the result of knowledge externalities. Firms, then, are intuitional mechanisms that internalize these information externalities.

Knowledge repositories and the capabilities theory of the firm

Building on Penrose,[24] G.B. Richardson develops a capabilities theory of the firm, where "activities had to be undertaken by organisations with appropriate capabilities. ... Firms would find it expedient ... to concentrate on similar activities. ... [Dissimilar activities] would be the responsibility of different firms."[25] Richard Nelson and Sidney Winter look at firms as collections of individuals who have firm-specific intangible assets which aggregate as the firm's capabilities.[26] Thus, Philippe Dulbecco and Pierre Garrouste conclude, "The firm is then quite logically understood, not as the place where the factors of production are combined, but as the place where capabilities are built and modified."[27] The idea that dissimilar activities are carried out by different firms is examined in more detail later in the chapter, but it is also the case that similar activities are carried out by different firms, each trying to develop similar capabilities. The fact that competingfirms are vying for the same customers by selling similar products makes the firm's boundary an important container of the firm's knowledge.

Why do the entrepreneurs who are building and modifying capabilities ensconce themselves in the rigid framework of the firm? The reason is that the structure of the firm acts as a boundary that contains the tacit knowledge of those who work within it. By containing this knowledge, the boundary of the firm also contains the profits that are produced by the firm's capabilities. Entrepreneurship generates innovation which produces profit, and those within the firm have an incentive to keep any knowledge they have developed within the firm's boundaries, so that the firm can profit from it. Richard Adelstein says that all Austrian theories of the firm have a common element in that they link "the emergence of firms to the discovery of new opportunities for profit and the entrepreneur's need for the close cooperation of others to exploit them."[28] In addition to exploiting them, they have an incentive to keep those opportunities within thefirm so that the profit they generate is not competed away.

Nicolai Foss emphasizes that firms are an institutional response to the need of entrepreneurs to coordinate economic activity in the face of disbursed knowledge.[29] Firms can stimulate the entrepreneurial process of discovery. But to profit from entrepreneurial innovation, the entrepreneur also needs to maintain control of the knowledge behind it. The boundaries of the firm help the entrepreneur to retain exclusive control of that knowledge. Once that knowledge escapes the boundaries of the firm, competition dissipates the profit the knowledge produces.

Ulrich Witt concludes that entrepreneurs need firms because some people want to take leadership positions to realize the profitable innovations they envision, while others prefer to be non-entrepreneurial employees, and the firm is an institution that sorts these types of people.[30] There is more to the story, however, because everyone within the firm's boundary can benefit from the knowledge of their fellow employees, and they have an incentive to keep that knowledge within the firm to maintain the firm's profitability. Entrepreneurs need firms because without the firm's boundary, entrepreneurial profits escape as they are seized by others and competed away. So the recognition that firms are knowledge repositories is both an extension of the capabilities theory of the firm and a necessary component of the entrepreneurial theory of the firm.

Inevitably, some knowledge escapes through reverse engineering and other factors that can be observed from the outside. Even being in close proximity outside the firm's boundaries – for example, being in the same city —can assist competitors to assimilate tacit knowledge. But firms have an incentive, not only to develop capabilities through sharing of knowledge within thefirm's boundaries, but also to keep that knowledge from escaping to competitors. When it does, the profits that accrue to innovators are competed away. The idea of reverse engineering to capture knowledge suggests that knowledge is embodied in the capital of a firm, and this will always be the case if one thinks of capital as a stock that produces a flow of income.[31] Physical capital takes knowledge to create, but it also takes knowledge to use. Consider capital goods as different as a wrench and a computer. Both take specialized knowledge to manufacture – knowledge that becomes embodied in the product – but both also take specialized knowledge to use: one cannot just hire "labor" to use wrenches and computers.

Richard Langlois and Paul Robertson, in a chapter titled "A Dynamic Theory of the Boundaries of the Firm," note, "In the long run, the spread of knowledge should lead to a tendency toward the generalized spread of capabilities,"[32] recognizing that the spread of knowledge outside the firm's borders gives competitors capabilities that were once specific to the originating firm, but without noting that this spread erodes the profitability of the firm's specific knowledge. Seeing firms as knowledge repositories is an extension of the capabilities theory of the firm, and explains why entrepreneurs want the firm's boundaries to protect its stock of specialized knowledge. Firms not only develop specialized capabilities, they have an incentive to keep the knowledge that maintains the proprietary nature of those capabilities.

As Peter Lewin and Steven Phelan note, "Once a potential profitis perceived by at least one person, the question then arises as to which organizational arrangement is best suited to its appropriation or renders it vulnerable to appropriation by others."[33] They go on to conclude that "Firms are formed in order to realize, and perhaps protect, the creation of value."[34] The realization of value is the recognition of the capabilities theory of the firm. The protection of value results because firms are knowledge repositories.

The supply chain: selling knowledge

Firms may benefit from revealing knowledge they have to other firms who are not direct competitors. In many industries, from autos to computers, firms that sellfinal products to consumers do not build the product from the raw materials from which it is composed. Rather, they purchase components from suppliers and assemble them. The component suppliers, in turn, may purchase some of their components from other suppliers, and so on until the supply chain reaches the raw materials from which the product is made.

If the knowledge of the supplier is not tacit knowledge, the purchaser may be able to produce the component in-house, which may explain why vertical integration occurs. But if the supplier has some tacit knowledge, the supplier can reveal that it has the knowledge to produce the component without the purchaser being able to obtain that knowledge. The supplier will reveal to potential customers that it has a stock of tacit knowledge, and will offer to sell the advantages it has as a result of that knowledge. The purchaser can see that the seller has valuable tacit knowledge, even though the purchaser is unable to obtain the knowledge. Thus, a supplier can maintain a viable business based on using its tacit knowledge.

The purchaser is in a position to benefit from this type of arrangement, especially if there are several possible suppliers. The purchaser can shop around to get the best component at the best price as potential suppliers compete among themselves. Meanwhile, the purchaser does not have to develop or maintain that knowledge base itself. For this reason, the common practice of manufacturers using multiple suppliers to supply a particular component makes sense. It makes even more sense when one recognizes the dynamic nature of the economy and how entrepreneurs are always looking for ways to make better products and lower their costs within it. The competing suppliers will look for innovations so that they can go to their customers and offer them higher-quality components at lower prices. Competing suppliers help improve product quality and lower costs as a byproduct of their entrepreneurial efforts to find profit opportunities.

If the purchaser decided to vertically integrate, and went to the expense of developing the knowledge in-house, that knowledge would become increasingly obsolete over time, and it may be less costly to purchase inputs from firms that specialize in maintaining and advancing particular types of knowledge that lie outside the firm's immediate areas of expertise. For example, in many (but not all) cases computer manufacturers buy their microprocessors from firms that specialize in making them, rather than trying to maintain current expertise in-house. To benefit from bringing suppliers' activities in-house through vertical integration, the purchaser would not only have to match the suppliers' current level of knowledge and expertise, but also would have to be committed to continually adding to its stock of knowledge to keep up with the market's innovation. Often, a better strategy is to rely on outside suppliers who have specialized knowledge in their areas.

Of course, there are good reasons for firms to vertically integrate. Langlois and Robertson offer a thorough discussion of vertical integration,[35] and Coase's well-known transaction cost explanation of the firm's boundaries also explains why firms may vertically integrate rather than rely on market transactions. The literature has addressed reasons for vertically integrating and bringing processes in-house. The arguments given in this section suggest that vertical integration may be desirable if it is built on proprietary knowledge the firm wants to keep in-house, but will work against the firm when other firms in the supply chain have proprietary information the firm in question does not, or even when all firms are on an equal informational footing. In a dynamic economy, there are advantages to being able to choose among suppliers.

Consider two companies producing consumer goods, one that produces its components in-house and the other that buys them from suppliers. The one that produces its components in-house does not have the advantage of competition among component suppliers that provide the incentive to improve the quality of the product and lower cost. Vertical integration takes away that advantage. Over time, it may be that the most advantageous component suppliers will change, so the firm buying components on the market has the flexibility to shift from one supplier to another to take advantage of any innovations in any firms. The firm that is not vertically integrated can take advantage of competing knowledge repositories, all of whom depend upon continuing innovation to maintain their profitability. The vertically integrated firm must count on its own divisions being able to match or exceed all competitors. The supply chain is selling the advantages of the knowledge it possesses, and has an incentive to continue innovating and adding to its stock of knowledge.

Information diseconomies

Larger firms may be able to take advantage of scale economies to give themselves a competitive advantage. However, there are diseconomies of scale that arise in knowledge repositories because individuals within firms can view fellow employees as competitors, which can inhibit the sharing of knowledge.

One reason the management of a large firm is a challenge is that it may be difficult to assign profits (and losses) to particular elements of the firm. In the aggregate, the firm incurs costs and collects revenues, with profit being the excess of revenues over costs. However, in a large firm it may be difficult to determine which components of the firm are producing the profits and whether some components are generating losses. Some components – the overall costs of managing the firm are an example – may be difficult or impossible to assign to particular revenue streams. One way this is sometimes handled is to divide the firm into divisions or profit centers and do separate accounting of profits and losses in each division or profit center.

A problem that arises with this type of organization is that the divisions can view themselves as competing with each other, so those with knowledge in one division may be reluctant to share their knowledge with those in competing divisions. Thus, the internal organization of the firm can erect the very knowledge barrier that the existence of the firm (as a knowledge repository) is designed to overcome. As Ivan Pongracic notes,[36] the profitability of large-scale firms requires that intra-firm institutional structures be developed to align the interests of the firm's owners and workers, and provide incentives for employees to contribute toward the firm'spro fits. Such structures are difficult to design, which places a limit on the size of the firm.

Large firms will have many employees who undertake the same functions, and they too may view each other as competitors. For example, consider a firm with many people in its sales force. More sales improve the profitability of the firm, so in this sense each salesperson benefits from increased sales. However, it is not uncommon for the sales of individual salespeople to be compared with each other. People who sell more will earn more, and may be in line for promotion to sales manager. (Of course, large firms have competing sales managers too.) Salespeople whose sales are low by comparison may lose their jobs. Thus, people within the firm can come to view others working for the same firm as competitors, which will inhibit their sharing of knowledge. Salespeople with good techniques, or who come up with good lines to attract customers, will keep them to themselves rather than share them. Again, the internal organization of the firm can erect a knowledge barrier that the existence of the firm is designed to overcome.

The same is true in manufacturing. A large firm with several plants that produce the same, or similar, components, can compare those plants with each other to gauge their efficiency. But if rewards go to the people at the more productive plants that can provide a disincentive for sharing knowledge among the plants. Again, the function of thefirm as a knowledge repository can be compromised.

Good management of the type described by Armen Alchian and Harold Demsetz means overcoming these barriers, making all employees believe that they are working on the same team and that they are not competitors, and keeping them from shirking, where shirking includes not sharing productive information with fellow employees.[37] But there is a limit to how successful this can be as the firm grows. If the firm has one salesperson, that person has no disincentives toward sharing any available information with others in the firm (such as, for example, informing management about product characteristics customers say they desire). As the sales force increases, it will become more difficult for mangers to create a team spirit among the employees. Thus, information diseconomies can create the same barriers within firms that exist between firms, which is a factor that limits the size of firms.

Eventually, it becomes more efficient to have smaller units to reduce these knowledge barriers. Smaller units can be produced by having firms divest themselves of divisions, which is a common occurrence, but can also occur when smaller more nimble competitors take market share from larger firms with less flexible management structures.

Division of knowledge and knowledge repositories

One way to view firms is as specialists in the division of knowledge, to use Hayek's phrase.[38] Some firms specialize in one type of knowledge while others specialize in other types. Consider, for example, the manufacture of a computer, where one firm produces the microprocessor, another produces the hard disk drive, and so forth. Each firm specializes in knowledge that the others do not possess, nor do they need to possess, because the market coordinates the knowledge of the various firms to produce the final product and bring it to market. Indeed, as Peter Lewin and Ludwig Lachmann note, there is a similar division of capital, which can occur within or among firms, in which capital becomes increasingly specialized, enhancing productivity and generating increasing returns.[39]

Taking this approach to the analysis of firms suggests a natural division that creates firms within a market. Firms specialize in a particular type of knowledge, which lies within the firm's boundaries and engages in market transactions to take advantage of the different types of knowledge held by others. Such an approach is consistent both with Coase's view of the division between firms and market transactions based on transaction costs and with Richardson's capabilities view of the firm. Knowledge outside the boundaries of the firm is more economically acquired by purchasing it from others who specialize in that knowledge, rather than developing it in-house. This approach illuminates the way in whichfirms cooperate with each other and coordinate their activities to bring goods and services to market. The firms within a supply chain are complements, not substitutes.

This division of knowledge approach to understanding firms does not explain why there are many firms that serve the same function. One may conjecture that there may be limits to managerial efficiency that would limit the size of firms, but viewing firms as knowledge repositories offers another reason. Firms organize so that owners and employees with specialized knowledge can pool that knowledge for their mutual benefit, reaping the profit from it, and exclude others who are outside the firm from that profit. The firm is an institution that enables entrepreneurs to capture and contain the profits from their innovations. Sharing knowledge within the firm provides a mutual benefit; allowing knowledge to escape outside the firm's boundaries reduces the firm's competitive advantage and places outsiders on a more equal footing. The role of profits and losses within the market gives people within the firm's boundaries an incentive to share knowledge among themselves, and to keep that knowledge from others who are not part of the firm.

This manifests itself in trade secrets, and non-compete agreements which specify that if employees leave a firm they are not allowed to take proprietary information with them, nor work for competing firms. Under the contemporary legal structure it also manifests itself in patents. While patented information does become public, the patent holder has a monopoly right to it, which also enables the patent holder to lease the patent or trade it for information held by other firms.

An overlooked reason for the existence of firms is that they act as knowledge repositories. A single entrepreneur may spot a profit opportunity, and believe that organizing afirm around that profit opportunity will enable the entrepreneur to sell the firm's output for more than the cost of the inputs, yielding a profit. If the entrepreneur's knowledge were easily observable, and if the entrepreneur's activities were easy to duplicate, that knowledge would be a common pool resource that would rapidly erode the entrepreneur'sprofit, rendering the establishment of a firm pointless. There is always risk in an entrepreneurial venture, because the entrepreneur is attempting to take advantage of a profit opportunity others have not exploited. Thus, there is no way to know from experience whether the entrepreneur's venture will succeed.

The entrepreneur will take a risk only if there is the potential for a return in the form of entrepreneurial profit. For this to happen the entrepreneur must possess knowledge that will be difficult for others to appropriate. Were it not the case, imitators would enter the entrepreneur's market as soon as profits appeared, eliminating the profit opportunity. The entrepreneur would suffer losses if the profit opportunity were only an illusion, but would gain no profit (because it would immediately be competed away) if the profit opportunity were real. There would be no payofffrom taking the risk of entrepreneurial action.

For entrepreneurship to pay off, the entrepreneur must have knowledge that remains within the knowledge repository of the firm, so cannot be imitated by competitors. Often, this can come solely from the entrepreneur's first-mover advantage. The entrepreneur can start a successful business, and as other businesses try to imitate the entrepreneur's success the entrepreneur acquires additional knowledge that allows them to innovate and stay ahead of rivals. The entrepreneur may also believe that the firm has engineering, design, or marketing talent that will enable it to maintain a lead over its rivals. This is an essential component of the capabilities theory of the firm. A good example at the beginning of the twenty-first century is Apple Computer, which has come out with a series of innovative goods and services (iMac, iPod, iTunes, iPhone, iPad) that have meant rivals are always trying to catch up. Before they can imitate one innovation, another appears on the market.

Entrepreneurs run the risk that those inside the firm's boundaries may leave, taking the knowledge they have acquired with them, and start competing firms.

While this can happen, employees who leave will lose the synergy produced by the continually developing tacit knowledge of their former colleagues. Unless they develop a substantial amount of tacit knowledge themselves, employees benefit from staying within the boundaries of their current firm. Departing employees may be on a relatively equal footing with their former employer when they leave, but unless their new firms can become knowledge repositories that rival their former firms, they will be unable to keep up in their markets. An example is former IBM engineer Gene Amdahl, who left IBM in 1970 when IBM mainframe computers dominated the market, to start a rival company manufacturing mainframe computers. Amdahl's first mainframe computer was sold in 1975. Amdahl's computer company did enjoy some success, but was bought by Fujitsu in 1997. IBM continues to be a dominant force in the computer industry in the twenty-first century, while Fujitsu is not among the major players in that industry.

One reason for forming a firm is to concentrate that knowledge into a knowledge repository so that the profits from the knowledge can be shared among the firm's owners, managers, and employees. The firm's boundary serves as a knowledge boundary. The firm's knowledge stays within that boundary to maintain the firm'sprofitability, and to prevent that knowledge from becoming a common pool, which would eliminate its profit potential. Firms are established to allow those within the firm to take advantage of the firm's knowledge repository, and to prevent those outside the firm from doing so.

Firms and institutions

Firms are institutions that are the product of the institutional framework within which they exist. Institutions determine the nature of firms. One example from earlier in the chapter are the institutions of accounting conventions. Without effective accounting procedures it would be difficult for firms to expand beyond one product line; or at least, the firm's owners would have to separate the revenues and expenditures of each product line to see whether they were covering their costs. As accounting has become more sophisticated, it has been better able to keep track of revenues and costs, and so has enabledfirms to expand to take advantage of economies of scale. Double-entry bookkeeping is a well-recognized innovation, and sophisticated methods of accounting for depreciation of capital assets, the accumulation of goodwill, and other accounting nuances support firms.

Following Jensen and Meckling,[40] if a firm is viewed as a nexus of contracts, contract law also has an important role to play. The institutions are partly a spontaneous creation of contract provisions that are to the mutual benefit of the contracting parties, but are also partly a created order of legal requirements. For example, laws requiring compensation rather than specific performance, and those that make labor contracts binding on employers but (for the most part) not binding on employees, affect the design of firms.

One of the most significant institutional developments affecting firms was the creation of the limited liability corporation. Rather than making a firm's owners liable for the actions of the firm, the stockholders in limited liability corporations are liable only up to the amount they have invested in the corporation. Without limited liability it would be much more difficult for corporations to finance their operations, which would limit the size of corporations. The point is that firms have to be viewed as more than just a nexus of contracts, and the role firms play in an economy is determined by an institutional structure that partly is a spontaneous order, but is also partly a result of government design.

Penrose observes that corporate research "enables at least the large firms to turn aside the process of 'creative destruction' and to thrive on the novelty that might otherwise have destroyed them."[41] But this holds true only if firms can make proprietary use of the discoveries of their research. If firms were not able to erect barriers to prevent knowledge they developed from flowing to their competitors, knowledge would become a common pool resource. All firms would be looking to take knowledge from that common pool, but there would be little incentive for people to contribute knowledge to the common pool. If it were available to everyone, the knowledge would be a public good, and those developing the knowledge would incur the expense of doing so, while sharing the benefitofthe knowledge with everyone else. Thus, economic progress depends on firms being knowledge repositories, so that those who go to the expense of developing knowledge are able to profit from doing so.

Knowledge is often tacit; that is, it belongs to specific individuals who have acquired it by experience, and who may not be able to communicate that knowledge to others even if they try. Even ordinary tasks such as riding a bicycle or hitting a baseball can only be learned through experience. One can explain to someone how to ride a bicycle, but no explanation is sufficient. The same is true of manufacturing operations, crafts, and management functions. Thus, even when competing firms "see" what their competitors are doing, by observing their production processes or seeing their products, the firm is still likely to hold some information in its repository that cannot easily be appropriated by its competitors.

Over time, competitors can see, imitate, and eventually acquire the knowledge the entrepreneurial firms they compete with have generated. That is why firms must continually be entrepreneurial to have continued success. By the time competitors have obtained the knowledge of their entrepreneurialcompetitors,entrepreneurialfirmswillhavebuiltontheirpre-vious knowledge to keep their lead, as the example of Apple Computer in the previous section illustrated. Other firms see and imitate (and sometimes even improve on) the features of new products, but by the time they do, the more entrepreneurial firms have built on those advances to offer even more improvements.

Because the profits from innovation and knowledge stay within the boundaries of the firm, firms have the incentive to develop new knowledge and bring new innovations to market, because those within the firm's boundaries profit from the development of that knowledge. Thus, the firm, as a knowledge repository, is an important economic institution that generates economic progress.

Conclusion

Economists have examined firms from two different perspectives. One perspective looks for reasons why firms exist as economic organizations; the other takes firms as a feature of the economy and explains how firms affect market outcomes. This chapter has considered both perspectives. Taking up where the previous chapter left off, the innovative entrepreneurship of firms is the foundation of economic progress and the generator of prosperity. Following Schumpeter, the entrepreneurial activity of firms can be seen as disruptive, but short-run disruptions open entrepreneurial opportunities to reallocate resources, so (following Kirzner) profit-seeking entrepreneurs reallocate resources to more valuable uses. The equilibrating entrepreneur depicted by Kirzner reallocates resources that were disrupted by Schumpeter's creatively destructive entrepreneur. The process of disruption and reallocation is the process of economic progress.

This leaves open the question about why entrepreneurs undertake their activities withinfirms. The theoryof thefirm has offered many explanations, and they are not mutually inconsistent, so firms may serve a number of roles. This chapter emphasizes one that has not been brought out in the literature. Firms act as knowledge repositories, because firms are profit repositories. The firm's boundary serves the dual function of providing an incentive to share tacit knowledge within the firm, for the benefitof its owners and employees, and preventing that knowledge from spreading to competitors, which would result in its value being dissipated. One reason to establish a firm is to share the benefits that come from combining the tacit knowledge of its employees. Entrepreneurs create firms to contain and capture the profits from their innovations.

Many other theories have been proposed for the existence of firms, and this analysis is not intended to replace them. The institution of the firm may serve many purposes. However, previous theories have not recognized the importance of the firm's boundary as a mechanism to prevent tacit knowledge from escaping to others outside the firm. In a dynamic and entrepreneurial economyfirms can only survive by improving their products and production methods, and some of the knowledge that can produce this type of innovation is tacit knowledge. Within the firm there is an incentive to share this tacit knowledge, whereas those within one firm have an incentive to keep this tacit knowledge from those in competing firms.

The existence of tacit knowledge in an entrepreneurial economy also helps explain why firms utilize the supply chains of other firms to produce components for their production. It also provides a reason why there are informational limits that constrain the size offirms. The literature has noted that agglomeration economies explain the existence of concentrations of economic activity, and explain why industries tend to concentrate geographically. The institutional structure of the firm reduces some of the barriers that inhibit agglomeration economies by creating an environment within which people have an incentive to make their knowledge available to others in the firm. Thus, the firm can be viewed as a manifestation of knowledge-based agglomeration economies. From this point, seeing the role of the firm, the next chapter looks at how an entrepreneurial economy generates progress and prosperity.

7 The trajectory of the economy

]7 The trajectory of the economy

Growth and progress

An equilibrium approach to economic analysis would suggest that economic forces keep an economy close to equilibrium. The economy can be subject to shocks – the term often used to describe exogenous disequilibrating forces – after which the equilibrating nature of the economy returns the economy to equilibrium. Pushing equilibrium analysis to the extreme, one might assume the economy is always in equilibrium. The static concept of equilibrium can be extended to a dynamic equilibrium in which the economy remains on an equilibrium trajectory as it grows over time. This is how the theory of economic growth developed over the twentieth century.

Growth theory depicts growth as increasing economic output. For the purpose of the theory, output is depicted as homogeneous. While a growth in output, giving individuals higher levels of income and consumption, is a component of economic progress, it is a minor component. The larger component is changes in the type of output being produced, meaning new goods and services, and improved versions of existing goods and services. Furthermore, growth, measured as increases in output, is dependent upon this qualitative aspect of progress, so depicting an economy as growing while leaving out the forces that create qualitative changes leaves out the engine that actually causes the growth.

Over the twentieth century incomes in the United States increased about sevenfold, which certainly increased people's standards of living. But the larger change was the change in the type of output, so that by the end of the century people could cross the continent in jet aircraft in a few hours rather than taking the train; they could communicate by telephone and email rather than by sending letters; horses became sources of recreation rather than means of transportation; and people could cook their meals in microwave ovens and wash their dishes in automatic dishwashers. Huge advances in healthcare technology extended lifespans and increased the quality of life. Huge changes also occurred in the workplace so that jobs became less physically demanding and less dangerous. By the end of the twentieth century people looked for jobs that gave them personal satisfaction beyond the incomes they earned. This would have been rare at the beginning of the twentieth century, when most jobs were simply a source of income to offset the disutility of work.

If there had been no change in the goods and services people consumed over the century, there also would not have been the substantial growth in the measured quantity of output. While people did tend to eat more at the end of the twentieth century than at the beginning, they did not eat seven times as much. People would not have had the demand for seven times as many horses; indeed, the demand for horses fell as people started using automobiles for personal transportation and tractors replaced farm animals. Output could only increase as much as it did because new and improved goods and services were available for people to consume. This aspect of economic progress has been vastly underappreciated in the economic theory of growth.

New goods and services, and new production methods that reduced the unpleasantness of work even as they increased labor productivity, are the result of entrepreneurship. Economic progress would not be possible without entrepreneurship, and economic growth would not be possible without economic progress. If one starts with even the high levels of income people in developed economies enjoy today looks back at growth rates of per capita income, it becomes apparent that income growth is a phenomenon only a few centuries old. Extrapolating backward from today's incomes, only a few centuries ago incomes would have been at a subsistence level prior to the Industrial Revolution. It was only the awakening of people's entrepreneurial instincts through the development of a capitalist economy that has produced economic progress, which is necessary for income growth.[1]

Growth theory

The twentieth-century theory of economic growth builds on the equilibrium framework for economic analysis discussed in Chapter 1. Using the neoclassical framework from Chapter 5, output, Q, is a function, f, of inputs capital, K, and labor, L, or Q = f(K,L). Growth is depicted by increases in Q, and the production function shows that this can be accomplished by increasing the inputs, K and L, or by increasing the productivity of production by improving the production function. One can envision Q*= g(K,L) > Q' = f(K,L), and growth can occur by shifting from production function f to production function g. Typically, this has been thought of as incorporating technological change.

As Robert Lucas insightfully argues, L should be viewed not just as the number of laborers, but rather as a measure of human capital.[2] Thus, looking at the production function, growth can occur by investments in physical and human capital, and by technological advances. In hindsight some evidence that this approach to economic growth leaves out some essential causal factors comes from economies that applied this model of growth most literally. For example, the former Soviet Union invested heavily in both physical and human capital, and their educatedworkforce engaged in a policy, directed by central planning, of incorporating technological advances into their production processes. Yet, despite their investments to increase K and L, and their policy of trying to shift production from Q* = f(K,L) to Q' = g(K,L), the Soviet Union collapsed as a result of its economic failures.

This central economic planning found support not only from communist governments, but from the leading architects of twentieth-century economic theory as well. Taking this framework for economic growth very literally, Paul Samuelson, in the ninth edition of his best-selling introductory textbook published in 1973 (the year of Ludwig von Mises's death), estimated that the Soviet Union's per capita income was about half the per capita income in the United States; yet because of its superior economic system he argued that the Soviet Union's economic growth rate was greater, and projected that it would catch up to the United States in per capita income perhaps as early as 1990, and almost surely by 2010.[3] Historical events have shown that the theory underlying Samuelson's forecast is somehow flawed or incomplete, yet that same foundation remains in neoclassical growth theory in the twenty-first century.

One problem with this framework was noted earlier. When analyzing the substantial increase in economic well-being that people in market economies have enjoyed over time, the major component in that increase is the new goods and services that people have to consume. While it is true that if one just reduces all of the heterogeneous production of the economy into a homogeneous measure, Q, measured Q has increased substantially, this method of analysis leaves out the fact that as aggregate income has grown the components that make it up change over time. The method of aggregation in growth theory depicts the process as the growth of a homogeneous measure of income without recognizing that this growth embodies the progress represented by changes in the components.

This is a fundamental shortcoming of the theory (and not a mere detail) because it is the change in the components of output that cause the increase in aggregate income, and in a framework that does not incorporate the change in the components the fundamental cause of growth is assumed away. If the model assumes away the fundamental cause of growth, it cannot explain what it has assumed away. For growth theory to accurately depict the nature of economic growth, it would have to account for economic innovation and changes in the composition of output, not just depict growth as increases in Q. In other words, it would have to model progress, not just income growth.

As impressive as it is that general equilibrium macro models can be calibrated to accurately describe the movements of economic variables, recall the discussion in Chapter 2 that drew a parallel between these models and Ptolemy's model of the universe as concentric spheres. The Ptolemaic model of the universe shows that a model can be calibrated to accurately replicate real-world movements in data and yet be completely inaccurate in its depiction of the actual underlying phenomena that cause the data movements. If the engine of economic progress is the new goods and production methods that are completely left out of a growth model that just shows growth as increases in aggregate output, one might question whether such growth theories really are descriptive of the underlying processes that cause growth, even if they are able to closely replicate movements in macroeconomic data.

Income could not have grown as much as it did in the twentieth century if the characteristics of output had not changed. People did not work so they could buy more of the same goods, but because new and improved goods were available to them. Had the characteristics ofoutput remained the same, total output would not have expanded as much as it did. Progress is the engine of economic growth and the economy grew because progress created improved output – as judged by the demands of those who consumed it. Income growth cannot be understood without incorporating the forces that led to innovations in the types of goods and services that people consume.

This type of innovation results in the creative destruction that lay at the foundation of Joseph Schumpeter's theory of economic development. If one envisions an economy prior to an innovation as in a competitive equilibrium, the disturbance caused by the innovation causes some activities that were previously profitable to become unprofitable, as people migrate from old products and markets to new ones. That creative destruction is welfare-enhancing and produces economic progress. If the creative destruction disturbs the existing way of doing things, it may be a move away from an existing Pareto optimum, further suggesting the problems with using Pareto optimality as a benchmark for welfare maximization. Movements away from Pareto optimality caused by innovative creative destruction are welfare-enhancing, in contrast with neoclassical welfare economics. When economic progress is seen as the cause of improvements in welfare, sometimes movements away from Pareto optimality are welfare-enhancing.[4]

By the end of the twentieth century the equilibrium framework had been extended, with Nobel laureate Robert Lucas in the lead, to incorporate macroeconomic fluctuations as equilibrium phenomena, and to integrate the theory of macroeconomic fluctuations with the theory of growth.[5] Depicted this way, the economy is always in equilibrium, and economic fluctuations are integrated into the economy's equilibrium growth path.

This integration of macroeconomic fluctuations with growth was also depicted by Joseph Schumpeter in his 1939 book, Business Cycles,[6] but because the Keynesian revolution in economics was underway when Schumpeter's book came out, it never entered the economic mainstream. While Schumpeter's book has in common with general equilibrium growth theory the idea that macroeconomic fluctuations are a part of the same process that produces economic growth, their underlying engines are different. Schumpeter depicts both macroeconomic fluctuations and growth as a product of the creative destruction of entrepreneurship, which requires the introduction of new and changing goods, whereas the general equilibrium approach depicts output as a homogeneous Q. Ultimately one would want to have an economic framework that explains both business cycles and growth, rather than different models to explain the two economic phenomena, because the Q that fluctuates with business cycles is the same Q that increases with growth. However, the homogeneous Q assumed in growth theory is inadequate to depict the actual process by which output grows.

Equilibrium, growth, and progress

Equilibrium models cannot describe economic progress because of the way that they depict competition, as an equilibrium end-state, rather than a process whereby competitors in a market are looking for competitive advantages. The previous chapter illustrated that firms must be entrepreneurial in order to remain viable in the market, and this continual entrepreneurship means continual changes in the activities of firms. Production processes, inputs, and the nature of outputs are always subject to change, and entrepreneurs cannot use them as a point of reference for planning current conditions, but rather must anticipate future conditions.

If one depicts growth as an equilibrium phenomenon represented by ever-increasing levels of Q, the underlying cause of increasing Q is left out of the analysis. The reason income goes up is that entrepreneurs bring new and improved goods and services to market, which gives rise to demands for those goods. Thus, to understand why economies grow, a framework which contains heterogeneous and continually changing goods must be employed. Entrepreneurial innovation must be an integral component of a model of growth because ultimately that is what drives it.

The neoclassical framework of equilibrium growth theory depicts technological progress as being the result of research and development, which is produced in the same way as any other manufactured input.[7] Inputs of capital and labor are combined to produce R&D, which then leads to technological advances. This depiction falls short in that it describes how new technologies are developed, but not how they enter the market to generate economic progress. Schumpeter made the distinction between invention and innovation, calling invention the technological discovery and innovation the incorporating of the advance into the production process.[8] Within the neoclassical framework, technological progress is represented simply as discovering a new technology, and there is no recognition of the challenges involved in moving technology from an inventor's laboratory into a marketable commodity. The neoclassical model of perfect competition is really a model of perfect stagnation. Profit-maximizing firms keep doing the same thing year after year, and keep getting the same normal profits from doing it.

In the real-world competitive environment, firms are always finding new and improved ways of doing things, as the previous chapter described, so firms cannot simply find the profit-maximizing strategy and continue to follow it in the future, because their competitors will always be improving, and the profit-maximizing strategy of today will generate continually declining profits, and eventually losses, in the future. Real-world competition is a world of continual change, where competitors must improve just to keep up with the competition. This competition leads to innovation, which generates economic progress. Incomes rise as a by-product of the entrepreneurial actions prompted by economic competition.

William Baumol's book, The Free-Market Innovation Machine, provides a good example of the way that innovation can be treated simply as an outcome of research and development.[9] Baumol depicts economic progress as a result of research and development that has become a routinized part of corporate bureaucracy. Baumol completely ignores the entrepreneurial function of implementing such advances in a profitable way, and as a result argues that entrepreneurship is less relevant now than it has been in the past. Meanwhile, Baumol, with co-authors Litan and Schramm, depicts an economy whose economic progress is driven forward by entrepreneurs who are, typically, not a part of the corporate bureaucracy of large firms.[10] So, it appears that the analysis of Baumol's book with Litan and Schramm argues against the thesis put forward by Baumol's Free-Market Innovation Machine.

Baumol, Litan, and Schramm argue that economic progress is generated by entrepreneurial firms which tend to be smaller and more nimble, and that bring innovative products to market, which generates economic progress. That idea is entirely consistent with the framework being developed here. Baumol's Free-Market Innovation Machine, in contrast, argues that research and development has been integrated into corporate bureaucracies, and has become a routine part of corporate activity, making entrepreneurial firms obsolete, or at least minor players in economic progress when compared with the routinized R&D-led progress of large corporate bureaucracies. Baumol here fails to distinguish between invention and innovation, as Schumpeter did, and so fails to recognize that what he describes in corporate R&D is the invention, not the innovation.[11]

There is little point in picking at inconsistencies in these two books. The main point here is to see the way that R&D is, in fact, treated as an input into the production process, much like the inputs of capital and labor, which then results in technological change. This analysis leaves out the way that inventions are transformed into innovations that actually result in economic progress. That was the subject of the previous chapter: entrepreneurship takes inventions and turns them into innovations. A theory of growth that leaves out entrepreneurship, and a theory of growth that does not take into account the innovations that lead to economic progress, cannot adequately describe the process of growth because it leaves out the proximate cause. Entrepreneurs are always looking for competitive advantages, and they have to in order to survive, because their competitors are also behaving entrepreneurially. One cannot understand economic growth except with the foundation of entrepreneurial firms, as described in Chapter 6.

Because models are simplified depictions of reality, no one model can be ideal for every application. The neoclassical model of competition is an excellent model for depicting the equilibrating nature of a market economy and the static efficiency properties embodied in that underlying equilibrium. However, that model leaves out fundamental elements of the economy that generate economic progress. Therefore, the equilibrium model that has served economics well for some purposes is inadequate for the purpose of depicting economic progress.

Some might argue that this analysis of the neoclassical competitive model is a straw man, and that economists recognize the competitive strategy of product differentiation and that firms must innovate to remain competitive. Of course this observation has been made before. However, the analysis is not a straw man, in that the dominant models in growth theory at the beginning of the twenty-first century are general equilibrium models that depict growth as increases in a homogeneous measure of output. Yes, in the abstract economists do recognize that entrepreneurs continually improve their products as a part of the competitive process, but even while recognizing this, growth theory is built on a framework within which growth is represented as an increase in a homogeneous measure of output. In fact, growth requires progress, and progress occurs through entrepreneurship, product differentiation, and the introduction of new and improved goods and services into an economy.

Equilibrium, entrepreneurship, and progress

Entrepreneurship is the engine of economic progress, and economic progress is an integral part of a market economy. The very notion of equilibrium paints an unrealistic picture of a competitive economy. Schumpeter's depiction of the process as creative destruction provides a more accurate picture of the competitive process.

Markets can be temporarily disrupted, as the equilibrium framework suggests, but the market process continually produces permanent dislocations that are not easily capturedwithin an equilibrium framework. The term equilibrium suggests that if it is disturbed the economy re-equilibrates by returning to equilibrium, that there are market forces to pull the economy back to the equilibrium state from which it was disturbed. This paints a fundamentally misleading picture of the market process, because while some disruptions may be of that type, the more significant disruptions change the underlying conditions so that, disturbed from one equilibrium, the economy will never return back to it. The economy never returns to its old equilibrium, because conditions are always changing, so the old equilibrium − if it even makes sense to use that term − no longer exists.

In this situation, the equilibrium framework loses much of its usefulness, because the underlying equilibrium is always changing, and entrepreneurs cannot simply account for new conditions and try to maximize profits under the new regime. Entrepreneurs must know (to remain profitable) that there is, in fact, no equilibrium, in the sense of some unique outcome toward which economic forcesarepulling the economy. Entrepreneurs react to current conditions, and the very act of their entrepreneurship changes the outcome toward which the economy is tending. Profits can be made by taking advantage of disequilibrium situations, as Kirzner[12] emphasized, but the notion that entre-preneursareequilibrating theeconomy ismisleading in thesensethatthereis no underlying stable equilibrium toward which the economy is tending.

Chapter 4 made the distinction between equilibrium, in the sense of general equilibrium versus equilibrium, as a reference to market clearing. The distinction is important because there is no reason why the action of market clearing would pull an economy closer to an equilibrium. Indeed, the connection between market clearing and equilibrium is only an artifact of equilibrium models. Markets can clear, in the sense that prices adjust so that the quantity supplied equals the quantity demanded, without any underlying equilibrium toward which the economy was tending. The fact that markets tend to clear says nothing about equilibrium, unless equilibrium is defined to mean market clearing and nothing more.

Neoclassical growth theory, building on Solow's foundation, depicts an equilibrium growth path, or trajectory along which the growing economy moves, but this too is misleading. While in hindsight one can see the path over which the economy has moved, the trajectory is not determinate, and depends crucially on the decisions made by entrepreneurs. But entrepreneurs can never know whether they are making optimal decisions or choosing the most profitable opportunities. By choosing one option they are foregoing others, and while they can observe the profitability of the options they actually chose they can never know what the outcome would have been had they made a different choice. Their choices change the economic environment, which in turn alters the future trajectory of the economy.

Cost, choice, and progress

The neoclassical framework, in which Q = f(K,L), lays out all options in front of the firm's decision maker, so that one can determine with mathematical precision the optimal course of action. Finding the optimal course of action is simply a matter of mathematical maximization. In the real world, taking one course of action means closing offothers. One can see, in hindsight, what the result was of a particular choice, but there is no way to compare the choice that was taken with what would have happened under a different course of action.[13]

In the framework of neoclassical profit maximization, firm managers can adjust their inputs and observe what happens to their costs and profitability, so within that framework, managers can move toward more profitable outcomes until they maximize their profits. Any slight movements away from profit maximization will lower profits, providing evidence to the firm's managers that they are in fact at the point of profit maximization. Looking at the firm's production function, profit maximization means choosing the optimal levels of inputs, and inputs are continuous variables that can be adjusted in small increments. If moving in one direction (say, increasing K) lowers profit, decision makers would then know to move back in the other direction. If it increased profit, decision makers would continue that move, marginally adjusting until profit no longer increased. This describes, from a procedural standpoint, the mechanism a firm's management could use to maximize profit, but when the production function is given even this marginal adjustment is not necessary because management can calculate the values that would result in profit maximization.

In the real world firms cannot observe the mathematical forms of their production functions, and even the strongest advocates of neoclassical price theory would not claim they could. But the equilibrium framework assumes firms do, in fact, maximize profit, and the example of iterating toward aprofit maximum illustrates how this could occur.

The actual decision framework that entrepreneurs face is different from this. Firms face discrete choices of many kinds, and there is no way to know, given that one choice is made, what the outcome would have been had the entrepreneur made a different choice. Looking at the production function, Q = f(K,L), entrepreneurs choose Q, the products they produce, and their characteristics. Choosing to produce one product means diverting resources from producing others. Entrepreneurs also choose f, the production function. If an entrepreneur is considering two production processes, f and g, choosing one means the entrepreneur will not observe the result that would have occurred with the other. Similarly, while the simple theory assumes homogeneous K and L, in reality different types of capital and labor will be required depending on the output produced, Q, and the production process, f.

Consider four examples: Henry Ford's decision to build automobiles on an assembly line, IBM's decision to produce the 360 operating system in the 1960s to help market computers to businesses, Steve Jobs's decision to start his own company selling personal computers, and Boeing's introduction of the 787 airliner.

In the young industry for automobiles, Ford looked at existing firms using a production technology such that Q = f(K,L) and conjectured that a different production process, Q = g(K,L), could produce automobiles at lower cost, and that at that lower cost there would be an increased market for his automobiles. Ford could not examine actual market data to support his conjecture, because no data were available. Automobiles were assembled by hand, and were expensive, so Ford would not have had any evidence on the actual cost savings that could be achieved through assembly line production, nor could Ford have had any evidence on how many automobiles he could sell at a lower price – the price elasticity of the demand for automobiles – because no automobiles were sold as inexpensively as Ford was able to sell his Model T.

In hindsight one can observe that Ford's conjectures were very profitable, but one cannot observe that they were profit maximizing. There may have been different features Ford could have offered in his cars to make them even more appealing to buyers, and there may have been different production methods Ford could have used to make them at even lower cost. Indeed, Ford – and his competitors – did modify their production methods over time, which did increase efficiency. This is the nature of entrepreneurship. An entrepreneur can gain an advantage in the market, as Ford did, but other entrepreneurs can imitate and improve on the advances made by their competitors. One can see that a particular choice results in profits, but one can never see that a choice not taken would not have resulted in even greater profits. In other words, one can see that firms earn profits, but one can never tell whether they are maximizing profits.

The whole entrepreneurial process is based on the presumption that existing firms are, in fact, not maximizing their profits, so that there are changes that can be made to product characteristics and production processes that would be more profitable than the status quo. One entrepreneurial action opens opportunities for additional entrepreneurship as firms attempt to keep up with, and surpass, their competitors.

At the beginning of the 1960s computers had single-tasking operating systems and computer programswere written to run on the computer. IBM's innovation with their very successful IBM 360 computer was to design a multitasking operating system, and to have programs written to run on the operating system, so that the operating system acted as an interface between the program and the hardware. Rather than compiling a program (if it was written in a higher-level programming language) and then having the compiled program run directly, the operating system handled the running of the program and allowed many programs to run simultaneously. This multitasking operating system was an innovation, but also a substantial risk to IBM because of the expense of developing it. Within IBM people referred to the 360 operating system project as "you bet your company" because it was so expensive some thought the company could go bankrupt if it failed.

Hindsight shows it was hugely successful, and the gamble IBM undertook gave them so much market share that the Justice Department sued them for monopolizing the industry. One can see in hindsight that the innovation was profitable, but it may be that different characteristics in the operating system might have made it even more desirable to consumers, or that different production methods might have made it less expensive to develop. One can see that it was profitable, but there is no way to tell whether it maximized profits, because there is no way to observe what would have happened had the company made different choices.

Similarly, Apple Computer was started by Steve Jobs and Steve Wozniak because they were unable to interest existing firms in the computer industry in their product. The experts in the industry viewed their personal computer as a toy with a limited market. So, they started their own company, and in hindsight it was a profitable decision. But again, there is no way to tell whether it was profit-maximizing, because one cannot observe what the outcome would have been had they made different choices. They decided to use a more expensive color monitor on the Apple II, their first successful commercial product, rather than monochrome, because Wozniak saw a demonstration of a color monitor and thought that color would enhance the desirability of their product more than its cost.[14] One can see that the actual product, with the color monitor, was profitable, but there is no way to tell whether it was more profitable than had the computer been less expensive and come with a monochrome monitor, because Apple did not take that option.

Both IBM and Apple looked at the computer industry as it existed, selling output Q, and conjectured that there was a product with different characteristics, Q', that could be produced with different production methods, to yield an output more profitable than what the industry was producing, so they chose to produce Q' = g(K,L) rather than Q = f(K,L). But choosing Q' over Q, and g over f, are discrete choices. Entrepreneurs can rarely iterate toward a profit maximum through marginal adjustment, and there are no data that can reveal whether the choices they were about to make would even be profitable, let alone profit maximizing. Many other entrepreneurs in the computer industry introduced what they hoped were innovations into the market, but went out of business because the changes they introduced were not profitable when stacked up against the options offered by their competitors.

Boeing's development of the 787 airliner can be represented similarly. Boeing decided, first, to make the fuselage out of composites rather than aluminum, partly to lower the weight of the aircraft and therefore increase its fuel efficiency. Boeing also decided to design the aircraft on computers prior to any manufacturing. This required Boeing to design its own software, because an aircraft had never been designed this way previously. Boeing also changed its assembly methods and relied more on subcontractors to supply the components. So, with different product characteristics and different production methods, Boeing chose Q' = g(K,L) over Q = f(K,L), and as before, one can observe the profits from the choices they made, but there is no evidence that their choices were profit maximizing, because there is no way to see what the result would have been had Boeing made different choices. Boeing made several discrete choices in deciding on how to produce the 787 and there was no way the choices could have been made as marginal adjustments.

The decisions these entrepreneurs made, in hindsight, were profitable, but there may have been additional options they did not take that would have been even more profitable, and many entrepreneurs undertake business activities that ultimately turn out to be unprofitable. They were wrong. Profitability means having revenues that exceed costs, but both revenues and costs are market determined, and depend on the activities of others in the market. If Henry Ford had not begun the assembly line production of automobiles, hand-assembled autos would have remained profitable, at least until another entrepreneur seized that profit opportunity that the hypothetically more cautious Ford passed up. This is true in the neoclassical framework as well, but it is obscured by the assumption that firms produce homogeneous products with given production functions. Even there, profitability means being able to match what one's competitors are doing. Profitability is always benchmarked by how one firm is doing relative to others. In an entrepreneurial economy, however, this means keeping up with the advances of one's competitors, not just finding the profit-maximizing formula and sticking with it.

Progress, profits, and entrepreneurship

If Henry Ford had not decided to pursue assembly line manufacturing for his automobiles, cars would have continued to have been manufactured by hand, and would have been more expensive. Old methods would have continued to be profitable until someone else seized the profit opportunity of assembly line production. If Steve Jobs had not decided to start his own computer company, hobbyists would still have been toying with computers, and eventually, undoubtedly, personal computers would have emerged as marketable commodities, but not as soon as actually happened when Jobs and his partner Steve Wozniak took their leap and formed Apple Computer. Meanwhile, the mainframe and minicomputer manufacturers would have enjoyed additional years of profitability. One can see, then, that whether it would have been profitable to manufacture automobiles by hand depended upon whether others were using assembly lines. The profitability of particular products and production processes depends upon the products and production processes of a firm's competitors.

Progress is built on the innovations of entrepreneurs, and entrepreneurship is built on the lure of profits. Profits provide the incentive for entrepreneurs to take risks, and they provide the indication that the entrepreneurial decision actually was welfare-enhancing. Decisions that are not result in losses. One might respond that if Ford had not started building automobiles on an assembly line, someone else would have, and if Jobs had not started his company to market personal computers, someone else would have, but the point is: someone has to do it, or the economy will remain in its evenly rotating state of perfect stagnation. Entrepreneurs take these risks and make these decisions, and when they do, the economy's underlying equilibrium, if it even makes sense to call it that, is permanently changed. The economy never goes back to an old equilibrium, or arrives at an equilibrium at all, because the underlying conditions are always changing, so a profitable business strategy today will become increasingly less profitable in the future, and always needs to be adapted to the changing underlying conditions.

Was Ford's strategy the profit-maximizing one? We can never know. We see that what he did was profitable, but we cannot know whether another strategy might have been even more profitable. Was the Apple II aprofit-maximizing product when it was introduced in 1977? We can never know, although we do know that it was profitable. As Nelson and Winter say, after discussing business strategies for discovering profit opportunities, "Our basic point is that firms cannot hope to find optimal strategies."[15] After the introduction of the Model T, or the Apple II, the directions of those industries permanently changed and there was no going back to an old equilibrium. The term equilibrium may not even make sense in this context.

Path dependence

When one looks beyond the concept of economic growth to see that it is but a component of economic progress, it becomes apparent that a theory of growth based on increases in inputs and technological change is inadequate to describe the actual process that takes place. While the dollar value of output does grow, that is but a by-product of the fact that the characteristics of output are changing, and production processes are changing, to allow ever more desirable products to be produced at ever-decreasing cost. What drives the process is not the increases in inputs, or even advances in technology, but the entrepreneurial innovation that brings new products and production processes to market.

Technological advances do not cause innovative products to be brought to market. Rather, innovation in markets leads to technological advances, as entrepreneurs have an incentive to find those technological advances, giving the scientists and engineers who work for entrepreneurs an incentive to find them. Similarly, innovative new products cause income growth, because people have an incentive to produce those products, and consumers have an incentive to earn the incomes to buy them. Entrepreneurship causes growth, and income growth could not take place without the economic progress generated by entrepreneurship.

Entrepreneurial advances are not typically marginal changes, and the results of introducing innovations into the market are by their nature uncertain. They must be, because entrepreneurs are speculating on the effects of something that does not currently exist. The successes are easy to see in hindsight – the visionary entrepreneurs like Ford and Jobs. The failures are not so visible, because their innovations disappeared from the market, submerged by losses. The innovations of successful entrepreneurs permanently change the nature of the market. The creative destruction described by Schumpeter means the old market conditions will never reappear, having been displaced by new products and production processes.

Future conditions depend upon current entrepreneurial innovations. There is a path dependence, as Brian Arthur suggests, and the decisions of entrepreneurs at one point in time permanently change the future direction of the economy.[16] As Nelson and Winter note, "In many technological histories the new is not just better than the old; in some sense the new evolves out of the old."[17] An entrepreneurial discovery at one point in time simultaneously closes off business strategies that formerly were profitable and opens up new opportunities for entrepreneurial discovery that previously did not exist. The economy does not grow along a predetermined path that can be predicted within some model. Rather, it progresses as the result of a multitude of entrepreneurial actions that steer the economy in directions that cannot be forecast in advance.

The economy's trajectory is determined by entrepreneurial decisions that change the course of economic development. The economy's trajectory is non-deterministic, because it depends on the decisions of entrepreneurs. Sometimes those decisions are right and, in hindsight, sometimes they are wrong. One can never say that those decisions are optimal, because to do so would require comparing them to what would have occurred had different choices been made in the past.

The impact of bad decisions by entrepreneurs is limited, because bad decisions result in losses, and the products and production processes that resulted from those decisions disappear from the economy. One can only identify entrepreneurial decisions as incorrect in hindsight, and it is possible that the decision resulted in losses only because of better entrepreneurial decisions of others that the mistaken entrepreneur did not foresee. A trivial example that clearly makes the point is someone who discovers a better way of making buggy whips just as the automobile industry takes off. Were it not for the entrepreneurial auto makers, the entrepreneurial buggy whip manufacturer could have had a very profitable business.

The decisions of profitable entrepreneurs, in contrast, permanently alter the direction of the economy. Because advances in one area lay the foundation for further advances, choices can have permanent effects. Technologies can be locked in. For example, if successful automobiles are a certain size that will determine the dimensions of roads, which will then have a further effect on automobile characteristics. A different decision early on may have produced very different trajectories. Choosing technology A over B, even if B would be lower-cost, enables others to build on technology A if the product using it is successful. So, products built around technology A continue to advance while no development takes place in technology B, and it can never catch up. If B actually is better, entrepreneurs will have a window of opportunity to spot that before A gains an insurmountable lead. Thus, the more open an economy is to entrepreneurial opportunity the less likely it is to get locked into a technology that is suboptimal.

One can conjecture about optimal and suboptimal economic trajectories in theory, but one can never know whether a trajectory is optimal in practice. Choosing one option closes off others, and there is no way to tell what the results would have been had those other options been taken. There is no such thing as an optimal growth path. More economic progress can be generated by a more entrepreneurial economy, but as the economy progresses in one direction, it moves away from trajectories that might have been generated had different entrepreneurial choices been made in the past.

Producing progress

There is not an optimal growth path for the economy that is lurking out there waiting to be discovered. Rather, the path that produces economic progress is revealed as a product of the process that produces it. Entrepreneurs have the incentive to exploit previously unexploited profit opportunities, and their attempts to do so reveal whether their actions contribute to economic progress. Profits indicate they do, losses indicate they do not. Successful innovations permanently change the economic environment, which creates additional entrepreneurial opportunities in that new environment. They are opportunities that exist only because of a prior entrepreneurial innovation. The path that progress takes cannot be mapped out ahead of time, because each innovation alters the future feasible options. Economic progress depends on entrepreneurial discovery, and is directed by it.

In the socialist calculation debate, discussed in Chapter 1, those who argued that central planning cannot work did so based on the idea that the signals of market prices are required to provide the information required by those in the economy to direct their activities. Knowledge is decentralized and often tacit, so that it cannot be passed from one individual to another, therefore the challenge lies in using this decentralized knowledge as effectively as possible. The information that is required is generated by the market, so that information could not even exist without a market allocation of resources. But information about potential profit opportunities is always uncertain, because it is based on a conjecture about future conditions that cannot be observed in the present. Thus, economic progress depends upon people being willing to take risks today based on information that is necessarily uncertain. Any impediments to entrepreneurial risk-taking stand in the way of the entrepreneurial decisions that generate economic progress.

Conclusion

Because economic theory, as it developed through the twentieth century, is based heavily on the concept of equilibrium, it is natural in extending this framework that a theory of economic growth should also be based on an equilibrium approach. Keeping in mind the methodological discussion in Chapter 2, all economic models assume some things away in order to focus attention on particular phenomena the model helps explain, and equilibrium models in economics assume away the fundamental causes of economic growth. Therefore, that framework must be expanded, or even set aside, to develop a theory of growth. Growth is the result of progress, and cannot be understood independently of the entrepreneurial aspects of the economy that introduce new goods, services, and production processes. If growth is simply seen as increases in income, or expansions in output without considering the changes in the nature of output, its fundamental causes are being left out of the model.

This is not an argument for the abandonment of neoclassical economics. To revisit an earlier analogy from Chapter 4, if one wants to understand how a cyclist can ride a bike from one location to another, one might look at the physics of bike riding to understand how a cyclist can remain balanced on two wheels while directing the bicycle where the cyclist wants it to go, or one can look at how the cyclist knows which route to follow, and what turns must be made to go from the origin to the destination. For some purposes it is sufficient to simply assume the rider can remain balanced on the bicycle and then map out the cyclist's route. When looking at the causes of prosperity, by analogy, one can assume that the equilibrating forces of the economy are at work, as illustrated by neoclassical price theory, but then set that framework aside. Assuming that there are forces that cause markets to clear, how are innovations introduced into an economy in such a way that they result in economic progress, of which economic growth is a by-product?

An equilibrium approach suggests that an economy moves away from equilibrium, due to shocks, and then moves back. The whole notion of equilibrium is the idea that if the existing state of affairs is disturbed, there are forces at work that re-establish the equilibrium state of affairs that existed prior to the disturbance. The economy is pushed away from equilibrium, to be followed by the activation of economic forces that pull the economy back. But that is not what happens. While some temporary disturbances may reverse themselves in this way, the most significant shocks to the economy are irreversible. Once the disturbance has occurred, conditions have permanently changed such that the economic state of affairs will never return to its earlier state. This argues for leaving the framework of equilibrium economics aside to examine the economic process.

What does an economy do: grow, or equilibrate? It does both, although Chapter 4 argued that it is more accurate to thinkof the equilibrating forces in the economy as market clearing rather than equilibrium – precisely because the economy will not return to its former state of affairs. To understand how economies produce prosperity, those market clearing forces can be assumed to be at work, to focus on the issue of longer-run importance regarding how economies generate progress. This is not to deny the equilibrating, or market clearing, forces in an economy, but rather to note that they are of secondary importance when one is trying to understand how continual innovation leads to economic progress.

Growth cannot take place without progress, so a theory of growth that assumes away the factors that generate progress cannot accurately depict either progress or growth. This chapter has argued that entrepreneurship is the engine of economic progress, and that entrepreneurial decisions are necessarily made under conditions of uncertainty. When an entrepreneur introduces an innovation into the economy, there is no way to determine its effect, because innovations go beyond previous experience, and there can be no data to reveal the effect of something that is beyond the realm of current experience. Certainly reasonable guesses can be made, for example, if similar innovations have occurred elsewhere in the economy. Henry Ford's innovation was to apply assembly line production to automobiles. That had not been done before, but assembly lines had been used to produce other products. Entrepreneurial decisions are not complete shots in the dark, but they also never have an outcome that can be predicted with certainty beforehand.

Economic progress occurs because of these decisions, but because making such decisions rests on the knowledge possessed by the entrepreneur, the conjectures the entrepreneur makes about likely outcomes of various alternatives, and the degree to which the entrepreneur is willing to bear the risks of making uncertain decisions, means that the choices made by entrepreneurs are not determinate, and consequently the future path of economic progress is not determinate either. Choosing one option over another, or choosing to play it safe rather than take a risk, will permanently alter the path of economic progress. The actual trajectory of the economy is path-dependent, and is determined by the decisions of entrepreneurs who weigh the risks and returns of various courses of action.

Growth theory, then, must be based on an entrepreneurial foundation. Understanding the causes of growth means understanding the forces that generate economic progress – the innovations in output and production processes – that cause increases in incomes and the quantity of output. The focus must be on understanding how economies are able to cultivate entrepreneurial innovation rather than on increases in inputs into the production process, or on technological improvements. Certainly, increasing the quantity and quality of human and physical capital is important to growth and progress, and introducing new technology into the production process is important. By these are by-products of economic progress. If there is a payoff to obtaining human capital, and investing in physical capital, then those inputs will increase because the people who control those inputs have an incentive to bring more of them into factor markets. If technological innovations can pay off in terms of increased profits, then people have an incentive to undertake research and development to improve technology, and have an incentive to find ways in which technological inventions can be transformed into market innovations. The factors that much of growth theory focuses on will be produced as a result of market forces in an entrepreneurial economy. The focus, therefore, should be on innovation and entrepreneurship, rather than on inputs and technology.

8 Economic welfare

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Theory and policy

The subject matter of economics has always revolved around how to design policies to improve economic well-being. Indeed, Adam Smith's complete title, An Inquiry Into the Nature and Causes of the Wealth of Nations, reveals his intention to explain how people are able to create wealth by working together well beyond what any individual could achieve alone. He begins by saying, "The greatest improvement in the productive powers of labour ... seem to have been the effects of the division of labour,"[1] and goes on to give the delightful example of the pin factory in which he argues that people are hundreds of times more productive when they specialize their labor, and increase their well-being by trading with others who specialize in different productive activities. Just by looking at the prosperity that characterizes the modern world – in those parts of the world that have a well-functioning market economy – there can be no doubt that economic welfare is advanced by the results of resource allocation within a market setting.

While economic welfare has always been the central subject matter of economics, modern welfare economics can trace its origins to A.C. Pigou's The Economics of Welfare, originally published in 1920. Pigou notes that there is a difference between economic and non-economic welfare, and limits his study to the former. Neoclassical welfare economics followed Pigou's lead in this regard, and this chapter will do the same. Pigou goes on to note, "What we wish to learn is, not how large welfare is, or has been, but how its magnitude would be affected by the introduction of causeswhich it is in the power of statesmen or private persons to call into being."[2] As a good student of Marshall, on the same page Pigou goes on to introduce a comparative statics element to his inquiry, noting that welfare economics would compare policies so that "it will tell us how total welfare will differ from what it would have been if that cause had not been introduced, ... and this is the information of which we are in search."

While the basic Pigouvian framework remains intact into the twenty-first century, welfare economics enjoyed major advances in the 1950s, based on the proof of the uniqueness and stability of competitive equilibrium.[3] Using competitive equilibrium as abenchmark for efficiency, welfare maximization occurs when marginal rates of substitution in consumption are equal to marginal rates of transformation in production for all goods so that resources are allocated Pareto optimally.[4] If those conditions are not met, the market fails, and the policy goal of neoclassical welfare economics is to design policies that move the economy to a Pareto optimal allocation.[5] The formalization of conditions for welfare maximization advances Pigou's ideas by defining more rigorously what is meant by awelfare maximum, but is completely consistent with Pigou's framework. This is the foundation of contemporary welfare economics, and this chapter offers a critical evaluation, and a suggestion for an alternative way to view economic welfare.

The first place to look to examine the foundations of welfare economics is at actual economic welfare in the real world. As much as Adam Smith marveled about the remarkable increase in the wealth of nations caused by the division of labor, modern economists must be all the more impressed by the wealth of nations in today's world. Wherever market economies have been allowed to operate, prosperity follows.[6] In the United States even those classified as poor at the end of the twentieth century had higher standards of living in many dimensions than the average American in 1970.[7] When one looks at the remarkable increase in economic well-being over the past 20, 50, or 100 years, few would argue that the reason we are better off is that we are closer to Pareto optimality. As the previous chapter discussed, welfare has improved over the years, decades, and centuries, because of economic progress. Yet Pareto optimality remains the benchmark by which economic welfare is measured. Policies that move the economy closer to Pareto optimality are welfare-enhancing, and once the economy arrives at Pareto optimality welfare is maximized.

Vernon Smith criticizes this neoclassical framework, saying "the microeconomic theory of the pre-1960s ..." is a "dead end." He continues, "Fortunately for the economy, but unfortunately for academic economics, this formulation of Pareto efficiency is not the problem that real markets and other allocative institutions attempt to solve."[8] Mier Kohn looks at the formal mathematical framework on which welfare economics is founded, which he traces back to Hicks[9] and Samuelson,[10] and argues, "the Hicks–Samuelson research program has done virtually nothing to assist in the formulation of economic policy." After giving some examples, Kohn goes on to observe:

This is not to suggest that economists as individuals have made no contribution.

However, their advice has relied more on economic common sense than on high theory. It is difficult to see how a 19th century economist, or even one from the 18th century, would have made a less useful policy advisor than a tooled-up modern theorist.[11]

Modern welfare economics, with its welfare-maximizing benchmark of Pareto optimality, is nearly irrelevant to actual economic welfare. While it is true that a Pareto improvement will increase welfare - by definition, because such a move makes some people better off, but nobody worse off - actual improvements in welfare are not the result of moving closer to a Pareto optimal allocation of resources. Rather, welfare has improved because of the remarkable economic progress that has occurred since the beginning of the Industrial Revolution.

How is welfare actually maximized?

The appropriate place to begin in formulating a foundation for welfare economics is to look at what one is trying to measure. Following Pigou's lead, welfare economics should begin by looking at factors that improve people's material well-being. And, if one takes as a point of departure people's well-being in the real world rather than some theoretical framework, in the real world welfare is improved through economic progress. People were better offat the end of the twentieth century than they were at the beginning, for example, not because the economy was any closer to Pareto optimality at the end of the century, as neoclassical welfare economics seems to imply, but because economic progress had enhanced people's opportunity sets. Increasing incomes allowed people to consume more of everything, and at least as important, economic progress has offered people the opportunity to consume new goods and improved versions of old ones that nobody had the opportunity to consume in the past.

Many of the transactions people can undertake today that would not have been in their opportunity sets in the past were made possible by transactions that had preceded them, as progress in one market enables progress in others. For example, supermarkets replaced corner grocery stores because the development of the automobile allowed supermarkets to attract customers from a wider area who were able to buy more each time they shopped because they could carry their purchases in their cars. Similarly, "big-box" retailers like Wal-Mart and Best Buy rely on automobile travel to draw customers from a wide geographical area who bring with them the means (their automobiles) to transport their purchases home. The development of the automobile led to the development of the supermarket and the big-box retailer, giving shoppers a greater variety of choices at lower cost. Internet shopping is a twenty-first-century advance that gives consumers even greater choices.

Economic progress has increased the efficiency of the production process, replacing hand production with assembly lines – sometimes under robotic control – and has produced a new array of goods and services that people could consume. Rather than traveling by horse and buggy, or by steam locomotive, people travel in air-conditioned automobiles and jet aircraft. They communicate via cellular telephones and over the Internet rather than by mailing letters. They watch television for news and entertainment, in addition to reading newspapers and magazines. And they consume more of everything because increases in productivity came along with improvements in the types of goods and services produced.

Looking at real-world economic welfare, it is apparent that welfare maximization is not accurately described as arriving at a Pareto optimal allocation of resources. Rather, welfare is maximized by creating an institutional environment that facilitates economic progress and gives entrepreneurs the incentive to introduce innovations into the economy that consumers value more highly than the status quo. Following Adam Smith, this enhances the division of labor, which Smith says is the source of "the greatest improvement in the productive powers of labour." Economic progress occurs as the division of labor increases as a result of an ongoing expansion in the availability of mutually agreeable exchanges. Innovators search for profit opportunities, understanding that they can benefit by offering others exchange opportunities they would prefer to the status quo. This suggests a process-oriented approach rather than the neoclassical outcome-based approach that depicts welfare maximization as a static outcome.

Welfare is improved by lowering impediments to mutually advantageous exchanges so that people can engage in transactions that were not previously possible and buy goods and services that were not previously available. The opportunity for such exchanges gives entrepreneurs the incentives to take risks and innovate to bring better products to market. As Smith notes, the entrepreneur "intends only his own gain," but is "led by an invisible hand" to "promote the public interest." Welfare is maximized by policies that enhance this process that generates economic progress.[12]

The benchmark of Pareto optimality

Pareto optimality has a number of drawbacks that make it inappropriate as a benchmark for welfare maximization, even within the neoclassical framework within which it was developed. It is a purely theoretical construct that has no real-world counterpart. It is unobservable, untestable, and there is no way to tell whether one is getting closer to a Pareto optimum.

Ever since Milton Friedman's famous essay[13] on positive economics, economists have placed a premium on theories that contain testable hypotheses: theories that can be falsified by looking at data. While positivism has had its critics,[14] if one actually wants to use Pareto optimality as a benchmark for judging whether a policy is welfare-enhancing, one must be able to observe the benchmark to see whether a policy change would move toward it. Pareto optimality cannot be observed, and there is no possible empirical test to reveal whether an economy is at a Pareto optimum, or close to one.

The theoretical construct of Pareto optimality requires major assumptions for its existence. A minor assumption is that the economy is in equilibrium, with no remaining mutually advantageous exchanges that are unmade. The major assumptions are behavioral assumptions that define the utility functions of individuals and the managerial behavior of firms. Utility functions must be transitive, exhibit diminishing marginal rates of substitution, and in a dynamic setting be stable (although the foundations of neoclassical welfare economics are built in a static setting where time is not an issue). Experimental and behavioral economists have called these assumptions into question,[15] and if people's utility functions do not conform with the neoclassical assumptions even if a competitive equilibrium exists it may not be Pareto optimal. The optimality result relies on the assumed utility functions underlying the Paretian framework.

Setting these problems aside and granting all the assumptions underlying the construction of a Pareto optimum, Lipsey and Lancaster, explaining the general theory of second best more than half a century ago, note that "if there is introduced into a general equilibrium system a constraint which prevents the attainment of one of the Paretian conditions, the other Paretian conditions, although still attainable, are, in general, no longer desirable." Therefore, "there is no a priori way to judge as between various situations in which some of the Paretian optimum conditions are fulfilled while others are not."[16] One would think that the general theory of second best, by itself, would be enough to dissuade economists from using Pareto optimality as a benchmark for judging social welfare. For any policy change to be demonstrated as an improvement in welfare, that change by itself would have to result in a Pareto optimum. From a practical standpoint few economists would argue that the world is one step away from Pareto optimality, but regardless, because Pareto optimality is an unobservable and untestable concept, there would be no way to tell in any event.

Even when analyzed on its own terms, Pareto optimality is a purely theoretical construct that does not hold up as a benchmark for evaluating welfare in the real world. But, as the previous chapter noted, it would not provide an accurate measure of improvements in welfare anyway, because actual improvements in people's material well-being come almost entirely from economic progress, not from wringing static inefficiencies out of the economy.

Welfare maximization as a process, not an outcome

If one were to cling to the use of Pareto optimality as a benchmark for welfare maximization, then once one reaches that Pareto optimal allocation of resources welfare is maximized and cannot be increased further. Simple observation of the way that material well-being has increased over the years, decades, and centuries reveals how poorly this conception of welfare maximization fits the real-world facts: welfare maximization is a process that has no theoretical maximum. Progress can keep occurring, and as it does, people's welfare can continue to improve. Welfare is maximized by those economic forces that create economic progress, and any movements toward Pareto optimality – while they may be beneficial, whether or not they can be observed – are mostly irrelevant to actual economic welfare. Not even the staunchest Paretian would argue that people's welfare was higher at the end of the twentieth century than at the beginning because the economy was closer to Pareto optimality.

One might even conjecture that the economy is further away from Pareto optimality now than a century ago. For one thing, wealthier people have more resources at their disposal with which to generate externalities. They can run their power lawnmowers early in the morning to disturb their neighbors, and drive their poorly-maintained automobiles, adding to air pollution. Also, in a more mobile society, people are less likely to have long-term relationships with their neighbors, lessening social pressures against the creation of externalities, and lessening social pressures against being a free rider. Furthermore, public policy has generated externalities. For example, prior to taxpayer-provided healthcare, smoking imposed costs on the smoker but did not generate significant external costs.[17] With taxpayer-provided healthcare, someone's ill health caused by smoking now imposes costs on all taxpayers. But, the conjecture that the economy is further from Pareto optimality today than a century ago is just a conjecture, and any observer can see that whether the conjecture is correct is all but irrelevant to people's actual material well-being.

In the Paretian framework, Pareto optimality is a welfare maximum, with the caveat following Samuelson[18] that if interpersonal utility comparisons can be made lump-sum taxes and transfers may further enhance welfare. This static notion of welfare maximization, in which once a welfare maximum is reached no further improvements are possible, is inconsistent with simple observation of the real world in which material well-being has continued to advance since the beginning of the Industrial Revolution, with no end in sight. Welfare maximization is not an outcome, it is an ongoing process, so welfare economics needs to shift its foundation to be consistent with a process-oriented view of welfare maximization in which welfare can continue to advance, and in which there is no end-state that could be used as a benchmark against which welfare can be judged.

Welfare is maximized in an environment that maximizes the opportunity for economic progress, which points in the direction of identifying welfare-maximizing policies. Welfare maximization means enabling economic progress.

Pareto optimality versus Pareto improvements

In contrast to Pareto optimality – an unobservable and irrelevant benchmark for welfare maximization – Pareto improvements do improve welfare and can be observed.[19] A Pareto improvement occurs if at least one person is made better offwithout making anyone else worse off. If welfare maximization is a process, not an outcome, Pareto improvements are more applicable to people's actual welfare than Pareto optimality, because exchange is a process that improves the welfare of those engaging in it.

Some assumptions need to be made to conclude that exchange improves welfare, but the assumptions are weak compared to the assumptions needed to identify a Pareto optimum. By making one assumption – people are able to judge their own well-being to determine when an exchange makes them better off - Pareto improvements can be observed every time people engage in an exchange. There is no need to assume that utility functions are transitive or that indifference curves exhibit diminishing marginal rates of substitution. With the assumption that people engage in exchange to improve their well-being, the act of exchange by itself demonstrates that welfare has been enhanced and, by the definition of a Pareto improvement, a Pareto improvement has occurred. Pareto improvements can be observed just by observing the individuals who are party to the exchange, whereas observing Pareto optimality would require knowledge about the marginal conditions for every individual and every production process in the economy.

Whereas Pareto optimality is essentially irrelevant to actual economic welfare, Pareto improvements cause improvements in welfare. However, one must look beyond simple exchange to see the relevance of Pareto improvements. While welfare is enhanced by exchange, the economic progress that raises today's level of welfare beyond what existed before is the creation of exchange opportunities that did not previously exist. For example, prior to 1870 nobody in the United States could exchange anything for a banana, because bananas were not available in the United States. A Brooklyn entrepreneur, Minor Keith, built railroads in Costa Rica and planted bananas by the tracks for the purpose of importing them into the United States. As a result, Pareto-improving exchanges were possible that could not have occurred before, giving people in the United States the opportunity to consume bananas. Henry Ford and Steve Jobs provide similar examples, allowing people who could not have purchased them before to buy automobiles and personal computers, respectively.

If people have the opportunity to make the same exchanges they made in the past, their level of material well-being will be maintained. If people have the opportunity to make exchanges that were unavailable in the past – and they actually make those exchanges – their material well-being will be improved. They have demonstrated that in their behavior. By choosing the new opportunities over those previously available to them, they reveal that they are better off.[20] Pareto improvements are relevant to welfare maximization, then, not so much in the static sense of analyzing how people can exchange to make themselves better off, but in the dynamic sense of creating new opportunities for exchange that were previously unavailable. This is the role of entrepreneurship in an economy. Entrepreneurs present new opportunities for consumers. The creation of those new opportunities constitutes economic progress, and economic progress is what enhances welfare.

Profits and welfare maximization

In neoclassical welfare economics, profit (beyond a normal profit) is inconsistent with welfare maximization. Profit is an indication of either monopoly, which is inefficient because too little of the monopolized output is produced, or of disequilibrium, which also results in an inefficient allocation of resources. In the competitive general equilibrium that is the benchmark for welfare maximization, profits are zero, and at least one type of profit (monopoly profit) is an indicator of market failure – that is, a failure to reach a welfare-maximizing Pareto optimum. In the neoclassical framework, profit is inconsistent with welfare maximization.

When one views welfare maximization as a process that generates economic progress, profits are necessary for welfare maximization. Welfare is improved through economic progress, as entrepreneurs discover more efficient methods of production, and new and improved goods and services that can be produced for consumers. Profit both provides an incentive for entrepreneurs to engage in this function, and provides an indicator of the success of the entrepreneurial activity.

Entrepreneurs necessarily face uncertainty when they introduce innovations into an economy. Changes in production methods do not always work as foreseen, and there is no way to look at economic data to tell whether innovations that are brought to market will prove profitable. Profits are revenues minus costs, and because a new product has never been on the market one can only speculate about the demand for it. Entrepreneurs undertake these uncertain ventures because they will profit if their judgment is correct. Profit serves the dual role of giving entrepreneurs an incentive to take a risk and introduce an innovation into the economy, and as an indicator of whether, after the fact, the innovation was welfare-enhancing. Aprofitable innovation indicates that welfare has been enhanced, because purchasers are willing to pay more for the output than it costs to produce. A loss indicates that welfare has been diminished, and that the entrepreneur should change course. Innovations are initiated in response to the lure of profits. As Schumpeter said, "Without development there is no profit, without profit no development."[21]

Consider the example of the introduction of the IBM 360 computer in the 1960s, which was described in the previous chapter. IBM dominated the computer industry after the introduction of the 360 because computer users preferred the IBM computer to those offered by competitors. Prior to the 360 the features it offered were not available to computer users at any price. The profits IBM made were an indicator of the value users placed on the 360 computer beyond what they could get with competing products. Profitswere a measure of the value created by IBM, not a measure of economic inefficiency, so indicated that welfare had been enhanced by IBM. A static model of monopoly would count those profits as indicating a welfare loss, but in fact they measured the welfare gain attributable to the introduction of a previously unavailable product.

When one views welfare maximization as a process that generates economic progress, profits are necessary for welfare maximization; yet, in neoclassical welfare economics profits are inconsistent with welfare maximization. This, by itself, illustrates why a Pareto optimal allocation of resources inhibits welfare maximization and is inconsistent with welfare maximization.

Welfare and public policy

A reexamination of the foundations of welfare economics might be interesting purely on theoretical grounds, but would be much more valuable if it brought with it policy implications, and it does. Welfare is enhanced through economic progress, and economic progress is generated through entrepreneurship. Profits serve both as the incentive for entrepreneurial activity and as an indicator that innovation has actually been welfare-enhancing. If innovators can combine resources in such a way that the cost of production is less than the revenue from selling what is produced, resource allocation is improved, based on the values buyers and sellers place on the resources and products that are marketed. This economic activity takes place through voluntary exchange, which public policy should protect and encourage.

In keeping with the process-oriented nature of welfare economics, welfare-maximizing policies are those that create an environment in which entrepreneurship and innovation are most likely to flourish. A growing literature is identifying and empirically verifying the characteristics of this environment.[22] This literature empirically supports the conclusion that protection of property rights, freedom of exchange, rule of law, low taxes and regulatory barriers, limited government, and access to sound money, are the key features that lead to economic progress. A reformulation of the foundations of welfare economics would place the foundation of economic welfare on the degree to which the institutional structure fosters entrepreneurship and economic progress. This means rewarding successful innovators through market-generated profits and penalizing those who use resources ineffectively through market-generated losses.

The Austrian school in the socialist calculation debate argued for the importance of markets in the rational allocation of resources. When one recognizes that entrepreneurship is the essential element to economic progress, and that economic progress is what generates increases in welfare, it follows that the private ownership of capital is essential to welfare maximization. Only with the private ownership of capital do those who make investment decisions have the right incentives to make risky entrepreneurial decisions, because those owners stand to reap the benefits that accrue to decisions that lead to economic progress, but must suffer the losses of decisions that, in hindsight, result in a misallocation of resources.[23] Thus, collective ownership of the means of production is inconsistent with the process of welfare maximization.

When one recognizes that improvements in welfare are the result of economic progress, and that progress is only peripherally related to the neoclassical benchmark of Pareto optimality, welfare maximization does not appear to be as precise a target as the neoclassical framework would suggest. Welfare maximization is an ongoing process, not an outcome, so welfare never approaches a maximum. Furthermore, one can never know, even in hindsight, whether welfare could have been higher had different decisions been made in the past. The results of choices actually made can be observed, but one can only conjecture about what might have happened had different choices been made.[24] Taking this process-oriented approach, it is not possible to compare the actual state of the world against some hypothetical welfare maximum.

From a policy perspective, that means that welfare-maximizing policies are designed to facilitate the process, not to generate some specific outcome. The process is facilitated by an institutional structure that provides all individuals with the incentive to make best use of the knowledge they possess in a manner that enhances the productivity of the economy. Thus, the implications of this reformulated welfare economics are about how institutions can be structured to facilitate entrepreneurship and innovation, leading to progress, not how the economy can be engineered to arrive at some precisely-defined goal.

The static welfare economics framework has been used to justify public policy since it came into being. It is the basis of antitrust policy based on the neoclassical theory of welfare losses due to monopoly,[25] the concept of corrective taxes levied on externalities, and public goods theory, formalized by Samuelson.[26] The notion of market failure as a failure of markets to achieve a Pareto optimal allocation of resources is not just some heuristic device. It is a framework that has been used to design public policy for more than a century.

Growth versus progress

Chapter 6 offered a critical assessment of general equilibrium growth models. Those models depict the economy as always in equilibrium, and depict growth as increases in the model's homogeneous output, Q. The problem with this, Chapter 6 noted, is that progress occurs because of changes in the characteristics of output produced, so the model assumes away the cause of what it hopes to explain. Another problem is that profits are necessary for innovations to occur in an economy, yet in equilibrium all profits are competed away. When analyzing the substantial increase in economic well-being that people in market economies have enjoyed over time - that is, when analyzing improvements in economic welfare – it is the new goods and services that people have to consume that are the major component. These goods and services are the result of entrepreneurial activity that leads to economic profits. The process of creative destruction, as Schumpeter described it, requires profits. Thus, if one views the economy at a Pareto optimum when an entrepreneur successfully introduces an innovation, the disruption moves the economy away from that Pareto optimum, which is necessary for economic progress.

Neoclassical welfare economics is based on the presumption that a Pareto optimal allocation of resources maximizes welfare, yet starting from this point, the creative destruction of entrepreneurship would move the economy from a Pareto optimal allocation to a non-optimal allocation, improving welfare as a result. A process-oriented view of welfare recognizes that Pareto optimality cannot be the benchmark for welfare maximization, and that welfare is maximized by economic progress, although it never reaches a maximum.

Theoretical foundations of welfare economics

While welfare economics traces its (modern) origins back to Pigou, Schumpeter's Theory of Economic Development provides a foundation more along the lines of identifying the actual causes of improvements in economic welfare. Schumpeter distinguishes invention from innovation. Invention refers to the technological advance, but technological advances by themselves do nothing to improve economic welfare. Innovation means bringing new products and production processes to market, leading not just to growth, but to progress. Innovation may be built on invention, but it is the innovation – not the invention – that leads to economic progress. Very likely, Pigou's ideas evolved into the still-current neoclassical welfare economics of the 1950s not because neoclassical welfare economics contained more insight than Schumpeter's approach, but rather because it could be directly built upon the mainstream general equilibrium framework popularized by Hicks and Samuelson, which was on the cutting edge of economic theory in the middle of the twentieth century.

The welfare economics of the 1950s developed in the way it did because it was using the increasingly mathematical tools of neoclassical economics. To quote an old saying, "When the only tool you have is a hammer, everything looks like a nail." The hammer of the neoclassical framework was used to build a welfare economics that defined a welfare maximum as a static Pareto optimum. Meanwhile, Schumpeter's framework was not so amenable to mathematical rigor, and suggested an indeterminacy in economic outcomes that was not so readily compatible with the more mathematically rigorous general equilibrium approach.

A look out of the window at the real-world economy suggests that it is better described by an evolutionary approach than an equilibrium framework, and while authors such as Armen Alchian,[27] Richard Nelson and Sidney Winter,[28] and Eric Beinhocker[29] have made some contributions toward an evolutionary framework for economics, their insights have not been applied to welfare economics, at least not directly. One major difference between an evolutionary framework and an equilibrium framework is that while an evolutionary system continues evolving, it does so in a path-dependent way and is not headed toward some deterministic outcome. Entrepreneurial decisions made today can change the future trajectory of the economy, as Chapter 6 noted, in contrast to an equilibrium framework where, regardless of what people decide today, market forces always pull the economy back to equilibrium. In an evolutionary setting there can be no benchmark, like Pareto optimality in the neoclassical framework, that indicates whether welfare is maximized.

Welfare maximization is a process, not an outcome, so welfare can never arrive at a maximum, even in theory, and there can be no benchmark to evaluate whether welfare is maximized. Welfare is maximized by creating conditions within which individuals are not impeded from undertaking entrepreneurial activities that lead to improvements over the status quo.

The cost of attaining Pareto optimality

Economic policy is frequently oriented toward allocating resources in a Pareto optimal way. If externalities exist, corrective taxes or regulatory constraints are designed to control them and, ideally, produce the optimal amount of the externality. If firms have monopoly power, policies are designed to regulate them to produce a competitive environment, or even to break up monopolies into smaller firms to generate competition. Implementing policy to improve welfare is costly, so if policies are adopted to further a Pareto optimal allocation of resources, they may lower prosperity, especially if those policies create disincentives for entrepreneurship.

This is clearest when the apparent market failure is monopoly power. While many firms with monopoly power gained that power as a result of government intervention – whether one is considering regulated utilities, or firms that have successfully lobbied for import restrictions, product regulations, or other barriers to competitors – some firms gain market power by being innovative and beating their competitors to market with products, or producing products customers prefer. Perversely, public policy protects those firms whose monopoly power came about through government privileges that circumvent the market process, while penalizing those firms whose profits are the result of innovation. Doing so reduces the incentive to innovate, which impedes economic progress.

Even when production generates externalities, it is not clear that the optimal policy is for government to regulate them. If externalities are the by-product of innovation, regulating them can stifle innovation. Imagine, for example, if in its infancy the aviation industry was prevented from generating noise that might disturb those in the vicinity of airports. At one time many jurisdictions had a requirement that automobiles had to be preceded by a flagman on foot. Enforcing such a requirement clearly would have a detrimental effect on progress in the automobile industry. Negative externalities are undesirable but, as Coase pointed out,[30] when they exist entrepreneurial people have an incentive to find ways to internalize them. Government intervention to mitigate externalities, or any other market failure, may exact a higher cost on the economy than it produces in benefits if the intervention stifles entrepreneurship and hinders progress.

Conclusion

People's well-being can be characterized across a number of different dimensions, and there is no reason not to look at indicators such as per capita income, life expectancy, capabilities, and inequality,[31] or even measures of happiness,[32] because those are the things economic actors are striving for as they make choices throughout their lives. However, welfare economics as defined by Pigou focuses on people's material wellbeing, and people's material well-being is maximized through economic progress. Because progress is an ongoing process, not an outcome, the focus of welfare economics naturally should turn to those economic institutions that are most conducive to the generation of progress.

Because welfare maximization is a process, not an outcome, welfare is never "maximized," if that means arriving at a welfare maximum. A process-oriented approach to welfare maximization more accurately describes real-world increases in welfare rather than the outcome-based neoclassical depiction of welfare maximization. One would be hard-pressed to describe the huge increase in economic welfare over the past century as a consequence of moving toward Pareto optimality, following neoclassical welfare economics. Rather, it is a result of the process of economic progress, consistent with a process-oriented approach to welfare economics.

If we date the beginning of modern welfare economics to the publication of thefirst edition of Pigou's book in 1920, welfare economics is now nearly a century old, and it is not unreasonable to re-examine its foundations in light of developments in economic analysis, and in light of what actually has improved real-world welfare. There are fundamental – and fatal – flaws in the foundations of neoclassical welfare economics. If economists could step back just far enough to admit that welfare is not maximized by allocating resources Pareto optimally, and that in important ways a Pareto optimal allocation is actually inconsistent with welfare maximization, economics can move away from what obviously is a flawed foundation for welfare economics.

If Pareto optimality is abandoned as a benchmark for welfare maximization, what would replace it? Much of this chapter is oriented toward showing that there is a coherent way to think about welfare maximization as a process, and that this line of reasoning points toward a reformulation of the foundations of welfare economics. In a process-oriented approach, there can be no benchmark for welfare maximization; rather, welfare economics must be oriented toward enhancing the process of economic progress. From a policy-oriented perspective, this means creating and maintaining institutions that allow entrepreneurship and economic progress, not fixing market failures. True market failures are worth addressing, but even here, the traditional taxonomy of market failure includes at least one factor – monopoly profits – that may turn out to be an indicator of economic progress and welfare enhancement, once the foundation of welfare economics is reformulated. If one is interested in quantifying welfare, it must be within the context of measuring the expanded qualitative and quantitative improvements in consumption opportunities that economic progress affords people, not comparing the status quo against the benchmark of Pareto optimality. Economic progress is what improves economic welfare. Any reformulation of welfare economics must start from that clearly observable fact.

9 Rent seeking in a creative economy

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One of the factors that works against economic progress is the potential for the appropriation of the gains from entrepreneurship through government. Government exercises a monopoly over the legitimate use of force over people in a society. This force can be used to maintain peaceful and productive interaction among those in an economy, but it can also be used to transfer resources from some people to others. There is always some public interest justification for such transfers, whether it is helping those on the bottom of the economic ladder to get the resources they need to increase their well-being and productivity, or to boost the productivity of those who are already productive by making them more competitive. But because government has the power to transfer resources from some to others, entrepreneurial individuals will see that they have the opportunity to use the force of government to have resources transferred toward them. Anne Krueger called the seeking of transfers, through taxation or through regulation, from some people toward themselves "rent seeking", and the name has stuck.[1]

Productive and unproductive entrepreneurship

The idea of rent seeking was introduced into the modern economics literature by Gordon Tullock, in his article "The Welfare Cost of Tariffs, Monopolies, and Theft."[2] Since Tullock's article, rent seeking has been discussed almost exclusively as the seeking of transfers through the force of government, but as the title of Tullock's article indicates, there is a strong similarity between using the force of government to engage in rent seeking and engaging in simple theft. One important difference is that in the case of theft it is a one-time activity, whereas if one can employ the force of government to engage in rent seeking the transfers can occur into the indefinite future. There is a clear advantage to the recipient of the transfer from establishing an institutional setting in which transfers will continue to occur into the future, rather than just engaging in a one-time transfer of resources through theft.

In general, people can generate income for themselves in two ways: they can engage in productive activity, or they can appropriate the production of others. The alternatives are: production or predation. In primitive societies, and to a degree in all societies, predation is undertaken through simple theft. Predators see resources in the possession of others and use force to take them for themselves. Simple theft has several drawbacks. First, people can be inclined to fight back to maintain possession of their resources, and violence squanders resources and has the potential to destroy much of the value of the resources predators would like to appropriate. Second, if predators try to use simple theft as a continuing source of income, producers will cease to engage in productive activities, knowing that what they produce will just be taken from them. Finally, as a result of both of these factors, predators must move on to find new victims, which is a costly and uncertain activity. A better arrangement, from the standpoint of the predators, is to set up an institutional structure whereby they can continue to appropriate the productivity of a group of people while getting the agreement of the productive class and keeping those people productive.

The institutions of government can serve this purpose. Through taxation and regulation, the institutions of government can require that resources continue to be taken from some and transferred to others. Entrepreneurial individuals will engage in productive activity and continue to bring innovations into the economy as long as they can receive the rewards from their productivity. If the burdens of government on the productive class grow too large, however, incentives change, and entrepreneurial individuals will find it more profitable to use the force of government to prey on others rather than to engage in productive activities themselves. Entrepreneurs will shift from productive to destructive activities. William Baumol has insightfully analyzed the institutional framework that leads to destructive entrepreneurship.[3] A productive society requires institutions that reward productive activity and that limit the returns people can get through rent seeking.

Establishing institutions that can limit the returns derived through rent seeking is not trivial. Douglass North, John Wallis, and Barry Weingast analyze societies at different levels of development and argue that rent seeking societies, in which those who prosper the most do so through their political power and connections rather than their productive activity, tend to be stable equilibrium institutions, and that it is unusual for societies to develop to the point where productivity is the main source of income and wealth for those at the upper end of the income distribution.[4] While their analysis is worth pursuing in some detail, their primary point for present purposes is that maintaining institutions in which entrepreneurial individuals have an incentive to engage in productive rather than predatory activities is a difficult undertaking.

Mancur Olson has given some indication why this is the case. He has argued that individuals have an incentive to organize into groups that can further their common interests,[5] and that over time these groups become increasingly influential in the political process.[6] As societies and their political systems mature, interest groups solidify their connections within government, and as they become more powerful they are increasingly able to use their political power to employ the force of government to transfer resources from others to themselves, through taxation and through regulation. As political transfers become increasingly profitable to politically connected groups, entrepreneurial individuals find their talents increasingly drawn away from productive activity toward destructive entrepreneurship: they look for ways to plunder others, using the force of government. Indeed, Anne Krueger, in her article that gave the name to rent seeking, lamented that the brightest and most talented people in India – which was the inspiration for her article – were employed not in productive activity, but were engaged in trying to secure transfers through government. In that particular case it was mainly though government restrictions on the economic activities of others, and government privileges to the rent seekers, but the same principle applies to all regulatory and financial transfers that are effected through the force of government.

Over time political connections solidify and rent seeking becomes more firmly entrenched in the structure of government. Thus, prosperous nations can decline as destructive entrepreneurship overtakes productive entrepreneurship and becomes the surest way to increase one's wealth. Theft can work, in the short term, as a transfer mechanism, but roving bandits cannot appropriate as much over time as can stationary bandits, who can continue to appropriate resources from those who are productive.[7] The challenge for those who are on the receiving end of the transfers is to design an institutional structure that allows them to continue receiving transfers from productive individuals without killing off their incentives to continue producing. They must come up with some sort of an arrangement whereby those who are producing believe that those who are taking a share of their production will leave them with enough after their predation that production will still be worthwhile. The arrangement will be more stable if those in the productive class are also led to believe that the transfers are justified: that they are not just pure theft. Philosophers have given a name to this type of arrangement. They call it a social contract.

The social contract and rent seeking

The social contract is a normative device which argues that people in a society are bound together by a social contract, and that they are obliged to abide by the rules of that contract. What those rules are varies from one social contract theory to another, but their common point is that everyone in society has a moral obligation to abide by the contract's rules. Thomas Hobbes argued that the rules of the social contract are set by the sovereign, whether that is a king, dictator, or democratic government, and that citizens are obligated to obey the sovereign's rules.[8] Hobbes reasoned that without a sovereign to lay down and enforce rules, society would devolve into anarchy, where life would be nasty, brutish, short, solitary, and poor, and would be a war of all against all. The social contract was the mechanism for avoiding such Hobbesian anarchy. Jean Jacques Rousseau argued that people have a common will that is expressed through democratic decision-making, so that what is democratically determined is, in fact, an expression of the common will. Rousseau says:

When in the popular assembly a law is proposed, what the people is asked is not exactly whether it approves or rejects the proposal, but whether it is in conformity with the general will, which is their will. When therefore the opinion that is contrary to my own prevails, this proves neither more nor less than that I was mistaken, and that what I thought to be the general will was not so.[9]

John Locke argued that individuals have natural rights, and that the role of government is to protect those rights.[10] Locke's social contract obligates citizens to respect the rights of others, and the government intervenes when that social contract is broken. In Locke's view, unlike the views of Hobbes and Rousseau, those in government could violate the terms of the social contract just as its citizens could, but the idea remains that there is a social contract that specifies rules that everyone is morally obligated to obey.

While the social contract theory of the state is centuries old, modern contractarians continue to support it. John Rawls has argued that terms people would agree to from behind a “veil of ignorance,” where people are ignorant of their own personal characteristics such as height, gender, race, intelligence, handicap, and all other personal characteristics, are a part of a social contract to which people are morally obligated to conform.[11] James Buchanan says that if one imagines going back to a state of Hobbesian anarchy and then renegotiating a social contract to create order out of chaos, the terms to which people would agree are the social contract.[12] The common element in all of these variants of the social contract theory of the state is that there is a set of rules, enforced by government, that people have a moral obligation to obey.

The degree to which people buy into this idea of a social contract, whether they use those exact terms or not, determines the degree to which government taxation and regulation is distinguished from simple theft. If people do buy into the idea, government mandates are more easily enforced, because people feel some moral obligation to abide by the government’s rules.[13] Indeed, one of the ideas underlying the philosophy of democratic government is that not only are all citizens able to participate in this form of self-government, but that participation is a sign of good citizenship. People can participate and further, normatively, people should participate. This means that rent seeking activities – that is, addressing those with government power to request that government undertake certain activities -is, under the social contract theory, not an immoral act of theft, but rather a virtuous act of democratic participation.

Rent seeking losses in a static framework

The idea of rent seeking has been developed in a static model, and within this model all rents that might accrue to rent seekers are dissipated by the resources that are used in rent seeking. The basic logic can be seen in a simple example. Assume that the government budget allows for a transfer of $100 to be allocated to some unspecified person or group. Seeing the possibility of getting the $100, people will devote resources to competing for the money. They may hire lobbyists, which uses up resources, create ad campaigns to argue why they deserve the money, take members of Congress out for dinner, and so forth. The rent seekers who are competing for the $100 would be willing to spend up to $100 to get that $100 prize, so the resources used in competing for the rent will completely dissipate the rent. There will be no net benefit.

The logic behind the complete dissipation hypothesis is that if competitors for the rent spend less than the amount of the rent, rent seeking will be profitable and more rent seekers will enter the competition, spending more resources until the rent is completely dissipated. If rent seekers are spending more than the total rent, rent seeking will be unprofitable, causing some rent seekers to use resources in other ways. An equilibrium exists when the costs incurred in rent seeking are equal to the expected benefits from engaging in rent seeking. Just as in a competitive industry following standard microeconomic theory, if rent seeking is profitable it will attract entry; if it is unprofitable it will lead to exit, until the expected benefits of rent seeking equal the costs of engaging in rent seeking.

Some rent seeking expenditures may be transfers rather than a use of potentially productive resources, but if so this merely pushes the rent seeking back a step. For example, if rent seekers used some of their rent seeking resources to pay bribes to politicians, this is a pure transfer that does not waste any resources. However, such transfers make the position of the recipient more valuable, so there will be rent seeking for those positions. For example, people will invest real resources into being elected into political office so they can be transfer recipients. Eventually, all of the resources from rent seeking are dissipated into nonproductive uses.

This static model of rent seeking presents a pessimistic view of the ability of government to allocate resources to produce any net value at all. Should the government consider any regulatory changes, those who stand to benefit, or have costs imposed on them, would expend resources to bend the regulation their way up to the point where the resources they expended would equal the value of the benefit provided to them by the regulation. Similarly, any increase in the government's budget would open the door to rent seeking by individuals and groups who would like some of that revenue spent to benefit them. As the government budget grows over time, the potential benefits to rent seeking grow along with it. That results in more rent seeking losses, and further provides an incentive for entrepreneurial individuals to engage in destructive rather than productive entrepreneurship.

In 1910, federal government spending in the United States was less than 3 percent of GDP. By 2010 federal spending was 25 percent of GDP. In 1910 there was relatively little to gain from engaging in rent seeking activity to try to capture a share of the federal budget. Entrepreneurs had more of an incentive to engage in productive entrepreneurship, introducing innovations into the economy. The larger share of government expenditures in 2010 created more of an incentive to shift entrepreneurial effort toward trying to gain a share of government spending, rather than introducing private-sector innovation.

Rent seeking and economic progress

The pessimistic conclusions that come from the static model of rent seeking do not necessarily apply in a dynamic setting. In a static setting rent seeking benefits, once institutionalized, appear to last into the indefinite future, making them very valuable. If a domestic firm, for example, can lobby Congress to place a tariff on the imports of its foreign competitors, this limits competition and provides some degree of monopoly profits for the domestic producer. A static model depicts this process as a snapshot in time, but the implication appears to be that month after month, year after year, the domestic producer can continue to receive these rents. That makes the rents very valuable, which means that rent seekers will have an incentive to expend substantial resources to obtain them.

In a dynamic setting rents that are created today will be eroded as time progresses. One reason is that legislation passed today can be reversed tomorrow. Another reason is that once legislation is passed, people can begin looking for ways to avoid the effects of the legislation. More significantly, in many economies economic progress can make the activity on which the rent is bestowed increasingly obsolete and irrelevant. Consider the implications of each of these three reasons why rents can be eroded over time.

In the first case, consider the potential to place a tariff on a firm's foreign competitors that would allow the domestic firm to raise its prices and increase its profits by \$100 a year. If the rent seeker views that rent as permanent, then, assuming a 10 percent discount rate, \$100 a year has a present value of \$1000, and the rent seeker would be willing to expend \$1000 in resources today to obtain that future flow of rents. Now, assume that flow of rents must be defended every year in order to remain in place. The rent is now only worth \$100 this year, and the firm would only be willing to expend \$100 to capture the rent. Next year a similar rent seeking activity would have to be undertaken to have the rent renewed. Few rents are certain, because those who are forced to make the transfers have an incentive to try to get the policy that creates the rent reversed. For example, customers exploited by a tariff who want to pay less for the domestic firm's output, or for imported competing output, may engage in rent seeking to remove the tariff. The more permanent any granting of a rent seeking benefit appears to the rent seeker, the more the rent seeker will be willing to pay to get it.

In the second case, even if the legislation providing the rent is viewed as completely permanent, there may be ways for entrepreneurs to avoid any restrictions and thereby erode the value of the rent. Consider two examples. In the 1980s the Reagan administration imposed "voluntary" quotas on the importation of Japanese automobiles. Most observers believed that "voluntary" quotas meant that if Japanese importers did not abide by them, those in government would take steps to make the quotas mandatory. Policies like this pose an ongoing threat to targeted firms. Voluntary quotas could be lifted next year, only to be reimposed in future years. One result has been that Japanese automobile manufacturers have opened manufacturing plants in the United States, so any restriction on imported automobiles would have a much more limited reach, making it less valuable for rent seekers to seek such a restriction.

As a second example, in the early 1990s the federal government imposed tariffs on the importation of computer memory chips, to protect domestic manufacturers of computer memory. The tariff was not imposed on products that contained memory chips, just imports of the chips themselves. The result was that domestic computer manufacturers moved their facilities to other countries so that the entire computer could be built overseas and the computer containing the memory could be imported to avoid the tariff on the memory chips. These examples are offered as illustrations, but they are obvious enough that they likely do not do justice to the creativity of entrepreneurs. When taxes and regulations are passed, entrepreneurial individuals can find many ways to avoid the costs imposed on them by those regulations, and as such opportunities are discovered, and the value of any restrictions passed for the benefit of rent seekers falls.[14]

In the third case, rent seekers may seek a subsidy for their products, tariff protection, or other regulatory protection, which enables the rent seeker to gain some rent. Over time, economic progress will render the economic activity protected by the legislation increasingly obsolete, so the benefit provided to the rent seeker will diminish with economic progress. For example, large-scale textile manufacturing began in Britain, migrated to the United States, and now is centered in the Pacific rim countries. Efforts to protect those domestic industries ultimately failed because economic progress continued to lower the cost of overseas producers to the point where any politically feasible protection was insufficient to maintain domestic profitability.

In a static framework, it appears that a subsidy or regulatory barrier that gives a producer a cost advantage should result in a permanent rent. In a dynamic framework, that rent offers the recipient a cushion that can result in costs creeping up to erode the rent. For example, a profitable firm may see its labor costs rise as unions are able to use that profit as a bargaining chip for higher wages. Meanwhile, competing firms, acting entrepreneurially, are finding ways to lower their costs to the point where their cost advantage is greater than the amount of rent that it is politically feasible for the protected firm to obtain. The benefit obtained by the rent seeker in the past sees its value eroded to the point where it becomes worthless. Over time, the rents obtained by rent seekers become worth less and less. For example, labor unions in the United States were able to strengthen their bargaining power in the 1930s and 1940s through rent seeking to get Congress to give them more bargaining power relative to the unionized firms that employed their workers. This led to a decline in heavily unionized industries such as railroads and auto manufacturing in which union workers were replaced by non-union workers, in the rail industry by trucking firms, and in autos by new competitors employing non-union labor.[15]

If potential rents, once established, can be long-standing, rent seekers have an incentive to expend more to get them. If potential rents are transitory, the incentives to use resources for rent seeking are reduced.

Institutions and rent seeking

When one takes into account factors that can change with the passage of time, the losses from rent seeking may not be as great as they appear in the static model, because any rents generated would have a limited value because of their transitory nature. How transitory will depend on the institutions under which the rents are sought. One of the factors considered in the previous section was the permanence of any legislation that would provide rents to a rent seeker. Consider a political regime run by a dictator with a strong hold on power. Rent seekers able to get some political favors from the dictator could anticipate that they would be long-lived. Thus, rent seekers would be inclined to devote substantial resources toward their rent seeking efforts. The rent seeking expenditures are a drain on the economy, but so are any anti-competitive privileges, such as monopoly franchises, tariffs, or other regulatory barriers that limit competition. In such a setting, rent seeking losses will be large, just as depicted in the static rent seeking model, because as time passes things remain much the same. A static model is more descriptive of a static world.

If a regime is extremely static, so that those in power have a limited group of supporters and there is no good way for rent seekers to compete for government favors, that may limit any rent seeking expenditures aimed at gaining favors from the incumbent regime. Outsiders would not be successful anyway. In this case rent seeking activities have the potential to turn from political activities toward violent ones. People may be inclined to engage in revolution to replace the incumbent government with their own group, in an attempt to capture the rents that accrue to those favored by the incumbents.

If the political process provides less protection for rent seekers, so that, for example, rents granted by one democratic regime could be voted away by the next one should the regime be overly generous in its grants of political favoritism, rent seeking will be reduced and rent seeking losses will be minimized. There are benefits to having a relatively stable legal system that can only be amended when there is a substantial consensus in favor of doing so. But these benefits do not extend to legislated privileges given to rent seekers. If those privileges appear tenuous, this will discourage rent seeking activity.

When one reflects on William Baumol's distinction between productive and destructive entrepreneurship, it is apparent that the institutions that best discourage rent seeking are those institutions that reward innovative private-sector activity, and provide no incentive for seeking politically generated favors. This is not a new insight. One can trace its roots back to Adam Smith, at least, and economic historians[16] and political economists[17] have made persuasive arguments about the benefits of market institutions. More recently, James Gwartney and Robert Lawson have undertaken a project[18] to measure empirically the degree to which institutions support market exchange and suppress destructive entrepreneurship. As they quantify it, smaller government, with lower taxes and expenditures, protection of property rights, rule of law so that the legal system treats everyone the same, low barriers to international trade, sound money, and minimal regulatory barriers, are the institutions that define a market economy, and a substantial amount of research using their measure shows that economic freedom, as they define it, is strongly correlated with higher income and income growth, and other indicators of well-being.[19] With good institutions, people have the incentive to engage in productive entrepreneurship that leads to prosperity, and those institutions discourage the destructive entrepreneurship of rent seeking. Solid economic research has identified those institutions that lead economic actors to engage in activity that produces prosperity.

Competition controls rent seeking

Within the static model of rent seeking, rents never disappear once they are acquired, because the model depicts only a timeless snapshot in time that shows the results of rent seeking. But it is apparent that in some circumstances rents are eroded more rapidly than in others. Consider two simplified theoretical frameworks which can be labeled steady state general equilibrium and Schumpeterian creative destruction. In the steady state general equilibrium framework, an economy arrives at general equilibrium and, once there, economic forces keep things as they are period after period. This corresponds with the conception of general equilibrium elegantly developed by Leon Walras[20] and promoted by John Hicks and Paul Samuelson,[21] and with the concept of the evenly rotating economy depicted by Ludwig von Mises and Murray Rothbard.[22] Because things remain the same in general equilibrium period after period, the rents accruing to rent seekers now last into the indefinite future, increasing their present value, and thus increasing the resources rent seekers will devote to acquiring those rents.

Within the Schumpeterian model of creative destruction,[23] economic activity that is profitable in the present will become increasingly less profitable in the future as the forces of competition make current products and current production processes increasingly obsolete. Entrepreneurs are always looking for better production methods and more desirable products and product characteristics to produce at lower cost, and to attract customers from their competitors. In this Schumpeterian vision of the economy competition controls rent seeking, because any rents that can be secured today lose their value in the future as the entrepreneurial forces of capitalism lead competitors toward improved products and improved production methods. Competition controls rent seeking by reducing the present value of those rents, because rents that are valuable today will see their value rapidly eroded in the future. The more entrepreneurial an economy, and the more rapid economic progress, the less value politically generated rents will have, because competition erodes their value.

The purpose of this comparison between the steady state general equilibrium framework and the Schumpeterian creative destruction framework was not to set up a straw man to criticize particular theories, but rather to consider the fact that these two theoretical frameworks resemble different economies to varying degrees. An economy that has little economic development, so that things are, in fact, much the same year after year, will have a closer resemblance to the steady state general equilibrium framework, whereas an entrepreneurial economy in which economic progress continues to introduce new products and production processes will more closely resemble the Schumpeterian creative destruction framework. Sometimes political institutions are designed to hinder economic progress for the purpose of preserving the rents that exist under the current regime.

For example, building codes that specify allowable materials, methods, and so forth, can stand in the way of economic progress, and yet provide rents to current producers. In some US cities plumbers are allowed to use PVC water lines, where in others building codes require that water lines are copper. Not only is copper more expensive, it is more difficult to install, so that jobs that might be do-it-yourself for some homeowners, if they could use PVC, will require the services of a plumber if the building code does not allow it. In many cases laws specify that products be manufactured a particular way, preventing economic progress and locking in those who specialize in the current process, and in many cases producers are granted monopolies, which again locks in the rents that accrue to the current producers.

The degree to which regulatory restrictions, import restrictions, and monopoly franchises exist varies from country to country, and even from city to city, but as these institutions become more prevalent they limit competition and push real-world economies to more closely resemble the steady state general equilibrium framework. Within this framework rent seeking is more profitable, which pushes entrepreneurs away from productive entrepreneurship toward destructive entrepreneurship. More competitive economies lower the value of rent seeking, so reduce the amount that takes place. The degree of competition is influenced by institutions, but competition by itself works to limit rent seeking because economic progress erodes the value of rents.

The creative destruction of an entrepreneurial economy limits rent seeking for two reasons. Not only does it lower the present value of any rents that could be obtained, it also raises the value of productive entrepreneurship by opening up an increasing number of new entrepreneurial opportunities. An entrepreneurial economy continually generates new entrepreneurial opportunities. An economy resembling the steady state general equilibrium model generates no new opportunities, by the very nature of its steady state.[24]

Institutions and progress

Economic progress requires a particular set of political institutions: institutions that have been in place for only a few centuries. Nevertheless, there is an inclination to take these institutions for granted, partly because they have been in place far longer than the lifetime of anyone currently alive. John Stuart Mill, who had a deep concern for people's economic well-being, argued that "The laws and conditions of production of wealth partake of the character of physical truths. There is nothing optional or arbitrary in them. . . . It is not so with the distribution of wealth. This is a matter of human institutions solely. The things once there, mankind, individually or collectively, can do with them as they like."[25] Mill's views here might be questioned on both counts: whether the laws of production are physical truths, and especially whether the distribution of wealth is unaffected by the conditions of production.

Certainly there are physical laws that govern production, but taking a long-term view the physical constraints on production that exist today will be considerably looser in the future because of the innovations brought by entrepreneurs. Consider that a Boeing 747, introduced in 1970, has over six million parts, while a Boeing 777, introduced in 1995, has only three million parts. The Boeing 777 has a passenger capacity similar to the early generation 747–100 and 747–200 models, but uses one-third less fuel and has maintenance costs that are 40 percent lower.[26] Both are long-haul aircraft used on the same routes, so the aircraft are interchangeable in use, and passengers will notice no significant differences when flying in one or the other aircraft. The big difference is on the production side, where the cost of production, and the operating costs once the aircraft is produced, have been dramatically reduced.

Advances in the production of electronics have been so stunning that they are obvious to all. The dramatic improvement in performance, along with reductions in cost, of computers, phones, televisions, and other electronic devices has been dizzying. This even applies for products as mundane as food: farmland yielded about 23 bushels of wheat per acre in the 1970s, an amount that had risen to around 31 bushels per acre by 2010 – an increase of 35 percent – largely because of improved wheat varieties and farming methods.[27] While in a narrow sense one might agree with Mill that production is constrained by physical laws, over the longer term the physical constraints on production can be modified so that a larger output of higher-quality, and previously unavailable, goods and services can be produced.

The increase in the productive capacity of resources does not just happen. It is the result of the innovations of entrepreneurs. If entrepreneurs have no incentive to innovate, then Mill's observation would seem to have much force. The laws of production are physical truths, and producers can only work within the constraints of the production functions they are given. But entrepreneurs do not accept the production functions they are given, they innovate to provide better production functions for themselves, as the examples just given illustrate. This only happens if institutions are favorable for entrepreneurial activity.

Mill is even more wrong in asserting that the distribution of wealth is solely a matter of institutions, because once things are produced they can be distributed in any way people decide. The incentive to engage in productive activity, and to take the risks involved in entrepreneurship and innovation, depend upon institutions that allow producers to keep the results of their productivity. If the institutional structure sets up a system whereby the production is taken from people who are productive and shared with everyone, that removes the incentive to be productive. At first glance one might equate this conclusion to an argument that productive people are selfish, and will not be productive if they have to share with others less fortunate than themselves. That view appears consistent with Mill's conclusion, in that once productive people produce things, society can allocate that production in any way they choose.

What this initial reaction fails to recognize is the risks that are inherent in production, and especially in innovation and entrepreneurship. When entrepreneurs innovate, they are doing things nobody has done before, and there is no way for them to know whether they will succeed. There are risks, and potential losses to be borne, but entrepreneurs believe that their innovations will succeed. Still, because they are risking something in trying, there is no incentive to try if they must bear the losses of failure, but any gains from success will be taken from them and redistributed to others. The laws of distribution are not arbitrary. If the institutions of distribution allow productive people to keep the rewards of their productivity, they will be more productive, and the types of economic progress illustrated by the examples earlier in this section will continue to occur. If the institutions of redistribution take from the productive to level the distribution of income, productivity will be stifled, and there will be less for everybody.

Mill's statement about the physical truths involved in production, and the arbitrary nature of the laws of distribution, made in the mid-nineteenth century, which shows a profound misunderstanding of the economic forces that produce prosperity, is common in opinions about economic policy in the twenty-first century. When public policy views the incomes of high-income people as available to be redistributed because the public policy toward redistribution is independent of the laws of production, the stage is set for the introduction of rent seeking institutions that stand in the way of entrepreneurship and block economic progress. An economy that moves from continual progress toward stagnation also moves toward destructive entrepreneurship and away from productive entrepreneurship, primarily because the creative destruction of an entrepreneurial economy is one of the main factors that limits the value of rent seeking.

The politics of progress

Most of the analysis in this chapter has been focused on the rent seekers, not the rent granters. The degree to which rent seeking can succeed, however, depends on the degree to which those who grant the rents have the power and willingness to do so. The power to grant rents depends upon the constitutional constraints placed on those who hold political power, and the budget they have which can be doled out in the form of rents. The willingness to grant rents depends on the degree to which rents can garner political support relative to other political activities. Political entrepreneurs will look at their opportunity sets to see whether rent-granting activities offer more potential benefit than other ways they can allocate their political capital.

One can see that the institutional structure is critical. The degree to which the rule of law is enforced, and the degree to which property rights are protected, determines in part the degree to which rents can be allocated by the political class. When these two components of economic freedom are met, the power of the political class to use the force of government to transfer resources from some people to others is limited. Similarly, when the government budget is large, the government controls more resources and has more power to bestow rents on people through budgetary allocations. These rents can come in the form of simple monetary transfers such as welfare payments and corporate subsidies.

They also come in the form of government contracts. Even if one fully supports the government's building of roads and operation of schools, the contracts to build the schools and the roads, and to maintain them, are sought by many, and in the competitive environment to obtain government contracts, making those awards to some rather than others benefits the recipients, and thus can provide political capital to the politicians who hand them out.

If there are institutional procedures that make the awarding of contracts merit-based, and based on objective criteria, those with political power will find themselves limited in their ability to grant rents. As government grows, many business enterprises increasingly rely on government contracts for their incomes rather than market-based sales. For example, Lockheed Aircraft was once a premier manufacturer of commercial aircraft from its founding in 1926 through to its last commercial aircraft, the L-1011, first produced in 1970 and manufactured untl 1984. Since then Lockheed has only manufactured military aircraft. Merged with Martin Marietta in 1995, the company's primary customer is the federal government, to which it sells missiles, space vehicles, and military airplanes. As a growing government budget generates more private-sector businesses that rely primarily on the government for their business, the distinction between market and government allocation of resources becomes less distinct, and not only do the opportunities for rent seeking increase, but an increasing share of the economy becomes dependent on having government decisions go their way to ensure their survival. There is a dependency on the political process that gives those with political power additional influence, and the ability to extract rents from those who depend on their decisions.

Politics is a very competitive endeavor, and those who get to the top must work hard and be entrepreneurial to beat their rivals in what often is a zero-sum game. In the private sector, where markets can grow, there is the opportunity for all rival firms to thrive; but in politics, and especially in electoral politics, when one candidate wins another loses. Political entrepreneurs must constantly be alert to opportunities to enhance their political power, to retain their current positions, and to put themselves in a position to compete for higher offices.

Within politics there are both productive and unproductive entrepreneurial opportunities. If rule of law and protection of property rights are important to the productivity of the economy, politicians have the opportunity to make changes that secure property rights and make the legal system more fair and objective, creating gains that can be shared throughout the economy, and at the same time increasing the political capital of the political entrepreneur. If regulations can be modified to protect the rights of some, or streamlined to lower the costs imposed on others, this is a productive political opportunity. Entrepreneurial politicians can act on those opportunities and generate gains for some without imposing losses on others.

Entrepreneurial opportunities also arise to transfer resources from some people to others, gaining the support of the recipients, even though perhaps losing the support of those on whom the costs are imposed. Within the political process people tend to put more weight on concentrated gains than they do on small losses, however, so a political entrepreneur stands to gain by imposing small costs on many people to provide a concentrated benefit to a few. Indeed, in an overly simple example, if there are ten equal-sized groups in an economy and the politician taxes everyone to provide a concentrated benefit to the first group, and also taxes everyone to provide a concentrated gain to the second, and so on for the third through tenth groups, the likely result is that all ten groups will view the politician as having providing benefits to them. As has often been observed, politics is the process by which politicians bribe people with their own money.

The opportunities for productive political entrepreneurship are limited by the inefficiencies that political remedies can address, whereas the opportunities for unproductive political entrepreneurship are unlimited. There is no limit to the number of favors that can be bestowed on one group of rent seekers at the expense of others. In addition, when political entrepreneurship generates efficiencies the benefits are widely dispersed, giving them the character of a public good, whereas when political entrepreneurs bestow private benefits on an interest group the politician gets full credit, and the benefits go to that politician. Because politicians have a limited amount of political capital that they can expend, they are often in a position where they must choose between productive and destructive entrepreneurship: that is, where they must choose between using their political capital to produce gains that will be widely shared, or gains that will go narrowly to themselves. As a result, incentives often push politicians toward destructive rather than productive political entrepreneurship.

Those living today have seen economic progress throughout their entire lifetimes, making it easy to take for granted, but the economic progress that is an accepted part of life today is less than 300 years old – a very brief period of human history indeed – and it has only occurred in places that have established the economic institutions of capitalism. To preserve those institutions, opportunities for rent seeking must be limited, which means there must be constitutional constraints on political actors that can be enforced through a balance of power. It is easier to describe the institutions that lead to prosperity than it is to implement those conditions in places that do not have them.

Conclusion

Rent seeking activities stand in the way of economic progress, because they divert resources from productive to destructive uses. In the static framework within which the concept of rent seeking was developed, rent seeking losses are potentially huge, and looking at the model one would be surprised that rent seeking losses do not consume the economy. Partly, they do not because of constitutional constraints that help control them. Often this is a major difference between prosperous nations, where rent seeking is to a degree controlled, and poor nations where prosperity is determined by how close an individual is to the center of political power. However, the static model of rent seeking leaves out factors that can erode the gains from rent seeking over time.

One of those factors is the impermanence of political decisions. In static political regimes where political decisions today can convey benefits that last into the indefinite future, the benefits to rent seeking are potentially large, and as a result rent seeking activity is also substantial. When political decisions are subject to re-evaluation in the future, rents generated today may be removed tomorrow, lessening the value of rent seeking. These are political considerations. Perhaps the biggest factor limiting the value of rent seeking in a dynamic economy is the creative destruction that makes today's economic activity increasingly obsolete as the innovations of entrepreneurs introduce new and improved products, and improved methods of production.

A competitive economy contains its own restraints on rent seeking, because even if an activity is subsidized, or competition limited through taxation or regulation, that protected activity will become increasingly less relevant as it is overtaken by economic progress. To keep up with the rest of the market, entrepreneurs must engage in the productive entrepreneurship that makes their output more valuable, rather than the destructive entrepreneurship that generates rents that will turn out to be short term.

One example is the railroad industry in the United States, which at one time was the major provider of long-distance transportation and freight. Heavy regulation limited competition, and the resulting profits of railroads were eroded by unions using their own rent seeking activities to increase union employment and wages. The result was that the limited passenger rail now available in the United States is government-provided, and trucks haul the vast majority of freight, both long distance and local. In a dynamic economy, the creative destruction of competition means that the rent seeking that can provide short-run gains makes the rent generating activity less competitive, and over time it is supplanted by the innovations brought along by entrepreneurs.

Rent seeking that can be profitable in a static, or stagnant, economy, will be less so in a dynamic competitive economy. Therefore, in a competitive economy the returns are higher to productive entrepreneurship, which pushes entrepreneurs toward productive rather than destructive entrepreneurship. There is a positive feedback mechanism, in that the productive entrepreneurship of a competitive economy produces greater opportunities for future productive entrepreneurship. A competitive economy contains within it the forces that are required to limit rent seeking. The challenge, then, is to maintain the institutions that support the competitive economy.

10 The coordination of economic activity

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The way in which the institutions of a market economy coordinate economic activity is nothing short of astounding. Consider that a computer designed in the United States might have a microprocessor manufactured in Israel, a hard drive manufactured in Taiwan, and might be assembled in China using other parts manufactured in other locations all over the world. This remarkable cooperative effort coordinates the economic activities of people throughout the world, even though those people whose coordinated activities go into building the computer will never see each other, and if they did meet may not be able to communicate directly with each other because they speak different languages. Yet the fact that they will never interact directly with each other does not hinder their abilities to coordinate their economic activities to see that computer constructed, from design, to manufacture, to sale, for the benefit of the final user – another person that none of those who had a hand in producing the computer are ever likely to interact with directly.

One example is remarkable enough, but when one considers the number of goods and services produced in an economy, the coordination of market activity is truly amazing. People are able to adjust their economic activities to various circumstances of which they have no direct knowledge, because the market system gives them the information they need to know. If a freeze in Florida damages the orange crop, people need to reduce their consumption of oranges, and higher prices for oranges will give consumers all the information they need to know, even if they know nothing about the freeze. As long as the market is left to function freely, economic forces are at work to make markets clear, so that the quantity supplied equals the quantity demanded in all markets. As Chapter 4 discussed, economists understand a great deal about the properties of the economy when all markets clear. Things are constantly changing, but as long as entrepreneurs are able to see and act on new information, markets will tend to continue to clear, so that the quantity supplied equals the quantity demanded in all markets.

As Chapter 1 noted, the development of economic ideas in the twentieth century focused on the properties of economic equilibrium; that is, the properties of an economy in which all forces which pushed prices to change had already worked themselves out. Keynesian macroeconomics leaves open the possibility that an underemployment equilibrium could exist, in which no jobs are offered despite there being unemployed people who would like to work at the current wage. The newer general equilibrium business cycle theory[1] depicts an economy where markets are always in equilibrium despite macroeconomic fluctuations. In equilibrium models, problems involved in coordinating the various economic actors so that their actions are consistent with one another are left out of the model, because markets are always in equilibrium. In equilibrium, price signals always convey an accurate picture of the relative scarcity of goods and services, because all prices are market clearing prices, again because the models assume an equilibrium. What happens if some market participants are poorly informed? Because they do not see all of the trades for a particular good or service, or all of the offers to buy and sell, they may agree to trades at non-market clearing prices. Trades like this could provide faulty information to other traders.

This is more likely to happen when economic activities have a long time-dimension. In a manufacturing process, for example, years may pass between an initial investment in a factory and bringing the first goods produced in that factory to market. Current transactions are made based on expectations of future prices: expectations that may prove to be wrong. The result is that economic activity is not coordinated because the plans of various economic actors are not consistent.

Market coordination: the consistency of plans

The heavy focus of twentieth-century economics on the properties of economic equilibrium led to some thought on how a market equilibrium is defined. In the static general equilibrium framework of Leon Walras, equilibrium meant that the quantity supplied equals the quantity demanded in all markets.[2] Walras described an iterative process whereby an auctioneer would call out prices and register the quantities supplied and demanded at each price. If the quantity demanded exceeded the quantity supplied, in the next iteration the auctioneer would raise the price; if the quantity demanded was less than the quantity supplied the price would be lowered. Eventually the auctioneer would iterate to a vector of prices that would allow all markets to clear. No trade would take place at the disequilibrium prices, so Walras's mechanism has the unrealistic feature that price information is generated even when no exchanges take place – which means there are no actual market prices to observe.

The twentieth-century neoclassical model, explained well by C.E. Ferguson,[3] one of the century's preeminent neoclassical economists, employed the assumption of perfect information to generate trades that always occurred at equilibrium prices. With perfect information, sellers could always avoid selling too cheaply, and buyers could avoid paying too much. The neoclassical model described by Ferguson, and the Walrasian framework, are static equilibrium models, so equilibrium means that market forces keep all prices at their market clearing levels, but only by unrealistic assumptions of perfect information, or a mechanism that generates price information without any actual exchange taking place.

Israel Kirzner takes a different approach, and defines equilibrium as a situation in which there are no unexploited profit opportunities.[4] In a neoclassical world of perfect information, Kirzner's concept of equilibrium is the same as the neoclassical concept. But people do not have perfect information, so Kirzner's view is that the economy is never at an equilibrium, and there is always the opportunity for entrepreneurs to find and exploit profit opportunities, bringing the economy closer to equilibrium. Unlike the neoclassical view of equilibrium, in Kirzner's framework market clearing and equilibrium are not the same things.

Yet another view, more congruent with the ideas presented in this book, is that economic equilibrium exists when the plans of all individuals in the economy are mutually compatible.[5] This view of the consistency of plans among individuals in the economy recognizes that economic decision making takes place through time by emphasizing the consistency of plans, and also points directly to factors that can upset economic coordination. As Chapter 4 noted, there is an argument to be made for using the term market clearing rather than equilibrium when thinking about the market forces that cause the quantity supplied to equal the quantity demanded in markets. If there really is an equilibrium, one would expect that, like the Walrasian auctioneer, eventually prices and quantities would iterate toward that steady state equilibrium where economic affairs remained the same throughout time, and the term equilibrium suggests that if anything disrupts the current state of affairs, equilibrating forces exist to pull the economy back to equilibrium. In some cases temporary disturbances do occur like that, but in many cases disruptions of the current state of affairs create new conditions, so that the old configuration of prices and quantities will never return. Unlike an equilibrium model, in which current conditions imply all future states of the world, everyone knows today that future events will unfold in ways that nobody could foresee in their entirety. This creates the uncertainty that causes people to make inconsistent plans.

The reason that the coordination of economic activity breaks down is because things change in ways that are not easy to foresee, and when they do people respond to that uncertainty by making plans for a future that each forecasts in a slightly different way, resulting in a decline in the degree of coordination of activities in the economy. One can foresee that things will be different in the future, just not how they will be different. The steady state view of an economy is obviously misleading, and even in the ordinary course of business unforeseen changes occur. Sometimes changes occur over a long enough time horizon that the adjustment is orderly, such as when computers replaced typewriters. People working as typists had time to adjust. Sometimes, as with the robotization of assembly lines and the building of new factories, there is some lead time, but this can be disruptive to some, especially when an old factory is replaced by a new one in a different location, and when the skills needed to do the new jobs are different from those needed to do the old ones. Cyclical real-estate markets can lead to construction booms and busts. Coordination breakdowns are not necessarily the result of shocks to an economy, like an oil embargo, or a tsunami, but occur as a regular part of the process of economic progress. In contrast to the assumption of perfect information, everyone's knowledge base is different, so everyone does not foresee the same future. As a result, people make plans that are mutually inconsistent.

The evolution of coordination

The challenge facing any economy is the coordination of everyone's plans. This coordination enables economic cooperation. Market equilibrium depicts the result of successful coordination. The quantity supplied equals the quantity demanded, and the general equilibrium framework shows the possibility of all markets clearing simultaneously, so that the quantity supplied equals the quantity demanded in all markets. Price adjustments maintain the equilibrium and allow for everyone's plans to be realized. One great accomplishment of twentieth-century equilibrium economics was to show that market forces pull the economic activity of individuals toward an outcome that coordinates everyone's plans. Given the vector of equilibrium prices (perhaps computed by the Walrasian auctioneer), everyone's plans are mutually consistent. Indeed, the market works so well in this respect that it is easy to take the coordinating functions of the market for granted.

The equilibrium framework depicts the coordinating function of the market taking place in an institution-free environment. People buy and sell at the market price without the support of the many economic institutions that facilitate exchange taking place. In fact, economic institutions play an important role in allowing this coordination of individuals' economic plans, and therefore, the extent to which people can cooperate in their economic endeavors. One would not want to anthropomorphize the economy, because the economy is not a thing, but rather an aggregation of individuals who are all making their own consumption and production decisions. However, because people in an economy act within a common institutional framework, in this sense one might think about an economy in the singular, as an aggregation of individual economic activities taking place in one particular institutional setting. Individuals make plans today with the hope of fulfilling them in the future, and institutions play a role in determining the degree to which they can succeed and the level of complexity that an economy can sustain.

Economic institutions vary from place to place, so which economy one trades in depends on one's location. More complex economies require a more complex institutional structure to support their economic activity. One can envision a society in which people get by with subsistence agriculture, growing crops mainly for their own consumption, and engaging in market exchange to get durable goods, farm implements, and other items they will not need on a daily basis. Contrast this with farming in an advanced economy, where large quantities of a single crop may be grown, most of it sold in the market, in exchange for most of what the farmer consumes. In the advanced economy, most people rely almost entirely on others for everything they consume, including food and other items they need on a daily basis. They rely so heavily on others for their survival, yet they hardly think about it because the coordination of economic activity works so well. How many residents of New York, Paris, London, or Tokyo, are concerned that the coordination of the market might break down, ceasing to bring them food, and leading to their rapid starvation? They take the market institutions that provide their sustenance for granted, because the market mechanism works so well that their concerns are about more trivial things, such as whether their favorite sports team will win its game, or how the plot will twist on their favorite television show.

Even when they are concerned about significant economic matters, like whether their company will make that sale that will get them their bonus, or whether the newly designed product they are selling will increase their market share, those economic concerns are about the things that will bring them their incomes, not whether the coordination mechanism of the market will fail, bringing them to starvation. Meanwhile, people who survive through subsistence agriculture run the risk of bad weather that can lead to poor harvests, and starvation. Because of the coordination of the market economy, reliance on others through the division of labor makes people more secure than if they rely on themselves for their security.

People are more interdependent in an advanced economy, but they are also more likely to be able to realize their plans, because market institutions spread risk and allow people to draw on past production for current consumption (through saving) or draw on future production for current consumption (through borrowing). In a primitive economy people could save by setting aside stocks of goods they produce for future consumption, with the attendant storage costs and possible spoilage and depreciation. In a modern economy people can save by setting aside purchasing power. They can provide resources to others by buying financial assets with the hope of earning a return on those assets, which they can then in the future turn into purchasing power. Enabling this possibility requires a complex and sophisticated institutional structure.

One can think about simple transactions, like buying fruit from a roadside stand. This requires relatively little in the form of economic institutions. The fruit vendor gives the buyer the fruit as the buyer simultaneously gives the money to the vendor. Even there, the medium of exchange may be supported by complex institutions, and those institutions can affect the degree to which people are able to realize their plans. The monetary theories of the business cycle discussed later in this chapter show how changes in the money supply can frustrate some economic plans, and should inflation get too high people's transactions may be at least partly driven by avoiding the holding of the medium of exchange, rather than achieving other economic goals.

The fruit transaction is a simple one. Consider someone who wants to build a house. The person will probably need a mortgage, and the lender will want assurance that the land on which the house is built has clear title. The lender will also make provisions to provide money to the builder as the construction progresses, to help guard against the builder taking the money and not building the house. The lender will also use the house as collateral for the mortgage, so that if the homeowner stops paying the mortgage, the lender can foreclose and take possession of the house. These economic institutions – title registration, title insurance, the banking system that provides funding, the court system that gives the lender some assurance of being repaid – allow complex transactions to take place, but models of market equilibrium tend to leave them out of consideration. As Hernando de Soto explains, these institutions are essential if an economy is going to advance beyond a primitive level of existence.[6]

A key factor pushing for increased complexity in institutions is the passage of time. When an entire transaction occurs at one point in time, such as buying fruit from a fruit stand, relatively simple economic institutions are all that are needed to allow the transaction. When the transaction takes place over an extended period of time, such as financing and building a house, more complex institutions are required. Even though the construction of the house may be completed in under a year – which is a complex undertaking in itself – the financing might extend to 30 years, so complex institutions are required when initiating a transaction today in which some of what is promised may not be delivered until several decades into the future.

This coordination mechanism has to build up over time. An entrepreneur decides that there is a profit to be made selling clothing to farmers so that they do not have to make their own. Over time, farmers find they get better clothing at a lower cost by buying it than making it. An entrepreneur opens a grocery store that sells a variety of foods, and over time the subsistence farmerfinds that by selling his own produce he can take the income and buy a better variety of foods at a lower cost than growing his own. This allows the farmer to specialize in growing things (or one thing) that the farmer has a comparative advantage in producing, and as this spreads, everybody is more productive, everybody is wealthier, and everybody is more dependent on the economic activities of others for their survival. People can only take those steps away from self-sufficiency when economic institutions are sufficiently established such that the coordination mechanism is stable and they are confident that it will not break down, taking away their means of survival. Modern economies have passed well beyond this point, so that people take these coordinating institutions for granted and do not even consider the possibility that they might fail to provide for not only their basic needs, but also for goods and services that would have been considered unobtainable luxuries only a few decades earlier.

The intertemporal coordination challenge

The coordination problem and its resolution can be seen in a simple supply and demand framework, where prices adjust so that the quantity supplied equals the quantity demanded and everyone's plans can be realized. Everyone can buy or sell as much of everything as they desire at the market-clearing price. When economic activity consists of a series of transactions that take place over time, differences in expectations, or unforeseen circumstances, can result in a situation where some people make plans that are frustrated. A real-estate investor might build an apartment building, taking out a mortgage and anticipating repaying it with the rent future tenants will pay. If it turns out that the investor has misjudged the future demand for those apartments, the investor may not be able to collect sufficient rent to pay the mortgage. Economic affairs do not always turn out as people expect, and people may engage in activities today anticipating future conditions that do not materialize.

This is true regardless of one's economic activities. Because of the entrepreneurial nature of a market economy, over time things change, and those changes affect everybody. Even if economic agents wanted to, they could not continue to do what they do today into the indefinite future. People must change their plans because they know that future conditions will be different, and the best strategy for both consumption and production today will not be best in the future. Of course, everybody knows this. Coordination can take place in the face of an uncertain future because people do not need to make inflexible plans. Recognizing the uncertainty of the future, people make their plans such that when the inevitable unforeseen changes occur, they can modify their plans to accommodate those changes. People plan for contingencies, thinking along the lines of "If this happens, I can do that; if something else happens I can change my plans to this alternative." Flexibility in planning facilitates coordination.

Flexibility has its limits, however. Businesses make capital investments that are not easily reversible. If homeowners lose their jobs, they may have to sell their homes to move to other locations where they can find the best job, but homes are not liquid investments and it may take time to sell one. Meanwhile, it may take time for unemployed workers to survey the job market to determine their best employment alternatives. While renters will not face this particular problem, they may inaccurately forecast future rents in their area and be priced out of their homes due to rising rents. Firms bring new products to market by forecasting the demand for them, but they can never know for sure what the demandwill be for a product that has never been on the market before. Much can change between the design and planning for a new product and the time when the product actually is available to be sold.

People do plan for contingencies so that they can respond when the uncertain future eventually arrives, but sometimes they must lock themselves into a particular set of actions that will be difficult or costly to change. Under ideal circumstances, under the rapid adjustment that takes place in equilibrium models, people adapt to economic changes as they occur and markets continue to clear. When the disruptions are greater than people's ability to adapt to them, coordination breaks down. People remain unemployed because they cannot find a job, or because they believe there are better alternatives than those they currently are aware of if they take more time to seek them out. Retail locations remain empty, and factories remain idle, as their owners look for more productive ways to employ those resources. And indeed, in some cases investments might have been made in anticipation of economic conditions that did not materialize, such that those resources do not have a positive value. An empty lot might be more valuable than an obsolete structure that would have to be torn down before productive use could be made of its location.

With perfect foresight people's economic planswould always be consistent with each other, and resources would never be underutilized. People would move from one job to another immediately, and there would be no unemployment. Capital would move from one use to another immediately, and there would be no underutilized capital, no vacant storefronts, and no empty houses and apartments. Because the future is uncertain, and people cannot know each others' economic plans – partly because those plans are often contingent, so people do not know with certainty their own plans – economic coordination is imperfect and resources are sometimes underutilized when compared with the imaginary world of perfect foresight.

One might believe, along the lines of Ronald Coase,[7] that the underutilization of some resources is a transaction cost that is inevitable in a complex economy. For example, one can walk into a grocery store, or any retail establishment, and see a large inventory of merchandise for sale. This temporarily unemployed inventory is there because sellers do not know what items buyers will want to buy that day, so they offer a variety of merchandise, some of which might go unsold, might have to be marked down below their cost, or might spoil and have to be discarded. This is not solely due to sellers not knowing buyers' intentions. Often buyers might enter a store unsure of what they want to buy, and leave having made purchases they did not anticipate before entering the store. Sellers need the inventory to attract the buyers, but they could cut their costs if they could figure out a way to reduce their ratio of inventory to sales. "Big-box" retailers have done this by increasing their volume of sales, putting smaller retailers out of business.

Similarly, manufacturers keep an inventory of parts for assembly, and "just-in-time" manufacturing lowers costs by having components delivered right when they are needed, reducing the manufacturer's inventory costs because there is less "unemployed" inventory waiting to be used. But if there are disruptions in the supply chain, firms using just-in-time inventory might have to stop their operations, while firms with an inventory of parts could keep producing. Just-in-time inventory methods require more coordination of economic activity, with the additional complexity being rewarded by lower costs – or at least, the firms using those methods hope so.

People can plan for many contingencies, but contingency planning is costly, and both firms and households rely on the coordination of the market to replace contingency planning. Just as firms try to minimize their holdings of inventory, households buy a limited amount of food, counting on their ability to visit the grocery store when they run low, and they fill their cars with gasoline only when their tanks get low, counting on gas stations to have gas on hand when they want to buy it. If economic coordination breaks down, people's economic plans will no longer be consistent and some plans will not be able to be realized. If gas stations have no gas, as might happen during a natural disaster, people will have to limit their driving. If people unexpectedly lose their jobs, they will have to rearrange their economic affairs as they look for new ones.

If the real world were as static as neoclassical equilibrium models, market adjustments would allow everyone's plans to be mutually consistent and the challenge of coordinating economic activity would be easily met. In a complex and dynamic economy, where the future is inherently unknowable, one challenge the economy faces is maintaining the coordination of everyone's economic plans. When that coordination breaks down and people's plans are frustrated, entrepreneurial individuals act on profit opportunities to reallocate resources and minimize the disruption. People adjust their plans and coordination is reestablished. Entrepreneurial activity acts to coordinate people's plans and once again make them mutually consistent. In a changing and uncertain world, complete consistency would be too much to hope for, but in an entrepreneurial world, people are continually updating their plans to make them more consistent with the plans of others.

Macroeconomic instability: the breakdown of coordination

In a completely static general equilibrium, characterized by perfect information and no changes in demands or production technology – including no entrepreneurship – the economy remains stable and all resources, including labor, remain fully employed. One of the theoretic triumphs of twentieth-century economics was the proof of the uniqueness and stability of general equilibrium.[8] Everybody's plans are mutually consistent and coordinated. Macroeconomic instability is the result of a breakdown of this coordination. People's expectations are not realized, so they must change their plans in the face of new conditions, and while adjustment takes place, resources are underutilized. A subset of this underutilization is labor unemployment. Underutilized resources represent a profit opportunity, and if entrepreneurs can act on those profit opportunities as rapidly as they arise, resources can be reallocated without economic disruption, and unemployment will be avoided. When the required reallocation of resources is greater than the ability of entrepreneurs to reallocate resources, the result is an economic downturn due to the unemployment of labor and other resources.

In a general equilibrium framework, where everyone has perfect knowledge, everyone expects the same future. In the real world, where the future is uncertain, different people have different expectations. Consistency of plans does not mean that everyone expects the same economic future, it means that people have incorporated enough flexibility into their plans so that when the actual future arrives they are able to adjust. Macroeconomic stability requires a continual adjustment of people's plans, because the future always evolves in ways that few people completely anticipate. Coordination of plans does not mean creating a world in which everyone's expectations are the same, it means creating an outcome where, despite people'sdiffering expectations, they are able to adjust to conditions as they develop. Macroeconomic instability occurs when the adjustments are slower than would be required to completely adapt to changes in real-world conditions.

Macroeconomic instability is a result of a breakdown of the coordination of people's economic plans, and that breakdown can have many causes. The next several sections consider some of these causes.

Monetary causes of instability

Coordination of economic activity relies on accurate price signals generated by markets. Anything that interferes with market prices results in a misallocation of resources, because the manipulated price does not reflect the underlying values held by market participants. In a static framework, price ceilings cause shortages, price floors cause surpluses, and taxes and subsidies distort the allocation of resources even though markets may still clear. In a dynamic framework people make plans based on current prices and their expectations for future prices. The most visible proponent of the idea that macroeconomic instability often has monetary causes is Milton Friedman,[9] who argued that while in the long run money is neutral in its real effects, in the short run, when the money supply grows faster than is generally expected, it can stimulate an increase in real output, and when the money supply grows slower than is generally expected it has a negative effect on real output.

Friedman's argument about the neutrality of money in the long run is an oversimplification, and Friedman would recognize that rapid inflation caused by excessive money growth would reduce the informational content of prices and interfere with economic calculations. But as a first approximation, Friedman's point is that people take account of real prices, not nominal prices, when making economic decisions. In the short run, people may not have a good forecast of changes in the value of money, so may not immediately recognize that changes in total spending in an economy are due to monetary rather than real forces. In the short run they may misperceive monetary changes and interpret them as real changes.

In the framework of the equation of exchange, where M equals the quantity of money, V equals the velocity of money (the number of times the average dollar is spent each year), P is a measure of the price level, and Q is a measure of real output, the equation of exchange says that M x V=P x Q. So, M x V is the quantity of money multiplied by the number of times the average dollar is spent, which equals total spending. The velocity of money is the inverse of the demand for holding money balances, which is likely to change slowly over time, as is real output, so assuming V and Q to be relatively constant, any significant changes in P must be the result of changes in M. In the long run, the assumption of constant Q would mean Q is unaffected by M; that is, changes in the money supply affect nominal variables, but not the real economy.

In the short run people will have difficulty perceiving whether a change in total spending, M x V, is a result of monetary rather than real causes. Individuals perceive spending on the things they are selling, and the prices of the things they are buying, so as Friedman argued, there will be long and variable lags between the time when the growth rate of the money supply increases and the price level responds by increasing. In the short run, if the demand for holding money balances is unaffected, which means that V is relatively constant, and if prices do not immediately increase in response to an increase in the growth rate of the money supply, then an increase in M will cause an increase in Q. That is, real output will temporarily rise, until P responds, at which time the effect on output will disappear. The opposite holds true for an unanticipated decrease in money growth. In the short run, before prices respond, that decrease will have a contractionary effect on real output.

Consider Friedman's monetary framework in the context of the coordinating function of the market mechanism. Prices signal the cost of purchasing in the market, and the benefit of selling, and Friedman's framework shows how current prices systematically represent the real costs and benefits of transactions. For example, a decrease in money growth, below what is generally expected, lowers M x V, that is, lowers total spending, but if prices are slow to adjust this is perceived as a reduction in total demand which leads to unemployment. In the short run, money is not neutral, but distorts prices so that people do not accurately perceive real costs, which interferes with the coordinating function of the market. Thus, Friedman's monetary framework shows how instability can have monetary causes.

The one price that has the biggest impact on macroeconomic instability is the interest rate, because it is the price of future purchasing power in terms of present purchasing power. People make intertemporal resource allocation decisions based on the costs and benefits they anticipate in the future. A lower interest rate increases the value of allocating resources into the more distant future; a higher interest rate shifts the balance toward the present. Ludwig von Mises developed this idea early in the twentieth century to show how the expansion of bank credit can lead to business cycles.[10] In an economic upturn banks are optimistic about business prospects and draw down their reserves to make more loans. This expansion of credit drives down interest rates and encourages more borrowing. But eventually, banks will not lower their reserves any further, which pushes interest rates back up. Some projects which would have been feasible at lower interest rates are no longer feasible, and as those projects run into problems the economic expansion stops. Banks become more pessimistic and want to build up their reserves, further pushing up interest rates and leading to a decline as banks reduce their lending. But eventually banks believe they are holding sufficient reserves, the economy bottoms out, and businesses once again invest as the outlook becomes more optimistic. Thus, the cycle starts again. Mises develops this endogenous theory of the business cycle caused by a cycle of credit expansion and contraction. The coordination of economic activity is disrupted by the interest-rate changes that are caused by credit expansion and contraction, not be real factors.

Mises's framework lays the foundation for the Austrian theory of the business cycle, and is the product of fractional reserve banking. When Mises was writing the world was on a gold standard, without government or central bank interference with the money supply, but Mises's framework extrapolates well to situations in which credit expansions and contractions are the result of central bank policies.

Friedrich Hayek further developed Mises's Austrian business cycle theory in the 1930s to explicitly take account of the effect of credit expansion, not only on aggregate investment but also on the type of investment.[11] Lower interest rates raise the profitability of longer-term investment relative to shorter-term investment, so not only is overall investment affected by interest rates but so is the structure of investment. Erroneous price signals cause people to invest in the wrong types of capital, resulting in malinvestment. Macroeconomics tends to look at economic aggregates, and in particular looks at aggregate investment and the aggregate capital stock. Hayek's contribution illustrates that when capital and investment are analyzed as aggregates, the aggregation assumes away the real effects that monetary changes have on the type of investment that takes place. When credit expansion leads to malinvestment, if capital is specificto certain activities it may never find employment.

Specific capital cannot just be shifted from one use to another, and because of this theories of economic instability that do not recognize the heterogeneity of capital will not fully recognize the degree to which monetary effects can disrupt economic coordination and lead to economic instability. When the time dimension of investment is taken into account, investments that, in hindsight, would not have been made, affect the structure of capital, resulting in overinvestment in some types of capital but insufficient investment in others. This is an integral part of Hayek's business cycle model. Because of this effect on durable capital, relative prices are also affected, so prices of goods and services will not be the same as they would have been had investments been made with perfect foresight. As Mailath et al. note, "current investments cannot be coordinated with future transactions. . . . the inability to condition investments on future prices leads to nontrivial unpredictability of prices and inefficient outcomes."[12]

Further, malinvestment is more likely when goods, including capital goods, do not trade at equilibrium prices. Many models assume all trades are at equilibrium prices, but in the actual economy they do not just jump from one market clearing price to another. Market participants must first observe excess supplies or demands, and adjust their prices in response. The market does not give any indication of what the new market clearing price will be, so if the disruption is large there may be many trades at non-market clearing prices as the market price iterates toward (and perhaps in some cases overshoots) the actual market clearing price.

A more recent development of Austrian school macroeconomics places heavy emphasis on the monetary causes of macroeconomic instability. Both Steven Horwitz and Roger Garrison emphasize the problems of coordination of economic activity and the way that inappropriate monetary policy can act as a destabilizing force.[13] While both authors emphasize the microeconomic foundations of macroeconomic instability, and stress the problems of intertemporal coordination of individuals' plans, they focus primarily on monetary causes of instability, following the lead of Hayek.

Fisher's equation and monetary instability

The equation of exchange referenced above is the creation of Irving Fisher, who had a more nuanced view of it than that discussed in the previous section.[14] On the left side of the equation, one can represent total spending as the quantity of money multiplied by the number of times the average dollar is spent each year, or M x V. Fisher represented the equation of exchange as an identity, so that total spending equals total receipts. Total receipts will equal the quantity of transactions on one type of item purchased multiplied by its price plus the quantity of another type of transaction multiplied by its price, and so on, or the sum of all types of transactions multiplied by their prices. So, where Ti represents type of Transaction, i, and Pi represents its price, total spending equals total receipts, or

$$M \times V = (T1 \times P1) + (T2 \times P2) + (T3 \times P3) + \ldots + (Tn \times Pn)$$

for an economy with n different types of transactions. If we can represent all prices with some price level, P, and aggregate all transactions into total transactions, Fisher's equation can be simplified to M x V=P x T. But Fisher did not start his analysis this way, and explicitly recognized that this short form of the equation of exchange would require aggregation from the long form that represented the sum of all transactions as a representation of total receipts.

The modern representation of the equation of exchange as M x V= P x Q, where Q represents the economy's real output, is not a tautology like Fisher'sM x V=P x T, because real output does not include all transactions. Sales of used goods are not included in Q but are included in T. Perhaps more significantly when looking at economic instability, Q does not include financial transactions. To include financial transactions, consider an intermediate step that assumes away purchases of used goods, and every other transaction in the economy except purchases or real output and purchases of financial assets, which can be represented as F. Thus, considering these two types of transactions, the equation of exchange can be simplified to

$$M \times V = (PQ \times Q) + (PF \times F).$$

Now consider the implications for the evaluation of monetary policy.

Often, the ease of tightness of monetary policy is judged by changes in prices, using an index such as the consumer price index (CPI), which would be PQ in the equation above. In the simpler equation, M x V=P x Q, this makes a great deal of sense, assuming that V and Q remain relatively constant over shorter periods of time. An excessive increase in M will then show up as price inflation. But in the more complicated equation, it is possible that any increase in M finds its way into an increase in PF. The price of financial assets rises as a result of excessively loose monetary policy, and the effect does not show up in PQ if the CPI is used as a measure of PQ. So, it appears that monetary policy is not overly easy, but only because the prices that easy monetary policy is pushing up are not included in the price measure that policy makers are using. One might view the "dot-com bubble" and rise in stock-market prices in the late 1990s as an example of this. The argument has been made that monetary policy was not too easy because there was only a modest increase in PQ, but when looking at an only slightly expanded Fisher-type equation, increases in the quantity of money could have been the fuel that allowed stock prices, represented by PF, to rise.

A similar analysis could be undertaken for the housing bubble in the 2000s.

Housing costs are included to a degree in the CPI, but not to the extent that the CPI's housing-cost measure incorporates the price of buying a new house. So, easy monetary policy could provide the liquidity to produce a rise in housing prices that would not show up in the CPI, which is a proxy for P. The simple equation of exchange, like any model, assumes some things away, and in this case the simplification of Q for Tcan eliminate the evidence that easy monetary policy is causing a distortion in relative prices – asset prices relative to the prices of goods and services, for example.

Looking at Fisher's equation of exchange in a framework slightly more complex than the simple M x V=P x Q shows ample possibility for monetary factors to interfere with the coordinating function of the market economy, leading to macroeconomic instability. Monetary phenomena are more likely to induce instability when one recognizes that exchanges may take place at non-equilibrium prices. Without a Walrasian auctioneer who can suspend trading until equilibrium prices are found, in the market trades can take place at disequilibrium prices and may only iterate to equilibrium prices over a prolonged period of time. If people had accurate knowledge of future prices, events like the stock market bubble in the 1990s and the housing price bubble in the 2000s would not have happened because buyers would have been reluctant to buy at prices that they could foresee would be substantially lower in the future. Because foresight is not as good as hindsight, people can make bad investments that, if they had better foresight, they would have avoided. This section and the previous one have illustrated how monetary factors can lead to malinvestment and, more generally, to making purchases and sales that might have been avoided had those engaged in the transactions been able to better forecast the future.

Market prices are not always equilibrium prices, and even when market prices do clear markets, they do not always represent an accurate forecast of future events. When the future is revealed and people adjust their behavior to the reality that is now before them, the market cannot always adjust fast enough to prevent instability and a breakdown of coordination.

Instability in aggregate demand

John Maynard Keynes proposed in his General Theory that macroeconomic instability was the result of fluctuations in aggregate demand and, in particular, in investment demand.[15] Keynes recognized the uncertainty of the future, and noted:

Most, probably, our decisions to do something positive, the full consequences of which will be drawn out over many days to come, can only be taken as a result of animal spirits – of a spontaneous urge to action rather than inaction, and not as the outcome of a weighted average of quantitative benefits multiplied by quantitative probabilities.[16]

Looking into an unknowable future, Keynes viewed investment demand as particularly volatile, and when investors as a group became more pessimistic, investment demand would fall and the effects would ripple through the rest of the economy as a multiplier effect, leading to macroeconomic instability.[17]

The forces of supply and demand may not be sufficient to stabilize the economy at full employment. While Keynes has had many interpreters and interpretations over the decades, one Keynesian line of reasoning about how a lack of coordination can persist is that when people are unemployed because of a macroeconomic disruption, they pull back their spending and do not register any effective demand through the market. The economy could move toward full employment if, in the aggregate, employers recognized that if they hired workers those workers would then earn the incomes that would enable them to purchase the goods the newly-hired workers would be producing.[18]

The element that this theory has in common with others in this chapter is that a change in expectations about the future undermines the coordination of economic plans generated by market forces. People who made plans based on their expectations of the future find that future is different than they expected it to be, in this case because of changes in the expectations of others. And once that coordination is disrupted, there may not be sufficient market forces to restabilize the economy and recoordinate people's plans. The Keynesian framework has been criticized on a number of grounds, a significant one being that it is an aggregate framework that does not show a clear relationship between the individual elements of the economy and the economic aggregates on which the theory is based.[19] However, it is built on the insight that people's expectations of the future are not always consistent and can change. And, when they do, it is possible for the economy to find itself in a situation where market signals are not sufficient to immediately reestablish that coordination of plans.

Exogenous shocks

The unclear connection between the microeconomic elements and economic aggregates in Keynesian economics led to the development of macroeconomic analysis in a general equilibrium framework.[20] Within this framework, macroeconomic instability arises as a result of exogenous shocks to the economy. Shocks can come from a number of sources. They may be a product of the physical environment, such as earthquakes, hurricanes, or droughts. They may be a product of the institutional environment, so that an unexpected change in monetary policy would be a monetary shock, and an oil embargo would be a resource shock.

General equilibrium models of macroeconomic instability might be criticized on several grounds. Would an event like the Great Depression really be best depicted as an economy in equilibrium? Further, to make models tractable they make a number of simplifying assumptions. Often, households in an economy are depicted as a "representative agent," so there is only one household, which means that coordination problems among households cannot arise. Also, as with most macro models, capital is an aggregate so that the type of malinvestment Hayek was concerned about is assumed away.[21] But as Chapter 2 noted, all models make simplifying assumptions, which leave some elements of the real world out of the model to focus on other elements.

These general equilibrium models focus on the impact of unexpected events – shocks to the economy – on macroeconomic stability. Consistent with the framework of this chapter, people do not view the future with perfect foresight, and when the future does not meet their expectations, they must change their plans. This change of plans disrupts the coordinating functions of the economy, leading to instability. Sometimes the resulting lack of coordination overwhelms the entrepreneurial elements of the economy that bring plans into mutual consistency. So, it does appear that the shocks incorporated into these models can be the result of economic stability, and can tax the coordinating functions of a market economy.

An entrepreneurial theory of the business cycle

Joseph Schumpeter noted the creative destruction that characterizes a market economy. Entrepreneurs are always introducing innovations into the economy, and these innovations disrupt the existing order as they create a new one. The economy never finds itself in equilibrium as depicted by the neoclassical framework, but rather is continually evolving as the disruptions of some entrepreneurs open up opportunities for others. In his Theory of Economic Development Schumpeter depicted this process of creative destruction as the engine underlying economic development, and Schumpeter extended this framework in Business Cycles to depict macroeconomic fluctuations as an integral part of the process of economic development.[22]

The role of entrepreneurship in generating economic progress is well-recognized, and when the disruptions caused by entrepreneurial innovations are greater than the ability of entrepreneurs to reallocate resources in reaction to those disruptions, the economy may temporarily find itself with unemployed resources, including unemployed labor, during the adjustment process. Entrepreneurial innovations are often unforeseen, in which case when they appear in the economy it may take time for entrepreneurs – typically, not the same entrepreneurs who were behind the initial disruption – to re-employ resources that have been dislocated by innovation. Thus, the same forces that drive economic progress can also interfere with the coordination of economic resources, resulting in business cycles. Business cycles are an integral part of the development process, following this Schumpeterian entrepreneurial theory of the business cycle.

Entrepreneurial innovation does not necessarily lead to business cycles. For one thing, entrepreneurs in an economy are often able to anticipate innovation. One example is "Moore'slaw, " which is the prediction by Gordon Moore that the number of transistors that can be economically placed in an integrated circuit will double every two years.[23] Thus, entrepreneurs building computers and other electronic devices, designing software, or undertaking other activities relying on integrated circuits anticipate this innovation when they design their own products. Writers of a computer program that would be ready for sale on the market in two years, for example, would anticipate that the computing power available to their program in a new computer would be approximately double what it is when their programming began. The software would likely not be competitive in the market if it did not anticipate this innovation. Another factor is that entrepreneurs are flexible and plan for various contingencies, so even if an unexpected innovation shows up on the market, entrepreneurs are likely to be in a position to adjust to it. Those who are most adaptable in this regard will profit, and find more resources under their control, than those who are less adaptable.

In many circumstances, Kirznerian entrepreneurship can reallocate resources in an economy as rapidly as Schumpeterian entrepreneurship destabilizes the economy. Sometimes, however, the disruption will be too great, and an economic downturn will result from the same forces that generate long-run economic progress.[24] Business cycles caused by the creative destruction of entrepreneurship are a part of the wealth-enhancing nature of economic progress. Still, the adjustment process can be prolonged by government interference, which can take many forms. There are often regulatory barriers that interfere with a reallocation of resources, which range from zoning, building permits, and other land-use regulations, to product specifications, licensing requirements, and government approval processes that take time to accomplish. There may be arguments in support of various government measures that impede resource reallocation, but such impediments are typically created at one point in time in response to a particular issue, and in a dynamic economy those impediments become increasingly irrelevant over time. Thus, as time passes, regardless of their initial merits, the costs associated with them increase while the benefits decrease.

The essential point is that some business fluctuations that result from entrepreneurial innovation are an integral part of the economic process that generates economic progress. Entrepreneurs are often able to reallocate resources rapidly enough to smooth fluctuations from this source, but not always, and sometimes government intervention is a factor that slows the adjustment process.

Conclusion

One of the things a market economy accomplishes is the coordination of people's economic activity so that people can realize their plans. Specialization and gains from trade can only take place when people are confident that if they do specialize and produce output for the consumption of others, that there will be others who are willing to specialize to produce for them, and that market transactions will enable the coordination of economic activity for the benefit of everyone. Without some confidence that this will be the case, people will be forced to produce for their own consumption, and will be unwilling to rely on the market activity of others to provide for their needs and wants. In advanced economies, people are so confident in the market economy that they take its coordinating functions for granted, and do not give a second thought to the fact that their very survival depends on the economic activities of many thousands of people they will never even meet.

Many things can disrupt this coordination, and the most complex part of the market's coordination of people's activities involves intertemporal coordination. Few people consider that when they eat their steak dinner, the rancher had to raise the cattle to provide the beef more than a year before the final consumer consumes the steak. Sometimes the plans of some people, like those who raise cattle, may not be consistent with the plans of others, such as those who will want a steak dinner in a year or two. The adjustment of prices to the forces of supply and demand helps to match the desires of buyers and sellers by clearing markets in the short run, but price declines for sellers, and increases for buyers, will be disappointments that will require adjustments elsewhere. People realize the future is uncertain, so their plans are flexible, with contingencies built in because they know the future will not evolve exactly as they expect at the present time. Contingency planning and market adjustments help to keep people's economic activities coordinated even when they have not foreseen the future perfectly, but sometimes the required adjustments are greater than the contingencies people have planned for. Sometimes people must make long-term commitments – when building a factory, for example – that are difficult to change once the commitment is made. When entrepreneurs are unable to adjust to new conditions rapidly enough to keep resources, including labor, employed, coordination can break down and macroeconomic instability is the result.

Instability can be caused by many factors. Keynesian economists have emphasized the volatility of investment demand and changes in expectations. Austrian economists have emphasized monetary changes that can affect interest rates and other prices, leading to malinvestment and abreakdown in the coordination of people's plans. Schumpeter noted that entrepreneurial innovations are often unexpected and disruptive. New classical economists refer to the causes of instability as shocks – unexpected changes in economic conditions. None of these views are wrong, and disruptions in the coordination of people's plans can come from many sources. Indeed, it is remarkable that market institutions can facilitate this intertemporal coordination at all when one considers the huge number of people involved. How could it be possible, for example, for an American saving for a child's education to provide funding to help an Indian entrepreneur finance a business on the other side of the world? The American can easily do this by buying shares in an Indian mutual fund. Economic and financial institutions lay the foundation for worldwide economic cooperation and coordination of literally billions of people's plans in the global economy.

Much goes right as a result of the market's ability to coordinate people's plans but, when disrupted, coordination can break down and resources will be temporarily unemployed. There is no single source of the disruptions, but the single resolution of them is the entrepreneurial insight that recognizes that resources could be better allocated. Entrepreneurs seeing profit opportunities help in coordinating people's economic plans, and many of the continual disruptions that cause people to change their plans are the result of entrepreneurial innovation that thwarts old plans even as it opens up new opportunities. When coordination is upset by entrepreneurial innovation, the fluctuations in economic activity are the short-run result of the same process that generates economic progress in the long run.

Even though they are intended to be stabilizing, monetary and fiscal policies can be disruptive, as the Austrian business cycle shows. Other government policies, ranging from regulations, subsidies, and unemployment compensation, can also interfere with the coordinating functions of the market, slowing the market's adjustment to disruptions and new conditions. Policies that enable coordination by allowing market forces to work best lay the foundation not only for long-run economic progress, but also for short-run stabilization in response to instability. The economy is always evolving in ways that nobody – including policy makers – can foresee, because its evolution is the result of the decentralized decisions of millions of individuals who are all making their own economic decisions in response to conditions that nobody can foresee perfectly.

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The coordination of economic activity can take place because institutions create a setting in which exchange can take place, and because people are able to obtain information about the economic activity of others, largely as a result of the same institutions that facilitate exchange. In the most basic exchanges, where one good is simultaneously bartered for another, little in the way of economic institutions is required to facilitate the exchange. Complicate the exchange only a little by having a good being purchased with money and a complex set of monetary institutions come into play. If the money is a full-bodied money, like a gold coin, monetary institutions can be relatively simple, but the institutional requirements grow tremendously if the transaction takes place with a check or a credit card.

Similarly, if the transaction takes place over time, or over a distance, more complex institutions are required. A buyer in the United States purchasing from a seller in China needs substantial institutional support for the exchange, to assure both parties that when they agree to a transaction the expectations of both parties will be met. If a developer wants to build a shopping center, or an individual wants to build a house, institutions must be in place to assure a mortgage lender that borrowed money will be repaid, that the owner has clear title to the land on which the construction will take place, that the borrowed money actually will go toward construction, and that the contractor actually will build the structure as payment is made from the lender. This straightforward example illustrates the complex institutions that must be in place for anything beyond the simplest exchange to take place.

These same market institutions provide a wealth of information to market participants, some of it in the form of information participants supply to each other in the form of product characteristics, availability, terms under which delivery can be made, and so forth. Some information is generated by the market, such as market prices and interest rates. Participants also get information by observing the products and transactions of other market participants. For example, when Apple introduced their iPhone in 2007, other firms could observe the characteristics of the product and the price at which it was selling, and many decided that it would be profitable to design and sell a competing product with similar characteristics.

This is a good example of market-generated information because when the iPhone was introduced many observers thought it would have limited success because it substituted a touchscreen interface for a physical keypad. Market-generated information showed that opinion to be incorrect, which led to a large number of competitors entering the market with touchscreen "smartphones." This information could only be generated by the market. Prior to its introduction, the iPhone's designers could conjecture on the demand for the product and the price they could charge, but nobody could know what the demand would be for a product that had never been marketed, and many expert commentators misjudged the demand for the product because they thought buyers would prefer a phone with a physical keypad.

Small differences in a product can make a big difference in its marketability. Apple followed its iPhone with the iPad, introduced in 2010, which was another big success that spawned many imitators. Hewlett-Packard introduced their tablet computer with their webOS operating system in 2011. Hewlett-Packard did not develop its webOS from scratch, but built it from the Palm operating system after buying the Palm company for $1.2 billion in 2009. The entrepreneurs at Hewlett-Packard saw a product they liked in the Palm operating system, and paid handsomely for it, believing that it would lay the foundation for a product that could compete with the iPad. Despite observing Apple's successful product, and so having the ability to imitate its features plus adding their own, Hewlett-Packard's tablet computer was withdrawn from the market within a year of its introduction.

Any discussion of entrepreneurship tends to focus on entrepreneurial successes, but for every Steve Jobs or Henry Ford there are many more entrepreneurs who thought they had a profitable idea but turned out to be wrong. As noted before, the Apple Newton handheld computer was a flop, even though the Palm Pilot succeeded in that same niche a few years later, and Henry Ford helped General Motors when he told customers they could have any color car they wanted, as long as it was black, before seeing General Motors gain market share by offering a choice of colors in their automobiles. Information provided by the market is not sufficient to determine whether an innovation really will be profitable, because the market provides no direct information about the viability of yet-to-be-introduced products.

Entrepreneurs need more than just market information. They need knowledge about the markets they trade in and the products they buy and sell. Knowledge is more than just a collection of information. That information must be placed in a context such that the holder of the information is able to understand what it means.[1] Simply knowing the price of a good adds little to knowledge. Is the price high or low by historical standards? Are there factors that might have temporarily increased or decreased the price? Can a potential buyer anticipate that the good will be more or less expensive in the future? To be able to answer these questions the individual needs to know more than just the market price, and an accurate answer will require historical price and other information, along with asufficient context of background knowledge. Knowledge adds to a collection of information a context in which the information can be understood and used.

Tacit knowledge and agglomeration economies

In a market economy, firms must continually innovate to keep up with the economic progress that has characterized capitalism since its origin in the Industrial Revolution. If firms are already making best use of the knowledge they have, to produce the best products at the lowest cost that they can, any improvement means coming up with new ideas. If entrepreneurship is the noticing of a previously unnoticed profit opportunity, how does one spot something that has not been noticed by anyone previously? Some people are more observant and more insightful than others, but in almost all real-world cases the person who spots a profit opportunity will be someone who has specialized knowledge related to that profit opportunity. An automotive engineer is unlikely to spot an opportunity to introduce a profitable new drug into the market, and a pharmacist is unlikely to spot an opportunity to introduce a profitable refinement to an internal combustion engine.

Even what may appear in hindsight as obvious profit opportunities may not be easy to spot beforehand. Revisiting some examples that were considered earlier, Ray Kroc was a seller of restaurant equipment who was surprised by a large order for milkshake mixers from a small restaurant in San Bernadino, California, named McDonald's. Because of his experience in the restaurant industry, Kroc saw a profit opportunity that even the owners of the restaurant did not, bought the restaurant in 1954, and turned McDonald's into a world-wide chain. The restaurant was in plain sight of its customers, its other suppliers, and of course, its founders, but it was Kroc who was the entrepreneur who saw the extent of the profit opportunity it represented. Pierre Desrochers describes several other examples of such entrepreneurial insight,[2] where a key element in the entrepreneurial action was some tacit knowledge produced by personal observation. One example Desrochers relates was a meeting Steve Wozniak, co-founder of Apple Computer, attended in which a color computer monitor was being demonstrated. Wozniak stated that if he had not seen the demonstration, the Apple II, Apple's first commercial product, would likely have had a monochrome monitor.

What the Kroc and Wozniak cases have in common is that both entrepreneurs had a specialized body of knowledge that enabled them to recognize an entrepreneurial opportunity that could be observed – but not recognized – by others. In both cases, it was first-hand observation that led the entrepreneurs to see the profit potential in the opportunity.

Much of the knowledge people have is tacit knowledge, that as Hayek explained, is specific knowledge of time and place that some people have, but that is difficult to articulate and pass along to others. Friedrich Hayek notes that everyone has "knowledge of the particular circumstances of time and place ... which use can be made only if the decisions depending on it are left to him or are made with his active cooperation."[3] This tacit knowledge is knowledge that cannot be articulated, verbally or in writing, because otherwise, to use Hayek's description, it would not depend on the active cooperation of the person with the knowledge. If the knowledge were not tacit, the person with the knowledge could pass it on to others who could then act in place of the person with the knowledge. This tacit knowledge is not textbook knowledge, or scientific knowledge, but rather knowledge that comes through experience but that is difficult to explain to others.

Corporations recognize the value of this knowledge and try to capitalize on it through mentoring. Individuals learn first-hand by observing those with experience that makes them more productive. Business schools have also recognized the value of knowledge generated through observation and embodied it in case studies, with the idea that their students can learn by studying actual events. However, despite heavy reliance on case studies in MBA programs the evidence is that they cannot teach students all there is to learn from observation, and as such businesses still rely on mentoring young employees with MBA degrees.

Hayek's description of tacit knowledge indicates how it can be acquired. Because it is knowledge of the particular circumstances of time and place, someone else who shares that time and place can absorb the knowledge by observing it in use. This explains the mechanism behind agglomeration economies. People in close proximity can absorb the knowledge of others by being in the same place at the same time, which is especially important when that knowledge is tacit – that is, not easily communicated by other means. What, exactly, constitutes the same time and place will vary depending on the knowledge. In some cases that knowledge might be contained within a firm's boundaries, and not easily acquired by those outside the firm, as discussed in Chapter 6. In other cases, like Silicon Valley for computers, New York for the financial industry, and Detroit for automobile production, the place is a larger geographic area. Firms in close proximity to each other can absorb some tacit knowledge held by others in their line of work, resulting in agglomeration economies.

People learn through experience, but like riding a bicycle or hitting a baseball, one cannot pass that knowledge to someone else without their also having the experience. Based on past experiences, businesspeople have ideas about how to best respond to the unique present situation, but those ideas come from a stock of past experiences that cannot be distilled into an instruction manual. Similarly, scientists and engineers build a stock of experience, enabling them to spot innovations that others cannot see, and often even the reasoning behind the entrepreneurial insight cannot be clearly articulated. Being nearby, and able to see that knowledge being used, facilitates its transmission.

Small differences can make the difference between profits and losses, and nobody can know ahead of time whether an innovation will be profitable. The brief history of handheld computing devices is a case in point. Apple was an early innovator, having brought their hand-held Newton, with handwriting recognition, to market in 1993, but it failed to be profitable. In 1996 US Robotics introduced the Palm-Pilot handheld computer, also with handwriting recognition, that was hugely successful and very profitable. In 2000 Palm was spun off into a separate company, which was bought by Hewlett-Packard in 2010, because Palm's products were eclipsed by other innovators, including Apple with its iPod, iPad, and iPhone. Products by Google and Microsoft, among others, entered this competitive market in 2011. The products are similar, but the differences among them will determine which will be profitable.

Nobody can know ahead of time which products, and which features, will ultimately succeed on the market, because it depends on consumer demand for them, and this demand cannot be observed before the products are available for purchase. Steve Jobs, when asked whether Apple relied on consumer opinion to determine what products they would bring to market, said "A lot of times, people don't know what they want until you show them."[4] Entrepreneurs must use their stock of specialized knowledge to make judgments about future conditions that have never been observed. Tacit knowledge plays a key role in keeping up with an ever-changing market. Were this not the case, experience would be worth no premium on the market, and boards of directors could hire recent college graduates with management degrees to run their firms, rather than pay substantial premiums to hire experienced people. This is true not only of the entrepreneurs who run the firms, but of every employee all the way down the line. One can marvel at the entrepreneurial insight of Ray Kroc, but anyone visiting a McDonald's can also see that, in general, experienced workers behind the counter are more productive, faster, and more reliable than new hires. One can show a new hire what to do, but through experience they become more proficient because they acquire tacit knowledge that can only be gained by experience.

Adam Smith made this same observation about entrepreneurs, noting:

A great part of the machines made use of in those manufactures in which labour is most subdivided, were originally the inventions of common workmen, who, being each of them employed in some very simple operation, naturally turned their thoughts towards finding out easier and readier methods of performing it.[5]

Echoing Smith's observation, Richard Florida says:

Japanese manufacturing firms leaped ahead of ours with their "creative factory" methods, which tap the intelligence of every worker on the factory floor to make continuous small improvements in the production process. U.S. firms – stuck in the Fordist system, whereby the engineers and top managers do all the thinking while the masses are paid to execute – nearly had their doors blown off. Many of our firms have now caught on to the new way.[6]

Tacit knowledge exists in every employee, and the challenge for the entrepreneur is to continue acquiring sufficient new knowledge to stay ahead of the continual economic progress of the firm's competition.

Tacit knowledge exists within the firm, but it also exists in other firms within the market. Sometimes these other firms are direct competitors, sometimes they are suppliers, and sometimes they are customers. By locating in an area where other similar economic activities are taking place, entrepreneurs are more likely to be in a position to observe and take advantage of the tacit knowledge of others. One reason is that there will be a pool of knowledgeable employees in the local job market. The job market in Silicon Valley has many people who have experience, and tacit knowledge, in the computer industry, and the job market in Detroit (still) has people who are experienced auto workers, designers, and engineers. Entrepreneurs can observe first-hand what their competitors, suppliers, and customers are doing in the local market, they can talk informally with others in the same industry, they are in a good position to hire people who have experience and tacit knowledge in the industry, and for all these reasonsfirms can be more successful when they locate in close proximity to others who are engaging in the same type of activity. That attraction creates agglomeration economies.

Static and dynamic agglomeration economies

One aspect of agglomeration economies is economies of scale. When automobile production centered in Detroit in the early part of the twentieth century this allowed more efficient production methods, which lowered the per-unit cost of output. Allyn Young, building on Adam Smith's idea of the division of labor, argued that increasing returns to scale is the foundation of economic progress.[7] Nicholas Kaldor, building on Young's insights, noted that static neoclassical economic models did not do a good job of depicting the economic progress that results from increasing returns to scale in production.[8] Adam Smith laid the foundation for this line of reasoning when he noted the increased productivity that comes with an increased division of labor. Smith's example of the pin factory, where individuals specializing in one small part of a larger manufacturing operation increase productivity by, perhaps, hundreds of times, shows the benefits of agglomeration economies.[9] The division of labor is limited by the extent of the market, Smith argued, so enlarging the extent of the market allows for a greater division of labor, which increases productivity and generates prosperity. By concentrating automobile production inDetroit ratherthan having automobiles locally built, the extent of the market is increased from one locality to an entire nation, and in some cases an entire world. The resulting agglomeration economies increase productivity and produce prosperity.

Economies of scale are important in a static sense, because they make people more productive, but another aspect of agglomeration economies is that they generate increased economic progress. This is more than just a static increase in productivity. This dynamic increase in economic welfare perpetuates itself through time. This is the creative destruction described by Schumpeter. Entrepreneurs try to gain a competitive advantage by introducing new and improved products, and by developing better and lower-cost methods of production. Because some entrepreneurs are innovating, all firms must continually update their production methods and product offerings, or they will fall behind those that do. Thus, in contrast to static equilibrium models, competitive markets are ever-changing, and are characterized by product differentiation and innovation that are the primary drivers of economic progress.[10] Firms have to continue to introduce innovations into their production methods and into the products they sell, or they will lose market share to innovators, until eventually the market for their output vanishes.

The neoclassical model of the competitive firm assumes that firms produce a homogeneous output through a given production technology, using given inputs. In the real world of economic competition, entrepreneurs realize that none of these are given. Competitive firms in this framework can do no better in the long run than just earn a normal rate of profit, but as Chapter 5 noted, firms with monopoly power can earn above-normal profit. So, the profit-maximizing strategy of a competitivefirm is to look for ways to gain some monopoly power to earn a higher profit. In fact, this is how competitive firms in the real world – as opposed to the textbook world – actually behave. They look for ways to differentiate their products, they look for improved production methods, and they look for different mixes of inputs to lower costs and enhance productivity.

To remain competitive, firms must continually innovate, improving their production processes and improving their products, to keep up with a dynamic market in which their entrepreneurial competitors are continually improving. However, firms are already doing the best they can with the knowledge they have on-hand. If entrepreneurs knew how to make a better product, they would already be doing it. If they knew how to produce more efficiently, at lower cost, they would already be doing it. How do they come up with the innovations that can help them keep up with an ever-improving marketplace? Partly, they can do this in-house, through their own formal or informal research and development efforts. Additional information can come from others in the same market who are innovating. They can watch the activities of their rivals, and they can get ideas from suppliers and customers. If a supplier of inputs has a better idea for the production of an input, that may lead to an improvement in the firm's production process. Some of this information will be plainly visible in the marketplace, but some will be tacit knowledge, which is best gained by being in close proximity to those with the knowledge. The sharing of this tacit knowledge is one factor that leads to agglomeration economies.

The creation of agglomeration economies

Because of the obvious advantages of agglomeration economies, policy makers try to foster them. They have offered tax incentives, development grants, low-cost sites, industrial parks, and many other incentives to try to lure economic activity to their areas. What policies actually work to lay a foundation for agglomeration economies?

A recent literature initiated by Richard Florida emphasizes the role of creative individuals in laying the foundation for economic prosperity – both their own prosperity and the prosperity of the areas in which they live.[11] Florida defines the creative class in economic terms, identifying them as people who earn their incomes through creative work. Florida observes that "the single most overlooked – and single most important – element of my theory is that every human being is creative. ... Tapping and stoking the creative furnace inside every human being is the great challenge of our time."[12]

Florida's idea complements Hayek's insight about individuals having specific knowledge of time and place. All individuals have that knowledge, following Hayek, and Florida's "challenge" is getting them to make best use of it. Hayek championed the market allocation of resources as a way of coordinating that knowledge of time and place, whereas Florida emphasizes an environment that attracts creativity, laying the foundation for agglomeration economies.

Florida emphasizes amenities that attract creative individuals, such as restaurants, museums, an active nightlife, and other cultural attractions. But amenities like this will be produced in places where there is a demand for them, by creative entrepreneurs who see them as profit opportunities. Despite the appeal of such amenities, policies will meet with more success when they are oriented toward attracting the creative population who will demand those amenities, rather than pursuing the idea that subsidizing such amenities will attract the creative class and lay the foundation for agglomeration economies. Florida's emphasis on creativity hits the mark, however, because entrepreneurship is the fundamental creative economic activity, and because the knowledge that lays the foundation for entrepreneurial activity is a product of creative work.

If Florida is correct that everyone is creative, then fostering agglomeration economies is a matter of enticing individuals to use their creativity in economically productive ways. While everyone has creative impulses, entrepreneurial innovation is necessarily characterized by uncertainty. Because entrepreneurial activity means doing something differently from the way it has been done before, there is no way to know whether an attempt at entrepreneurship will succeed. Past experience can be a guide, if the entrepreneur has observed similar innovations in the past, but it cannot provide direct evidence on something that has never occurred before.

When Henry Ford decided to manufacture automobiles on an assembly line, he could observe other goods being manufactured on assembly lines, but could only conjecture on how well assembly line production would work for automobiles. Similarly, Ford's idea that he could produce automobiles at lower prices and attract a larger number of buyers was also a conjecture. He could not observe the actual elasticity of demand for automobiles at the price he planned to sell them, because they never had been sold that cheaply, so any conjecture was an extrapolation beyond actual experience and real-world observation.

Boeing's 787 Dreamliner composite-body aircraft – an innovative design change from the aluminum-body airliners that preceded it – was brought to market in 2011, more than two years behind schedule. Smaller composite-body aircraft had been produced before, and Boeing has extensive experience in aircraft design and manufacture, but it is apparent that they could not foresee the manufacturing problems they would face in this project. Such uncertainty is necessarily a feature of entrepreneurship, because entrepreneurial activity means doing something that has not previously been done. Similarly, while they did take advance orders for the aircraft, it was also not possible when they announced it to know what the demand would be for an aircraft that had never been sold, although the firm's position as a leading aircraft manufacturer did put them in a good position to have more information about both cost and production issues than other firms, because of the tacit knowledge within the company's boundaries.

The primary issue in creating an environment for agglomeration economies is in creating an environment where people are willing to take the risks involved in being entrepreneurial. People will be more entrepreneurial when they are able to reap the benefits of their creative activity. Cultivating the creative class means creating an environment in which creative people keep the profits from their entrepreneurial actions, and also one in which they bear the losses from entrepreneurial attempts that are unsuccessful.

Subsidizing a firm to build a facility in a particular area, through tax breaks or providing a site at an attractive price, provides an incentive to build the facility, but will not necessarily produce a facility that is worth more than it costs, nor will it necessarily bring with it the entrepreneurship and innovation that lays the foundation for agglomeration economies. Governmental economic development organizations rarely make the distinction between facilities that provide jobs to an area and facilities that engage in entrepreneurial activity. Government-provided economic incentives may be sufficient to bring a manufacturing facility to an area, for example, that provides manufacturing jobs, while the innovative and entrepreneurial functions of the company remain elsewhere. The facility will provide employment to some, but will not bring in the kind of creative work that produces knowledge spillovers that foster agglomeration economies.

The division here may not be this clear-cut. If for example, factory workers are encouraged to use their on-the-job knowledge to suggest improvements in the processes at the facility, they could play a role in fostering innovation. Furthermore, having skilled workers of any type in an area means other firms looking for the same skill set can move into the area and have a pool of potential employees. The management in any facility will have to be engaged in some independent and creative decision-making, no matter how centralized the firm's overall management structure. The general point, however, is that subsidizing unproductive activities uses up more in resources than it generates in benefits, and does not produce prosperity.

If the entrepreneur takes all the risks in the entrepreneurial activity, then the individual who has the information behind the innovation has the incentive to see that the benefits are worth the costs. Needless to say, because of uncertainty entrepreneurs are often wrong, but that does not detract from the necessity of providing the appropriate incentives. Entrepreneurial individuals will tend to move to locations where they can reap the rewards of the entrepreneurial risks they take, and all individuals in an area have an incentive to be more entrepreneurial when the institutions in that area place few barriers in the way of acting entrepreneurially, and enable them to keep the profits from successful ventures they undertake.

Consider examples of agglomeration economies from throughout history, from the beginning of the Industrial Revolution in Britain, to the rise and fall of the automobile industry in Detroit, to the rise of the electronics industry in California's Silicon Valley. Those agglomeration economies were not created by government policies underwriting them, but rather by government remaining out of the way and not impeding the entrepreneurs who were generating economic progress. One can point to some apparent successes of government-directed entrepreneurial efforts, such as the industrial policies pursued by Japan and South Korea in the second half of the twentieth century. To analyze them in detail would digress from this chapter's subject, but note that the Japanese industrial policy that appeared so successful from the 1950s to the 1980s led to what has been referred to as Japan's lost decade of the 1990s. One might question whether South Korea, which began its industrialization in the 1970s, is following Japan's path toward stagnation and slower growth.[13] Examples like this notwithstanding, the often-celebrated agglomeration economies of Silicon Valley, Austin, and Boston, occurred independently of government intervention, and without their governments anticipating those outcomes.

One can point to the concentration of universities in Boston as a causal factor, and government in North Carolina has tried to replicate that with some success in their research triangle. Silicon Valley has a similar concentration of universities, and the University of Texas in Austin has undoubtedly contributed to Austin's success. These observations strongly suggest that there are advantages in concentrations of human capital, which makes sense when one considers agglomeration economies as taking advantage of knowledge spillovers. However, many centers of higher education have not produced similar results.

Institutions and agglomeration economies

One can call into question policy formulas designed to produce agglomeration economies on the grounds that, first, many agglomeration economies established themselves with only the passive policy of laissez faire, and second, that many government policy attempts to attract agglomeration economies have failed. The best hope for establishing agglomeration economies is to lay the foundation through limited government, protection of property rights, rule of law, and low barriers to trade.[14]

From a policy perspective, agglomeration economies are important because they imply that even when public policies are homogeneous across jurisdictions, economic activity, innovation, and entrepreneurship will not be uniformly distributed in space. Agglomeration economies imply that creative people benefit from clustering together, so two cities might be identical from the public policy standpoint and yet one is able to attract entrepreneurs and innovative activity while the other sees an outmigration of creative individuals. Why? Because creative people want to benefit from agglomeration economies, so cities with clusters of entrepreneurial activity start with an advantage in attracting more entrepreneurs. One cannot hope to replicate Boston, or Silicon Valley, by imitating the public policies of those areas. Agglomeration economies must begin with a nucleus of creative activity in place for those agglomeration economies to be cultivated. But, everybody has the potential to be entrepreneurial, so the cultivation of entrepreneurship starts by reinforcing the entrepreneurial impulses of an area's existing population. Agglomeration economies then build by making it easy and attractive for other entrepreneurial individuals to migrate in.

Agglomeration economies are a spontaneous order

When considering policies that foster agglomeration economies, it is important to recognize that agglomeration economies are the product of a spontaneous order. The benefits of agglomeration economies exist precisely because they cannot be foreseen. People cluster together because of the advantages of sharing tacit knowledge, and that sharing leads to the innovation that produces economic progress. Because that knowledge is decentralized, as Friedrich Hayek emphasized, no one individual can foresee the direction in which the aggregated knowledge will lead. Entrepreneurs must always be alert to new ideas, to remain on the forefront of an ever-evolving economy. As Joseph Schumpeter said, "The essential point to grasp is that in dealing with capitalism we are dealing with an evolutionary process. ... Capitalism, then, is by nature a form or method of economic change and not only never is but never can be stationary."[15] In a market economy, success can never mean finding a successful formula and sticking with it. Entrepreneurs always need to be picking up new ideas and moving forward, and being located in the midst of other innovators gives them the best chance of remaining profitable in a continually evolving economy.

Because agglomeration economies produce a spontaneous order, fostering them means creating conditions that make it inexpensive and easy for people to act on their creative impulses. Creativity, by its very definition, provides unexpected results. Economic planning, in the sense of planning for a particular outcome, is an impediment to creative activity. Yet despite this recognition, government planning with the hope of fostering economic prosperity is common at all levels of government. At the local level, many cities offer tax breaks, industrial development grants, and other enticements to attract specific businesses to their area. At the national level, economic planning goes by the name of industrial policy. In these examples of planning, government picked and supported particular firms, giving them economic assistance with the goal of making them competitive in world markets.

The spontaneous order of the market often produces unexpected outcomes. Who would have foreseen, in the 1960s and 1970s when IBM dominated the computer market, that Microsoft, founded in 1975, would be the dominant firm in that market 20 years after its founding? If US industrial policy had supported IBM in the 1960s and 1970s the way Japan supported Sony, and Korea supported Samsung, the advantages given to the dominant firm might have snuffed out emerging competitors like Microsoft and Apple. Similarly, because of Japanese and Korean industrial policy, the world economy may never see entrepreneurial firms like Microsoft and Apple emerge in Japan and Korea. The dominant firms have too many government-supported advantages over entrepreneurial start-ups. The firms that never came into existence remain unseen.[16] People can see the successes of IBM, Sony, and Samsung. They will never miss – because they cannot see – the entrepreneurial start-ups that would have been created in Japan and Korea had they not been inhibited by those countries' industrial policies. Because agglomeration economies produce unforeseen results due to their combining of the tacit knowledge of a large number of individuals, planning stifles entrepreneurship and creativity. The spontaneous order of the market cultivates creativity.

The challenge any economy faces, whether we are talking about a city or a nation or the global economy, is coordinating the economic activities of all its participants. Everybody has some specific knowledge that others not only do not know, but cannot know without sharing their specific time and place. People have tacit knowledge which allows them to know how to accomplish particular tasks, even though they may not be able to articulate that knowledge to pass it on to others. A market economy makes best use of that knowledge and coordinates the activities of everyone, for the benefit of everyone.

Cities are concentrations of knowledge. Because much knowledge is tacit, being in close proximity with those who have complementary tacit knowledge can enable people to observe and learn from them. For some knowledge, being in the same city means being in the same place, as Hayek uses the term. The concentration of tacit knowledge that gives rise to agglomeration economies better facilitates the best use of that knowledge. Individuals are more likely to find their best employment match when that match is in the same city because, first, it is easier to change jobs, and second, it is more likely that information about the job match will be available because of the proximity. This creates agglomeration economies, making those locations more attractive for others with complementary tacit knowledge. Agglomeration economies give entrepreneurial cities an advantage in maintaining their creative edge. This makes it difficult for less entrepreneurial cities to join their ranks, but also creates localized rents that can be exploited by governments. These locational rents give rise to predatory government, which over time erodes the advantages that support agglomeration economies.

People tend to take a short-sighted view of the factors that produce prosperity. Agglomeration economies give creative cities like New York and San Francisco an edge, but not necessarily a permanent one. Fifty years ago Detroit was prosperous as a result of the agglomeration economies produced by the automobile industry, and it would have taken quite a visionary to foresee not only the rise of agglomeration economies in other cities as Detroit fell, but also to foresee the computer industry overtaking the auto industry as the anchor of many new agglomeration economies. The best way to be a part of the next wave of agglomeration economies is to provide an environment conducive to entrepreneurial activity, and reap the rewards that will be the result of an unplanned order. Cities can create an environment that will cultivate entrepreneurship, but they cannot plan out an outcome that is of necessity the result of bottom-up individual activity.

Agglomeration economies are difficult to relocate

Agglomeration economies create concentrations of wealth that are geographically bound. One company could choose to leave Silicon Valley, but it could not take the advantages of the Silicon Valley agglomeration economies with it, and the entire Silicon Valley could not migrate to another more entrepreneur-friendly location. Partly, this is due to the physical infrastructure that is tied to the agglomeration economies. Indeed, some physical attributes are a part of the geography rather than the built environment. One cannot move a harbor or a navigable river, for example. Once constructed, moving a factory is also difficult. The owners cannot simply let the factory wear out, because physical capital must be continually upgraded to remain competitive. If the firm is expanding, it could choose to locate a new facility in a different location, but doing so might give up some advantages of the agglomeration economies in the old location. There are, however, advantages to having facilities in different locations – to reduce shipping costs to geographically disbursed customers, for example – which could lay the foundation for a move from an overly taxed and regulated jurisdiction to a less restrictive environment.

Some businesses will be more mobile than others. A firm involved in heavy manufacturing will have firm-specific and difficult-to-move capital, whereas a service-oriented firm may be able to rent office space and move at low cost when its lease is up. A move to a different geographic area will be complicated by the fact that all the firm's employees may not want to move, for family reasons or other reasons of geographic preference. Furthermore, moving away from the agglomeration economy severs those benefits, so any institutional costs a government imposes on firms in one location – the taxes, regulation, and other institutional impediments – must be weighed against the benefits of remaining to take advantage of the agglomeration economies. Residents of New York are heavily taxed, for example, but those in the financial industry stay, even though they could easily move when their leases are up, because of the advantages of being in close proximity to the rest of the industry.

Reductions in transportation and communication costs, and the growth in the production of services relative to manufacturing, will make it easier for agglomeration economies to relocate. Of course, an entire local economy is not going to pick up and move. In addition to the reasons given above, relocation is a bottom-up individual decision, not a top-down decision. But as new agglomeration economies develop, firms will find it feasible to migrate some of their operations from old ones. Financial firms operating on a global scale can locate themselves in New York or London, but also can shift some of their operations from one city to another in response to institutional changes. Moving is not necessarily an all-or-nothing decision. Thus, Charlotte is becoming a financial center. As China develops, Shanghai has the potential to compete with Hong Kong as a financial center, so agglomeration economies can migrate, although slowly.

The fall of agglomeration economies

Agglomeration economies have sufficient geographic roots that they cannot just pick up and move. This geographic immobility gives governments the power to extract some of the economic benefits of the agglomeration economies in the form of taxes and regulatory burdens. People can obtain income in two ways: they can engage in productive activity to generate income for themselves, or they can plunder income and wealth from those who have produced it. Government receives its revenue through taxation: that is, by using its coercive power to require that those who have resources tender some of those resources to the government. This creates the institutional structure for entrepreneurial individuals to use the force of government to obtain transfers for their benefit. As William Baumol noted, if the institutional structure prevents some individuals from using force to plunder others, entrepreneurs have an incentive to engage in productive activity, and to keep the profits from their productivity.[17] If the institutional structure provides mechanisms whereby some people can engage in plunder and transfer wealth from productive individuals to themselves, then entrepreneurial people will engage in destructive entrepreneurship. They will find it more privately profitable to work to obtain political power and favoritism in the legal system, to claim a share of what other people produce, rather than be producers and see their production taken from them.

Capitalist institutions limit the ability of people to engage in destructive entrepreneurship, but over time, if production is tied to a given geographic location, that provides the incentive for those with political power in those locations to turn that power to their advantage by using it to command the resources of productive individuals. Those with political power can use it to enrich themselves directly, by taking a share, and indirectly, by using the proceeds to buy the political support of others. Mancur Olson describes a process whereby, as political systems mature, interest groups gain increasing power, and increasingly are able to use that power to direct the force of government to provide themselves with benefits.[18] Thus, the wealth of income-generating agglomeration economies is transferred away from productive individuals toward those with political power. Agglomeration economies generate locational rents, and those rents can be taxed away.

In autocratic governments the plundering class is limited to the rulers and those who keep them in power, such as the military, the police, and the judges. In democracies the group that needs to be supported to maintain power is larger, because democratic leaders need to maintain a majority to retain their power.[19] The productivity of the agglomeration economy provides the resources which those with political power can target to provide benefits to themselves and those whose support allows them to maintain their power. Thus, the above-normal returns that should accrue to entrepreneurs in the agglomeration economy are taxed and regulated so that eventually all of the advantages of their entrepreneurship can be taxed away.

Because of their ties to a specific geographic location, the entrepreneurs working in agglomeration economies cannot just move away. However, because of the creative destruction that accompanies economic progress, the economic benefits that accrue to a particular set of activities will decline over time. Unless that entrepreneurial energy continues to produce innovations, the productivity of the agglomeration economy will decline until it is overtaken by agglomeration economies in other jurisdictions. Agglomeration economies do not just get up and move. Rather, their decline tends to come from two different sources. First, as noted in the previous section, multiple agglomeration economies often exist in the same industry. If one location loses some of its advantages relative to another, firms in that industry can expand in the more advantageous location, and eventually the center of innovation will move from its previous location to a new one. Second, as industries mature, advances become rarer, and it becomes easier for firms away from the center of activity to match the productivity of those in an overly taxed agglomeration economy.

In the early twentieth-century, when there was rapid innovation in the automobile industry, geographical proximity to other producers made a bigger difference. Meanwhile, as that industry matured, new industries became more innovative. In part, automobile manufacturing moved and became more disbursed, but in part the electronics industry rose as the leader in innovation. Silicon Valley replaced Detroit as the center of innovation not because it stole Detroit's industry, but because it created a new industry that could hardly have been imagined when Detroit was an industrial leader. This process continues to be repeated, as formerly high-tech industries lose that status through the spread of technology. The Industrial Revolution began in Britain largely as a result of the development of the textile industry, which migrated into, and then out of, the United States, and is now in the Eastern Pacific primarily.

As another example, Los Angeles was once a manufacturing center, and home to many automobile manufacturing plants. California saw its last large-scale automobile manufacturing facility close in 2009, and as of 2012 California has no major automobile manufacturing facilities. High taxes, burdensome regulations, and a unionized workforce imposed sufficient institutional costs to cause a substantial industry, with its associated agglomeration economies, to relocate.

Because they are geographically bound, agglomeration economies provide a target for political entrepreneurs, who can use the power of government to transfer income and wealth from productive individuals to themselves and their political supporters. Institutions change, and when the conditions of economic freedom that provide the environment that allows agglomeration economies to be born gives way to higher tax and regulatory burdens, entrepreneurs move to other less burdensome locations. The agglomeration economy cannot pick up and move, but as economic progress marches on the agglomeration economy becomes increasingly less productive and entrepreneurial. Meanwhile, other agglomeration economies arise where institutional conditions are more favorable.

Conclusion

Agglomeration economies are spontaneous orders that are produced by the sharing of the tacit knowledge of creative and entrepreneurial individuals who are in close proximity to one another, sharing the same "time and space," to use Hayek's terminology. Other factors contribute to agglomeration economies, to be sure. Geographic factors, such as natural harbors, and being in close proximity to major markets and to suppliers, can be significant, but these were not the factors that made Manchester a major textile manufacturing center around 1800, or that made Detroit a major automobile manufacturing center in the mid-twentieth century, or that made California's silicon valley the nucleus of the computer industry's development in the second half of the twentieth century. In those cases, agglomeration economies occurred because those locations had institutions that were supportive of markets and capitalism, that imposed low tax and regulatory burdens on economic activity, and that allowed entrepreneurs to reap the benefits of their productive activities. The tacit knowledge produced by innovative individuals was more easily transferred to those in close geographic proximity, giving everyone in that area an advantage over people in other areas. Imitators could not hope to keep up – except in other areas that also had concentrations of those activities that produced their own agglomeration economies.

Agglomeration economies produce advantages to those in a specific geographic area, and those advantages are tied to the area. In the short run, agglomeration economies are not mobile, and that creates an opportunity for those with political power in that area to use that power to transfer income and wealth from productive individuals to themselves and to those whose support those with political power rely on to maintain their power. Over time, the advantages that were going to entrepreneurs as a result of their innovative activity become increasingly captured by the political class, reducing the advantages of innovation and entrepreneurship. As economic progress renders the past innovations of the agglomeration economy increasingly obsolete, the productivity of the agglomeration economy fades as other locations with institutions more favorable to innovation overtake the fading agglomeration economies of the past. Thus goes the rise and fall of agglomeration economies.

Some economies of scale are static, in the sense that large-scale economic activity can have lower unit costs than smaller-scale activity. Some are dynamic, in the sense that they generate entrepreneurial innovation and increase economic progress. These dynamic economies of scale occur because of the production and use of knowledge. The knowledge that drives economic progress is often tacit, and can only be conveyed when people sharing it are in close proximity to each other. One result of agglomeration economies is that some geographic locales exhibit more economic progress, and more prosperity, than others. Institutional differences across locales account for a substantial part of the differences in the creation of agglomeration economies, and as a result, prosperity, in different locations. Good institutions foster agglomeration economies and generate prosperity.

For hundreds of years, before the Industrial Revolution began in Britain, China was more technologically advanced than Europe, but, as Robert Heilbroner notes, China's tradition-based economic institutions were not conducive to economic progress.[20] While China enjoyed the benefits of static agglomeration economies, its institutions prevented dynamic agglomeration economies that are generated by entrepreneurship supported by the production of knowledge. The Industrial Revolution began in Britain, rather than in China or in other parts of Europe, because Britain had economic institutions conducive to the creation of dynamic agglomeration economies.

Economic progress in Britain was concentrated in areas such as London and Manchester, where agglomeration economies allowed manufacturing to advance because of localized knowledge. In the twentieth century, institutional advantages shifted to the United States, which had lower tax and regulatory barriers to entrepreneurship than Europe. In the first part of the century Detroit became the center of automobile manufacturing. It is no coincidence that the "big three" US auto manufacturers of Ford, General Motors, and Chrysler – and lesser manufacturers such as Studebaker and Packard – were located in close proximity to each other. Because agglomeration economies are geographically based, once established they are subject to exploitation, and institutional changes eroded the advantages that Detroit had. Partly, this was due to the tax and regulatory costs imposed on the industry by Michigan's government, and partly institutional changes were imposed by the United Auto Workers union, which had the support of the federal government.[21]

Detroit was never alone as the only area in which agglomeration economies existed in auto production. Advances were made in Britain, Germany, and other locations. As innovation in the automobile industry slowed in Detroit, that created an opening for Tokyo to emerge as a competing location that generated agglomeration economies in automobile manufacture. Meanwhile, new industries emerged as innovation leaders. The Silicon Valley region of California has been at the forefront of the computer revolution for half a century, with competing agglomeration economies in that industry in Austin and Boston. It appears that California's high tax and regulatory costs imposed on businesses there may lead Silicon Valley to a fate similar to Detroit.

The pattern is: favorable institutions lead to innovation and agglomeration economies. Governments impose tax and regulatory costs on innovators to appropriate some of the wealth generated by agglomeration economies. Old agglomeration economies lose their advantages and fall behind as new agglomeration economies take their place. Because of the nature of economic progress, successful agglomeration economies cannot remain successful by just continuing their old ways. Progress means that what is successful today will become increasingly obsolete in the future, so continued success can occur only with continued innovation to stay ahead of other innovators. Institutions need to foster innovation and entrepreneurship for agglomeration economies to endure, and too often those with political power are able to impose costs on agglomeration economies for their short-term benefit. The long-term cost is that those agglomeration economies become increasingly less competitive, and are replaced by agglomeration economies elsewhere.

The success of agglomeration economies often means not just keeping the lead in a particular industry, but rather creating an environment such that as new industries emerge, they are able to thrive. Silicon Valley overtook Detroit not by producing better automobiles, but by laying the foundation for a new industry. In a world characterized by economic progress, maintaining the benefits of an agglomeration economy always means maintaining an environment within which knowledge is continually generated, and entrepreneurship is continually supported to apply that knowledge for productive purposes.

12 The evolution of the economy

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Differentiation, selection, replication

The development of economic analysis throughout the twentieth century has been focused on better understanding the properties of economic equilibrium. In microeconomics the Pareto optimal unique and stable competitive general equilibrium has been depicted as the welfare-maximizing benchmark for economic efficiency, and in macroeconomics the policy goal has been to design policies that arrive at an equilibrium with full employment and low inflation. The Keynesian macro framework in particular admits the possibility that the economy could rest at an underemployment equilibrium, and macroeconomic analysis in general sets the policy goal of finding policies that can lead the economy to that full employment low inflation equilibrium. Similarly, from a microeconomic perspective, factors keeping an economy from a competitive equilibrium are referred to as market failures which, if corrected, produce the welfare-maximizing Pareto optimal allocation of resources. Toward the end of the twentieth century, macroeconomic models began investigating economic growth, but even then in an equilibrium framework.[1] A major part of core economic theory rests on the idea that there are market forces pulling the economy toward a stable equilibrium, and that left free of outside disturbances that equilibrium will persist.

This framework for analysis is analogous to the foundation of physics in which physical systems do tend to arrive at a stable equilibrium. One of the popularizers of this equilibrium framework, Alfred Marshall, was cautious about its general applicability, and often suggested that the economy is more analogous to biological processes than physical processes. Marshall says:

The Statical theory of equilibrium is only an introduction to economic studies; and it is barely even an introduction to the study of progress and development of industries which show a tendency toward increasing return. Its limitations are so constantly overlooked, especially by those who approach it from an abstract point of view, that there is a danger in throwing it into definite form at all.[2]

Elsewhere, referring to economic development, Marshall says that "In this matter economists have much to learn from the recent experiences of biology: and Darwin's profound discussion of the question throws a strong light on the difficulties before us."[3]

Marshall is perhaps best known for establishing the partial equilibrium framework for the development of microeconomic theory, so it is interesting to see that he starts his chapter "On Markets" by saying:

A business firm grows and attains great strength, and afterwards perhaps stagnates and decays; and at the turning point there is a balancing or equilibrium of the forces of life and decay ... And as we reach to the higher stages of our work, we shall need ever more and more to think of economic forces as resembling those which make a young man grow in strength, till he reaches his prime; after which he gradually becomes stiff and inactive, till at last he sinks to make room for other and more vigorous life. But to prepare the way for this advanced study we want first to look at a simpler balancing of forces which corresponds rather to the mechanical equilibrium of a stone hanging from an elastic string, or of a number of balls resting against one another in a basin.[4]

Alfred Marshall, the father of twentieth-century equilibrium economics, says that the equilibrium framework is a first step toward understanding the forces underlying the market economy, and that an "advanced study" would want to look beyond this physical analogy to see the economy in an evolutionary setting, where a biological analogy is more appropriate than a physical one. The paragraph above contains elements of Schumpeter's creative destruction, in which existing firmsfind themselves past their prime and are displaced by young and vigorous firms. Taking Marshall at his word, the development of the equilibrium framework in twentieth-century economics serves as a foundation to prepare the way for a more advanced study of a continually evolving economy. Economics as a discipline has learned much about the properties of economic equilibrium over the twentieth century, to the point where many theoretical developments offer more insight into the nature of abstract models than of real-world economic processes.

In an economy characterized by economic progress there is no equilibrium in the sense of a state of affairs which, if disrupted, economic forces work to return the economy to. Markets can be temporarily disrupted, as the equilibrium framework suggests, but the market process continually produces permanent dislocations that are not easily captured within an equilibrium framework. Disruptions are often caused by entrepreneurial actions that permanently change the underlying environment, so that an economy evolves and will never return to its former state of affairs. This is the creative destruction identified by Schumpeter. As Schumpeter observed, "The essential point to grasp is that in dealing with capitalism we are dealing with an evolutionary process. ... Capitalism, then, is by nature a form or method of economic change and not only never is but never can be stationary."[5] The very notion of equilibrium paints an unrealistic picture of a competitive economy. Schumpeter's depiction of the process as creative destruction provides a more accurate picture of the competitive process.

Evolution versus equilibrium

The term equilibrium suggests that if it is disturbed the economy reequilibrates by returning to equilibrium; that there are market forces that pull the economy back to the equilibrium state from which it was disturbed. This paints a fundamentally misleading picture of the market process, because while some disruptions may be of that type, the more significant disruptions change the underlying conditions so that, disturbed from one equilibrium, the economy will never return back to it. The economy never returns to its old equilibrium, because conditions are always changing so the old equilibrium no longer exists. Not only does the economy not return to its former state, the direction in which the economy adjusts to disturbances is determined by the nature of the disturbance, so that there is a path dependence to the economic adjustments that occur.[6] Even when a dynamic model accounts for the passage of time, the notion of an equilibrium growth path does not hold up when one realizes that how the future unfolds depends upon the choices that people make in an uncertain environment today.

In this situation, the equilibrium framework loses much of its usefulness, because even when shoehorned into the equilibrium framework, the underlying equilibrium is always changing, and entrepreneurs cannot simply account for new conditions and try to maximize profits under the new regime. To remain profitable, entrepreneurs must know that there is, in fact, no equilibrium, in the sense of some unique outcome toward which economic forces are pulling the economy. Entrepreneurs react to current conditions, and the very act of their entrepreneurship changes the outcome toward which the economy is tending. Profits can be made by taking advantage of disequilibrium situations, as Kirzner emphasized,[7] but the notion that entrepreneurs are equilibrating the economy is misleading in the sense that there is no underlying stable equilibrium toward which the economy is tending.

As Chapter 4 noted, market forces tend to clear markets, but in an economy characterized by economic progress it is misleading to think that if markets clear, that action of market clearing is pulling an economy closer to equilibrium. Indeed, the connection between market clearing and equilibrium is only an artifact of equilibrium models. Markets can clear, in the sense that prices adjust so that the quantity supplied equals the quantity demanded, without any underlying equilibrium toward which the economy was tending. The fact that markets tend to clear says nothing about equilibrium, unless equilibrium is defined to mean market clearing and nothing more.

In hindsight one can see the path over which the economy has moved has not been a determinate trajectory, and depends crucially on the decisions made by entrepreneurs. But entrepreneurs can never know whether they are making optimal decisions or choosing the most profitable opportunities. By choosing one option they are foregoing others, and while they can observe the profitability of the options they actually chose, they can never know what the outcome would have been had they made a different choice.[8] As Richard Nelson and Sidney Winter say, after discussing business strategies for discovering profit opportunities, "Our basic point is that firms cannot hope to find optimal strategies."[9]

There is a path dependency in economic development, as Brian Arthur suggests, and the decisions of entrepreneurs at one point in time permanently change the future direction of the economy.[10] An entrepreneurial discovery at one point in time simultaneously closes offbusiness strategies that formerly were profitable and opens up new opportunities for entrepreneurial discovery that previously did not exist. The economy does not "grow" along a predetermined path that can be predicted within some model. Rather, it progresses as the result of a multitude of entrepreneurial actions that steer the economy in directions that cannot be forecast in advance. It cannot be forecast in practice, but it also cannot be forecast in theory, because of the path dependence that makes the future dependent on choices that people make under uncertainty. The information to make the "optimal" entrepreneurial choice is never available, because entrepreneurs are making decisions for which no past information exists. Economic progress is not an equilibrium phenomenon.

Evolution: differentiation, selection, replication

The idea of analyzing economic phenomena in an evolutionary framework is not new. Alfred Marshall argued that an evolutionary framework would be appropriate for advanced economic analysis, as already noted. Armen Alchian wrote an insightful article pushing an evolutionary framework,[11] and Nelson and Winter's book referenced in the previous section has received some attention. Eric Beinhocker presents an interesting analysis that combines evolutionary economics and complexity theory to look at economic progress.[12] Despite these examples,[13] evolutionary models in economics remain outside the mainstream of economic research.

The general evolutionary framework describes a process whereby: (1) differentiation among individuals occurs; (2) those more fit for the environment are selected; and (3) those characteristics that are selected are amplified through replication. There is much insight in using this evolutionary framework as a foundation for economic analysis, just as there is much insight in neoclassical equilibrium models, and evolutionary economics gives credit to Schumpeter, Hayek, Kirzner, and the Austrian school more generally for providing fundamental ideas for that line of reasoning.[14] But evolutionary economics and complexity theory leave out some crucial elements, particularly in that crucialfirst step of differentiation.

While biological analogies can be insightful, they leave out the key fact that differentiation in the economy comes from the deliberate choices of individuals. In nature random mutations may occur, but in the economy changes in production processes and in goods and services that are produced occur because entrepreneurs decide that there may be a better way of doing things. Nelson and Winter build on the work of Herbert Simon[15] to move beyond utility maximizing individuals toward a more realistic depiction of individual decision making, but many essential entrepreneurial aspects of the economic environment are left out of this analysis. As suggested in the previous section, many things affect how the degree to which individuals in an economy will be entrepreneurial.

The second step in the evolutionary framework is also fertile for economic analysis. While in a pure market economy profits are the signal of fitness, in a mixed economy profitable strategies are not always welfare-enhancing, so in this step the institutional framework within which economic activity takes place deserves some scrutiny.

The third step, in which successes from the second step are amplified, is also ripe for analysis, because amplification in one area opens up profit opportunities in others while at the same time closing off profit opportunities in some currently profitable activities. This is the creative destruction described by Schumpeter.

There is another feature of biological evolution that deserves some notice in the context of equilibrium economics. One can conceive of a species or an ecosystem evolving without any notion that there is some end-state toward which this evolution is leading. As change continues, one change leads to another, and the ecosystem in the future is different from the ecosystem today. Extending this biological analogy to economics, there is no reason why the creative destruction of a competitive economy should be leading toward some end-state, or equilibrium. As in biology, one change leads to another and the future is different from the present. That does not imply that there is some determinate end-state, or equilibrium, toward which change is leading, or that there is dome determinate trajectory within which change takes place. Biological systems and economic systems are evolving, not equilibrating.

Differentiation

In biological evolution differentiation occurs within a species as genetic mutations alter individual characteristics, and as different genetic combinations result in different characteristics within individuals. Differentiation occurs randomly and is unplanned. In economies differentiation occurs because individuals make the conscious decision to do something differently. There may be an element of chance in the decision as, for example, an entrepreneur may observe a profit opportunity that previously has gone unnoticed. But for any differentiation to occur the entrepreneur must make the decision to act on that profit opportunity – to do something different from what has been done before. In the neoclassical competitive equilibrium, firms take as given their product characteristics, their production processes, and the inputs that are available for them to use. Entrepreneurs take none of these as given, and entrepreneurial innovation may involve changing product characteristics, even to the extent of introducing new products, changing production processes, and perhaps changing the types of inputs that are used in production. Any of these changes require a conscious decision by the entrepreneur, and the motivation for making changes is that the change will result in an increase in the firm's profit. So, unlike biological evolution, where changes just happen, an economy evolves when an entrepreneur deliberately decides to innovate, and the motivation for the innovation is the potential profit the innovation could earn.

How do entrepreneurs spot profit opportunities?

Israel Kirzner emphasizes the act of entrepreneurship as a costless spotting of a profit opportunity, as a result of alertness.[16] While that may be descriptive of the entrepreneurial act itself, actions supporting entrepreneurship are much more complex, and the people who spot entrepreneurial opportunities are likely to be those who have put themselves in a position to see them, and who are more than just alert: they are actively looking for profit opportunities. As Adam Smith noted, "Men are much more likely to discover easier and readier methods of attaining any object, when the whole attention of their minds is directed towards that single object."[17] Friedrich Hayek talks about the specific knowledge of time and place individuals have, following up on Smith's insight.[18] Entrepreneurial opportunities are not just spotted at random. They are spotted by those who are in a good position to spot them and who have the specialized knowledge to recognize them for what they are. Fritz Machlup notes the importance of tacit knowledge that an entrepreneur cannot explain, or even fully understand, noting that the entrepreneur

would simply rely on his sense or his 'feel' of the situation. There is nothing very exact about this sort of estimate. On the basis of hundreds of previous experiences of a similar nature, the businessman would 'just know,' in avague and rough way.[19]

Through research and development, businesses create a specific time and place for their employees that make entrepreneurial insights more likely, but the entrepreneurial act can never be an automatic or routine part of doing business. So while the entrepreneurial act may be a spontaneous noticing of a profit opportunity that nobody has seen before, entrepreneurs typically do hard work to create an environment in which entrepreneurial opportunities are more likely to be spotted. Entrepreneurs can create an environment in which profit opportunities are more likely to be discovered, but the entrepreneurial act of identifying a profit opportunity is never an automatic act. Profit opportunities are often missed because they are not observed, even though they are in plain sight.

William Baumol suggests that corporate R&D has essentially routinized entrepreneurial activities that used to be more spontaneous,[20] but Baumol does not appear to have distinguished invention from innovation the way Schumpeter did. While entrepreneurs can create an environment within which profit opportunities are more likely to emerge and be spotted, the actual recognition of a profit opportunity remains as entrepreneurial as ever. Corporate R&D has routinized the process of invention, but the innovation of transforming an invention into a profitable product or production process lies outside the corporate routine. In an example that has been referenced before, Xerox invented the graphical user interface for computers, but was never able to incorporate this invention into a profitable product. It was Steve Jobs and Bill Gates who took Xerox's invention and brought innovative and profitable products to market. Science and engineering can be used to create inventions, but entrepreneurial innovations always must go beyond scientific knowledge, because nobody can know whether a product that has never been produced will be profitable, or even if an investment to make an product less expensive will sell enough to pay for the investment. Henry Ford brought assembly line production to automobiles to produce them at lower cost, but he could only guess at the demand for automobiles at the lower price for which he intended to sell them, because automobiles had never been sold that inexpensively before. There is always an element of uncertainty in entrepreneurship.

Many businesses (and individual product lines within large corporations) prove unprofitable despite the optimism that brought them into being. Many profit opportunities may lie in plain view, but still unseen by most, because the entrepreneur must have: (1) the information about the opportunity; (2) a knowledge base within which to place this specific information; and (3) the wisdom to sort out profitable opportunities from those that have some flaw that will keep them from actually returning profits.[21] As Baumol noted, "One can, indeed, describe what entrepreneurs used to do, but simply by virtue of having been reported in detail such an act is transformed into one that is no longer entrepreneurial."[22] It is too easy to focus on the entrepreneurial successes – like

Henry Ford and Bill Gates - without recognizing the many who tried to be entrepreneurial but failed.

The differentiation stage of economic evolution relies on entrepreneurs observing and acting on new and improved production processes and new and improved products that can be brought to market. Far from being a simple act of observing something that has not been noticed before, entrepreneurs work to put themselves in a position where they can spot profit opportunities, by observing what is profitable in their lines of business, and by creating an internal environment, perhaps with R&D, within which profit opportunities are more likely to be seen by them than by others. Profit opportunities are not just lying around waiting to be discovered. Entrepreneurs are looking for them, and they are likely to be acted on soon after they come into existence.

The origins of entrepreneurial opportunities

This raises the question of how entrepreneurial opportunities come into existence. Israel Kirzner defines entrepreneurship as the noticing of profit opportunities that have previously gone unnoticed. Read uncritically, this seems to draw a parallel with the biological analogy: in that the same way that mutations happen to occur in nature, entrepreneurs happen to notice profit opportunities. But, as just noted, entrepreneurs are actively looking for profit opportunities, and once they arise they tend to be acted on rapidly.

To provide a brief taxonomy, entrepreneurial opportunities come from: (1) factors that disequilibrate the market; (2) factors that enhance production possibilities; and (3) entrepreneurial activity that creates additional entrepreneurial opportunities.[23] This third factor – entrepreneurial activity that generates new entrepreneurial opportunities – is by far the most significant factor in economic progress. While at first it might appear that when an entrepreneur observes and acts on a profit opportunity that reduces the stock of profit opportunities available to others, in fact the opposite is true, and entrepreneurial activity creates many more profit opportunities than it uses up. This observation opens up a potential line of research on the self-reinforcing nature of entrepreneurship to economic progress.

The first source of entrepreneurial opportunities, factors that disequilibrate the market, calls to mind entrepreneurship in the way that Israel Kirzner discusses it, because Kirzner emphasizes the equilibrating role of entrepreneurship. Following this line of reasoning, it is easy to see that when markets are out of equilibrium, a profit opportunity exists that will move markets toward equilibrium. However, this takes a very static view of economic activity. Markets are out of equilibrium, so entrepreneurial action re-equilibrates them. In the real-world economy things are always changing, and the factors that disturb the existing state of affairs are often factors that change underlying conditions such that the old configuration of prices and quantities will never be re-established. That is why Chapter 4 put forward the argument that economic forces lead markets to clear, but to think of that situation in which all markets clear as an equilibrium does not fit the facts of a dynamic economy.

The second source of entrepreneurial opportunities, factors that enhance production possibilities, has more of a dynamic element, in that innovations in products and production processes do permanently alter the structure of the economy and open opportunities for entrepreneurial profits that build on those innovations. In addition, when production possibilities are enhanced, incomes rise, which enables people to buy more, further opening entrepreneurial opportunities. Thus, it is easy to see why factors that enhance production possibilities create entrepreneurial opportunities.

The third source of entrepreneurial opportunities is the activity of entrepreneurs themselves. When an entrepreneur discovers and acts on a profit opportunity, this increases profit opportunities available to other entrepreneurs (and, of course, to the entrepreneur who introduces the innovation). At first glance it might appear that there is a fixed stock of entrepreneurial opportunities, and as entrepreneurs discover them they are used up. However, the creative destruction ushered in by one entrepreneurial innovation will not only make some formerly profitable activities unprofitable – one often-used example is the reduction in the market for buggy whips after the introduction of the automobile – it opens up new profit opportunities because of the changed structure of the economy.[24] The introduction of the automobile, for example, brought with it opportunities for gas stations, for mechanics, and for after-market products to enhance people's cars. It brought opportunities for builders of roads and garages. The introduction of the transistor created the opportunity to create the microprocessor, which opened up the opportunity to build personal computers, which created the opportunity to develop the computer mouse to use with the graphical user interface, which opened the opportunity to make infrared and radio frequency mice to cut the cord, and to develop other pointing devices such as trackballs and touchscreens.

Most of the entrepreneurial opportunities in an economy are produced as a result of prior entrepreneurial innovations. One can see, then, that an economy that is stagnant, or that has the characteristics of a neoclassical general equilibrium, will have little innovation and entrepreneurship. Because entrepreneurship creates additional entrepreneurial opportunities, a less entrepreneurial economy offers entrepreneurs fewer opportunities. Breaking out of the status quo is risky, and there is no guaranteed return. This points to the question of what causes entrepreneurs to be entrepreneurial.

Why are entrepreneurs entrepreneurial?

This very important question can be broken down into two parts. First, what do entrepreneurs do to increase the likelihood that they will, in fact, spot profitable opportunities? This question is nearly a restatement of the earlier section on how entrepreneurs spot profit opportunities. Second, what features of the economic environment make it more likely that entrepreneurial opportunities will (2a) exist, and (2b) be spotted?

Ludwig von Mises suggests the evenly rotating economy as a point of reference for understanding the role of entrepreneurship.[25] Equivalently (for current purposes) think of an economy in a state of general equilibrium. Under these conditions in which economic activity continues in equilibrium as it has in the past, imagine the situation for a potential entrepreneur who believes he has spotted a profit opportunity. Of course, he may be wrong – we have already noted the importance of recognizing entrepreneurial failures along with entrepreneurial successes – and in an equilibrium setting, what are the chances that what at first appears to be a profit opportunity actually would yield profits if it were to be acted on? This situation is like the well-known economist joke:[26] Economist 1: "Look, there's a $20 bill on the sidewalk." Economist 2: "Couldn't be. If it was, someone would have already picked it up." If the economy appears to be in an equilibrium state, it is unlikely that what at first appears to be a profit opportunity will actually yield profits if acted upon. So in this situation: (1) people will not be very alert to the spotting of profit opportunities; and (2) even if someone does see what appears to be a profit opportunity, the person is unlikely (or at least less likely) to act on it, because of the risk that some unanticipated problem will actually cause action to lead to a loss. Why be alert to looking for something that likely does not exist, and even if it appears to exist is likely an illusion?

A simple recognition of this situation leads to a host of policy implications, because economic policy – from antitrust policy to the regulation of product characteristics and quality – is based on the equilibrium framework within which entrepreneurship is absent. If neoclassical competitive equilibrium is the benchmark by which policy outcomes are measured, it is likely that the resulting policies will be anti-entrepreneurial, simply because entrepreneurship is not a part of the benchmark by which desirable policies are measured.

In an entrepreneurial economy, people tend to be more entrepreneurial for several reasons. First, entrepreneurship generates more entrepreneurial opportunities. Second, the success of existing entrepreneurs reveals the possibility for profitable entrepreneurship (in contrast to the evenly rotating economy). It provides more than just information; it provides a knowledge base from which potential entrepreneurs can judge how profitable an innovation might be. Third, it provides information not only on the existence of profit opportunities, but on how others have acted on them successfully in the past.

Entrepreneurship will also create institutional changes within an economy to facilitate entrepreneurship. Venture capital firms are a visible example. Without entrepreneurship, no venture capital would be available to aid the potential entrepreneur. In an entrepreneurial economy, institutions arise to support entrepreneurship and enhance the success probabilities of entrepreneurs. The institutional features that help support entrepreneurship offer an interesting area of inquiry, as well as the impediments that public policy creates as obstacles to these institutions.

The previous chapter noted the existence of agglomeration economies, which directly create a more entrepreneurial environment. Agglomeration economies enable individuals to be more entrepreneurial, but because of the entrepreneurial environment they create they also require individuals to be more entrepreneurial. In a more entrepreneurial economy, not only do individuals have the opportunity to be more entrepreneurial, survival requires more entrepreneurship.

If there is relatively little entrepreneurship in an economy, so that, as in an evenly rotating economy, things remain the same (or almost the same) year after year, there is less of an incentive to be entrepreneurial, because once a person has found a successful economic strategy it can be continued year after year and still bring the same success. But in an entrepreneurial economy, characterized by change rather than stability, last year's successful strategy will not necessarily be successful next year, and will grow increasingly less fit for the economic environment as time goes on. Thus, an entrepreneurial environment requires entrepreneurial behavior to survive, whereas a stable steady-state environment does not.

This points to a major shortcoming for using an equilibrium framework to analyze economic progress. The economic environment since the Industrial Revolution has been entrepreneurial, requiring entrepreneurial firms that must constantly seek ways to improve what they are doing, even if what they are now doing is very successful. Entrepreneurial strategies are risky, because changing a successful formula may result in losses. But non-entrepreneurial strategies are fatal, because non-entrepreneurial firms will be crowded out by those entrepreneurial firms that make successful innovations.

Robert Heilbroner characterized economies as traditional, command, and market.[27] Within Heilbroner's taxonomy, the equilibrium framework of twentieth-century economics applies more to the traditional and, in the framework of Oskar Lange and Fred Taylor (advocates of central planning),[28] command economies. Equilibrium is not descriptive of the market economy, which is characterized by economic progress.

Compare this equilibrium framework with a dynamic and entrepreneurial economy characterized by substantial innovation. This setting provides both a carrot and a stick to entrepreneurs. The carrot is that entrepreneurs can see others innovating and profiting from opportunities they seize, which means that, first, they see the possibility of profits from entrepreneurship, because it is happening to others, and second, the entrepreneurial actions of others continually generate new profit opportunities in the marketplace. The stick is that the creative destruction of an entrepreneurial economy means that businesspeople who fail to be entrepreneurial will find themselves left behind by those who are. So, entrepreneurship is necessary for survival in this environment, but it is also encouraged. Entrepreneurship generates more entrepreneurial opportunities.

Selection

Selection is the survival of the fittest, and in an evolutionary economyfitness is determined by profits. Firms that generate profits survive, and those that generate losses die out. Profit produces income and can be accumulated to produce wealth for the firm's owners, so the owners of a firm have an incentive to make profits, and as aby-product they have an incentive to keep the firm alive. Through retained earnings firms can accumulate profits so that if they have a period in which they take losses, they can live offtheir accumulated past profits, just as animals can store calories as fat, and live offaccumulated calories when food is scarce. Similarly, multiproduct firms can sell some product lines that are sold at a loss if they have others that generate offsetting profits, just as animals living offone source of food might have access to another if their preferred source becomes scarce. Firms that are dependent on a single source of profit are riskier than those that are more diversified. As a spectacular example, the investment bank Lehman Brothers was highly leveraged and depended upon a continual inflow of loans for financing, and when they were unable to obtain further financing in October 2008 went out of business over a weekend.

The biological analogy holds up pretty well if one views profits in economic evolution as analogous to food in biological evolution. Firms need profits like plants and animals need food. In the ecosystem plants and animals are also subject to predation, so survival depends on both getting sufficient food and being able to avoid predators. The entities analogous to predators in an economy might be pirates, bandits, and perhaps government, although the government relationship is somewhat complicated. The analogy is weakened because in advanced economies predators in the form of pirates and bandits are fairly well controlled. And indeed, the institution that controls this type of predation the most is government. So, while government may engage in some predatory activities, it also protects economic actors from predators. The relationship here is more parasitic. Government gets its income through the forcible taxation of economic activity, but in exchange protects those engaging in that economic activity from other predators. There is an exchange of protection for tribute.[29]

In a market economy firms earn profits as a result of voluntary exchange. If the price at which a good or service is sold is greater than the cost of the inputs to produce the good or service, the result is profit, and that profitis a measure of the amount by which the exchange increases value in an economy. The buyer values the output by that much more than it cost the seller to produce it. However, not all profit in real-world economies is produced through voluntary exchange. Sometimes governments provide direct subsidies to producers, so the revenue of the seller is the price paid by the purchaser plus the subsidy. The subsidy was generated through coercive taxation, and it may be that the value of the good sold is less than the cost of production plus the subsidy. Government policy can produce profits by methods more indirect than direct payments to sellers. Regulation, taxes, or tariffs on competitors, and prohibitions on the selling of competing products, are examples of cases in which some profit comes to firms as a result of government coercion rather than through voluntary exchange.

Government interference in a market economy may be defended as a part of government's role in the protection of property rights. For example, if the government shuts down one business because the business is violating a patent given to another business, one might view that action as protecting the right of the patent holder, or as preventing the patent violator from engaging in voluntary exchange. More generally, Gary Becker and Donald Wittman argue that democratic decision-making in general produces efficient policies because the process weighs the interests of people on all sides of an issue.[30] The issue will be considered further in this chapter, but at this point note that some profits are the result of voluntary exchange and some are the result of government interference in the market.

One might think of the genetic engineering of biological species, or selective breeding, as analogous to government interference in the marketplace if one considers the direct manipulation of the genetic composition of other species by humans to be analogous to government intervention into markets. Natural selection is overridden as one species deliberately and directly manipulates the characteristics that lead to survival in another. It is likely that few if any dachshunds could survive in the wild, but because they have characteristics that many people like, this species that has been produced through selective breeding proliferates. Similarly, in the early 2000s solar-energy firms were supported through government subsidies, and tax-preparation firms profit from calculating customers' income tax payments only because the government compels people to pay income taxes.

Whether profit comes purely from market forces, from government interference, or from some combination, the selection process in economic evolution is profit driven. Firms that make profits survive and those that do not die out.

Replication

The replication of successful innovation is perhaps the most straightforward stage of an evolutionary economy. Replication occurs through the growth and expansion of successful businesses, and through the creation of new businesses that imitate the successful innovations of others. The degree to which replication can take place will depend on the institutional environment within which the innovation takes place, and government-imposed institutions play a major role.

Patent and copyright laws often prevent replication in the form of new firms copying the innovations of existing firms. Other regulatory restrictions may also play a role. For example, zoning laws may prevent the siting of some businesses, licensing requirements may shut out some potential entrants, and in some cases regulators will not allow competitors to enter a market unless the potential entrants can show there is a need demonstrated by the fact that existing businesses cannot satisfy the existing demand. Imperfect substitutes may arise in these cases, such as gypsy cabs in New York City, where the city has placed a cap on the number of taxicabs that can be licensed. In industries such as the fashion industry, fashions are not covered by patent or copyright laws and companies remain in business producing less expensive knock-offs of the latest fashions as soon as they are released by the top fashion houses. Whether an industry is more similar to the New York cab industry or to the fashion industry often depends on the degree to which governmentally imposed institutions restrict entry.

Market forces also restrict entry, so even without government restrictions potential entrants may not have the expertise – whether technological, managerial, or marketing – to enter the market, or may not have the financial strength to enter and compete against a larger or better-established firm. Barriers to entry arise from market forces as well as government restrictions.

Similarly, replication through the expansion of an existing business depends on institutions. The limited liability corporation has provided a financial vehicle through which firms can grow larger much faster than would be possible without it. With limited liability the owners of a firm are liable for losses only up to the amount they have invested in the firm. This enables firms to raise money through stock sales to people who would not be willing to invest if their personal assets beyond their investment in the firm were at risk. In contrast, partners in a partnership are fully liable for their business losses, so if the business has liabilities in excess of the partnership's assets, the partners will find themselves legally liable. With the limited liability corporation, much more money is available for financing through many more potential investors.

The economic landscape would look considerably different were it not for the limited liability created by government. It enables small investors to hold a diversified portfolio without risking anything beyond their investments, and enables large investors to take a chance on entrepreneurial ventures that look promising. The venture capital industry would not be possible were it not for limited liability corporations.

Even more fundamental institutional aspects determine the degree to which replication is facilitated. Rule of law, which means that everyone is treated the same under the law, and protection of property rights, are essential, because without them innovators run substantial risks that if their innovations are successful the profits they earn will be taken by others.[31] The replication stage of economic evolution seems straightforward, in that profitable ideas tend to multiply and spread throughout the economy, but the degree to which this happens depends upon how conducive the economic environment is to this spread, and government-enforced institutions determine this to a large degree.

Profit as an indicator of progress

A biological system evolves over time, but there are no normative implications that arise from evolutionary changes, and one would not argue that the system that has evolved today is better or worse than what existed previously. Evolution does not make a biological system better or worse – just different. Among those who do pass judgments there seems to be a status quo bias, in that not only is a reduction in the numbers of some species viewed as undesirable, the increase in the populations of others who have found a more profitable niche in the ecosystem is also often viewed negatively. Even recognizing such judgments, from a biological standpoint one would not say that the result of evolution is an improvement in a biological system, but rather that evolution just keeps happening.

In contrast, economic evolution appears to bring improvement – economic progress. As with biological evolution, there may be some nostalgia for the good old days, but one would be hard-pressed to deny the huge economic benefits that have been generated by economic progress since the beginning of the Industrial Revolution. Even prior to that, one would have to recognize the advances of civilization in places like ancient Greece, Rome, China, Central America and elsewhere as progress, and normatively as beneficial to mankind. People's lives are better, in that they are healthier, they live longer, work is more enjoyable, and people have substantially higher levels of material well-being. Prior to the twentieth century the history of mankind was the search for enough calories to survive, and obesity was a sign of wealth because only the wealthy could afford enough calories to overeat. At the beginning of the twenty-first century, in high-income countries the bottom 10 percent of the income distribution on average weighs more than the top 10 percent. While one would not want to downplay the misfortunes of the poor, one remarkable indicator of economic progress is that in capitalist economies the poor can afford to overeat.

If one makes the claim that economic evolution results in progress that improves people's well-being so that, in contrast to biology, economic evolution creates a future that is not just different from the past but better, what is the mechanism by which this takes place, and how can progress (as opposed to mere change) be demonstrated? Profit is a measure of the benefit of economic progress. Eventually, profits are competed away, so the measure is transitory.

When an entrepreneur introduces an innovation into the market, if it increases well-being a profit will be earned. Considerfirst the straightforward case in which the innovation is a less costly method of manufacturing an existing product. Assume, to take the simplest case, that the price of the product remains unchanged, as would be the case for a firm that corresponds with the neoclassical perfectly competitive firm. The decline in the cost of production is the reduction in the opportunity cost of the resources used in the production process, and for finding a lower-cost method of production the firm in question keeps the difference between the old cost of production and the new cost of production as profit. The increase in profit is a measure of the extra value produced for the economy – it places a dollar value on the economic progress that the innovation brings.

If the firm faces a downward-sloping demand curve, the profit-maximizingfirm will want to sell more after the cost reduction, so will lower its price. In this case, the profit is an indicator of economic progress, but it is smaller than the actual benefit from progress, because some of the gain goes to purchasers in the form of a lower price. Similarly, if one looks at the industry over time, other firms may be able to imitate the innovating firm's innovation, so their costs will also fall, and the ensuing competition will compete the profit away. There will be a transitory profit to the innovating firm which will disappear over time as other firms move to adopt the same innovation. As time goes on the profit – the indicator of progress – goes away, but the benefit from the innovation remains. Initially, it goes mostly to the innovator, but the competitive process means the innovation will become diffused through the industry and the profit will dissipate as the entire benefit of the innovation goes to consumers. The benefit shifts from profit on the supply side of the market to consumer surplus on the demand side. The profit is an indicator of progress, but as a measure it underestimate the progress.

Now consider a new product that is brought to market, or a change in an existing product. If this introduction brings a profit to the entrepreneur, the profit is an indicator of economic progress, because it shows that after accounting for the cost of production, purchasers place a higher value on the innovation than they did on the previously existing alternatives. Not all new products, or changes, are profitable. If the new product results in a loss, that shows that the cost of production is higher than the value to consumers, and the loss also signals to the seller that the new product should be discontinued. If the new product results in a profit, that indicates that not only is the new product a change, in the evolutionary sense, but also an improvement in people's well-being, as indicated by their own willingness to pay for it. The example is symmetrical to the cost-reducing innovation just discussed, in that the profit to the innovating entrepreneur shows that purchasers place a higher value on the product than the opportunity cost of producing it, and the profit indicates that not only has a change taken place, that change has brought with it welfare-enhancing progress.

Also, as with the cost reduction example above, the profit is an indicator of progress but not a measure of progress because some of the benefit of the innovation may go to purchasers. This is true initially, and increasingly becomes the case as time goes on and competitors in the marketplace imitate the innovation. Eventually the profit is competed away, but the benefit to consumers not only remains but increases as the benefit shifts from producer surplus to consumer surplus.

Economic evolution, like biological evolution, is characterized by continual change, but in the case of economic evolution that change can be characterized as economic progress, and one can say that the future is better than the past. The profit that comes with entrepreneurial innovation is an indicator of that progress, because it shows that the value of the innovation is greater than the opportunity cost. The economy evolves, and that evolution is characterized by progress. Casual observation shows that progress has occurred and that people are better off because of it, and the profit earned by entrepreneurial innovators is the indicator that shows that change actually does bring with it progress.

Selection: exchange or predation?

The idea just put forward, that profit is an indicator of progress, was developed within the framework of a market economy, where profitis earned through voluntary exchange. Some profit comes to firms as a result of being able to forcibly transfer resources from others to themselves. As straightforward examples, if firms can convince government to grant them monopolies, or enact tariffs to protect them from foreign competitors, their profits will increase not because of innovations they have introduced into the marketplace but because they have changed the institutional framework so they can charge more for what they sell.[32] In this case the selection stage of the evolutionary process may favor predatory over productive activity, and profit may represent a reduction in well-being.

William Baumol has emphasized the role of institutions in directing entrepreneurial activity toward productive rather than predatory activity,[33] with protection of property rights and the establishment of rule of law being two key elements that direct entrepreneurship toward productive activity. In politics, there can be opportunities to modify institutions to enhance productivity, as Gary Becker and Donald Wittman have noted.[34] However, while such opportunities may sometimes exist, it may be difficult to put together a majority coalition to implement them – meanwhile, opportunities for predatory political entrepreneurship always exist. Regardless of the status quo, it is always possible to assemble an alternative that would transfer resources from a minority to a majority, which would be politically profitable.[35] As a simple three-person example, it is always possible to propose taxing A $10 to transfer $5 to both B and C, gaining the support of B and C.

Seeing the possibility, it is always possible that political entrepreneurs could engage in increased transfer activity from productive individuals to non-producers, which would produce no economic progress although it may generate profits for non-producers. One would expect offsetting reductions in profits, or perhaps losses, for producers, still making aggregate profit a possible indicator of progress. A broader point is that when looking at ongoing transfers which are a part of the status quo, if predation (transfer activity) does not increase, then profit resulting from market activity is an indicator of economic progress. As Gordon Tullock pointed out, as with profits in the market any profit from transfer activity is only transitory because people compete for those profits just as they compete for profits in any market-place.[36] Thus, if new profit is the product of voluntary exchange then that is an indicator of economic progress. If it comes from predation, transfers, or restrictions on economic activity, that may not be the case.[37]

How does the evolutionary model hold up?

Formal economic analysis has developed largely along the lines of physics rather than biology, emphasizing equilibrium states rather than evolutionary processes.

In many ways the biological analogy seems to be more descriptive of an economy than the physical analogy. At the same time there are clear differences between biological evolution and economic evolution. The primary difference is that differentiation in biology is an unplanned process, the result of mutation and new genetic combination, whereas in economics differentiation is the result of conscious plans made by entrepreneurs. While there is no conscious motivation behind biological differentiation, entrepreneurs are motivated by the opportunity for profit, and profit underlies economic progress. Even entrepreneurs who claim, completely sincerely, that their motivation is not profit, can only engage in entrepreneurship as long as the result if profitable. Continued losses weed unsuccessful entrepreneurs out of the market, whereas profits provide the resources for successful entrepreneurs to continue to innovate.

The biological analogy seems to fit well, but the physical model of equilibrium states also lends insight into economic processes. What neoclassical theory calls equilibrium is best viewed as a state of affairs in which the plans of all economic actors are coordinated such that all plans can be realized, not an end-state toward which economic activity is gravitating. This view of equilibrium as mutually consistent plans rather than a state of rest provides a foundation for moving beyond the equilibrium framework, as Alfred Marshall recommended. The issue then becomes how the market mechanism works to allow individuals to make plans that, given the plans of others, actually can be realized.

Ludwig von Mises, when discussing the evenly rotating economy as a theoretical construction to aid in the understanding of the real-world economy, says:

In order to grasp the function of entrepreneurship and the meaning of profit and loss, we construct a system from which they are absent. This image is merely a tool for our thinking. It is not the description of a possible and realizable state of affairs. ... For it is impossible to eliminate the entrepreneur from the picture of a market economy. ... In eliminating the entrepreneur one eliminates the driving force of the whole market system.[38]

Mises goes on to say:

We want first of all to analyze the tendency, prevailing in every action, toward the establishment of an evenly rotating economy; in doing so, we must always take into account that this tendency can never attain its goal in a universe not perfectly rigid and immutable, that is, in a universe which is living and not dead.[39]

In other words, market forces are always at work to maintain market-clearing prices and quantities, but these forces are not pulling the economy toward a static equilibrium. The evolutionary framework is more descriptive of the process that determines an economy's trajectory over time.

The subject matter of economics has always looked at individual behavior with the idea of aggregating the activities of individuals to understand the aggregate economy. While economics has its models of behavior, rational and otherwise, psychology is the branch of science that studies the behavior and choices of individuals. Models of individual behavior in economics are a means to an end: an economic model says that if people behave a certain way, and interact within a certain institutional framework, then there will be certain aggregate results from everyone's individual behavior. Within the neoclassical framework, the result is an equilibrium with certain properties. By changing the assumptions about behavior and institutions, the properties of the resulting equilibria will change. The purpose of modeling the behavior of individuals and firms, and the way they interact through institutions, is to generate insights about the aggregate economy.

This has always been the case. Adam Smith was undertaking an inquiry into the wealth of nations, not the wealth of individuals, and the eighteenth-century question about wealth creation was how aggregate wealth grew, not how individuals could become rich. Similarly, the dismal forecast of Malthus and Ricardo about most people being destined to remain at a subsistence level of income was a conclusion about the aggregate economy. They recognized that even as most people remained pressed to a subsistence income, some individuals could become wealthy. Questions about the distribution of income have always been about aggregate outcomes. Economics has always looked at the behavior of individual elements in an economy as a means toward the end of understanding the implications for aggregate economic activity.

Understanding the way the economy tends to equilibrate – that is, the way that economic forces tend to aid in coordinating the activities of participants in the economy – is important, but additional development of the equilibrium framework adds no additional insight because, in fact, the economy does not tend toward an equilibrium. Rather, it is characterized by constant change, and the choices that individuals make today affect where the economy will be heading in the future. It is not an equilibrium process, so additional development of theory in that framework, while it may reveal more about the properties of equilibrium models, will not reveal more about real-world economic activity, and may actually be misleading.

The foundations for this line of inquiry were laid by Adam Smith, who began his treatise with a discussion of the division of labor and the way that it continually increases productivity and increases the wealth of nations. This notion of economic progress was lost in twentieth-century equilibrium economics – even when its subject was growth. The evolutionary framework has much to recommend it, not necessarily as a replacement for equilibrium economics but rather as a way of understanding economic progress.

Recall the analogy from Chapter 4 of trying to understand how an individual could ride her bicycle from her house to her friend's house. One type of answer to that inquiry was to explain how she was able to maintain her balance on the bicycle, shifting her weight as it moved so as to maintain upright, and the need to place her feet on the ground when the bicycle was stopped to keep from falling. The second type described how she planned her route of travel, what turns she made, and any obstacles she had to avoid to complete the trip. The first process of maintaining upright on the bicycle is completely different from the second process of finding a route to complete the trip. Similarly, the process by which a market economy maintains market-clearing prices and quantities is completely different from the second process of generating economic progress over time. This analogy illustrates why it is entirely reasonable to use a different theoretical framework to understand market-clearing processes and processes that generate economic progress over time.

Equilibrium economics is analogous to understanding how the cyclist can remain balanced on two wheels, but that lends little insight into understanding where the bicycle is going. To extend the analogy, we cannot learn anything more about where the bicycle is going by gaining a deeper understanding about how the cyclist can keep the bike balanced on two wheels. Similarly, we already know a great deal about the properties of economic equilibrium, and learning more about it will not lend any further insight into the fundamental issues of the causes of economic progress.

Conclusion

An economy simultaneously moves forward and maintains a tendency toward equilibrium (that is, the coordination of individual plans). Equilibrium economics lends much insight into the market-clearing forces in the economy that enable individuals to coordinate their plans, but while neoclassical growth theory uses the same tools that explain the equilibrating tendencies of the economy to also explain its progress, after some analysis one can see that these are two separate phenomena, and that the features of the economy that are essential to economic progress are assumed away in equilibrium economics. Progress is not an equilibrium phenomenon, and the discipline must move beyond equilibrium economics to understand economic progress.

Of course, there are other issues in economics besides progress, but economic progress is the key feature of a market economy, and – the neoclassical model of competition notwithstanding – a static market economy with no progress is unimaginable. Progress is a part of the market process, because market participants are always looking for ways to lower their costs and to make their output more desirable to consumers. While some efforts to do so succeed and others fail, the market system rewards the successes and penalizes the failures, which reinforces the success and creates imitators who amplify the successes. Imitators will always be trying to catch up to innovators, so the most successful response to innovation is further innovation, creating more improvement. Meanwhile, in other markets, even those that are very distant from the innovators, innovations will open more opportunities for further innovation. As entrepreneurs act on those opportunities, the economy continues to evolve, characterized by continuing progress, not equilibrium.

Because progress is such a fundamental characteristic of a market economy, it would seem that even when examining economic issues far removed from growth and progress, a framework in which those characteristics are present would be more amenable to analysis and would shed light on real-world economic processes. The challenge, then, is to build on the insights of Hayek, Schumpeter, and others to advance economic theory beyond the equilibrium notion, which after all, is not a very accurate representation of an actual economy.

The largest potential gains from moving beyond equilibrium economics lie in the policy arena. One can see, from contributors such as Oskar Lange and Fred M. Taylor in the socialist calculation debate that taking an equilibrium approach to economics leads to potentially disastrous policy conclusions.[40] Sanford Ikeda updates the debate, showing that its arguments remain relevant to economic policy after the collapse of the Berlin Wall.[41] The evolutionary approach to economic analysis appears to open some promising avenues for providing a better understanding of how a market economy produces progress and prosperity.

13 Producing prosperity

]13 Producing prosperity

The economic progress that capitalism has produced since the beginning of the Industrial Revolution has been so astounding that it surely would have appeared impossible to even the most farsighted people prior to its occurrence. Imagine explaining to someone in 1700 that in a few hundred years people of average means would be able to fly from one continent to another in a few hours. Imagine explaining to that person that you could talk to someone anywhere in the world through a small hand-held device people could carry with them wherever they went. While the technological marvels by themselves would be incredible, the widespread increases in the average standard of living would be at least as remarkable. Advances in healthcare mean that it is no longer rare for people to live into old age. Advances in agricultural productivity mean that for the first time in history poor people (in capitalist economies) are not threatened with starvation. Average people in the twenty-first century have luxuries well beyond what even the most privileged had a few hundred years ago. What prince in 1700 would not have gladly given up his horse-drawn “coach and six”[1] for an air-conditioned automobile with a sound system that would play his favorite music on command and that, seated in a luxurious seat, he could command to travel at high speed using only small body movements? That economic progress is remarkable, and yet we take it for granted because it has been a part of everyone’s life who is alive today, for their entire lives.

Prior to the Industrial Revolution, economic progress was so slow that people would not have noticed it in their lifetimes. The economic world in which they were born was the economic world in which they died. People would produce and consume the same goods, using the same production methods, and while individual fortunes could rise and fall, per capita incomes on average would be essentially unchanged from century to century.

Now, look at the development of economic analysis through the twentieth century, with an emphasis on equilibrium outcomes. The equilibrium framework from within which economic analysis describes the economy is more descriptive of the pre-Industrial Revolution economy, in which economic activity remained essentially the same over time than the post-Industrial Revolution economy that has been characterized by continual economic progress. Models that depict an economy in a steady state, or in a general equilibrium, are leaving out that part of a market economy that has been most responsible for the remarkable increase in prosperity in every economy that has adopted capitalist institutions.

The equilibrium approach to economic analysis

The particular approach economists took to developing their field in the twentieth century is understandable on several different levels. The sheer elegance of the mathematical models that advanced scientific knowledge of the physical world have attracted economists, partly for aesthetic reasons, partly because they appear to make economic analysis more scientific, and also partly because of the academic culture that places a premium on research and publication. In the nineteenth century many of the giants of economic analysis, ranging from David Ricardo to Karl Marx to John Stuart Mill, were not academics. In the twentieth century, academic institutions took over the advancement of economic ideas, and often rewards have come to researchers as much for advancing the sophistication of the models as for adding insight to the understanding of the economy. The relationship between the development of economic modeling and the understanding of the real-world economy has often been a loose one, as economic modeling has become increasingly abstract.[2]

Another more practical reason for focusing on equilibrium is that for much of the twentieth century a major question was whether an economy could be expected to maintain an equilibrium in which all its resources were being used productively. With the worldwide Great Depression in the 1930s it appeared, as Keynes depicted,[3] that an economy could come to rest at an underemployment equilibrium, and that government fiscal policy – the use of taxes and government expenditures – could help to stabilize the economy. Full employment returned only with the onset of World War II, which within a Keynesian framework looked like a substantial increase in the government component of aggregate demand (if one ignores all of the death and destruction that came along with it). With the 1930s dominated by the Depression and the 1940s by war, people naturally feared that after the war the economy might slide back into a depression. It happened in the 1930s after a prosperous decade in the 1920s, and it appeared that nobody had a good understanding of how that could happen. On into the 1980s macroeconomics was dominated by the question of what caused economic downturns, and what economic policies could be developed to mitigate them. The question was couched in an equilibrium framework, because the policy goal was how economic policy could be used to guide the economy to a full employment–low inflation equilibrium. The policy goal was to reach that equilibrium state.

The macroeconomic rationale for focusing on equilibrium states had a solid basis in fear: the fear that without understanding the nature of a full employment equilibrium and the forces that could prevent the economy from getting there, the world could find itself sliding into a second Great Depression. If economic policy could maintain full employment and low inflation, that would be a desirable outcome. The microeconomic rationale for focusing on equilibrium states seems more tenuous. Perhaps the best rationale would be that equilibrium analysis can provide a foundation for understanding how markets clear. However, that top-down equilibrium methodology provided the foundation for welfare economics to identify many problems that could, in theory, keep a market economy from allocating resources as efficiently as possible. Part of the theme of this book is that the focus on equilibrium models of economic welfare not only does not properly identify the source of people's economic well-being, but lays a foundation for the creation of economic policies that work against progress and prosperity. Why? Because those equilibrium models assume away the elements of the economy that produce prosperity.

Even worse, development economics also worked within the dominant equilibrium paradigm, leading economists to argue in favor of government planning rather than the development of markets to push underdeveloped economies to develop.[4] Development economics tended to put nations into either the category of developed or the category of less developed, and the policy goal was to get the less developed economies to be like the developed ones. That mindset did change toward the end of the twentieth century as economic growth theory focused on growth in the more developed economies along with the less developed.[5] Still, economic growth was depicted as an equilibrium phenomenon, leaving out the fundamental causes of prosperity.

There are ready explanations for why economic analysis has developed a framework that is designed to better understand the properties of economic equilibrium than to analyze the factors that produce prosperity. Furthermore, the focus on economic equilibrium has yielded useful insights that will remain a core part of economic knowledge. Regardless, one cannot hope to understand the causes of progress and prosperity by using models that assume those fundamental causes away. So, there is good reason to move beyond those equilibrium models – as Alfred Marshall argued we should – to develop economic analysis that can shed more light on the causes of prosperity. Economists should do this partly for their own understanding, but also because models that do not take into account the causes of prosperity can be, and have been, used to develop public policy that hinders economic development and makes everybody poorer.

Economists should consider how much more there is left to learn about the actual equilibrating forces in the real-world economy. The point is not that there is nothing else left to learn, but rather that the development of equilibrium models has reached the point where much contemporary research has more to say about the properties of the model being used than about the properties of the actual economy. This development is not merely worthless – it leads economic analysis toward policy conclusions that are counterproductive.

Entrepreneurship and prosperity

At the beginning of the twenty-first century economic analysis is showing signs that it places a renewed emphasis on understanding the causes of prosperity. That emphasis was at the center of economic analysis in the eighteenth century. Adam Smith's title, An Inquiry Into the Nature and Causes of the Wealth of Nations, indicates his view of the key question in economics, but over the centuries the emphasis of economic analysis drifted away from that key question. Chapter 1 gave an overview of the way in which economics moved away from analyzing the factors that produce prosperity, toward the equilibrium approach that views markets and economies at a steady state. One shortcoming of the equilibrium framework is that it depicts economic activity as the result of the balancing forces of supply and demand, without analyzing the underlying institutions that allow market transactions to take place.

The prosperity that much of the world enjoys in the twenty-first century, and which is spreading to more and more locations, is the result of economic progress that has brought with it new and improved goods and services, and innovations that have lowered the cost of production for many goods and services. Those innovations were introduced into the economy by entrepreneurs, and entrepreneurship has produced progress and prosperity. Entrepreneurship does not automatically happen, and entrepreneurial individuals do not automatically generate progress, however. William Baumol has made the insightful conjecture that every society has people who are equally entrepreneurial, but that differences in institutions steer those entrepreneurial impulses in different directions.[6] In societies with market-friendly institutions, people's entrepreneurial instincts drive them to look for ways to produce goods and services that will satisfy the desires of others, because there is profit in doing so. In societies where people get ahead more by political connections, people's entrepreneurial instincts push them to look for ways to acquire and solidify political power. People use political power to appropriate the property of others, using taxation and regulation to transfer resources from others to themselves. The first type of entrepreneurship is productive, using Baumol's taxonomy; the second type is destructive.

Productive entrepreneurship is a result of people looking for ways that they can gain through market exchange, which relies on everyone who is party to a transaction perceiving that it is to their mutual advantage. In other words, productive entrepreneurs enhance their well-being by finding ways to create value for those with whom they transact. Destructive entrepreneurship is a result of people looking to produce a gain for themselves by forcibly transferring wealth from others. This is destructive partly because if the value one produces faces the threat of being taken by others it reduces any incentive for productive activity. It is also destructive because people will have to devote resources toward protecting anything of value that they already have.[7] In a society where property rights are not completely protected, people devote resources toward protecting their property by buying locks, by building fortified walls around their property, by hiring guards, and similar activities that divert resources from productive to protective activities. In a society with high taxes or high regulatory barriers, they devote resources toward finding loopholes in the rules and regulations that will bring profit to them, but that add nothing to the total product of their society.[8] Resources that could have been used productively are instead wasted on looking for ways to obtain income transferred from the productivity of others.

The role of institutions

If economic analysis is to be more geared toward understanding how economies produce prosperity, that means understanding how people's entrepreneurial instincts can be channeled into productive activity. In a supply and demand framework, economic analysis depicts suppliers on one side of the market exchanging with demanders on the other, but the degree to which these transactions can occur depends on the economic institutions within which they operate. For some types of exchanges relatively little institutional support is required. If one stops by a roadside fruit stand to buy some fruit, the fruit changes hands at the same time as the money, and the exchange can take place with minimal institutional support. Even in an exchange like this, however, there must be some security of property rights to ensure that the fruit is bought, not stolen. Also, there must be institutions that support money as a medium of exchange; otherwise, transactions will be much more complicated if transactions have to be made through barter.

When one considers more complex transactions, such as a developer building a shopping center, much more in the way of institutional support is required. There must be some system of registration for land ownership, so that the developer can be assured that nobody in the future will challenge the developer's right to build on that property. Even here, developed economies will require title insurance to get a mortgage to finance the property, so that in case some challenge to the title of the property does arise, any losses from the title challenge will be insured. The mortgage itself requires sophisticated financial institutions. All of these economic institutions support transactions that, in a simpler framework, assume the institutions away and only depict suppliers providing something in exchange for payment from demanders. Without the institutional support, much of this supplying and demanding would not be feasible.

Economists are taking the institutional foundations required for prosperity more seriously at the beginning of the twenty-first century than they were during most of the twentieth. Twentieth-century equilibrium economics produced high-powered mathematical general equilibrium models that were essentially institution-free, never specifying what institutions were in place to allow suppliers to transact with demanders, and to channel economic activity into voluntary exchange rather than allowing people to obtain resources through force and fraud. Increasingly, economists have come to recognize that certain institutional requirements are necessary for progress and prosperity, and that when they are missing, nations will remain poor.[9] Empirical evidence indicates that countries that protect property rights, that have rule of law (where everyone is treated the same under the law), that have low levels of taxation, regulation, and government spending, that have stable money, and that allow freedom of trade, have higher incomes and higher income growth rates than those that do not.[10] Economists have a fairly good idea about the institutional structure that produces prosperity, but all too often economic analysis takes place without recognizing the importance of the institutional framework.

While economists have a good idea about the general institutions that produce prosperity, the effects of specific institutions have not been analyzed in as much detail. The modern corporate form, and the limited liability that comes with it, surely has had a substantial impact on the nature of economic institutions and on the trajectory of the economy. Patent and copyright law also has had a substantial impact, although there is debate on whether the impact has been positive or negative. Surely the availability of patent and copyright protection has steered resources toward producing patents and copyrights. Some argue that this encourages creative activity while others argue it produces monopoly power, inefficiency, and rent seeking. Economists know the institutional framework is important, and there is some movement toward trying to get a better understanding of the impacts of specific institutions on progress and prosperity.

Economic progress is a dynamic phenomenon

The bulk of economic analysis takes place within an equilibrium framework. Often it is a static equilibrium framework, but even when a time dimension is brought in the analysis remains within a deterministic equilibrium setting where, once the initial conditions are stated, the entire time-path of the economy can be derived. The real-world economy does not work this way. There is a path dependence that results from the entrepreneurial actions of individuals within an economy, and entrepreneurial decisions made at one point in time determine the economy's future trajectory. This theme has been developed throughout this volume, and when applied it suggests some fundamental changes in the way that economists depict economic activity. It suggests that the equilibrium framework within which twentieth-century economics has developed is more accurately described as depicting market-clearing forces than equilibrium, in the sense that the most significant disruptions of economic activity permanently change the nature of the economy so that, when disturbed, markets do not return to their former state. There is no equilibrium – rather, there is a trajectory of progress.

It suggests that entrepreneurship is the key economic function of firms, not management, and that an important role that firms fill is the creation of an institution that allows an entrepreneur to capture the profits from entrepreneurial innovations. It suggests that welfare economics, as it has developed through the twentieth century, has little to do with the actual economic welfare people enjoy, and that the policy prescriptions of welfare economics can be counterproductive because they view welfare in a static sense, whereas actual increases in welfare come from the dynamic economic progress that increases people's standards of living. It suggests that market economies produce prosperity by coordinating all of the decentralized and often tacit knowledge that everyone in the economy possesses,[11] and that because of this agglomeration economies are an important source of economic progress. It suggests that an evolutionary framework works better for understanding how prosperity is produced than a static equilibrium framework.

One important policy conclusion that comes from this line of analysis is that within a market framework profit is an indicator of economic progress. In the neoclassical equilibrium framework profit is a sign of economic inefficiency, because it arises due to either monopoly or disequilibrium. In an evolutionary framework, if purchasers place a higher value on the innovations introduced by entrepreneurs than on what they could purchase before the innovation, then they are willing to pay for the innovation and profit is an indicator of the value that entrepreneurship adds to the economy. However, it is an indicator but not a measure because, first, some of the value is captured by consumers in the form of consumer surplus and, second, the profit erodes as other producers imitate and improve on successful innovation, competing the profit away. The benefitfromthe innovation remains, but over time is shifted from the seller'sprofit into consumer surplus. Public policy is often made within a framework that depicts profit as undesirable, but within a market setting that is not true. When profit is a result of special interest regulation and legislation, then it can be unproductive, but in those cases the profit is a result of a public policy measure.

A focus of economic analysis on factors that produce prosperity suggests moving beyond the equilibrium framework to one that takes account of the innovative and entrepreneurial forces in the economy that are responsible for the high – and ever-increasing – standard of living that capitalism has generated. This volume has suggested how that move can take place. Indeed, much of the groundwork has already been done, even though the mainstream of economic analysis has not picked up on it.

Conclusion

Adam Smith, often referred to as the father of economics, was focused on the factors of the market economy of his day that produced prosperity, and the title of his famous book, An Inquiry Into the Nature and Causes of the Wealth of Nations, clearly indicates his subject matter. Smith had a clear understanding of the factors that produce prosperity, and remains known for his advocacy of limited government and freedom of exchange. Economic analysis has advanced substantially in sophistication and rigor since Smith wrote, but one might question whether economists today, despite all their sophisticated models and techniques, have any better understanding of the economic policies that produce prosperity than Smith didwhen he published The Wealth of Nations in 1776.[12]

Smith argued that, in a market economy, individuals pursuing their own interests are led by an invisible hand to produce an outcome that is best for everyone. Smith's argument is persuasive when one looks around the world and sees that nations that have adopted the types of institutions Smith recommended have become wealthy, while those that have not, have not. Not everyone would agree with Smith's proposition about the invisible hand. Some would argue that the economic prosperity produced by market economies has been possible only because economic activity has been regulated and limited by government, and has been assisted by public sector production that has provided the infrastructure, both physical and social, within which a market economy operates. While some argue that prosperity would be enhanced by a more limited government that would stand out of the way of the entrepreneurial engine of capitalism, others argue that the greed induced by market incentives pushes capitalism in a counterproductive direction, and that more government intervention would enhance prosperity.

The question goes beyond just more or less government, and extends to what specific policies the government should pursue in an economy where most goods and services are produced in the private sector. Do antitrust laws help or hinder prosperity? What types of regulations on financial markets (if any) will enhance prosperity, and what types of regulations will stand in the way of prosperity and progress? These are the types of policy questions that economic analysis has always sought to answer.

Economists use models as a framework for analysis to answer these questions. All models are simplifications of reality, but if a model is to be useful in answering these questions, the model must accurately depict the factors that produce prosperity. This volume argues that the equilibrium framework for economic analysis that was refined and developed in the twentieth century often leaves out the factors that produce prosperity, and the result is that policy conclusions drawn from those models often stand in the way of prosperity and continued economic progress. Economic analysis, which has always at its foundation been an examination of factors that improve the material well-being of people, must take into account the factors that produce prosperity for its theoretical conclusions to actually point toward policies that produce prosperity.

While much economic analysis has left out the key elements that underlie economic progress, that is not universally true, and this volume has referenced several strands of economic analysis at the beginning of the twenty-first century that have the potential to pull economic analysis toward a clearer understanding of the factors that produced prosperity. This volume organizes those ideas, starting with methodological foundations, then looking to the behavior of individuals and firms, and then aggregating these components to look at the nature of a dynamic economy that is characterized by progress, not equilibrium. Analyzing economic activity within this framework provides a clearer picture of the factors that produce prosperity, and provides a framework that gives policy makers a better chance of designing economic policies that enhance prosperity rather than standing in its way.

Notes

]Notes

1 Producing prosperity: The subject matter of economics

1 John Stuart Mill, Principles of Political Economy, Vol. 1 (New York: D. Appleton and Company, 1870), p. 267.

2 Ludwig von Mises, Human Action, 3rd rev. ed. (Chicago, IL: Henry Regnery Company, 1966), p. 1.

3 Mises, Human Action, p. 2.

4 See, for example, Robert B. Ekelund, Jr., and Robert F. Hebert. A History of Economic Thought and Method, 4th ed. (New York: McGraw-Hill, 1997), pp. 40–41, for a "mercantilist manifesto" of policies to accomplish mercantilist goals.

5 Randall G. Holcombe, "Equilibrium Versus the Invisible Hand," Review of Austrian Economics 12, no. 2 (1999), pp. 227–43, gives a more detailed discussion of Smith's ideas, contrasting them with the twentieth century idea of equilibrium.

6 Thomas Robert Malthus, An Essay on Population (New York: E.P. Dutton, 1914 [orig. 1798]).

7 David Ricardo, The Principles of Political Economy, 3rd ed. (London: J.M. Dent, 1912 [orig. 1821, 1st ed. 1817]).

8 Ricardo, Principles of Political Economy, p. 72.

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17 Kenneth J. Arrow and Gerard Debreu, "Existence of an Equilibrium for a Competitive Economy," Econometrica 22, no. 3 (July 1954), pp. 265–90.

18 Holcombe, "Equilibrium Versus the Invisible Hand," cited earlier.

19 John Maynard Keynes, The General Theory of Employment, Interest, and Money (New York: Harcourt Brace and Company, 1936).

20 John R. Hicks, "Mr. Keynes and the 'Classics'; A Suggested Interpretation," Econometrica 5 (April 1937), pp. 147–59.

21 Robert E. Lucas, Jr., "On the Mechanics of Economic Development," Journal of Monetary Economics 22 (1988), pp. 3–42.

22 Paul M. Romer, "Endogenous Technological Change," Journal of Political Economy 98, no. 5, part 2 (October 1990), pp. S71–S102; and "Increasing Returns and Long-Run Growth," Journal of Political Economy 94 (1986), pp. 1002–37.

23 Douglass C. North, Institutions, Institutional Change, and Economic Performance (Cambridge: Cambridge University Press, 1990).

24 For a non-exhaustive list of examples, see Omar Azfar and Charles A. Cadwell, eds., Market-Augmenting Government: The Institutional Foundations for Prosperity (Ann Arbor: University of Michigan Press, 2003); W. Michael Cox and Richard Alm, Myths of Rich and Poor: Why We're Better Off Than We Think (New York: Basic Books, 1999); James Gwartney, Randall Holcombe, and Robert Lawson, "Economic Freedom, Institutional Quality, and Cross-country Differences in Income and Growth," Cato Journal 24, no. 3 (Fall 2004), pp. 205–33; Gwartney, Holcombe, and Lawson, "The Scope of Government and the Wealth of Nations," Cato Journal 18, no. 2 (Fall 1998), pp. 163–90; Gwartney, Lawson, and Holcombe, "Economic Freedom and the Environment for Economic Growth," Journal of Institutional and Theoretical Economics 155, no. 4 (1999), pp. 643–63; Randall Holcombe, Entrepreneurship and Economic Progress (London: Routledge, 2007); David S. Landes, The Wealth and Poverty of Nations: Why Some Are So Rich and Some So Poor (New York: W.W. Norton, 1998); Joel Mokyr, The Gifts of Athena: Historical Origins of the Market Economy (Princeton, NJ: Princeton University Press, 2002); Mokyr, The Lever of Riches (Oxford: Oxford University Press, 1990); Stephen Moore and Julian L. Simon, "The Greatest Century That Ever Was: 25 Miraculous Trends of the Past 100 Years," Cato Institute Policy Analysis no. 364 (December 15, 1999); Mancur

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32 Robert E. Lucas, Jr., "On the Mechanics of Economic Development," Journal of Monetary Economics 22 (1988), pp. 3–42.

33 Paul M. Romer, "Endogenous Technological Change," Journal of Political Economy 98, no. 5, part 2 (October 1990), pp. S71–S102; and "Increasing Returns and Long-Run Growth." Journal of Political Economy 94 (1986), pp. 1002–37.

34 See, for examples, David S. Landes, The Wealth and Poverty of Nations: Why Some Are So Rich and Some So Poor (New York: W.W. Norton, 1998); and Joel Mokyr The Lever of Riches (Oxford: Oxford University Press, 1990).

35 Murray N. Rothbard, For a New Liberty (New York: Macmillan, 1973); and David Friedman, The Machinery of Freedom: A Guide to Radical Capitalism, 2nd ed. (La Salle, IL: Open Court Press, 1989).

36 Randall G. Holcombe, "Government: Unnecessary but Inevitable," The Independent Review 8, no. 3 (Winter 2004), pp. 325–42.

37 There are perhaps too many arguments along these lines to even cite them in a footnote. The literature on the instability of the macroeconomy was substantially buoyed by John Maynard Keynes, The General Theory of Employment, Interest, and Money (New York: Harcourt Brace and Company, 1936). While Keynes's specific ideas do not have as much currency in the twenty-first century as they did in the twentieth, the general idea that government-designed macroeconomic policy is necessary for macroeconomic stability remains. Inefficiencies in resource allocation due to other market failures were laid out by Francis M. Bator, "The Anatomy of Market Failure," Quarterly Journal of Economics 72, no. 3 (August 1958), pp. 351–79. Arguments against this market failure view are found in Randall G. Holcombe, Public Policy and the Quality of Life (Westport, CT: Greenwood, 1995), but any introductory economics textbook will still recite these market failure arguments for government intervention.

38 Keynes, The General Theory, p. 383.

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40 See William Easterly, The White Man's Burden: Why the West'sEfforts to Aid the Rest Have Done So Much Ill and So Little Good (New York: Penguin Press, 2006).

2 Economic models: A framework for analysis

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3 (1981), pp. 377–87.

4 William J. Baumol and Alan S. Blinder, Economics: Principles and Policy, 2nd ed. (New York: Harcourt Brace Jovanovich, Inc., 1982). James M. Buchanan, "What Should Economists Do?" Southern Economic Journal 30, no. 3 (January 1964), pp. 213–22, also uses a map analogy, but for a slightly different purpose.

5 John Maynard Keynes, The General Theory, Chapter 2.

3 The market process

6 A.W. Phillips, "The Relation Between Unemployment and the Rate of Change of Money Wage Rates in the United Kingdom, 1862–1957," Economica 25 (November 1958), pp. 283–99.

7 Paul A. Samuelson and Robert M. Solow “Analytical Aspects of Antiinflation Policy,” American Economic Review 50 (1960), pp. 177–94.

8 Robert E. Lucas, Jr. “Econometric Policy Evaluation: A Critique,” pp. 19–46 in Karl Bruner and Alan H. Meltzer, eds., The Phillips Curve and Labor Markets (Amsterdam: North-Holland, 1976).

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10 Friedman, “The Methodology of Positive Economics,” p. 8.

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12 Donald N. McCloskey, “The Rhetoric of Economics,” Journal of Economic Literature 21, no. 2 (June 1983), pp. 481–517; and The Rhetoric of Economics (Madison: University of Wisconsin Press, 1985).

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14 Finn E. Kydland and Edward C. Prescott, “Time to Build and Aggregate Fluctuations,” Econometrica 50 (November 1982), pp. 1345–70. See also Edward C. Prescott, “Theory Ahead of Business Cycle Measurement,” Federal Reserve Bank of Minneapolis Quarterly Review 10 (Fall 1986), pp. 9–22; and Thomas Cooley and Edward C. Prescott, “Economic Growth and Business Cycles,” in Thomas Cooley, ed., Frontiers of Business Cycle Research (Princeton, NJ: Princeton University Press, 1995), pp. 1–38, for a further discussion of calibration of macro models.

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3 Mises, Human Action, p. 258.

4 Mises, Human Action, p. 248.

5 Murray N. Rothbard, Man, Economy, and State, with Power and Market, Scholar's ed. (Auburn, AL: Ludwig von Mises Institute, 2004), Chapter 5.

6 Ronald H. Coase, "The Problem of Social Cost," Journal of Law & Economics,

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7 Douglass C. North, Institutions, Institutional Change, and Economic Performance (Cambridge: Cambridge University Press, 1990), p. 3.

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21 Kohn, "Value and Exchange," pp. 308–9.

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24 Perhaps the prototypical example of this neoclassical utility maximization is found in C.E. Ferguson, Microeconomic Theory, rev. ed. (Homewood, IL: Richard D. Irwin, 1969). For more contemporary expositions of this paradigm, see David Besanko and Ronald R. Breautigam, Microeconomics, 2nd ed. (Hoboken, NJ: John Wiley & Sons, 2005); Edgar K. Browning and Mark A. Zupan, Microeconomics: Theory and Applications, 8th ed. (Hoboken, NJ: John Wiley & Sons, 2003); Robert H. Frank, Microeconomics and Behavior, 5th ed. (Boston, MA: McGraw-Hill Irwin, 2003); Robert S. Pindyck and Daniel L. Rubinfeld, Microeconomics, 6th ed. (Upper Saddle River, NJ: Prentice Hall, 2005); and Hal R. Varian, Intermediate Microeconomics: A Modern Approach, 6th ed. (New York: W.W. Norton, 2003). The point is, this is the standard starting point for the development of the equilibrium approach to economics.

25 Herbert A. Simon, "A Behavioral Model of Rational Choice," Quarterly Journal of Economics 69, no. 1 (February 1955), pp. 99–118.

26 See, for examples, Daniel Kahneman, "Maps of Bounded Rationality: Psychology for Behavioral Economics," American Economic Review 93, no. 5 (December 2003), pp. 1449–75; and Daniel Kahneman, Jack L. Knetsch, and Richard H. Thaler, "Anomalies: The Endowment Effect, Loss Aversion, and Status Quo Bias," Journal of Economic Perspectives 5, no. 1 (Winter 1991), pp. 193–206.

27 Smith, Vernon, "Microeconomic Systems as an Experimental Science," American Economic Review 72, no. 5 (December 1982), pp. 923–55; and "Economics in the Laboratory," Journal of Economic Perspectives 8, no. 1 (Winter 1994), pp. 113–31.

28 George A. Ackerlof, "The Missing Motivation in Macroeconomics," American Economic Review 97, no. 1 (March 2007), pp. 5–36.

29 Milton Friedman, Essays in Positive Economics (Chicago, IL: University of Chicago Press, 1953).

30 Mises, Human Action, pp. 18–19.

31 Mises, Human Action, p. 20.

4 Market clearing forces in the economy

1 Adam Smith, An Inquiry Into the Nature and Causes of the Wealth of Nations (New York: Modern Library, 1937), p. 55.

2 Alfred Marshall, Principles of Economics, 8th ed. (London: Macmillan, 1920), p. 367.

3 Marshall, Principles of Economics, p. 499.

4 Smith, The Wealth of Nations, p. 56.

5 Smith, The Wealth of Nations, pp. 57–8.

6 Marshall, Principles of Economics, p. 346.

7 Marshall, Principles of Economics, pp. 378–9.

8 William Stanley Jevons, The Theory of Political Economy, 2nd ed. (London: Macmillan and Co., 1879), pp. 179–80. Emphasis in original.

9 Carl Menger, Principles of Economics (New York: New York University Press, 1976), pp. 66–7.

10 See http://money.cnn.com/2003/06/17/pf/autos/convertibles, accessed October 21, 2008.

11 See www.autoblog.com/2008/10/13/gm-closing-wisconsin-suv-plant-early-doors-closed-by-end-of-yea/, accessed October 21, 2008.

12 www.nytimes.com/2006/01/05/international/americas/05sharks.html, accessed October 21, 2008. From that article: "If you go to any reef around the world, except those that are really protected, the sharks are gone," said Ransom Myers, a marine biologist at Dalhousie University in Halifax, Nova Scotia. "Their value is so great that completely harmless sharks, like whale sharks, are killed, for their fins."

13 These concepts of equilibrium are discussed in more detail in Randall G. Holcombe, Entrepreneurship and Economic Progress (London: Routledge, 2007), Chapter 4.

14 Friedrich A. Hayek, "The Use of Knowledge in Society," American Economic Review 35 (1945), pp. 519–30.

15 Israel Kirzner, Competition and Entrepreneurship (Chicago, IL: University of Chicago Press, 1973).

16 Kirzner, Competition and Entrepreneurship.

5 The innovative nature of firms

1 Ronald Coase, "The Nature of the Firm." Economica 4 (1937), pp. 386–405.

2 Michael C. Jensen and William H. Meckling, "The Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure," Journal of Financial Economics 3 (1976), pp. 305–60. See also Steven N.S. Cheung, "The Contractual Nature of the Firm," Journal of Law and Economics 26, no. 1 (April 1983), pp. 1–22.

3 Brian Loasby, "Explaining Firms," in Nicolai J. Foss and Peter G. Klein, eds., Entrepreneurship and the Firm (Cheltenham: Edward Elgar, 2002).

4 The role of the firm's managers in ensuring the efficient use of inputs is discussed by Armen A. Alchian and Harold Demsetz, "Production, Information Costs, and Economic Organization," American Economic Review 62 (1972), pp. 777–95.

5 Pierre Garrouste, "Knowledge: A Challenge for the Austrian Theory of the Firm," in Nicolai J. Foss and Peter G. Klein, eds., Entrepreneurship and the Firm (Cheltenham: Edward Elgar, 2002), discusses the interaction between knowledge and incentive problems in the theory of the firm.

6 The distinction between managerial and entrepreneurial theories of the firm is discussed by Donald J. Boudreaux and Randall G. Holcombe, "The Coasian and Knightian Theories of the Firm," Managerial and Decision Economics 10 (1989), pp. 147–54.

7 When I took my first economics course, the textbook, Campbell R. McConnell, Economics: Principles, Problems, and Policies, 3rd ed. (New York: McGraw-Hill, 1966), pp. 24–5, said there are four factors of production: land, labor, capital, and entrepreneurship. By the time I got to graduate school in economics, all books, articles, etc., said $Q = f(K,L)$, and there were only two factors of production left: labor and capital. I think it is fair to ask what happened to those other two factors of production!

8 Michael E. Porter, Competitive Strategy (New York: Free Press, 1980).

9 Institutions matter, and as William J. Baumol, "Entrepreneurship: Productive, Unproductive, and Destructive," Journal of Political Economy 98, no. 5, Part 1 (October 1990), pp. 893–921, notes, welfare-maximizing behavior on the part of firms holds only when institutions allow for the free exchange and protection of property rights, and do not constrain how individuals may allocate the resources they own.

10 Robert S. Pindyck and Daniel L. Rubinfeld, Microeconomics, 6th ed. (Upper Saddle River, NJ: Prentice Hall, 2005), p. 265.

11 Pindyck and Rubinfeld, Microeconomics, p. 282. Emphasis (italics and boldface) in this and all other quotations that follow are in the original quotation.

12 David Besanko and Ronald R. Breautigam, Microeconomics, 2nd ed. (Hoboken, NJ: John Wiley & Sons, 2005), p. 305.

13 Pindyck and Rubinfeld, Microeconomics, p. 283.

14 Pindyck and Rubinfeld, Microeconomics, p. 285.

15 Besanko and Breautigam, Microeconomics, p. 335.

16 Besanko and Breautigam, Microeconomics, p. 414.

17 Besanko and Breautigam, Microeconomics, p. 412.

18 Besanko and Breautigam, Microeconomics, p. 351.

19 Hal R. Varian, Intermediate Microeconomics: A Modern Approach, 6th ed. (New York: W.W. Norton, 2003), p. 345.

20 Jeffrey M. Perloff, Microeconomics, 3rd ed. (Boston: Pearson Addison Wesley, 2004), p. 267.

21 Joseph A. Schumpeter, Capitalism, Socialism, and Democracy, 2nd ed. (London: George Allen & Unwin, 1947), p. 82.

22 Pindyck and Rubinfeld, Microeconomics, p. 439.

23 Perloff, Microeconomics, p. 470.

24 Varian, Intermediate Microeconomics, p. 454.

25 Andreu Mas-Colell, Michael Dennis Whinston and Jerry R. Green, Microeconomic Theory (New York: Oxford University Press, 1995), p. 396.

26 Edgar K. Browning and Mark A. Zupan, Microeconomics: Theory and Applications, 8th ed. (Hoboken, NJ: John Wiley & Sons, 2003), p. 314.

27 Robert H. Frank, Microeconomics and Behavior, 5th ed. (Boston: McGraw-Hill Irwin, 2003), p. 507.

28 Pindyck and Rubinfeld, Microeconomics, p. 359.

29 Varian, Intermediate Microeconomics, p. 426.

30 Frank, Microeconomics and Behavior, p. 406.

31 Although this is not how neoclassical economics describes competition, it is consistent with the view Israel Kirzner takes on competition in Competition and Entrepreneurship (Chicago, IL: University of Chicago Press, 1973).

32 Murray N. Rothbard, Man, Economy, and State, with Power and Market, Scholar's ed. (Auburn, AL: Ludwig von Mises Institute, 2004), p. 729. See also his extended discussion of monopolistic competition, pp. 720–38.

33 Rothbard, Man, Economy, and State, pp. 987–8. See also pp. 976–88 for a longer discussion.

34 Friedrich A. Hayek, The Constitution of Liberty (Chicago, IL: University of Chicago Press, 1960), pp. 42, 44.

35 Dominick T. Armentano, Antitrust and Monopoly: Anatomy of a Policy Failure, 2nd ed. (New York: Holmes and Meier, 1990), p. 27.

36 Frank M. Machovec, Perfect Competition and the Transformation of Economics (London: Routledge, 1995), p. 185.

37 Mises, Human Action, p. 378.

38 Kirzner, Competition and Entrepreneurship, p. 115. See also the longer discussion on pp. 112–25.

39 Kirzner, Competition and Entrepreneurship, p. 115.

40 Kirzner, Competition and Entrepreneurship, p. 116.

41 Kirzner, Competition and Entrepreneurship, p. 127.

42 This theory was introduced by Joan Robinson, The Economics of Imperfect Competition (London: Macmillan, 1933); and Edward Chamberlin, The Theory of Monopolistic Competition (Cambridge, MA: Harvard University Press, 1933).

43 Peter G. Klein, "Entrepreneurship and Corporate Governance," Quarterly Journal of Austrian Economics 2, no. 2 (1999), pp. 19–42.

44 David Besanko, David Dranove, Mark Shanley and Scott Schaefer, Economics of Strategy, 3rd ed. (Hoboken, NJ: John Wiley & Sons, 2004).

45 Besanko and Breautigam, Microeconomics, p. 335.

46 Besanko et al., Economics of Strategy, p. 424.

47 Carl Shapiro and Hal R. Varian, Information Rules: A Strategic Guide to the Network Economy (Boston, MA: Harvard Business School Press, 1999).

6 The role of firms in the market

1 Frederic E. Sautet, An Entrepreneurial Theory of the Firm (London: Routledge, 2000).

2 Michael C. Jensen and William H. Meckling, "The Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure," Journal of Financial Economics 3 (1976), pp. 305–60.

3 Ronald H. Coase, "The Nature of the Firm," Economica 4 (1937), pp. 386–405; Armen A. Alchian and Harold Demsetz, "Production, Information Costs, and Economic Organization," American Economic Review 62 (1972), pp. 777–95; and Oliver E. Williamson, The Economic Institutions of Capitalism. (New York: Free Press, 1985).

4 Edith Penrose, The Theory of the Growth of the Firm (Oxford: Oxford University Press, 1959).

5 David A. Harper, "Institutional Conditions for Entrepreneurship," in Peter J. Boettke and Sanford Ideka, eds., Advances in Austrian Economics (Stamford, CT: JAI Press, 1998) pp. 241–75; and Entrepreneurship and the Market Process: An Enquiry into the Growth of Knowledge (London and New York: Routledge, 1996).

6 Israel Kirzner, Competition and Entrepreneurship (Chicago, IL: University of Chicago Press, 1973) defines the entrepreneurial act as noticing a profit opportunity, and states that acting on it goes beyond the basic entrepreneurial act because it uses other factors of production. So the concept of entrepreneurship here is in the spirit of Kirzner, but goes beyond his definition. The reasoning is that the profit only comes after the opportunity is acted upon.

7 Richard N. Langlois, "The Entrepreneurial Theory of the Firm and the Theory of the Entrepreneurial Firm," Journal of Management Studies 44, no. 7 (November 2007), pp. 1107–24.

8 Friedrich A. Hayek, “The Use of Knowledge in Society,” American Economic Review 35 (1945), pp. 521–2.

9 Pierre Desrochers, “Geographical Proximity and the Transmission of Tacit Knowledge,” Review of Austrian Economics 14, no. 1 (2001), pp. 25–46. The purpose of this paragraph is not to argue that Hayek had this transmission mechanism for knowledge in mind, but rather to clarify that as the term is used here, tacit knowledge is knowledge that is not easily articulated, but that can be absorbed when others observe it being used. Hayek was considering a different issue, which was the way knowledge is transmitted through the price system. Hayek, “The Use of Knowledge in Society,” p. 526, says, “We must look at the price system as such a mechanism for communicating information if we want to understand its real function” but he was not arguing that the price system is the only method for communicating information, or that it is the best mechanism in every instance.

10 This argument applies to other types of knowledge aswell, such as knowledge the firm holds as trade secrets. However, if the trade secret is easily explained, firms might share the knowledge with others and protect it with a nondisclosure agreement. If the knowledge is tacit, there would be more reason to place those with whom it was to be shared within the firm’s boundaries, so that the knowledge could be transmitted more easily to them. Meanwhile, the boundary of the firm prevents tacit knowledge from being passed on to others outside the firm, who are not in such close proximity.

11 Joseph A. Schumpeter, The Theory of Economic Development (Cambridge, MA: Harvard University Press, 1934).

12 This idea is developed further in Randall G. Holcombe, “The Origins of Entrepreneurial Opportunities,” Review of Austrian Economics 16, no. 1 (2003), pp. 25–43, and in Chapter 6 of Holcombe, Entrepreneurship and Economic Progress (New York: Routledge, 2007).

13 Kirzner, Competition and Entrepreneurship, p. 127.

14 In personal discussion with Kirzner, he has told me that he believes that the entrepreneurs he depicts and those depicted by Schumpeter are doing the same things, and that there is no difference between Kirznerian and Schumpeterian entrepreneurship. However, Kirzner does make a valid point in the quoted passage, which he discusses at greater length in his 1973 book, so despite Kirzner’s objection to this terminology there does seem to be some value in using it.

15 These two types of entrepreneurship are discussed in more detail in Randall G. Holcombe, “Entrepreneurship and Economic Growth,” Quarterly Journal of Austrian Economics 1, no. 2 (Summer 1998), pp. 45–62; and in Holcombe, Entrepreneurship and Economic Progress.

16 Different concepts of equilibrium are discussed in Randall G. Holcombe, “Equilibrium Versus the Invisible Hand,” Review of Austrian Economics 12,

no. 2 (1999), pp. 227–43, and Holcombe, "Does the Invisible Hand Hold or Lead? Market Adjustment in an Entrepreneurial Economy," Review of Austrian Economics 19, nos. 2/3 (June 2006), pp. 189–201.

17 Peter Lewin, "The Firm, Money, and Economic Calculation," American Journal of Economics and Sociology 57, no. 4 (October 1998), pp. 499–512.

18 Even here some caveats are in order. The product may have externalities that are not included in its cost, and the value of the product is measured as the value it has to its purchasers. Some products that have value measured this way, such as cigarettes, are nonetheless viewed as detrimental to economic welfare.

19 Smith, The Wealth of Nations (New York: Modern Library, 1937), p. 9.

20 Friedrich A. Hayek, "The Use of Knowledge in Society," American Economic Review 35 (1945), pp. 519–30.

21 See Pierre Desrochers, "Geographical Proximity and the Transmission of Tacit Knowledge," Review of Austrian Economics 14, no. 1 (2001), pp. 25–46, for an insightful discussion of the way that agglomeration economies arise because of the ability to absorb this tacit knowledge held by others in the observers' proximity.

22 This idea is explored further in Randall G. Holcombe, "Firms as Knowledge Repositories," Review of Austrian Economics (forthcoming).

23 This is consistent with Edith Penrose, The Theory of the Growth of the Firm (Oxford: Oxford University Press, 1959). See also Frederic E. Sautet, An Entrepreneurial Theory of the Firm (London: Routledge, 2000), Chapter 3.

24 Edith Penrose, The Theory of the Growth of the Firm.

25 G.B. Richardson, "The Organisation of Industry." The Economic Journal 82, no. 327 (September 1972), p. 895. See also G.B. Richardson, "Adam Smith on Competition and Increasing Returns," in Andrew S. Skinner and Thomas Wilson, eds., Essays on Adam Smith (Oxford: Clarendon Press, 1975), pp. 350–60.

26 Richard R. Nelson and Sidney G. Winter, An Evolutionary Theory of Economic Change (Cambridge, MA: Belknap, 1982).

27 Philippe Dulbecco and Pierre Garrouste, "Towards an Austrian Theory of the Firm," Review of Austrian Economics 12, no. 1 (1999), p. 57.

28 Richard Adelstein, "Knowledge and Power in the Mechanical Firm: Planning for Profit in the Austrian Perspective," Review of Austrian Economics 18, no. 1 (2005), p. 59.

29 Nicolai J. Foss, "Austrian Insights and the Theory of the Firm," Advances in Austrian Economics 4 (1997), pp. 175–98; and "The Use of Knowledge in Firms," Journal of Institutional and Theoretical Economics 155, no. 3 (October 1999), pp. 458–86.

30 Ulrich Witt, "Do Entrepreneurs Need Firms? A Contribution to a Missing Chapter in Austrian Economics," Review of Austrian Economics 11, nos. 1/2 (1999), pp. 99–109.

31 This idea is developed in Peter Lewin and Howard Baetjer, "The Capital-based Theory of the Firm," Review of Austrian Economics 24, no. 4 (December 2011), pp. 335–54. They trace the idea to Ludwig M. Lachmann, Capital and Its Structure (London: G. Bell and Sons, 1956). See also Nicolai J. Foss, and Peter G. Klein, Organizing Entrepreneurial Judgment: A New Approach to the Firm (Cambridge: Cambridge University Press, 2012), who argue that entrepreneurs work within firms and that a complete theory of entrepreneurship must include the structure of the firm within which entrepreneurship takes place. The heterogeneous capital under control of the firm is a key factor behind the entrepreneurial judgment that determines the firm's activities.

32 Richard N. Langlois and Paul L. Robertson, Firms, Markets, and Economic Change: A Dynamic Theory of Business Institutions (London: Routledge, 1995), p. 43, (italics in the original).

33 Peter Lewin and Steven E. Phelan, "An Austrian Theory of the Firm," Review of Austrian Economics 13, no. 1 (2000), p. 70.

34 Lewin and Phelan. "An Austrian Theory of the Firm," p. 74.

35 Langlois and Robertson, Firms, Markets, and Economic Change, Chapter 7.

36 Ivan Pongracic, Jr., Employees and Entrepreneurship: Spontaneity and Coordination in Non-hierarchical Business Organization (Aldershot: Edward Elgar, 2009).

37 Armen A. Alchian and Harold Demsetz, "Production, Information Costs, and Economic Organization," American Economic Review 62 (1972), pp. 777–95.

38 Friedrich A. Hayek, "Economics and Knowledge," Economica n.s. 4 (1937), pp. 33–54, reprinted in Friedrich A. Hayek, Individualism and Economic Order (London: Routledge and Kegan Paul, 1949).

39 Peter Lewin, Capital in Disequilibrium: The Role of Capital in a Changing World (London: Routledge, 1999) pp. 130–2; and Ludwig M. Lachmann, Capital and Its Structure,pp.79–84.

40 Michael C. Jensen and William H. Meckling, "The Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure," Journal of Financial Economics 3 (1976), pp. 305–60.

41 Penrose, The Theory of the Growth of the Firm, p. 115.

7 The trajectory of the economy: Growth and progress

1 Oliver Jean Blanchard and Stanley Fischer, Lectures on Macroeconomics (Cambridge, MA: MIT Press, 1989), pp. 1–2.

2 Robert E. Lucas, “On the Mechanics of Economic Development,” Journal of Monetary Economics 22 (1988), pp. 3–42.

3 Paul A. Samuelson, Economics, 9th ed. (New York: McGraw-Hill, 1973), p. 883.

4 Within a general equilibrium framework, if the economy is in equilibrium and then a Schumpeterian entrepreneur disturbs that equilibrium, producing creative destruction, the economy moves from a Pareto optimum to a situation that is not Pareto optimal, which generates economic progress and iswelfare-enhancing. The caveat here is that the economy may not be resting at a Pareto optimum, so one cannot know whether the change moves away from Pareto optimality. Because Pareto optimality is non-observable, one can never, in fact, say that an actual economy is moving closer to, or further from, a Pareto optimum. Chapter 8 discusses the welfare implications of this in more detail.

5 Robert E. Lucas, Jr., “An Equilibrium Model of the Business Cycle,” Journal of Political Economy 83, no. 6 (December 1975), pp. 1113–44; and Lucas, “On the Mechanics of Economic Development,” Journal of Monetary Economics 22 (1988), pp. 3–42.

6 Joseph A. Schumpeter, Business Cycles: A Theoretical, Historical, and Statistical Analysis of the Capitalist Process (New York: McGraw-Hill, 1939).

7 See, for example, Paul M. Romer, “Endogenous Technological Change,” Journal of Political Economy 98, no. 5, part 2 (October 1990), pp. S71–S102; and Paul M. Romer, “Increasing Returns and Long-run Growth,” Journal of Political Economy 94 (1986), pp. 1002–37.

8 Schumpeter, The Theory of Economic Development.

9 William J. Baumol, The Free-Market Innovation Machine: Analyzing the Growth Miracle of Capitalism (Princeton, NJ: Princeton University Press, 2002).

10 William J. Baumol, Robert E. Litan, and Carl J. Schramm, Good Capitalism, Bad Capitalism, and the Economics of Growth and Prosperity (New Haven: Yale University Press, 2007).

11 I wrote to Baumol about the seeming inconsistency in his two books, and he responded without acknowledging the inconsistency. As this was personal correspondence, not published research, it might indicate that he was kind enough to reply to me but did not give the reply a great deal of thought. At any rate, he did not have a ready explanation for the inconsistency.

12 Kirzner, Competition and Entrepreneurship.

13 This is the motivation for James M. Buchanan’s Cost and Choice: An Inquiry in Economic Theory (Chicago, IL: Markham, 1969).

14 This is reported by Pierre Desrochers, “Geographical Proximity and the Transmission of Tacit Knowledge,” Review of Austrian Economics 14, no. 1 (2001), pp. 25–46.

15 Richard R. Nelson and Sidney G. Winter. An Evolutionary Theory of Economic Change (Cambridge, MA: Belknap, 1982), pp. 255–6.

16 W. Brian Arthur, "Competing Technologies, Increasing Returns, and Lock-In by Historical Events," Economic Journal 99 (March 1989), pp. 116–31.

17 Nelson and Winter, An Evolutionary Theory, pp. 255–6.

8 Economic welfare: Theory and policy

1 Adam Smith, An Inquiry Into the Nature and Causes of the Wealth of Nations (New York: Modern Library, 1937 [orig. 1776]), p. 3.

2 A.C. Pigou, The Economics of Welfare, 4th ed. (London: Macmillan, 1962 [1st ed. 1920]), p. 12.

3 Kenneth J. Arrow and Gerard Debreu, "Existence of an Equilibrium for a Competitive Economy," Econometrica 22, no. 3 (July 1954), pp. 265–90.

4 The fundamentals of welfare economics are well explained by Francis M. Bator, "The Simple Analytics of Welfare Maximization," American Economic Review 67 (1957), pp. 22–59; and J. de V. Graaf, Theoretical Welfare Economics (Cambridge: Cambridge University Press, 1957).

5 The concept of market failure is explained by Francis Bator, "The Anatomy of Market Failure," Quarterly Journal of Economics 72, no. 3 (August 1958), pp. 351–79. It may be possible to further increase social welfare if redistribution would lower the utility of those from whom resources were transferred less than it would add to the utility of the recipients of those resources, following Paul A. Samuelson, "Social Indifference Curves," Quarterly Journal of Economics 73, no. 1 (February 1956), pp. 1–22.

6 See, for support of this idea, Joel Mokyr, The Lever of Riches (Oxford: Oxford University Press, 1990); and David S. Landes, The Wealth and Poverty of Nations: Why Some Are So Rich and Some So Poor (New York: W.W. Norton, 1998).

7 See W. Michael Cox and Richard Alm, Myths of Rich and Poor: Why We're Better Off Than We Think (New York: Basic Books, 1999) for support for this idea.

8 Vernon L. Smith, "Economic Theory and Its Discontents," American Economic Review 64, no. 2 (May 1974), p. 321.

9 John R. Hicks, Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory (Oxford: Clarendon Press, 1939).

10 Paul A. Samuelson, Foundations of Economic Analysis (Cambridge: Harvard University Press, 1947).

11 Meir Kohn, "Value and Exchange," Cato Journal 24, no. 3 (Fall 2004), p. 306.

12 Smith, The Wealth of Nations, p. 423.

13 Milton Friedman, Essays in Positive Economics (Chicago, IL: University of Chicago Press, 1953).

14 See, for examples, Imre Lakatos, The Methodology of Scientific Research Programmes, vol. 1 (Cambridge: Cambridge University Press, 1978); Bruce J. Caldwell, Beyond Positivism: Economic Method in the Twentieth Century (London: George Allen & Unwin, 1982); Donald N. McCloskey, The Rhetoric of Economics (Madison: University of Wisconsin Press, 1985); and Randall G. Holcombe, Economic Models and Methodology (New York: Greenwood, 1989).

15 For examples, Daniel Kahneman, Jack L. Knetsch, and Richard H. Thaler, "Anomalies: The Endowment Effect, Loss Aversion, and Status Quo Bias," Journal of Economic Perspectives 5, no. 1 (Winter 1991), pp. 193–206; Daniel Kahneman, "Maps of Bounded Rationality: Psychology for Behavioral Economics," American Economic Review 93, no. 5 (December 2003), pp. 1449–75; and Smith, "Economic Theory and Its Discontents," pp. 320–22.

16 R.G. Lipsey and Kelvin Lancaster, "The General Theory of Second Best," Review of Economic Studies 24, no. 1 (1956), pp. 11–12.

17 Second-hand smoke can be considered an external cost, of course, but can be limited by clearly defined property rights to determine where someone can smoke, and at any rate is more limited than the healthcare costs described in the next sentence. The externality of second-hand smoke can spread throughout an entire room. The externality generated by publicly funded healthcare can spread throughout an entire nation. And, if transaction costswithin a room are low, then following Ronald H. Coase, "The Problem of Social Cost," Journal of Law and Economics, 3 (1960), pp. 1–44, bargaining among individuals can internalize the externality of second-hand smoke.

18 Paul A. Samuelson, "Social Indifference Curves," Quarterly Journal of Economics 73, no. 1 (February 1956), pp. 1–22.

19 Jeffrey M. Herbener, "The Pareto Rule and Welfare Economics," Review of Austrian Economics 10, no. 1 (1997), pp. 79–106.

20 This is the idea behind revealed preference, and the idea behind what Murray N. Rothbard, "Toward a Reconstruction of Utility and Welfare Economics," in Mary Sennholz, ed., On Freedom and Free Enterprise: Essays in Honor of Ludwig von Mises (Princeton, NJ: Van Nostrand, 1956), pp. 224–62, calls demonstrated preference. While there is some similarity in the ideas also, criticisms of Rothbard by Roy E. Cordato, Welfare Economics and Externalities in an Open-Ended Universe: A Modern Austrian Perspective (Boston, MA: Kluwer Academic Publishers, 1992); David L. Prychitko, "Formalism in Austrian School Welfare Economics: Another Pretense of Knowledge?" Critical Review 7, no. 4 (Spring 1993), pp. 567–92; and Bryan Caplan, "The Austrian Search for Realistic Foundations," Southern Economic Journal 65, no. 4 (April 1999), pp. 823–38, are well-taken.

21 Joseph A. Schumpeter, The Theory of Economic Development (Cambridge, MA: Harvard University Press, 1934), p. 154.

22 A good taxonomy – and a good data source – is found in James Gwartney and Robert Lawson, Economic Freedom of the World: 2007 Report (Vancouver, BC: Fraser Institute, 2007).

23 Kirzner, Competition and Entrepreneurship, (Chicago, IL: University of Chicago Press, 1973) would clearly separate the function of the entrepreneur from that of the investor, but Murray N. Rothbard, Man, Economy, and State, with Power and Market, scholar's ed. (Auburn, AL: Ludwig von Mises Institute, 2004), p. 64, indicates that entrepreneurship consists not only of the entrepreneur spotting a profit opportunity, but also acting on it so that he has "invested accordingly."

24 James M. Buchanan, Cost and Choice: An Inquiry in Economic Theory (Chicago, IL: Markham, 1969).

25 But see Dominick T. Armentano, The Myths of Antitrust: Economic Theory and Legal Cases (New Rochelle, NY: Arlington House, 1972) for a critique of antitrust policy.

26 Paul A. Samuelson, "The Pure Theory of Public Expenditure," Review of Economics and Statistics 36 (November 1954), pp. 387–9; and "A Diagrammatic Exposition of a Theory of Public Expenditure," Review of Economics and Statistics 37 (November 1955), pp. 350–6. For a critique of public goods theory, see Randall G. Holcombe, "ATheory of the Theory of Public Goods," Review of Austrian Economics 10, no. 1 (1997), pp. 1–22; and "Why Does Government Produce National Defense?" Public Choice 137, nos. 1/2 (October 2008), pp. 11–19.

27 Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," Journal of Political Economy 58, no. 3 (June 1950), pp. 211–22.

28 Richard R. Nelson and Sidney G. Winter, An Evolutionary Theory of Economic Change (Cambridge, MA: Belknap, 1982).

29 Eric D. Beinhocker, The Origin of Wealth: Evolution, Complexity, and the Radical Remaking of Economics (Boston, MA: Harvard Business School Press, 2006).

30 Ronald H. Coase, "The Problem of Social Cost," Journal of Law and Economics, 3 (1960), pp. 1–44.

31 See, for example, Amartya Sen, Inequality Reexamined (Cambridge, MA: Harvard University Press, 1992).

32 Bruno S. Frey and Alois Stutzer, Happiness and Economics: How the Economy and Institutions Affect Human Well-being (Princeton, NJ: Princeton University Press, 2001).

9 Rent seeking in a creative economy

1 Anne O. Krueger, "The Political Economy of the Rent-Seeking Society," American Economic Review 64 (June 1974), pp. 291–303.

2 Gordon Tullock, "The Welfare Cost of Tariffs, Monopolies, and Theft," Western Economic Journal 5 (June 1967), pp. 224–32.

3 William J. Baumol, Entrepreneurship, Management, and the Structure of Payoffs (Cambridge, MA: MIT Press, 1993); and "Entrepreneurship: Productive, Unproductive, and Destructive," Journal of Political Economy 98, no. 5, Part 1 (October 1990), pp. 893–921.

4 Douglass C. North, John Joseph Wallis and Barry R. Weingast, Violence and Social Orders: A Conceptual Framework for Interpreting Recorded History (Cambridge: Cambridge University Press, 2009).

5 Mancur Olson, The Logic of Collective Action: Public Goods and the Theory of Groups (Cambridge, MA: Harvard University Press, 1971).

6 Mancur Olson, The Rise and Decline of Nations (New Haven: Yale University Press, 1982).

7 Dan Usher, The Welfare Economics of Markets, Voting, and Predation (Ann Arbor: University of Michigan Press, 1992), explains how, in democratic government, interest groups use the political process for predatory purposes, acting as stationary bandits.

8 Thomas Hobbes, Leviathan (New York: E.P. Dutton, 1950 [orig. 1651]).

9 Jean Jacques Rousseau, The Social Contract, Or Principles of Political Right. Translated by G.D.H. Cole. www.constitution.org/jjr/socon.htm. 1762. Note that in this quotation "people" is expressed as singular rather than plural. Although this is a translation, the translation accentuates the idea of the general will of one people, rather than a group of individuals.

10 John Locke, Two Treatises on Government (Cambridge: Cambridge University Press, 1967 [orig. 1690]).

11 John Rawls, A Theory of Justice (Cambridge, MA: Belknap, 1971).

12 James M. Buchanan, The Limits of Liberty: Between Anarchy and Leviathan (Chicago, IL: University of Chicago Press, 1975).

13 This idea is discussed and defended in more detail in Murray Edelman, The Symbolic Uses of Politics (Urbana: University of Illinois Press, 1964). Not everybody buys in to the idea, however. For one who does not, see Murray N. Rothbard, The Ethics of Liberty (Atlantic Heights, NJ: Humanities Press, 1982).

14 Meanwhile, politicians have an incentive to look for ways to cater to constituents who want to escape predatory policies, as Peter H. Aaronson, "Electoral

Competition and Entrepreneurship," Advances in Austrian Economics, 5 (1996), pp. 183–215, notes.

15 For a discussion, see Randall G. Holcombe and James D. Gwartney, "Unions, Economic Freedom, and Growth," Cato Journal 30, no. 1 (Winter 2010), pp. 1–22.

16 For examples, David S. Landes, The Wealth and Poverty of Nations: Why Some Are So Rich and Some So Poor (New York: W.W. Norton, 1998) and Joel Mokyr, The Gifts of Athena: Historical Origins of the Market Economy (Princeton, NJ: Princeton University Press, 2002).

17 For examples, see Milton Friedman, Capitalism and Freedom (Chicago, IL: University of Chicago Press, 1962); Friedrich Hayek, The Road to Serfdom (Chicago, IL: University of Chicago Press, 1944); and Joseph Schumpeter, Capitalism, Socialism, and Democracy, 2nd ed., (London: George Allen & Unwin, 1947).

18 James Gwartney and Robert Lawson, Economic Freedom of the World: 2007 Report (Vancouver, BC: Fraser Institute, 2007).

19 One study that surveys much of this literature is Niclas Berggren, "The Benefits of Economic Freedom: A Survey," The Independent Review 8, no. 2 (Fall 2003), pp. 193–211. Much research has been done since this survey article was published that continues to support the merits of market institutions.

20 Leon Walras, Elements of Pure Economics; or The Theory of Social Wealth, translated by William Jaffe (Homewood, IL: R.D. Irwin, 1954 [orig. 1874]).

21 John R. Hicks, Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory (Oxford: Clarendon Press, 1939); and Paul Anthony Samuelson, Foundations of Economic Analysis (Cambridge, MA: Harvard University Press, 1947).

22 Ludwig von Mises, Human Action, scholar's ed., (Auburn, AL: Ludwig von Mises Institute, 1998); and Murray N. Rothbard, Man, Economy, and State, with Power and Market, scholar's ed., (Auburn, AL: Ludwig von Mises Institute, 2004).

23 Joseph A. Schumpeter, The Theory of Economic Development (Cambridge, MA: Harvard University Press, 1934); and Schumpeter, Capitalism, Socialism, and Democracy.

24 This idea is pursued in more detail in Randall G. Holcombe, "The Origins of Entrepreneurial Opportunities," Review of Austrian Economics 16, no. 1 (2003), pp. 25–43; and Randall G. Holcombe, Entrepreneurship and Economic Progress (London: Routledge, 2007).

25 John Stuart Mill, Principles of Political Economy (London: John W. Parker, West Strand, 1848), Book II, Chapter I.

26 www.airliners.net/aircraft-data/stats.main?id=107, accessed January 7, 2011.

27 http://agrilife.org/today/2010/12/22/new-wheat-variety/, accessed July 27, 2012.

10 The coordination of economic activity

1 See Robert E. Lucas, Jr., “An Equilibrium Model of the Business Cycle,” Journal of Political Economy 83, no. 6 (December 1975), pp. 1113–44, for the pioneering work in this literature.

2 Leon Walras, Elements of Pure Economics; or The Theory of Social Wealth, translated by William Jaffe (Homewood, IL: R.D. Irwin, 1954 [orig. 1874]).

3 C.E. Ferguson, Microeconomic Theory, rev. ed. (Homewood, IL: Richard D. Irwin, 1969).

4 Israel M. Kirzner, Competition and Entrepreneurship (Chicago, IL: University of Chicago Press, 1973).

5 This view is along the lines of Friedrich A. Hayek, “Economics and Knowledge,” Economica n.s. 4 (1937), pp. 33–54, reprinted in F.A. Hayek, Individualism and Economic Order (London: Routledge and Kegan Paul, 1949); Frank H. Hahn, Equilibrium and Macroeconomics (Cambridge: Cambridge University Press, 1984); and Peter Lewin, “Hayekian Equilibrium and Change,” Journal of Economic Methodology 4 (1997), pp. 245–66.

6 Hernando de Soto, The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else (New York: Basic Books, 2000).

7 Ronald H. Coase, “The Problem of Social Cost,” Journal of Law & Economics, 3 (1960), pp. 1–44.

8 Kenneth J. Arrow and Gerard Debreu, “Existence of an Equilibrium for a Competitive Economy,” Econometrica 22, no. 3 (July 1954), pp. 265–90.

9 Milton Friedman, “The Role of Monetary Policy,” American Economic Review 58, no. 1 (March 1968), pp. 1–17.

10 Ludwig von Mises, The Theory of Money and Credit (New Haven, CT: Yale University Press, 1953 [orig. in German, 1912]).

11 Friedrich A. Hayek, Monetary Theory and the Trade Cycle (New York: Augustus M. Kelley, 1966 [orig. 1933]); Prices and Production, 2nd ed. (New York: Augustus M. Kelley, 1935); and The Pure Theory of Capital (Chicago, IL: University of Chicago Press, 1941). Hayek’s ideas are explained in a contemporary framework, and with reference to Keynesian and monetarist ideas, by Roger W. Garrison, Time and Money: The Macroeconomics of Capital Structure (London: Routledge, 2001).

12 This point is made by George J. Mailath, Andrew Postlewaite and Larry Samuelson, “Sunk Investments Lead to Unpredictable Prices,” American Economic Review 94, no. 4 (September 2004), p. 896.

13 Steven Horwitz, Microfoundations and Macroeconomics: An Austrian Perspective (London: Routledge, 2000); and Roger W. Garrison, Time and Money: The Macroeconomics of Capital Structure (London: Routledge, 2001).

14 Irving Fisher, The Purchasing Power of Money: Its Determination and Relation to Credit, Interest, and Crises (New York: Macmillan, 1911).

15 John Maynard Keynes, The General Theory of Employment, Interest, and Money (New York: Harcourt Brace and Company, 1936).

16 Keynes, The General Theory, p. 161.

17 A model along these lines is presented by Ana Fostel and John Geanakoplos, "Leverage Cycles and the Anxious Economy," American Economic Review 98, no. 4 (September 2008), pp. 1211–44. They present facts related to emerging economies, but their framework is more general.

18 See, for examples, Axel Leijonhufvud, On Keynesian Economics and the Economics of Keynes (New York: Oxford University Press, 1968); and Robert W. Clower, "The Keynesian Counter-revolution: A Theoretical Appraisal," in F.H. Hahn and F. Brechling, eds., The Theory of Interest Rates (New York: Macmillan, 1965), Chapter 5, pp. 103–25.

19 See, for example, the work in Edmund S. Phelps et al., Microeconomic Foundations of Employment and Inflation Theory (New York: W.W. Norton & Company, 1970).

20 The pioneering article along these lines is Robert E. Lucas, Jr., "An Equilibrium Model of the Business Cycle," Journal of Political Economy 83, no. 6 (December 1975), pp. 1113–44.

21 Jonas D. Fisher, "The Dynamic Effects of Neutral and Investment-specific Technology Shocks," Journal of Political Economy 114, no. 3 (June 2006), pp. 413–51, presents a model in which there are shocks specific to investment and shocks that more generally impact the economy, going part-way toward Hayek's emphasis on investment. But Fisher has only one type of homogeneous investment in his model, so does not allow for the more complex malinvestment depicted in Hayek's framework.

22 Joseph A. Schumpeter, The Theory of Economic Development (Cambridge, MA: Harvard University Press, 1934), lays the foundation for the macroeconomic framework he presents in Business Cycles: A Theoretical, Historical, and Statistical Analysis of the Capitalist Process (New York: McGraw-Hill, 1939).

23 Gordon E. Moore, "Cramming More Components onto Integrated Circuits," Electronics 38, no. 8 (April 19, 1965).

24 See, for example, Etienne Wasmer and Phiippe Weil, "The Macroeconomics of Labor and Credit Market Imperfections," American Economic Review 94, no. 4 (September 2004), pp. 944–63, who argue that credit market frictions create labor instability, referencing Schumpeter.

11 Agglomeration economies and economic progress

1 For further discussion of the relationship between information and knowledge, see Randall G. Holcombe, "Information, Entrepreneurship, and Economic Progress," in Roger Koppl, ed., Advances in Austrian Economics: Austrian Economics and Entrepreneurial Studies 6 (Amsterdam: JAI, 2003), pp. 173–95.

2 Pierre Desrochers, "Geographical Proximity and the Transmission of Tacit Knowledge," Review of Austrian Economics 14, no. 1 (2001), pp. 25–46.

3 Friedrich Hayek, "The Use of Knowledge in Society," American Economic Review 35 (1945), pp. 521–22

4 Jobs said this in a BusinessWeek interview in 1998. This sounds a lot like John Kenneth Galbraith's dependence effect in The Affluent Society (New York: Houghton Mifflin, 1958).

5 Adam Smith, An Inquiry Into the Nature and Causes of the Wealth of Nations (New York: Modern Library, 1937 [orig. 1776]), p. 9.

6 Richard Florida, Cities and the Creative Class (New York: Routledge, 2005), p. 194.

7 Allyn Young, "Increasing Returns and Economic Progress," Economic Journal 38 (December 1928), pp. 527–42.

8 Nicholas Kaldor, "The Irrelevance of Equilibrium Economics," Economic Journal 82 (1972), pp. 1237–55.

9 Smith, The Wealth of Nations, pp. 4–5.

10 See Randall G. Holcombe, "Product Differentiation and Economic Progress," Quarterly Journal of Austrian Economics 12, no. 1 (Spring 2009), pp. 17–35, for a more detailed discussion.

11 The initial book laying the foundation for his ideas is Richard Florida, The Rise of the Creative Class: And How It's Transforming Work, Leisure, Community, & Everyday Life (New York: Basic Books, 2002). He has followed up this book with more work, including The Flight of the Creative Class (New York: Harper-Collins, 2005), and Cities and the Creative Class (New York: Routledge, 2005).

12 Florida, Cities and the Creative Class, pp. 3–4.

13 Randall G. Holcombe, "South Korea's Economic Future: Industrial Policy, or Economic Democracy?" Journal of Economic Behavior and Organization (forthcoming) analyzes the South Korean case in more detail, and concludes that South Korean industrial policy (like the similar Japanese policy) is not a model for sustainable economic growth.

14 The institutions summarized in James Gwartney and Robert Lawson, Economic Freedom of the World: 2007 Report (Vancouver, BC: Fraser Institute,

2007) provide a good guideline.

15 Joseph A. Schumpeter, Capitalism, Socialism, and Democracy, 4th ed. (London: George Allen & Unwin, 1954), p. 82.

16 Frederic Bastiat, “What Is Seen and What Is Not Seen,” in Selected Essays on Political Economy (Irving-on-Hudson, NY: Foundation for Economic Education, 1995).

17 William J. Baumol, “Entrepreneurship: Productive, Unproductive, and Destructive,” Journal of Political Economy 98, no. 5, part 1 (October 1990), pp. 893–921; and William J. Baumol, Entrepreneurship, Management, and the Structure of Payoffs (Cambridge, MA: MIT Press, 1993).

18 Mancur Olson, The Rise and Decline of Nations (New Haven, CT: Yale University Press, 1982).

19 See Bruce Bueno de Mesquita, Alastair Smith, Randolph M. Siverson, and James D. Morrow, The Logic of Political Survival (Cambridge, MA: MIT Press, 2003); and William A. Niskanen, Autocratic, Democratic, and Optimal Government (Cheltenham: Edward Elgar, 2003).

20 Robert L. Heilbroner, The Making of Economic Society (Englewood Cliffs, NJ: Prentice-Hall, 1962).

21 See Randall G. Holcombe and James D. Gwartney, “Unions, Economic Freedom, and Growth,” Cato Journal 30, no. 1 (Winter 2010), pp. 1–22, for a discussion of the United Auto Workers in this context, and also the impact of government-mandated union costs on the decline of other unionized industries in the United States and around the world.

12 The evolution of the economy: Differentiation, selection, replication

1 This work has its foundation in Robert M. Solow, “A Contribution to the Theory of Economic Growth,” Quarterly Journal of Economics 70 (1956), pp. 65–94. A seminal article in general equilibrium growth theory is Robert E. Lucas, Jr., “On the Mechanics of Economic Development,” Journal of Monetary Economics 22 (1988), pp. 3–42.

2 Alfred Marshall, Principles of Economics, 8th ed. (New York: Macmillan, 1920), p. 461.

3 Marshall, Principles of Economics, p. 50.

4 Marshall, Principles of Economics, p. 323.

5 Joseph A. Schumpeter, Capitalism, Socialism, and Democracy, 3rd ed. (London: George Allen & Unwin, 1950), p. 82.

6 See W. Brian Arthur, “Competing Technologies, Increasing Returns, and Lock-in by Historical Events,” Economic Journal 99 (March 1989), pp. 116–31;

Paul A. David, Technical Choice, Innovation, and Economic Growth (Cambridge: Cambridge University Press, 1975); and Paul A. David, "Clio and the Economics of QWERTY," American Economic Review 75, no. 2 (May 1985), pp. 332–7.

7 Israel M. Kirzner, Competition and Entrepreneurship (Chicago, IL: University of Chicago Press, 1973).

8 This is the theme of James M. Buchanan, Cost and Choice: An Inquiry in Economic Theory (Chicago, IL: Markham, 1969).

9 Richard R. Nelson and Sidney G. Winter, An Evolutionary Theory of Economic Change (Cambridge, MA: Belknap, 1982), p. 255.

10 Arthur, "Competing Technologies."

11 Armen A. Alchian, "Uncertainty, Evolution, and Economic Theory," Journal of Political Economy 58, no. 3 (June 1950), pp. 211–22.

12 Eric D. Beinhocker, The Origin of Wealth: Evolution, Complexity, and the Radical Remaking of Economics (Boston, MA: Harvard Business School Press, 2006).

13 These examples are illustrative, but note that there is a research program in evolutionary economics, and a Journal of Evolutionary Economics that was established in 1991.

14 Nelson and Winter, An Evolutionary Theory, p. 41.

15 Herbert A. Simon, "A Behavioral Model of Rational Choice," Quarterly Journal of Economics 69, no. 1 (February 1955), pp. 99–118; and Herbert A. Simon, "Theories of Decision Making in Economics," American Economic Review 49, no. 3 (June 1959), pp. 253–83.

16 Israel Kirzner, Competition and Entrepreneurship.

17 Adam Smith, An Inquiry Into the Nature and Causes of the Wealth of Nations (New York: Modern Library, 1937 [orig. 1776]), p. 9.

18 Friedrich Hayek, "The Use of Knowledge in Society," American Economic Review 35 (1945), pp. 519–30.

19 Fritz Machlup, "Marginal Analysis and Empirical Research," American Economic Review 36, no. 4 (September 1946), p. 535.

20 William J. Baumol, The Free-market Innovation Machine: Analyzing the Growth Miracle of Capitalism (Princeton, NJ: Princeton University Press, 2002).

21 See Randall G. Holcombe, "Information, Entrepreneurship, and Economic Progress," in Roger Koppl, ed., Advances in Austrian Economics: Austrian Economics and Entrepreneurial Studies 6 (Amsterdam: JAI, 2003), pp. 173–95; and Randall G. Holcombe Entrepreneurship and Economic Progress (London: Routledge, 2007), Chapter 5.

22 William J. Baumol, Entrepreneurship, Management, and the Structure of Payoffs (Cambridge, MA: MIT Press, 1993), p. 15.

23 This taxonomy is discussed in Randall G. Holcombe, “The Origins of Entrepreneurial Opportunities,” Review of Austrian Economics 16, no. 1 (2003), pp. 25–43; and Holcombe, Entrepreneurship and Economic Progress, Chapter 6.

24 For a fuller depiction, see Randall G. Holcombe, “Entrepreneurship and Economic Growth,” Quarterly Journal of Austrian Economics 1, no. 2 (Summer 1998), pp. 45–62.

25 Ludwig von Mises, Human Action, 3rd revised ed. (Chicago, IL: Henry Regnery Company, 1966), p. 248.

26 Mancur Olson, Jr., “Big Bills Left on the Sidewalk: Why Some Nations are Rich, Others Poor,” Journal of Economic Perspectives 10, no. 2 (Spring 1996), pp. 3–24, begins his article with this joke.

27 Robert L. Heilbroner, The Making of Economic Society (Englewood Cliffs, NJ: Prentice-Hall, 1962).

28 Oskar Lange and Fred M. Taylor, On the Economic Theory of Socialism (Minneapolis: University of Minnesota Press, 1938).

29 For an elaboration of this idea see Randall G. Holcombe, The Economic Foundations of Government (New York: New York University Press, 1994); and Randall G. Holcombe, “Why Does Government Produce National Defense,” Public Choice Public Choice 137, nos. 1/2 (October 2008), pp. 11–19.

30 Gary S. Becker, “A Theory of Competition Among Pressure Groups for Political Influence,” Quarterly Journal of Economics, 98 (1983), pp. 371–400; and Donald A. Wittman, “Why Democracies Produce Efficient Results,” Journal of Political Economy, 97 (1989), pp. 1395–424; and Donald A. Wittman, The Myth of Democratic Failure (Chicago, IL: University of Chicago Press, 1995).

31 The institutional features that lead to economic prosperity are described and quantified in James Gwartney and Robert Lawson, Economic Freedom of the World: 2007 Report (Vancouver, BC: Fraser Institute, 2007).

32 These are examples of rent-seeking activities, following Gordon Tullock, “The Welfare Cost of Tariffs, Monopolies, and Theft,” Western Economic Journal 5 (June 1967), pp. 224–32; and Anne O. Krueger, “The Political Economy of the Rent-Seeking Society,” American Economic Review 64 (June 1974), pp. 291–303. See also Krueger’s The Political Economy of Policy Reform in Developing Countries (Cambridge, MA: MIT Press, 1993).

33 William J. Baumol, “Entrepreneurship: Productive, Unproductive, and Destructive,” Journal of Political Economy 98, no. 5, Part 1 (October 1990), pp. 893–921; and Baumol, Entrepreneurship, Management, and the Structure of Payoffs.

34 Becker, "A Theory of Competition Among Pressure Groups for Political Influence"; Wittman, "Why Democracies Produce Efficient Results"; and Wittman, The Myth of Democratic Failure.

35 See R.D. McKelvey, "Intransitivities in Multi Dimensional Voting Models and Some Implications for Agenda Control," Journal of Economic Theory 12, no. 3 (June 1976), pp. 472–82. Also on political entrepreneurship see Randall G. Holcombe, "Political Entrepreneurship and the Democratic Allocation of Economic Resources," Review of Austrian Economics 15, nos. 2/3 (June 2002), pp. 143–59.

36 Gordon Tullock, "The Transitional Gains Trap," Bell Journal of Economics 6 (Autumn 1975), pp. 671–8.

37 The caveat comes from the possibility that restrictions imposed by government could serve to protect rights and encourage productive economic activity. This line of reasoning could be taken further, but would be a detour from the primary aim here, which is to examine the way that markets produce prosperity.

38 Ludwig von Mises, Human Action, 3rd rev. ed. (Chicago, IL: Regnery, 1966), pp. 248–9.

39 Mises, Human Action, p. 250.

40 Lange and Taylor, On the Economic Theory of Socialism.

41 Sanford Ikeda, Dynamics of the Mixed Economy: Toward a Theory of Interventionism (London: Routledge, 1997).

13 Producing prosperity

1 This is a reference to Adam Smith, The Wealth of Nations (New York: New American Library, 1937), p. 56, where Smith says "Avery poor man may be said in some sense to have a demand for a coach and six; he might like to have it; but his demand is not an effectual demand, as the commodity can never be brought to market in order to satisfy it." Smith's observation is interesting in light of the fact that most poor households at the beginning of the twenty-first century in the United States have at least one automobile.

2 This argument is made by Meir Kohn, "Value and Exchange," Cato Journal 24, no. 3 (Fall 2004), pp. 303–39. See also Leland B. Yeager, "Austrian Economics, Neoclassicism, and the Market Test," Journal of Economic Perspectives 11, no. 4 (Fall 1997), pp. 139–65, for a discussion of the differences between markets in an economy and the marketplace for ideas.

3 John Maynard Keynes, The General Theory of Employment, Interest, and Money (New York: Harcourt Brace and Company, 1936).

4 See Peter T. Bauer, Dissent on Development: Studies and Debates in Development Economics (Cambridge, MA: Harvard University Press, 1972) for a critique of this line of analysis and a support of market-based development policies.

5 See Robert E. Lucas, Jr., "On the Mechanics of Economic Development," Journal of Monetary Economics 22 (1988), pp. 3–42; and Paul M. Romer, "Increasing Returns and Long-run Growth," Journal of Political Economy 94 (1986), pp. 1002–37.

6 William J. Baumol, "Entrepreneurship: Productive, Unproductive, and Destructive," Journal of Political Economy 98, no. 5, Part 1 (October 1990), pp. 893–921.

7 See the insightful and pathbreaking article by Gordon Tullock Gordon, "The Welfare Cost of Tariffs, Monopolies, and Theft," Western Economic Journal 5 (June 1967), pp. 224–32.

8 This was the subject of Anne O. Krueger's article, "The Political Economy of the Rent-seeking Society," American Economic Review 64 (June 1974), pp. 291–303. She notes that in India at the time she waswriting, some of the best and brightest people there devoted their work toward finding profitable loopholes in the regulations, which added nothing to the value of the Indian economy.

9 See, for examples, Joel Mokyr, The Lever of Riches (Oxford: Oxford University Press, 1990); and David S. Landes, The Wealth and Poverty of Nations: Why Some Are So Rich and Some So Poor (New York: W.W. Norton, 1998).

10 See James Gwartney and Robert Lawson, Economic Freedom of the World: 2007 Report (Vancouver, BC: Fraser Institute, 2007) for empirical measurements of those economic institutions.

11 While all of the ideas in this volume have antecedents, this idea in particular is associated with Friedrich Hayek: see his "The Use of Knowledge in Society," American Economic Review 35 (1945), pp. 519–30.

12 Meir Kohn, "Value and Exchange," Cato Journal 24, no. 3 (Fall 2004), pp. 303–39, asks that very question.

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